Unified Theory Based on Time-Wave Dynamics (The P-model)

Part III-B: Quantum Verification Applications

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This paper, within the quantum volume of the P-model, treats verification results and differential predictions based on public primary data. Theoretical definitions, the Pψ connection, the placement of electromagnetism, the adopted scope of the Born rule, the effective description of measurement, and the Bell-selection entry point are taken to be canonical in Part III-A.

Abstract

This paper (Part III-B), while retaining the minimal assumptions fixed in Part III-A, reanalyzes Bell tests, interference, quantum clocks, nuclear physics, condensed-matter/thermal systems, the weak interaction, and BBN on a single input/output interface and fixes both the cross-domain summary of Pass / Watch / Reject and the associated differential predictions. The aim is not to declare a replacement of quantum mechanics, but to present a reproducible verification package that specifies which assumption can fail against which primary dataset.

1 Introduction

Part III-B is the verification-side document that was split out of Part III-A in Ver.1.1 in order to separate the chapter that states the theory from the chapter that decides acceptance or rejection by verification. After confirming the theoretical premises and scope of applicability in Part III-A, the reader of this paper can focus on the data, results, differential predictions, and discussion from Section 3 onward.

To preserve reference stability during the transition period, section numbers remain consecutive across Part III-A and Part III-B. Accordingly, this paper begins with Section 3, continuing directly from Section 2 of Part III-A.

2 Data and Methods

This paper starts from primary sources (public primary data / official distributions) and fixes the stored data, processing conditions (window rules, outlier rules, and so on), and output filenames. Primary sources (URLs / access dates) are collected in the appendix "Data Sources (Primary Sources)." This prioritizes an explicitly reproducible state in which the inputs and processing conditions (windows, thresholds, outlier rules, pairing, and so on) are stated clearly, so that the same inputs reproduce the same outputs without arbitrary target selection or updates.

Metrics are aggregated into the verification summary (scoreboard), with statistical uncertainty (covariance) separated from systematics (procedure dependence). For Bell tests in particular, sensitivity to selection knobs (coincidence window / trial-based definition / event-ready offset / etc.) is fixed as a systematic, the statistical side is evaluated with bootstrap and related procedures, and the result is reduced to a falsification-condition package. The key point is not to move the frozen quantity from Part I (β) afterward to enforce consistency, but to leave as output where the fixed mapping actually reaches a decision.

2.1 Common Rejection Procedure

Each verification in this paper fixes falsifiability in the following common format.

- **Input**: Data used (primary / already estimated), dependent assumptions, and source (with access date).
- **Frozen**: Frozen parameters and the grounds on which they were frozen (which primary constraint fixed them, and when).
- **Statistic**: Evaluation statistic (for example RMS, χ^2, logL, ΔAIC, slope). ΔAIC is defined as

  $$\Delta\mathrm{AIC} \equiv AIC_{\mathrm{baseline}} - AIC_{\text{P-model}}$$

  where positive values indicate support for the P-model.
- **Reject**: Rejection threshold (for example, exceeding 3σ, $\Delta\mathrm{AIC} < -10$ where the baseline is strongly preferred, or exceedance including systematic error).

From this point on, every test is written in this format so that it is reproducibly rejectable as a protocol rather than as a discussion.

The frozen list (frozen values + source sha256) is not enumerated in the main text; the authoritative source is the manifest / JSON ledger in Part IV (Verification Materials).

3 Results

3.1 Verification Summary (Scoreboard)

The verification summary (scoreboard) is a consolidated summary of the results obtained under fixed processing conditions, and it is used in the main text as the entry point to each section. Section 4 is the results chapter that fixes the quantum-side (premise-verification) items one by one and tracks both the procedure-dependent points and the falsification conditions on a single input/output interface.

Objective: Provide an overview of the aggregate results of the verifications frozen in this paper (input, processing, metric, and rejection condition), and serve as the entry point to Sections 4.2-4.11.

Input: The verification summary (scoreboard) and the scoreboard figure that gives an overview of the state of all rows.

Processing: Fix, on a single input/output interface, both the statistics needed for acceptance/rejection in each topic (z/σ/Δ and so on) and the procedural systematics (sweeps), and aggregate them into the verification summary (scoreboard).

Metric: The "difference / metric" column of the verification summary (statistic + uncertainty, or procedure sensitivity) is treated as authoritative.

Results:

The judgment labels are handled as Pass, Watch, and Reject.

Status	Count	Summary
Pass	6	Consistency under the frozen conditions is confirmed for the main agreement items (quantum clocks / precision QED / part of matter-wave interference).
Watch	12	Procedure dependence or lack of precision is rate-limiting (Bell-selection systematics, suspended verification boundaries in condensed-matter / thermal tests, watch handling for planned rows, and so on).
Reject	3	Items not consistent under the current minimal assumptions are fixed on the rejection side (including "entry-point Pass but overall Reject" and "Reject in practice because the difference is not detectable").
Reference	0	There are no rows treated as reference-only in the present snapshot.

Note: The counts above are governed by the quantum-scoreboard JSON aggregation of $status = ok/mixed/ng$. The section map in 4.1.1 is a correspondence table that includes operational rows, planned rows, and compound-judgment rows (for example "Pass (Watch)" and "Theory Pass / Numerical Reject"), so its population does not match the count table exactly.

3.1.1 Item Map (Section Map)

Item	Judgment	Section
Bell (public primary data; selection sensitivity)	Reject (entry-point Pass; selection-dependent)	4.2
Gravity × quantum interference: COW	Reject (difference not detectable)	4.3
Gravity × quantum interference: atom interferometer	Pass	4.3
Quantum clocks (redshift)	Pass	4.3
Matter-wave interference	Pass	4.4
Matter-wave interference (precision)	Pass	4.4
Gravity-induced decoherence	Watch	4.5
Quantum interference of light	Watch	4.6
Vacuum / precision QED	Pass	4.7
Nuclear physics (effective potential: np scattering)	Reject	4.8.1-4.8.3
Nuclear physics (wave-interference mapping: deuteron / He-4)	Pass	4.8.4-4.8.6
Atomic / molecular systems (reference values: H I / He I / H2 / HD / D2 / D_0 / lines)	Watch	4.9.1
Condensed matter (Si lattice constant / Cp / ρ / α)	Watch	4.9.2-4.9.6
Statistical mechanics / thermodynamics (transport / blackbody)	Watch	4.10.1-4.10.3
BBN (Big-Bang nucleosynthesis: He-4 derivation)	Pass	5.5.1
Nuclear-force wave interference: differential predictions and precision requirements	Operational	5.4
Weak interaction (Route A: electroweak transplant)	Pass (Watch)	5.5
Weak interaction (Route B: P-native replacement V-A theory)	Theory Pass / Numerical Reject	5.5, 5.5.3

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Item	Judgment	Section
Bell (planned): time tags	Pass	4.2, 5.1
Bell (planned): cov+sys	Watch	4.2, 5.1
Quantum (planned)	Watch	4.11, 5.1

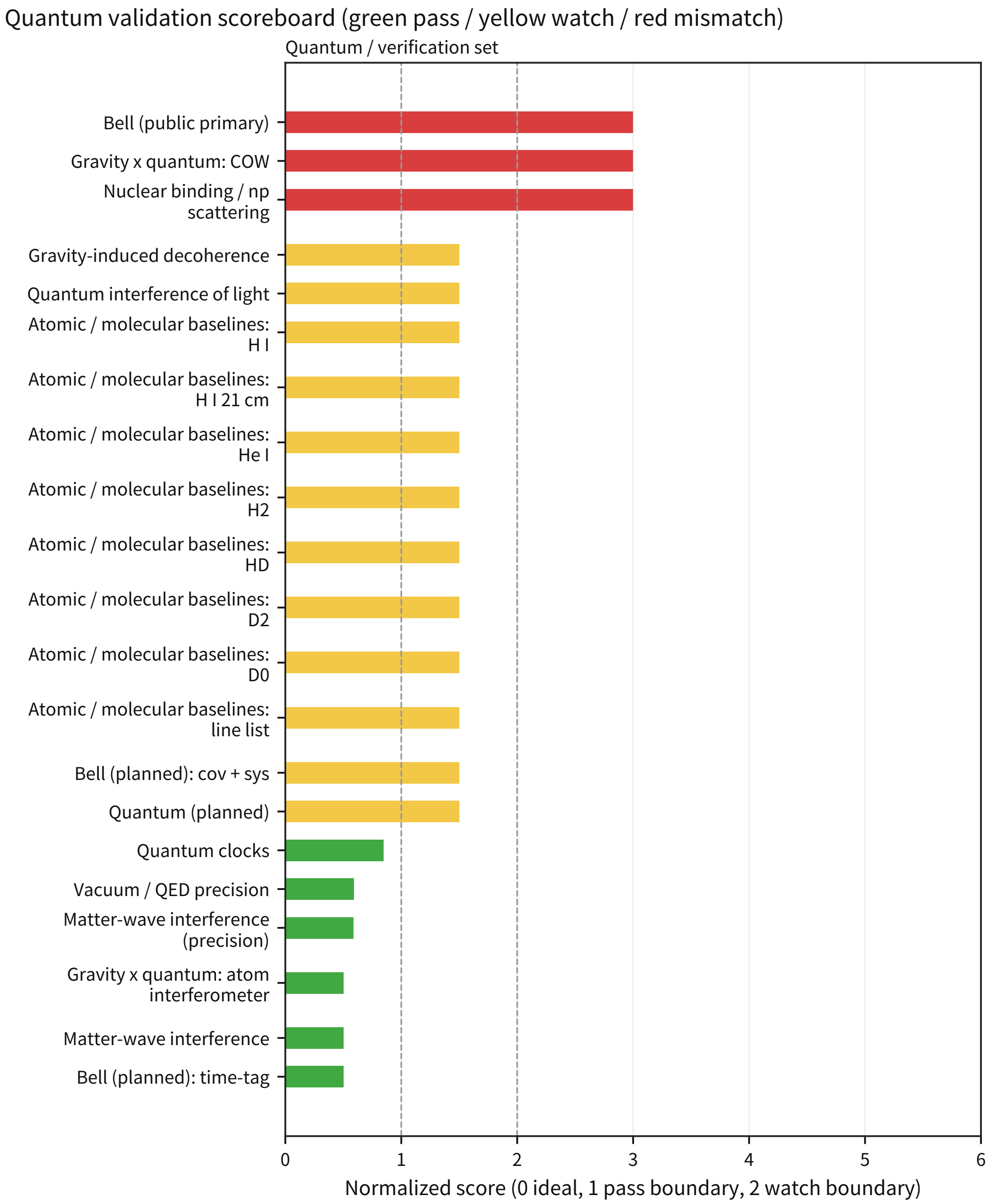

Figure 1: Gives a color-coded overview of all rows in the verification summary under "Pass / Watch / Reject / Reference."

Supplement: the colors here are not the conclusion itself; they are intended to visualize which systematics (sys) and precision gaps need to be tightened next. For the Route B weak-interaction line, `Theory Pass /Numerical Reject` is shown explicitly so that the existence of the theoretical mechanism and the failure of the numerical audit are treated separately.

Note: The scoreboard is simplified, so the types of metrics differ across categories. Acceptance or rejection is governed by the "Metric / Reject" definitions of each section. Displayed bar lengths are normalized scores in the judgment band; the raw measured values are tracked through the metric values of each row (corresponding to score_raw). The implementation key for score_raw is fixed in the Part IV verification ledger (the quantum-scoreboard JSON), and is tracked through `rows[].score` (display-normalized score) and `rows[].metric` (raw-metric summary).

The point reached in this paper is not a rederivation of quantum phenomena. Rather, using public primary data, it quantifies the entry points at which the observation procedure (selection) can move the statistics, and puts them in a state that can be connected to falsification. The following sections summarize the quantum tests item by item.

3.2 Bell Tests (Time Tags / Trial-Based Analysis)

Objective: Quantify, against public primary data, the sensitivity to selection procedures (coincidence window / trial-based definition / offset and so on), and fix it in a form that connects directly to the falsification conditions (under which procedure the hypothesis is rejected).

Input: NIST belltestdata (photon time tags), Delft (event-ready; trial table), Weihs 1998 (photon time tags), and Christensen et al. 2013 (Kwiat group; photon time tags; CH). The list of primary sources is collected in the appendix "Data Sources (Primary Sources)." [2][3][4][5][6]

Processing:

- NIST: GPS PPS alignment + click delay (sync reference) + coincidence-window sweep (greedy pairing). In parallel, perform trial-based aggregation (sync × slot) using the build (hdf5) product.
- Delft: Sweep the event-ready window start (offset), and evaluate the number of valid trials and CHSH.
- Weihs: Evaluate CHSH |S| sensitivity through a coincidence-window sweep.

Metric: For NIST and Weihs, freeze the setting dependence of click delay as **delay amount (Δmedian; ns) + residual (z)** (with KS shown in parallel as a proxy metric). In addition, evaluate CH J_{prob} for NIST and CHSH S (fixed variant) for Weihs / Delft.

Results:

Dataset	Selection knob	Proxy timing-bias metric (A/B)	Variation range of statistic (sweep)	Reference value
NIST	window / trial definition	$\Delta median$=158 ns (z≈34.8) / 217 ns (z≈43.9); KS=0.0287 / 0.0411	CH J_{prob}: $-1.41 \times 10^{-3} \to 2.10 \times 10^{-3}$; trial best: 6.52×10^{-6}	—
Weihs 1998	window	$\Delta median$=0.083 ns (z≈5.54) / 0.095 ns (z≈5.53)	CHSH $\lvert S\rvert$: 2.48→2.90	—
Delft 2015	event-ready offset	—	CHSH S: 1.71→2.50	2.422±0.204
Delft 2016 (Sci Rep 30289; combined)	event-ready offset	—	CHSH S: 1.10→2.54	2.346±0.184

In addition, the selection systematics are decomposed into a 15-item "catalog of error sources," and the ratio to the statistical error σ_{stat} (bootstrap and related estimates), $|\Delta\mathrm{stat}|/\sigma_{\mathrm{stat}}$, is fixed as the **operational systematic-error budget**. Here, Δstat is the range by which a statistic (S, J_{prob}, Δmedian, and so on) can move due to the choice of procedure (window / offset / correction / etc.), and $|\Delta\mathrm{stat}|/\sigma_{\mathrm{stat}} > 1$ means that procedure dependence (sys) is larger than the statistical uncertainty.

Error source (15 items)	median $\lvert\Delta\mathrm{stat}\rvert/\sigma_{\mathrm{stat}}$	max $\lvert\Delta\mathrm{stat}\rvert/\sigma_{\mathrm{stat}}$
offset	3.88	104.09
coincidence window (width / drift)	3.13	385.34
detector efficiency drift	0.35	0.62

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Error source (15 items)	median $\lvert\Delta\text{stat}\rvert/\sigma_{\text{stat}}$	max $\lvert\Delta\text{stat}\rvert/\sigma_{\text{stat}}$
polarization correction	0.30	1.35
threshold	0.25	3.11
transmission loss	0.25	0.25
setting switch timing	0.24	3.00
dark count	0.20	1.00
accidental correction	0.20	77.91
event-ready definition	0.20	0.20
dead time	0.19	0.20
environmental variation	0.18	0.86
trial definition	0.15	77.54
clock synchronization	0.15	0.18
electronic noise	0.079	0.485

(How to read the table) The two dominant items above (offset / coincidence window) are not simply instrument errors; they are selection knobs that decide which pairs are regarded as the same event. If the sweep range is taken widely, they can produce fluctuations far exceeding the statistical uncertainty σ_{stat}. Accordingly, if one optimizes the window/offset after seeing the result (S, J_{prob}), there is a risk that the very "violation" with $S > 2$ appears to be procedure-induced (the freedom of a posteriori optimization).

By contrast, once the freeze (natural window) is fixed by a prior rule and the effect is estimated as local drift around that point, the contribution falls to the order of $O(10^{-3})$. For example, in Weihs 1998, near the natural window (around window = 25 ns), $d|S|/d(\text{window}) \approx -5.18 \times 10^{-3}$ ns^{-1}. Using drift_half≈0.251 ns for the recommended window gives a representative estimate of only $\Delta|S| \approx 1.3 \times 10^{-3}$ (whereas the full-sweep value $\Delta|S| \approx 0.42$ indicates the size of the selection freedom).

For the CHSH series, the contribution from the "instrument / correction" side after excluding the selection items such as window / offset (dead time, detector-efficiency drift, dark count, threshold, clock sync, ...) remains at about $\Delta|S| \approx 0.05 - 0.14$ in L2 aggregation even when operational proxy metrics are included (smaller than the typical margin $|S| - 2 \approx 0.3 - 0.6$). The L1 (linear sum) upper bound is more conservative because it ignores correlations, and can reach about 0.4 for Delft (the worst case in which the contributions enter simultaneously with the same sign).

Dataset	Δabs(S) (instrument / correction; L2)	Δabs(S) (instrument / correction; L1)	Notes
Weihs 1998	0.049	0.081	Many metrics are derived directly from the time tags, so dependence on proxy metrics is relatively small.
Delft 2015	0.143	0.421	Some items have no direct sweep in the event-ready protocol, so conservative proxy metrics are included.
Delft 2016	0.133	0.398	Same as above (combined run).

In addition, for the same data, the effect of **implementation differences in pairing / estimators** (differences in convention) on the statistic is fixed as $\Delta_{\text{impl}}/\sigma_{\text{boot}}$ at the frozen window (or frozen point), with the operational gate $\Delta_{\mathrm{impl}}/\sigma_{\mathrm{boot}} < 1$. The purpose of this cross-check is not optimization, but to close the point as a systematic by confirming that implementation differences do not change the conclusion.

Dataset	Implementation-difference cross-check (frozen point)	$\Delta_{\mathrm{impl}}/\sigma_{\mathrm{boot}}$
Weihs 1998	greedy pairing vs mutual nearest-neighbor	0.086
NIST 2015	greedy pairing vs mutual nearest-neighbor	0.021
Kwiat/Christensen 2013	trial-based reference-point consistency (internal consistency)	0.000

As a concrete CHSH S example, in Weihs 1998 the coincidence window alone can move $\Delta|S| \approx 0.423$ ($|\Delta S|/\sigma_{\mathrm{stat}} \approx 16.7$). In Delft, the event-ready offset sweep shows $\Delta S \approx 0.788$ in 2015 ($|\Delta S|/\sigma_{\mathrm{stat}} \approx 3.88$) and $\Delta S \approx 1.44$ in 2016 ($|\Delta S|/\sigma_{\mathrm{stat}} \approx 7.84$). These are comparable to or larger than the margin $S - 2$ (for example, 0.422 when 2.422 ± 0.204), quantitatively showing that the treatment of the selection systematic becomes dominant when discussing whether S is "violated."

The same examples, summarized in terms of $S_{\mathrm{frozen}} \pm \sigma_{\mathrm{stat}}$ at the frozen point (natural window) and the min-max sweep range, are as follows.

Dataset	Selection freeze	$S_{\mathrm{frozen}} \pm \sigma_{\mathrm{stat}}$	S sweep (min→max)	$S_{\mathrm{frozen}} - 2$	min-2	ΔS	$\lvert\Delta S\rvert/\sigma_{\mathrm{stat}}$
Weihs 1998	window width (25 ns)	2.553±0.025	2.477→2.900	0.553	0.477	0.423	16.7
Delft 15	ER offset (-30 ps)	2.431±0.203	1.711→2.499	0.431	-0.289	0.788	3.88
Delft 16	ER offset (-30 ps)	2.321±0.184	1.100→2.542	0.321	-0.900	1.44	7.84

Likewise for CH (J_{prob}), NIST crosses from negative to positive under the window sweep alone, and Kwiat/Christensen 2013 also shows window-dependent variation in J_{prob} (though it does not cross 0 within the present sweep range).

Dataset	Selection freeze	$J_{\mathrm{frozen}} \pm \sigma_{\mathrm{stat}}$	J_{sweep} (min→max)	max(J)	ΔJ	$\lvert\Delta J\rvert/\sigma_{\mathrm{stat}}$
NIST	window width (1000 ns)	$(-6.99 \pm 0.091) \times 10^{-4}$	$-1.41 \times 10^{-3} \rightarrow 2.10 \times 10^{-3}$	2.10×10^{-3}	3.51×10^{-3}	385
Kwiat 2013	window width (2812.5 ns)	$(-8.12 \pm 3.94) \times 10^{-5}$	$-1.26 \times 10^{-4} \rightarrow -2.88 \times 10^{-6}$	-2.88×10^{-6}	1.23×10^{-4}	3.13

As an additional audit, for the Kwiat delay-signature watch the three sweeps window / offset / accidental-like were reevaluated under the same procedure. At the reference window, $\max|z| = 0.781$, and even over the full sweep the maximum is only $\max|z| = 2.73$ (window raw), so it does not exceed $z = 3$. Accordingly, the Kwiat delay signature is kept outside the main gate for the fast-switch class and retained as a non-primary watch indicator for the selection systematics.

Thus, even when only the window sweep is considered, Weihs keeps $\min(|S|) > 2$ on this window grid, whereas Delft crosses 2 in the offset sweep and NIST crosses 0 in J_{prob}. This paper does not assert whether a "violation" exists or not. Instead, it freezes the natural window (freeze) together with the systematic width Δ_{sys}, and connects the result to the later rejection conditions (5.1).

Supplement: instrument-side items such as dead time and detector-efficiency drift typically remain at about ΔS = 0.001-0.07 for CHSH (using proxy metrics), and often do not by themselves fill the margin in $S - 2$. By contrast, the selection definitions themselves, such as the coincidence window (Weihs) and the event-ready offset (Delft), can move ΔS by about 0.4-1.4 as in the examples above. Therefore, any discussion of $S > 2$ is dominated by which procedure was used to freeze window/offset (the natural window), and whether the sweep was treated not as optimization but as an **evaluation of the systematic width Δ_{sys}**.

Relation to Existing Criticism: Adenier (2007) pointed out that fair sampling, which is often left implicit in EPR/Bell experiments (the assumption that detection / acceptance is independent of the measurement setting), is not automatically supported by the experimental data and that the acceptance rule can distort correlation estimates. The event-by-event simulation family of De Raedt and collaborators shows that, even with local rules, combining time tags with coincidence decisions can yield CHSH-like "quantum violations" purely through dependence on window width or delay processing (the coincidence loophole). Larsson (2014) organized the loopholes such as detection, locality, memory, and coincidence time, and provided a framework for assessing which ones an experiment closes and which can still remain. What these lines of criticism share is the emphasis that the selection / coincidence-counting procedure can enter the statistics. [8][9][10][11]

The purpose of this paper is not to claim conceptually that loopholes exist. Rather, it is to fix, using public primary data, how much selection knobs (window / offset / threshold and so on) can move the statistics (S, J_{prob}, Δmedian, and so on) in a form that supports cross-dataset comparison on a single input/output interface. Instead of constructing an event-by-event generative model or theoretical analysis, the method here is to reanalyze the public time-tag / trial-based data directly and present covariance (bootstrap) separately from systematics (sweep width). Accordingly, the conclusion is not "whether Bell violation is artificial," but to freeze the degree of procedure dependence numerically so that it can be connected to the later Reject threshold.

Therefore, this paper does not newly claim the existence of loopholes; its aim is to fix the quantitative sensitivity to selection using public primary data and connect it to the falsification condition (under which procedure the hypothesis is rejected).

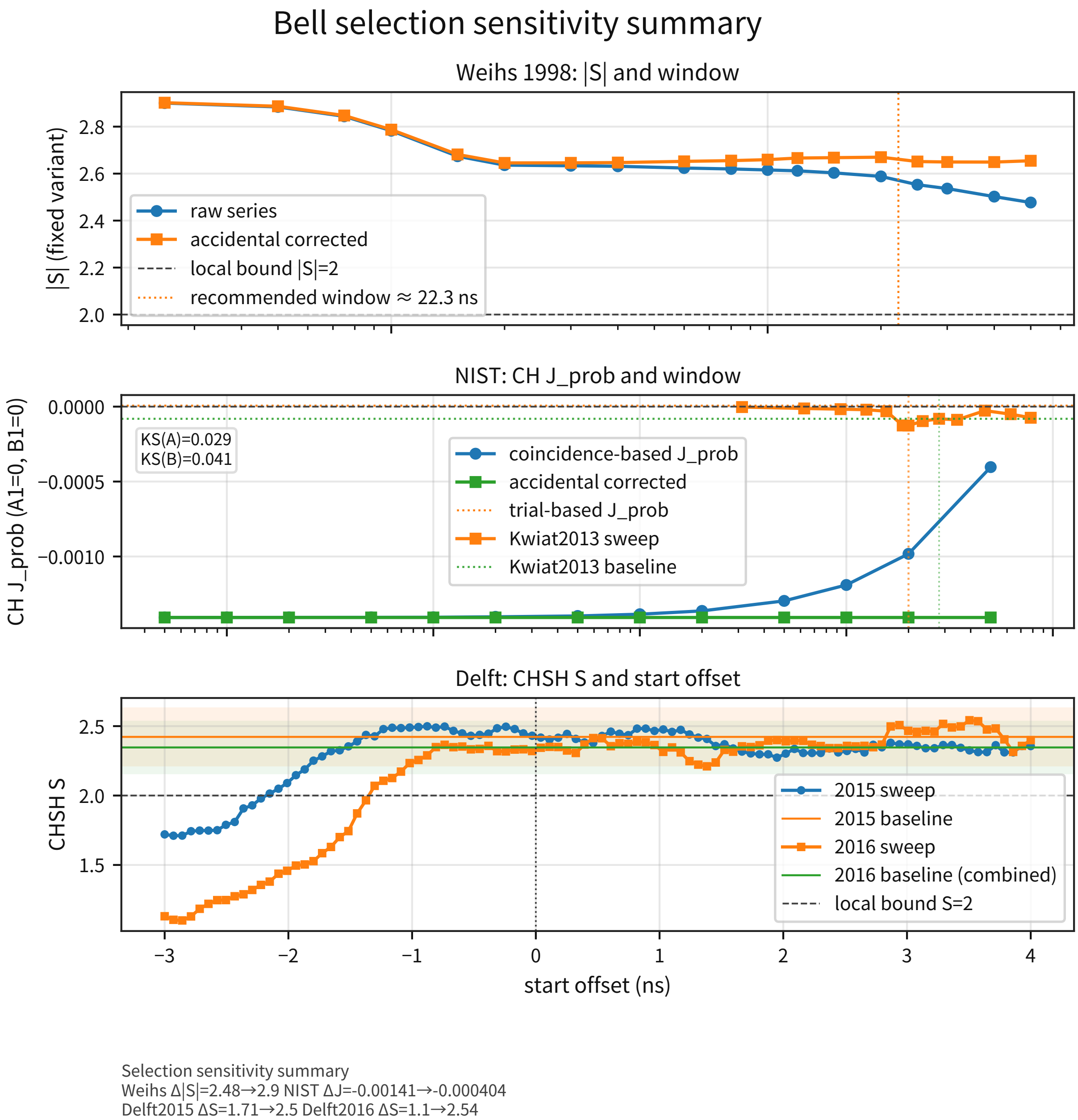

Figure 2: Cross-compares the variation range of the statistics against selection knobs (window / offset / threshold and so on).

Supplement: read this as the main figure giving an overview of the "procedure dependence (sys)" in this section.

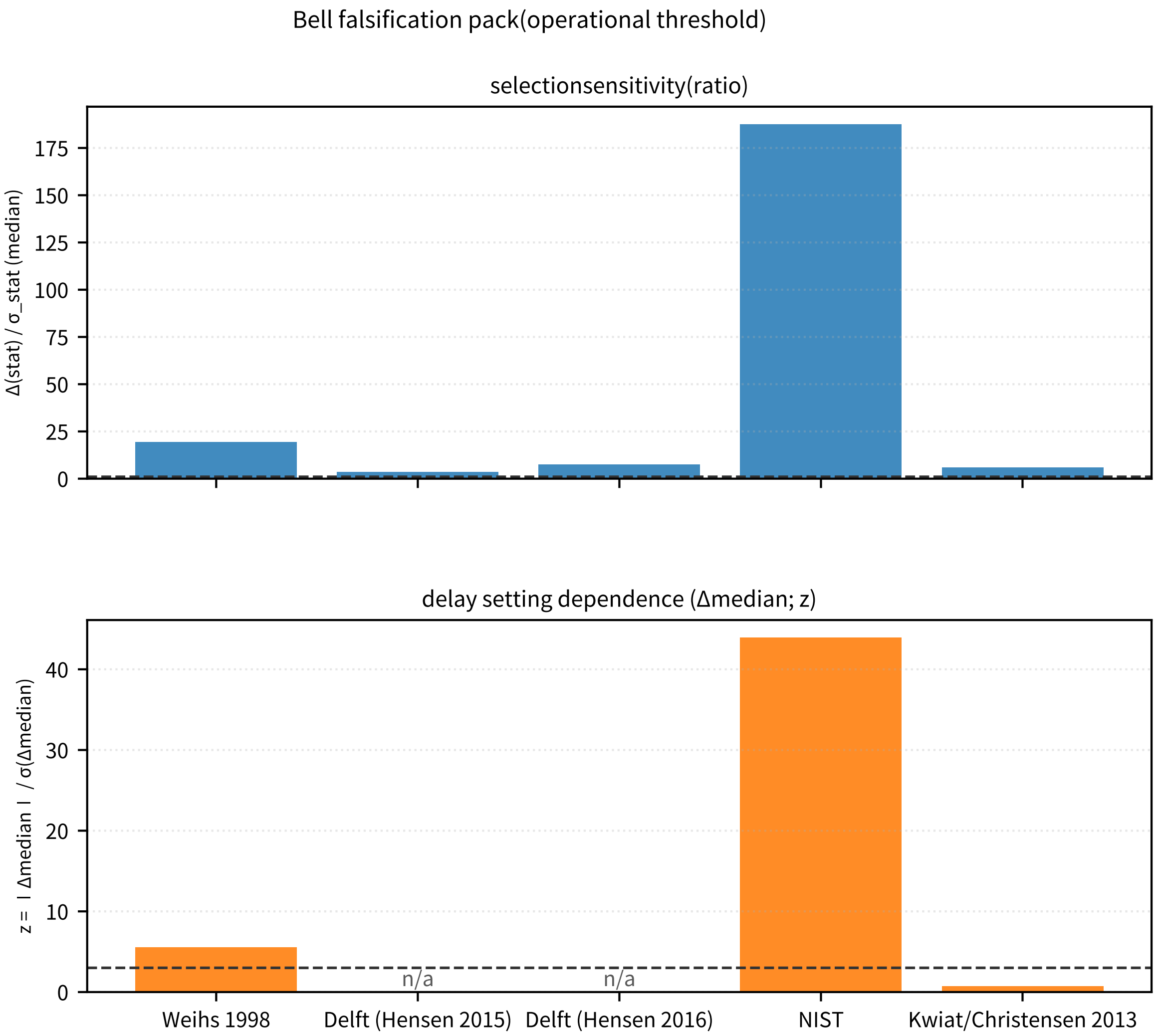

Figure 3: Places the selection sensitivity ($\Delta(\mathrm{stat})/\sigma_{\mathrm{stat}}$) and the setting dependence of delay (Δmedian; z) on the same axis across datasets.

Supplement: in this paper, ratio $=1$ and z $=3$ are fixed as operational thresholds, connected to the frozen point of the procedure and the reject rule, so that one can immediately judge whether the selection entry point is sufficiently large and whether a delay signature appears consistently under fast switching.

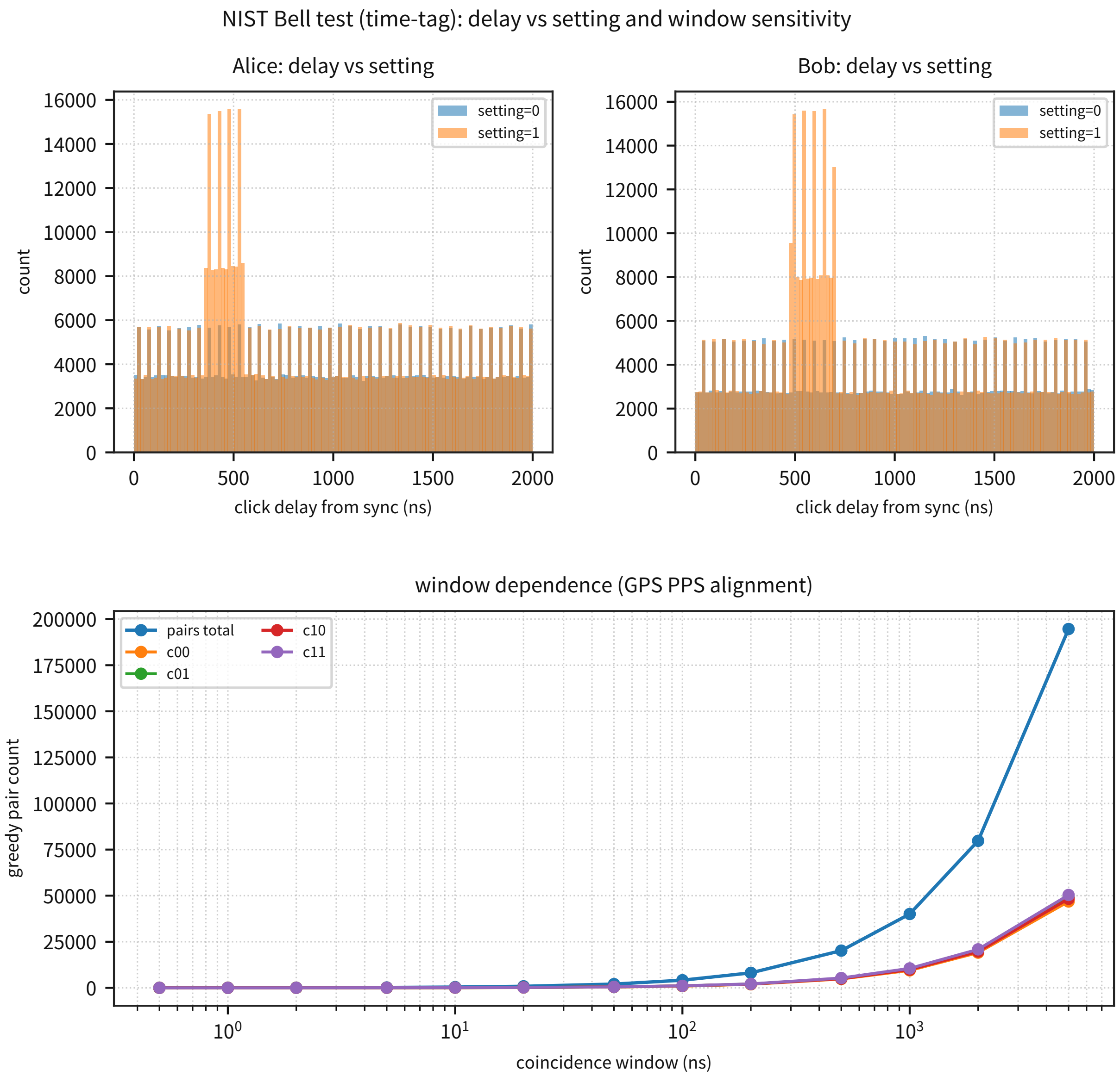

Figure 4: Visualizes the click-delay distribution (setting dependence) in the NIST time tags.

Supplement: this is a diagnostic figure for reading Δmedian (ns) and the residual (z). The Bob→Alice offset and the KS statistics (A/B), which had been embedded below the figure, are now tracked in fixed outputs external to the main text (metrics JSON).

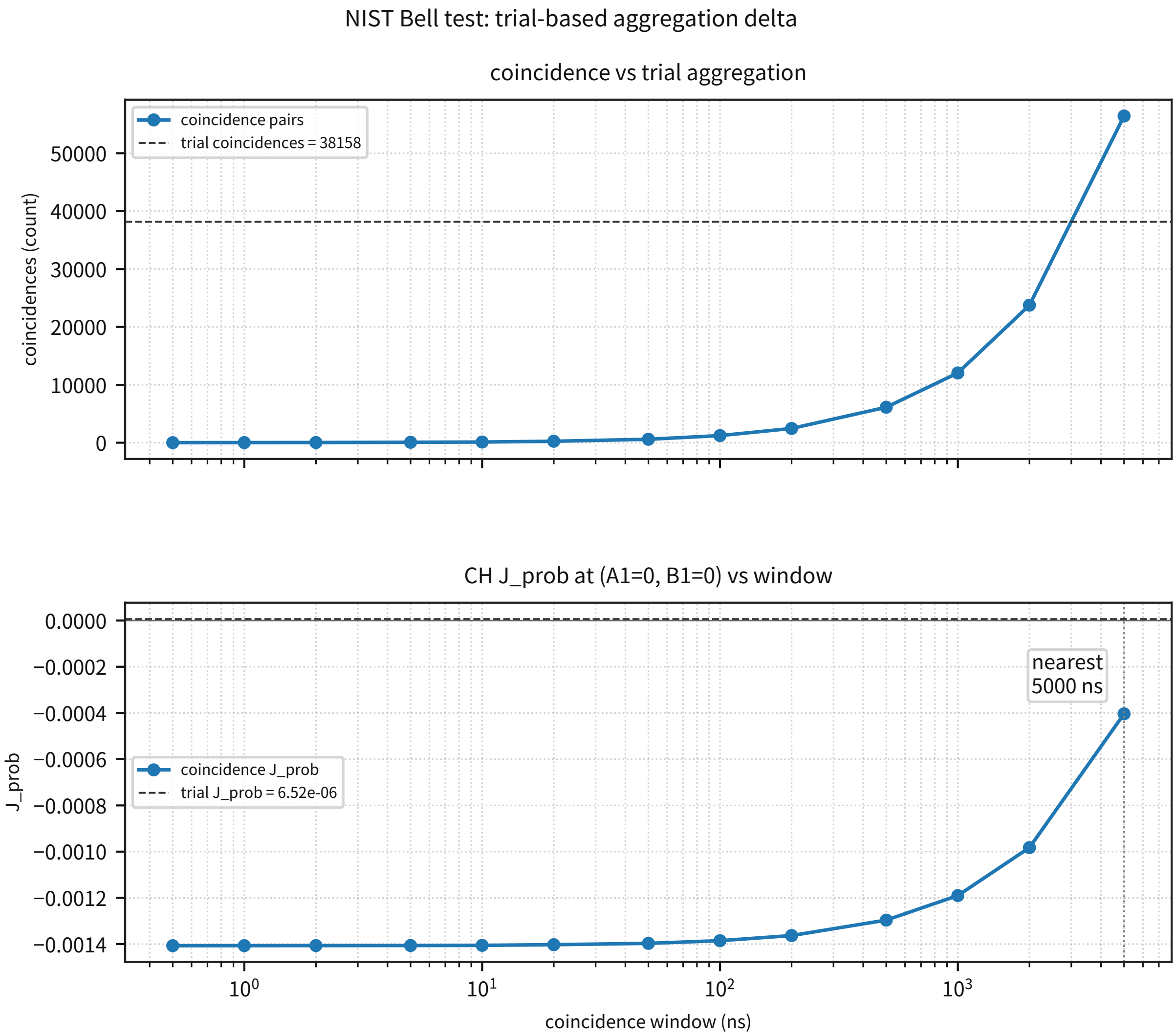

Figure 5: Compares the difference between trial-based and coincidence-based aggregation for NIST on the same data.

Supplement: this is the figure that checks the entry point at which the trial definition can change the sample population. The number of syncs used, the setting map, and the best J_{prob}, which had been embedded below the figure, are tracked in fixed outputs (metrics JSON).

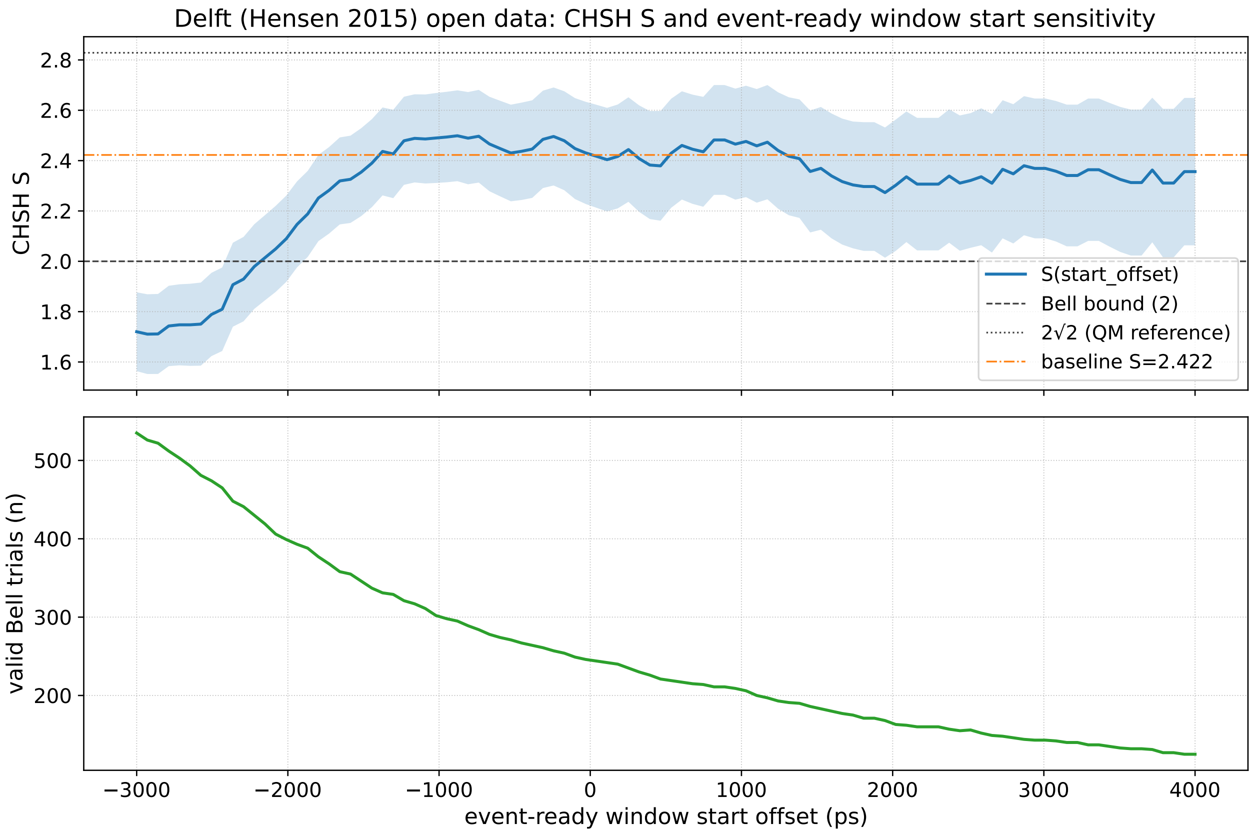

Figure 6: Shows the sensitivity of CHSH S to the offset sweep in Delft 2015 (event-ready).

Supplement: it is used not to overturn the event-ready protocol, but to fix the systematic width of the procedure dependence.

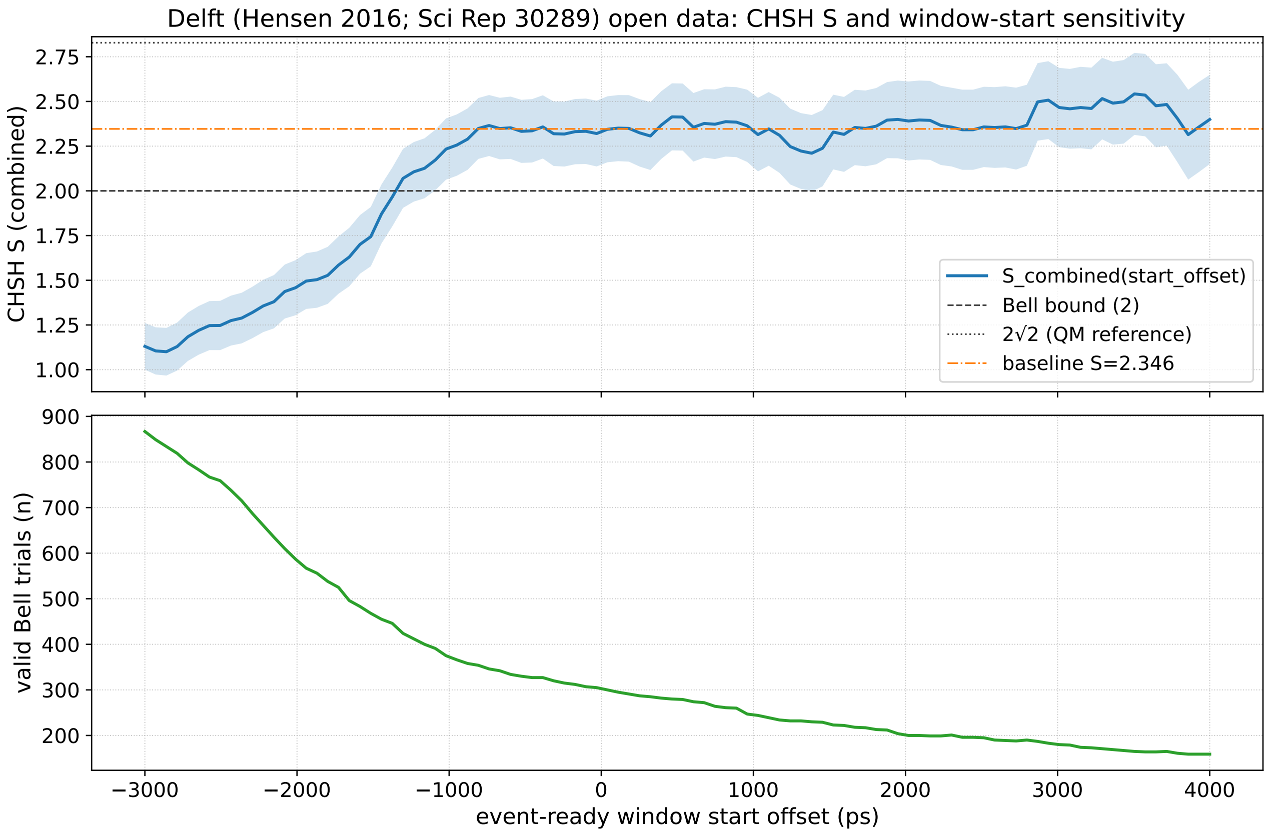

Figure 7: Shows S against the offset sweep in Delft 2016 (Sci Rep 30289).

Supplement: compare the robustness (systematic width) across multiple runs and combined conditions.

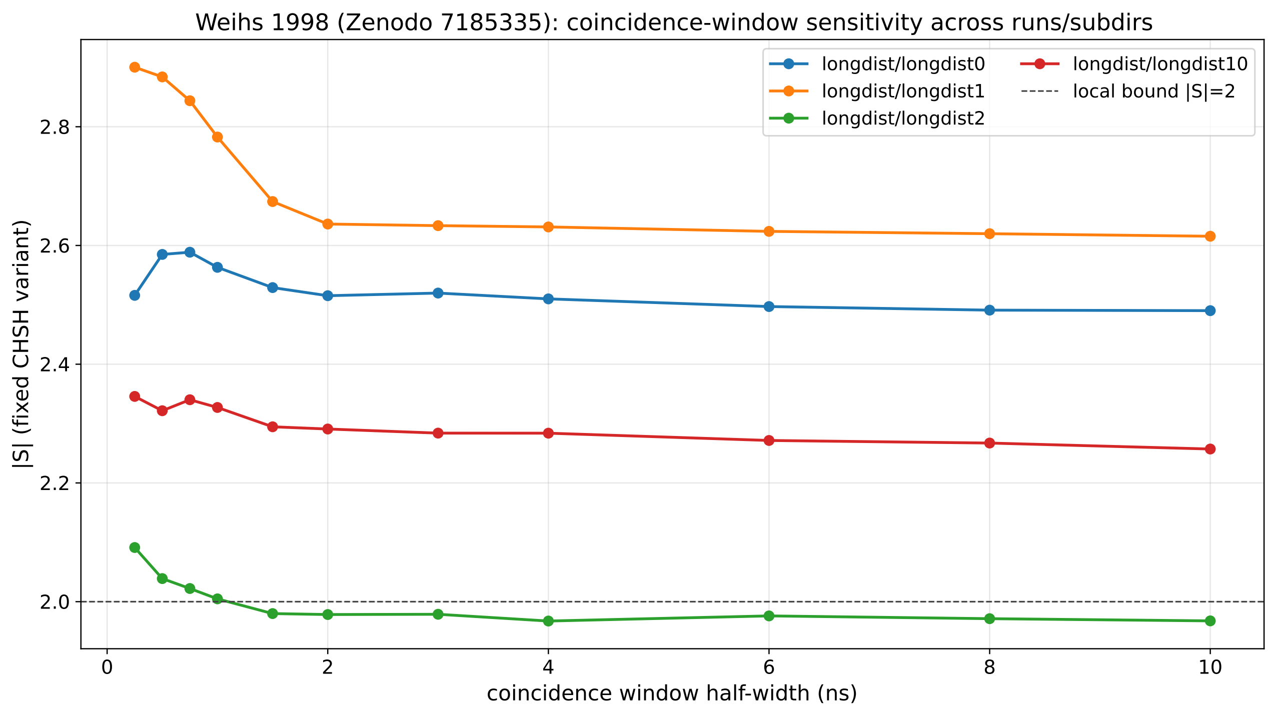

Figure 8: Cross-compares the coincidence-window sweep of Weihs 1998 across multiple conditions.

Supplement: read the window-dependent systematic width (ΔS) separately from the statistical error.

Note: (Positioning) What is fixed here is the location at which selection enters and its sensitivity; the paper does not claim a rederivation of all predictions of quantum mechanics. This is an entry point to falsification.

Vienna 2015 (Giustina et al. 2015) is excluded from the analysis here because its raw time tags are not public. If they are released, the same input/output interface can incorporate them later; at present the cross-dataset integration over the five datasets above is already sufficient.[7]

(Event-ready / natural window / a posteriori optimization) The event-ready protocol of Delft (Hensen 2015) is a protocol designed to close the detection and locality loopholes simultaneously, and this paper does not claim to overturn it. The offset/window sweeps are performed not to lower S arbitrarily (a posteriori optimization), but to visualize as a systematic **how much the acceptance rule (selection) can move the statistical population**. At the same time, to avoid looking like a posteriori optimization, the "natural window" is **recommended objectively from the data**. For NIST, the recommended value is the time-tag half-window that matches the trial-based coincidence total (checked against an independent pipeline). For Weihs / Kwiat, a constant background is estimated in the off-peak region (far bins) of the Δt histogram, and the recommended value is the minimum half-window at which "signal = peak integral - background $\times$ window width" saturates, based on the statistical definition accidental $\approx$ rate_A$\times$rate_B$\times$window (with drift taken into account). The movement of the statistics under expansion / contraction away from the recommended window is counted as systematic error through the sweep width. The recommended window (natural window), its reference point (freeze), the statistical covariance (bootstrap; N$\geq$1000), and the sweep ledger including accidental subtraction are fixed outputs for each dataset, and the reject rule (3σ) for judging deviation from the reference point is frozen at the same time.

Operational Freeze (Frozen / Reject / Next): After freezing, for each dataset, the natural window (recommended window $\rightarrow$ grid snap), the reference point (freeze), the sweep width $\Delta_{\rm sys}$, the statistical error $\sigma_{\rm stat}$ (bootstrap), and the delay signature (Δmedian; z), the selection-origin hypothesis is rejected under the operational condition in 5.1 when either (i) $|\Delta\text{stat}|/\sigma_{\rm stat} < 1$ (procedure dependence no larger than statistics) or (ii) the delay signature satisfies $z < 3$ and, under the same procedure, $|S| > 2$ (or $J_{\rm prob} > 0$) is reproduced at 3σ. The next step is to maintain the same input/output interface across the existing five datasets within the range not blocked by data release, and to increase the robustness of the rejection decision by updating the comparison between $\sigma_{\rm total}$ and $\Delta_{\rm sys}$ (selection sensitivity), including knobs such as threshold.

3.3 Gravity × Quantum Interference (COW / Atom Interferometer / Quantum Clocks)

Objective: Quantify whether interference and quantum clocks in the weak terrestrial field can detect, at experimental precision, the P-model-specific mapping difference ($\exp(-x)$ versus $\sqrt{1-2x}$), and fix both the current verdict "difference not detectable at present precision" and the future verification conditions.

Input: Representative conditions and published precision values from the primary literature for COW (neutron interference), atom-interferometer gravimetry, and optical-lattice-clock chronometric leveling.

Processing: Fix the representative conditions (COW: H, v_0, atom interferometer: T, clock: Δh), and compute the phase / redshift / frequency ratio. The difference is evaluated as the difference between the static-clock mapping $\exp(-x)$ and the reference mapping $\sqrt{1-2x}$, and converted into the 1σ precision required for 3σ detection.

Equation: With the definition of this paper $\phi \equiv -c^2 \ln(P/P_\infty)$ and $x \equiv GM/(c^2 r)$ for a point mass M, the static-clock mapping is

$$\ln\left(\frac{P}{P_\infty}\right) = x, \qquad \left(\frac{d\tau}{dt}\right)_{\mathrm{P}} = \frac{P_\infty}{P} = e^{-x}, \qquad \left(\frac{d\tau}{dt}\right)_{\mathrm{GR}} = \sqrt{1-2x}.$$

The frequency ratio (redshift) between two radii $r_{\mathrm{low}} < r_{\mathrm{high}}$ is

$$1 + z = \frac{(d\tau/dt)_{\mathrm{high}}}{(d\tau/dt)_{\mathrm{low}}}$$

defined by this expression, and the difference in z between the P-model and GR, $\delta z = z_P - z_{\mathrm{GR}}$, is compared. In the weak field ($x \ll 1$ and $|z| \ll 1$), the difference between exp and $\sqrt{\cdot}$ appears at $O(x^2)$, giving the approximation

$$\frac{\delta z}{z_{\mathrm{GR}}} \approx -(x_{\mathrm{low}} + x_{\mathrm{high}}) \tag{1}$$

where the sign indicates that the P-model gives a slightly smaller z than GR.

Metric: The differences $\delta\phi$ / $\delta(\Delta f/f)$ / δz and the 1σ precision required for 3σ detection (statistical + systematic).

Results:

Experiment	Representative condition	Main value	Primary source
COW (neutron interference)	H=3 cm, v0=2000 m/s	phase amplitude ≈ 11.16 cycles	[30][31]
Atom-interferometer gravimeter	T=0.4 s	gravity-term phase ≈ 2.31×10^7 rad	[32]
Optical-clock leveling	Δh≈399 m	$z = 0.85\sigma$; $\epsilon = (5.67 \pm 6.68) \times 10^{-4}$	[33]

The P-model vs GR difference scale (representative values) is:

System (representative)	Observable	P−GR difference	1σ precision required for 3σ detection
COW (H=3 cm)	phase	$\delta\phi \approx -9.75 \times 10^{-8}$ rad	relative $\lesssim 4.64 \times 10^{-10}$
Atom interferometer (T=0.4 s)	phase	$\delta\phi \approx -3.22 \times 10^{-2}$ rad	relative $\lesssim 4.64 \times 10^{-10}$
Optical-clock leveling (Δh≈399 m)	Δf/f	$\delta(\Delta f/f) \approx -6.05 \times 10^{-23}$	absolute $\lesssim 2.02 \times 10^{-23}$
Ground↔ISS (400 km; reference)	Δf/f	$\delta(\Delta f/f) \approx -5.54 \times 10^{-20}$	absolute $\lesssim 1.85 \times 10^{-20}$
Solar potential (0.3 AU↔1 AU; example)	Δf/f	$\delta(\Delta f/f) \approx -9.85 \times 10^{-16}$	absolute $\lesssim 3.28 \times 10^{-16}$
Strong-field example (neutron star: surface↔∞)	z	$\delta z \approx -4.72 \times 10^{-2}$	absolute $\lesssim 1.57 \times 10^{-2}$

In the additional differential audit, the weak-field mapping rows above and the dephasing row of 4.5 were reused to freeze the P-model-specific differential predictions into a five-row aggregate table and row-specific 3σ gates. The aggregate results are as follows.

Aggregate row	Frozen difference	1σ precision required for a 3σ decision	Current / required	Current position
dephasing structural parity entry	$(k_B T_{\text{env}}/\chi_P)_{\text{parity}} = 3.27 \times 10^{-21}$	1.09×10^{-21}	measurement undefined	entry-only; no matched inferred row
atom interferometer: P-model-specific local arm	$\Delta\phi_{\text{diff}} = -2.72 \times 10^{-4}$ rad	3.92×10^{-12} fractional phase	2.55×10^{2}	P-model-specific but not yet detectable
atom interferometer: reference apex envelope	$\Delta\phi_{\text{diff}} = -1.08 \times 10^{-1}$ rad	1.55×10^{-9} fractional phase	6.44×10^{-1}	inside the detectable range for the current proxy metric, but not P-model-specific
optical clock main 399 m	$\delta(\Delta f/f) = -6.05 \times 10^{-23}$	2.02×10^{-23} abs $\Delta f/f$	1.43×10^{6}	not detectable in Earth leveling as it stands
optical clock solar lever arm	$\delta(\Delta f/f) = -3.89 \times 10^{-14}$	1.30×10^{-14} abs $\Delta f/f$	2.23×10^{-3}	within current clock precision if the geometry is realized

Supplement: at present, only two rows enter the gate region, the reference-apex row and the solar-lever-arm row. The main bottleneck on the P-model-specific side is the atom-interferometer local β/light row, while the most promising future geometry is the solar 0.05 AU→1 AU row. The structural-parity row is retained as an entry-fixing row for differential prediction because there is not yet an inferred quantity in the same σ_z / τ_{free} regime.

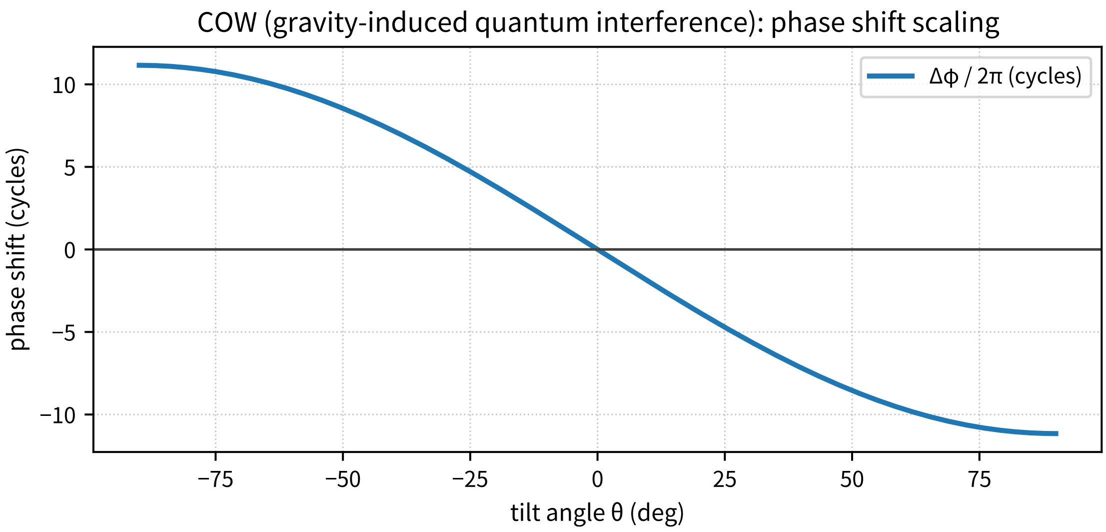

Figure 9: Shows the representative phase (gravity term) for COW (neutron interference).

Supplement: this figure is used to confirm the absolute phase scale (cycles). The note previously embedded under the figure is moved into the main text, and the formula $\Delta\phi = -mgH^2 \sin\theta/(\hbar v_0)$ together with the fixed values of H and v_0 are managed in the settings table.

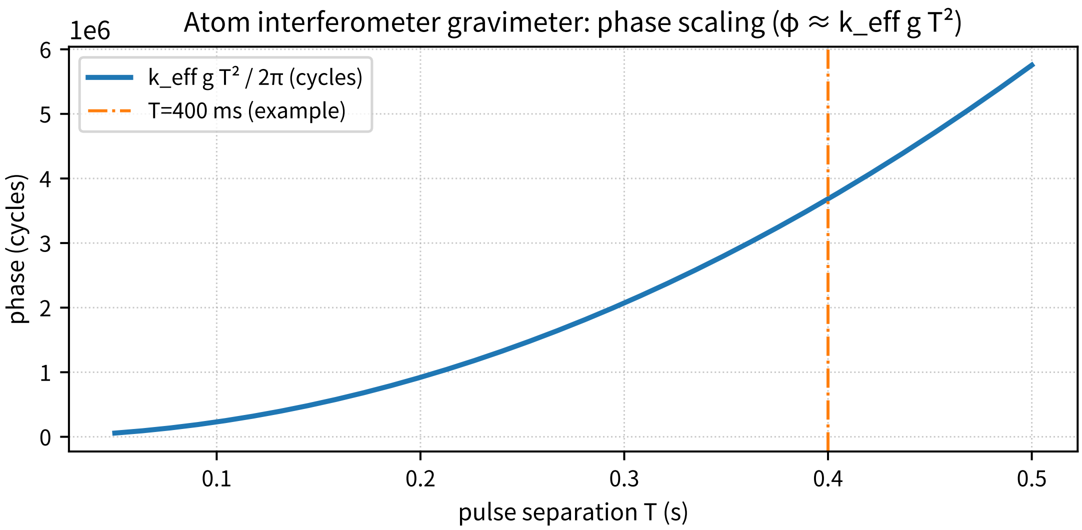

Figure 10: Shows the representative phase (gravity term) for the atom-interferometer gravimeter.

Supplement: before entering the differential discussion, this figure fixes the phase scale (rad). The note previously embedded below the figure is moved into the main text, and the gravity phase is evaluated as $\phi_g = k_{\text{eff}} g T^2$ with $k_{\text{eff}} = 4\pi/\lambda$.

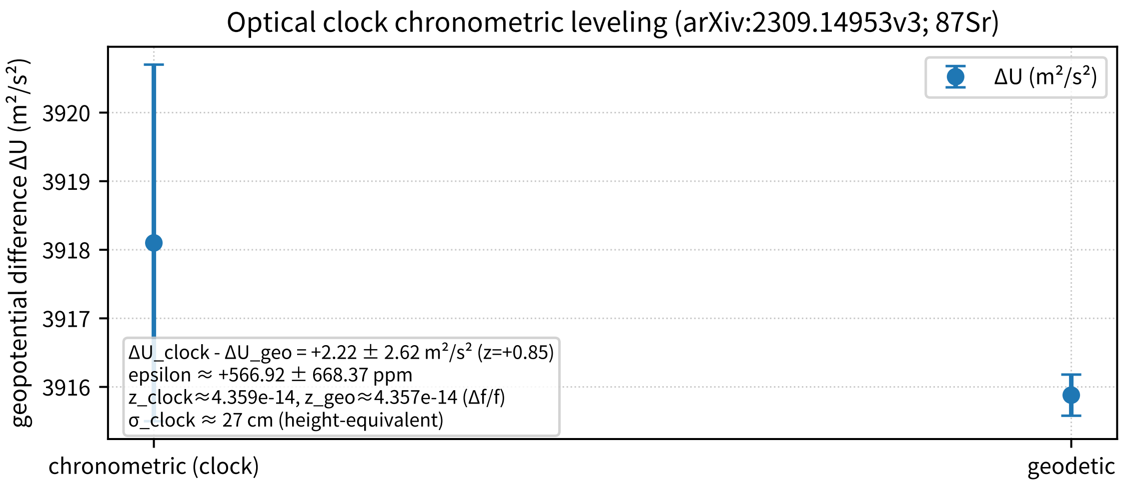

Figure 11: Shows the representative point of optical-clock height comparison (chronometric leveling).

Supplement: confirms consistency with $\epsilon = 0$ (the P-model minimum) in the weak field.

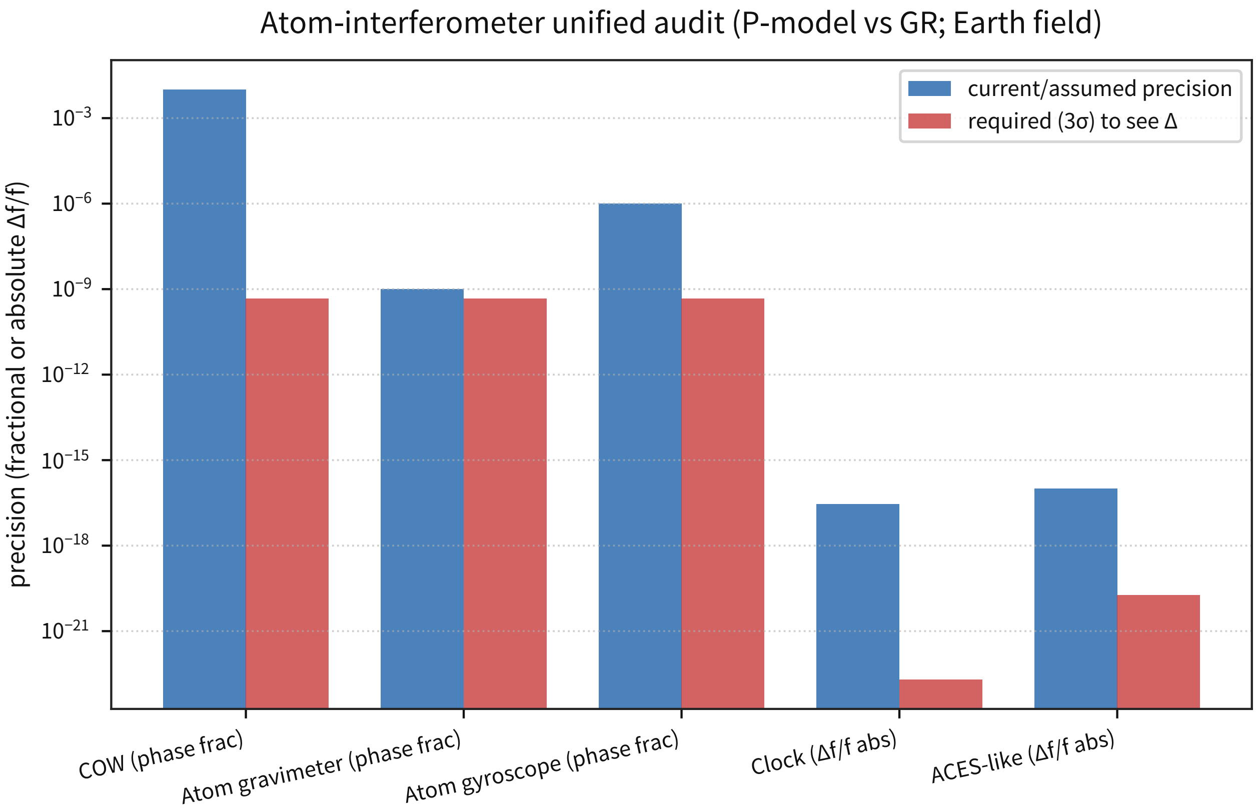

Figure 12: Shows the P-model vs GR differential prediction (required precision) together with representative experimental precision.

Supplement: this figure quantitatively visualizes "the difference is not detectable at present precision." The note previously embedded below the figure is moved into the main text, and the comparison is fixed using the P-model mapping $d\tau/dt = \exp(-x)$ and the GR mapping $d\tau/dt = \sqrt{1-2x}$ with

$x = GM/(c^2 r)$.

Discussion: The P-model potential is defined as $\phi_P \equiv -c^2 \ln(P/P_\infty)$. Writing $P/P_\infty = 1 + \epsilon$ in the weak field (with $|\epsilon| \ll 1$) gives

$$\phi_P = -c^2 \ln(1+\epsilon) = -c^2 \left[\epsilon - \frac{\epsilon^2}{2} + \frac{\epsilon^3}{3} - \cdots\right].$$

The Newtonian point-mass potential is

$$\phi_N = -\frac{GM}{r}, \qquad x \equiv \frac{GM}{c^2 r} = \frac{|\phi_N|}{c^2}.$$

Here, x_{low} and x_{high} denote $x = GM/(c^2 r)$ evaluated on the low-altitude side (r_{low}) and high-altitude side (r_{high}), with $r_{\text{low}} < r_{\text{high}}$ (hence $x_{\text{low}} > x_{\text{high}}$). If one identifies $\epsilon \approx x$ at leading order (the weak-field identification), the higher-order difference becomes

$$\Delta\phi \equiv \phi_P - \phi_N = c^2 \left[\frac{x^2}{2} - \frac{x^3}{3} + \cdots\right], \qquad \left|\frac{\Delta\phi}{\phi_N}\right| = \frac{1}{2}x + O(x^2).$$

However, under the unified convention of this repository, the Poisson solution for a point mass adopts $\ln(P/P_\infty) = x$ **(that is,** $P/P_\infty = e^x$**)**. In that case, $\phi_P = -c^2 \ln(P/P_\infty) = -c^2 x = \phi_N$, so the **difference in the conservative sector is 0**. Accordingly, the formula above is used as an estimate of the nonlinear higher-order scale (about $x/2$) that would be missed if one linearized $P/P_\infty - 1$ by $\ln(1+\epsilon) \approx \epsilon$. In the weak field, detectability actually appears through the difference between the **clock mappings** used in this section, $\exp(-x)$ and $\sqrt{1-2x}$ ($O(x^2)$; relative scale approximately $x_{\text{low}} + x_{\text{high}}$).

The comparison between current experimental precision and the correction scale in the weak field (representative values) is as follows:

Experiment	Typical x ($= |\phi_N|/c^2$)	Current precision (1σ; representative)	P-model correction (representative)	Detectability
COW (neutron; H=3 cm)	6.95×10^{-10}	relative phase $\approx 10^{-2}$ (assumed)	$\delta\phi \approx -9.75 \times 10^{-8}$ rad (relative $\approx 1.39 \times 10^{-9}$)	not yet reachable (needed for 3σ discrimination: relative $\lesssim 4.64 \times 10^{-10}$)
Atom interferometer (T=0.4 s)	6.95×10^{-10}	relative phase $\approx 10^{-9}$ (assumed)	$\delta\phi \approx -3.22 \times 10^{-2}$ rad (relative $\approx 1.39 \times 10^{-9}$)	not yet reachable (needed for 3σ discrimination: relative $\lesssim 4.64 \times 10^{-10}$)
Optical clock (Δh≈399 m)	6.95×10^{-10}	$\sigma(\Delta f/f) \approx 2.89 \times 10^{-17}$	$\delta(\Delta f/f) \approx -6.05 \times 10^{-23}$	not yet reachable (needed for 3σ discrimination: $\lesssim 2.02 \times 10^{-23}$)
ACES/PHARAO (ground↔ISS 400 km; reference)	(6-7) $\times 10^{-10}$	$\sigma(\Delta f/f) \approx 2.0 \times 10^{-16}$ (published specification)	$\delta(\Delta f/f) \approx -5.54 \times 10^{-20}$	reference only (outside the numerical-audit target; future 3σ discrimination would require $\lesssim 1.85 \times 10^{-20}$)

In the weak terrestrial field ($x \approx 10^{-9}$), the P-model mapping difference becomes extremely small at $O(x^2)$, so it is difficult to say that existing interference experiments and clock comparisons already function as direct differential detectors (the required precision remains extremely demanding even for the representative values of this section). By contrast, the long lever arm of the solar potential (deep-space clocks) and the gravitational redshift of strong-field objects can produce larger differences, making them candidates for connecting the differential prediction of this model to observation (though in the latter case astrophysical systematics tend to dominate).

On the difference induced by β (light propagation): in the derivation of Mueller 2007, k_{eff} is determined by optical dispersion (laser wavenumber), so the model $n(P) = (P/P_\infty)^{2\beta}$ can enter the phase through the laser-phase term. In local experiments, however, constant terms (common mode) are canceled or absorbed, so the differential effect is strongly suppressed by the potential difference within the interferometer. With frozen β (Cassini) and the representative value $T = 0.4$ s, the adopted model (local difference) gives $\delta\phi_\beta \approx 4.16 \times 10^{-14}$ rad, and the 1σ precision required for 3σ detection is estimated as $\lesssim 1.39 \times 10^{-14}$ rad.

Reject: Under each representative condition, if the observed redshift / phase deviates from the P-model static-clock mapping (e^{-x}) by more than 3σ of the total uncertainty (statistical + systematic), this minimal mapping is rejected.

Frozen: The representative conditions of COW / atom interferometer / quantum clocks are fixed, and the P−GR differential scales ($\delta\phi$, $\delta(\Delta \mathrm{f}/\mathrm{f})$, $\delta \mathrm{z}$) together with the "1σ precision required for 3σ detection" are frozen in table form. In addition, the Pikovski structural parity row, the atom-interferometer differential proxy metric, the optical-clock differential observable, the five-row aggregate table, and the row-specific 3σ gates are fixed in gravity_induced_decoherence_pikovski_comparison_metrics.json, atom_interferometer_pikovski_phase_difference_metrics.json, optical_clock_differential_frequency_metrics.json, and gravity_quantum_differential_prediction_table_metrics.json. **Reject**: The 3σ gates above (redshift / phase deviating from $\exp(-x)$) are frozen as the operational rejection condition; unless they are met, no claim of differential detection is made. **Next**: This differential audit is closed on the artifact side, and from here on updates are limited row by row. The differential prediction will be updated when data appear that can directly hit one of the row-specific gates in the table above, whether through a long-baseline clock in the solar potential, the local β/light channel of atom interferometry, or a dephasing inference in the same regime.

3.4 Matter-Wave Interference (Electron Double Slit / de Broglie Precision)

Objective: Confirm, with representative experiments, the scale of matter-wave interference (the de Broglie wavelength and interference fringes), and freeze the consistency of the independent constant α as a z-score so that it connects directly to the rejection condition.

Input: A representative electron double-slit setup and the value of α derived from atom recoil (interference), compared against an independent measurement.

Processing: For electrons, compute the de Broglie wavelength from momentum and evaluate the angular scale of the interference fringes. For α, convert the difference from the independent measurement into a z-score and freeze it.

Equation:

$$\lambda = \frac{h}{p}, \qquad p \simeq \sqrt{2mE}, \qquad z = \frac{\alpha_{\text{obs}} - \alpha_{\text{ref}}}{\sigma}.$$

Metric: The de Broglie wavelength and fringe scale, together with the z-score for α consistency (rejection condition: $|z| > 3$).

Results:

Topic	Representative condition	Main value	Primary source
Electron double slit	600 eV	de Broglie wavelength 50.05 pm; fringe spacing 0.179 mrad	[34]
de Broglie precision (α)	atom recoil $\rightarrow \alpha$	$z = 0.59\sigma$; effective $\epsilon = (-5.33 \pm 9.08) \times 10^{-9}$	[35][36]
Cross-channel I/F (electron / atom / molecule)	integrated	number of channels = 4; atom α-consistency z = 0.59; molecular scaling $z_{\max} = 2.64$; atom precision-gap median = 7.18×10^5	integrated fixed output (combined primary sources)

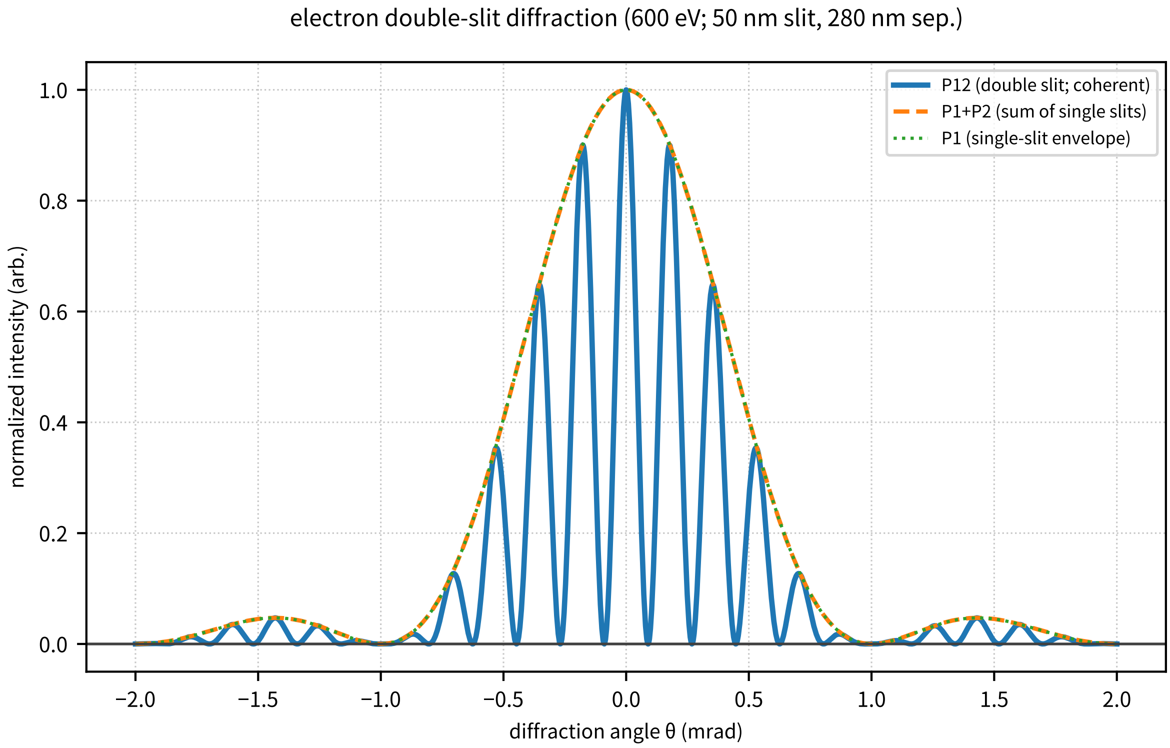

Figure 13: Shows the representative fringe spacing and de Broglie wavelength for the electron double slit.

Supplement: this figure is used as the entry point for fixing the scale of matter-wave interference. The note formerly placed under the figure is moved into the main text, and the 600 eV values of λ_e, $\Delta\theta \approx \lambda/d$, and the envelope zero $\theta \approx \lambda/a$ are listed here.

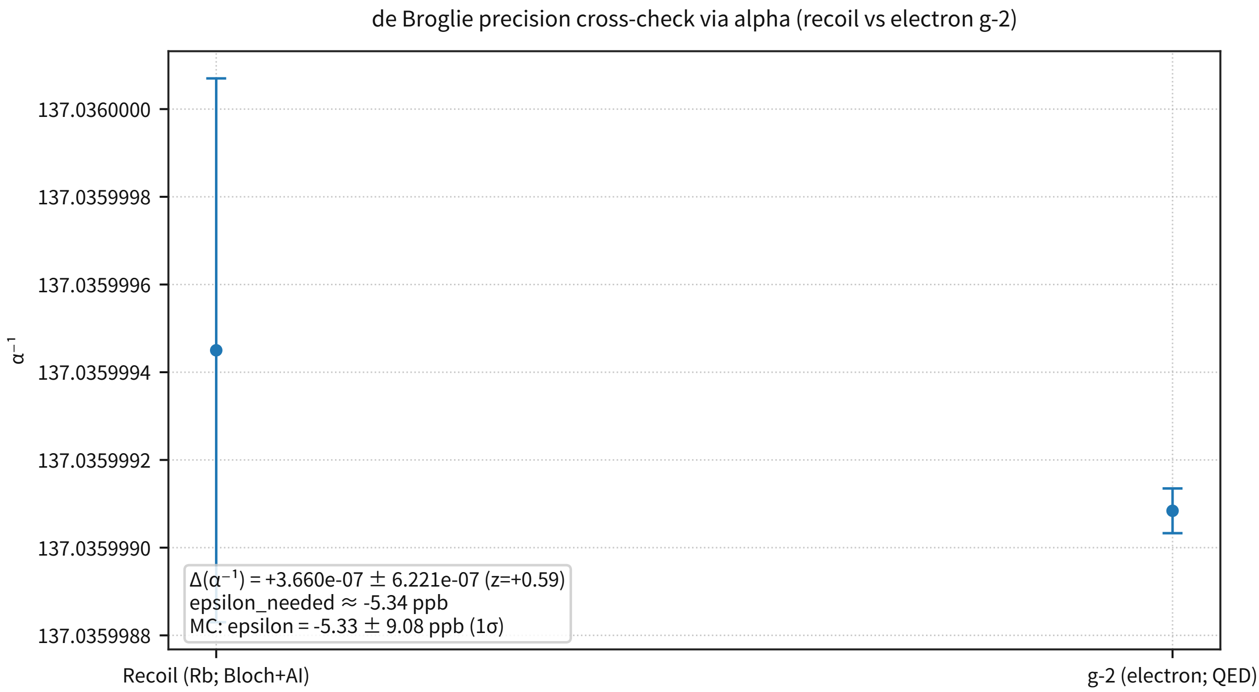

Figure 14: Shows the consistency between α derived from atom recoil (interference) and the independent measurement (g-2).

Supplement: fixes the z-score of the effective ϵ and connects it to the rejection condition ($z > 3$).

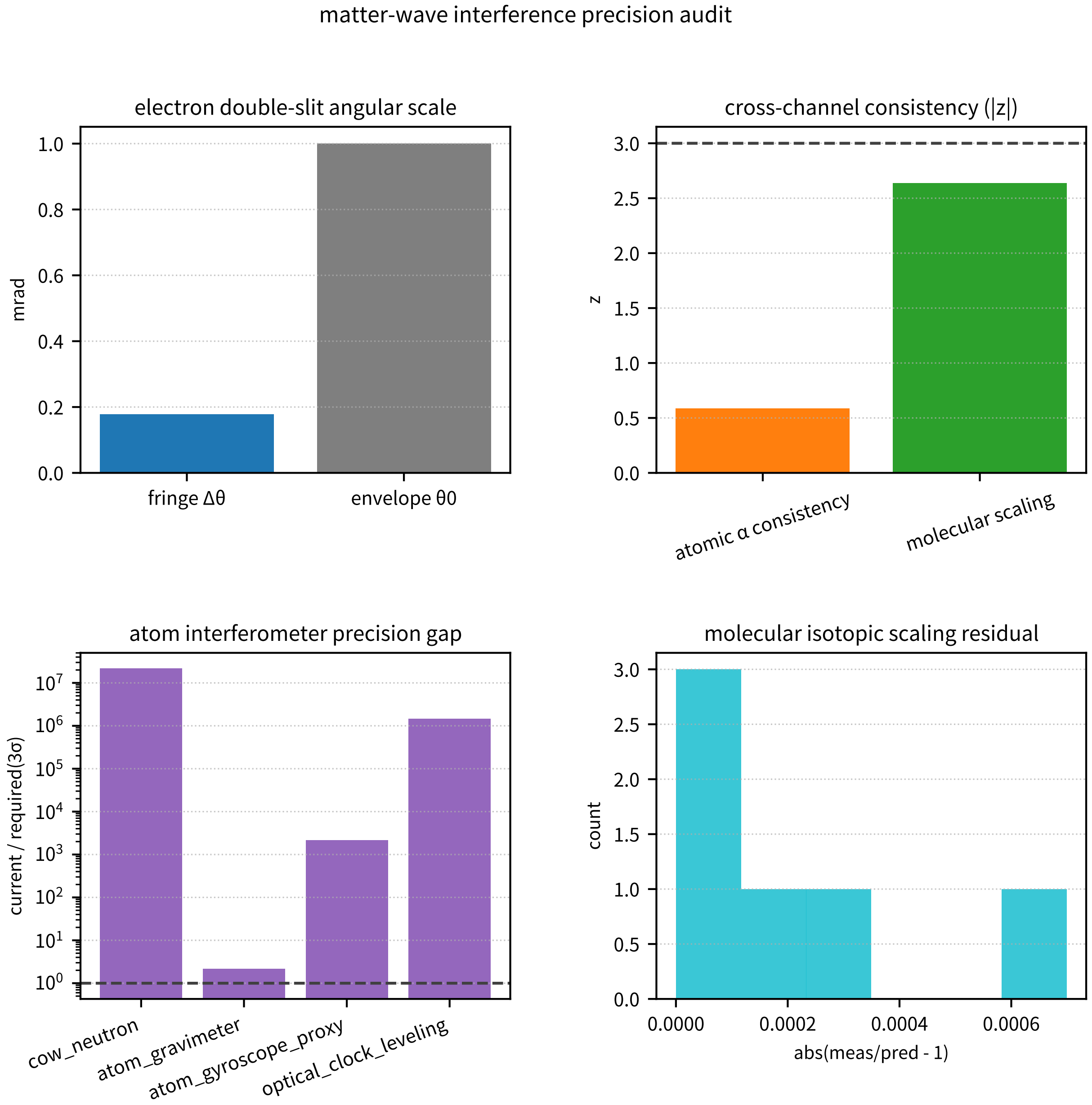

Figure 15: An integrated audit figure that displays, on the same input/output interface, electrons (fringe scale), atoms (α consistency and the precision gap in gravitational interference), and molecules (reduced-mass isotope scaling).

Supplement: the cross-channel metrics fixed in this section (channels_n=4) are treated as the entry point for the later comparison with optical interference (HOM / squeezed light).

Frozen: The representative fringe scale of electron interference and the independent cross-check of α derived from atom recoil (z-score) are fixed, and an entry point is created for audit indicators that cross electrons / atoms / molecules (channels_n, z_max, and so on). **Reject**: If α shows $|z| > 3$ between independent measurements, or if the interference-fringe scale deviates from the de Broglie prediction by more than 3σ of the total uncertainty, this minimal consistency is rejected. **Next**: Add molecular interference (large molecules / temperature dependence) and isotope systematics, and update the cross-channel indicators (channels_n, z_max) so that the contrast with optical interference (4.6) becomes sharper.

3.4.1 Tunneling Effect (Rectangular-Barrier Transmission)

Objective: Apply directly, to the transmission problem of a static rectangular barrier, the Schr envelope equation fixed in Part III-A §2.5.1, and thereby fix that tunneling is obtained mechanically from the established wave equation rather than from a separate axiom.

Input: The nonrelativistic envelope equation of Part III-A,

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m_*}\nabla^2 + m_*\phi\right)\psi$$

and the one-dimensional static barrier

$$V(x) \equiv m_*\phi(x) = \begin{cases} V_0 & (0 < x < a), \\ 0 & (\text{otherwise}) \end{cases}$$

are used. The representative benchmark is fixed by two electron cases: electron_stm_like ($E = 1.0\,\mathrm{eV}$, $V_0 = 4.0\,\mathrm{eV}$, $a = 0.50\,\mathrm{nm}$) and electron_thin_barrier ($E = 0.8\,\mathrm{eV}$, $V_0 = 2.0\,\mathrm{eV}$, $a = 0.30\,\mathrm{nm}$).

Processing: Take the stationary solution $\psi(x,t) = \psi(x)e^{-iEt/\hbar}$, and set the wave functions in regions I / II / III to

$$\psi_I = e^{ikx} + r\,e^{-ikx}, \qquad \psi_{II} = Ae^{\kappa x} + Be^{-\kappa x}, \qquad \psi_{III} = t\,e^{ikx}$$

respectively. Here

$$k = \frac{\sqrt{2m_*E}}{\hbar}, \qquad \kappa = \frac{\sqrt{2m_*(V_0 - E)}}{\hbar}$$

holds. Impose the continuity conditions on ψ, ψ' at $x = 0, a$, and cross-check the closed-form transmission coefficient against the four-variable continuity-matrix implementation. The WKB estimate

$$T_{\mathrm{WKB}} \approx e^{-2\kappa a}$$

is shown only as a diagnostic value.

Equation: The exact transmission for a rectangular barrier is

$$T_{\mathrm{exact}} = \left[1 + \frac{V_0^2\sinh^2(\kappa a)}{4E(V_0 - E)}\right]^{-1} \tag{2}$$

given by this expression. For the same barrier, agreement between $T_{\mathrm{matrix}} = |t|^2$ obtained from the continuity matrix and T_{exact} is used as the minimal audit of the implementation. The benchmark table below is intended to confirm implementation agreement for this transmission-coefficient formula.

Metric: The primary metric is

$$\Delta T_{\mathrm{impl}} \equiv |T_{\mathrm{matrix}} - T_{\mathrm{exact}}|$$

and the operational gate is fixed as $\Delta T_{\mathrm{impl}} \leq 10^{-12}$. As auxiliary metrics, record κa and $T_{\mathrm{exact}}/T_{\mathrm{WKB}}$ in order to diagnose how far WKB can be used as an approximation.

Results:

case	E (eV)	V_0 (eV)	a (nm)	κa	T_{exact}	T_{matrix}	T_{WKB}	Judgment
e-STM-like	1.0	4.0	0.50	4.437	4.20×10^{-4}	4.20×10^{-4}	1.40×10^{-4}	pass
e-thin barrier	0.8	2.0	0.30	1.684	1.24×10^{-1}	1.24×10^{-1}	3.45×10^{-2}	pass

Summary metric	Value
$\max \Delta T_{\mathrm{impl}}$	$< 10^{-15}$
continuity-matrix gate	pass
WKB role	diagnostic only

The reproducible outputs are fixed as the tunneling-benchmark artifacts of Part IV.

Discussion: The important point in this implementation is that tunneling does not require a new degree of freedom. If the Schr envelope equation fixed in Part III-A is applied directly to a static barrier, one recovers a transmission coefficient that is isomorphic to standard quantum mechanics. Therefore, the P-model-side difference shifts away from "the tunneling phenomenon itself" and into the outer modeling layer: which barrier is chosen as $V(x) = m_* \phi(x)$, or whether the barrier is dynamic or includes backreaction.

Note: This section is the minimal implementation of a rectangular barrier and does not claim that individual systems such as α decay, Josephson junctions, or STM images have been derived from first principles. Those cases require individual modeling of barrier shape, dissipation, internal degrees of freedom, and boundary conditions, so the next stage should extend them case by case while preserving the input/output interface fixed here.

3.5 Gravity-Induced Decoherence (Visibility Reduction)

Objective: Fix, using representative parameters, the scale of ensemble dephasing (visibility reduction) induced by gravitational redshift, and make explicit that under weak-field laboratory conditions the effect is extremely small and is not itself the claimed differential signal.

Input: A Pikovski-type dephasing model and representative evaluations of the parameters (σ_z, t, σ_y).

Processing: Assume a height-distribution width σ_z, and compute and freeze the time scale $t_{1/2}$ at which $V = 0.5$ is reached, together with the equivalent σ_y.

Metric: The visibility V and the required condition on the additional time-structure noise (fractional rate noise) σ_y.

Results: For height-distribution widths $\sigma_z = 0.1/1/10$ mm, the time scale corresponding to $V = 0.5$ is respectively **4.0e4 s / 4.0e3 s / 4.0e2 s** (equivalently, σ_y increases by factors of 10). [37][38][39][40][41]

Reject: Under weak-field laboratory conditions, if the observed visibility reduction exceeds the gravity-induced dephasing estimate of this section by more than 3σ of the total uncertainty, then the hypothesis of gravity-induced dephasing alone is rejected (the dominant cause must then belong to another systematic class).

Note: (Distinction from collapse) The "decoherence" treated in this section is visibility reduction (dephasing) as a time-delay / phase-diffusion effect, and is a different hypothesis from the Penrose-Diósi type "gravity-induced spontaneous collapse (objective collapse)." [42][43] Accordingly, constraints from underground experiments and similar studies on collapse models cannot automatically be applied to the model of this section (though if a collapse-type additional assumption is introduced in the future, its constraints will have to be organized separately). As a representative example, underground experimental constraints have been discussed for Diósi-Penrose-type verification.[44] Given the numerical scale in this paper ($V = 0.5$ at the level of $10^2 \ldots 10^4$ s), the effect is extremely small under weak-field laboratory conditions, and no correction is required that would immediately invalidate existing interference experiments (a safety statement).

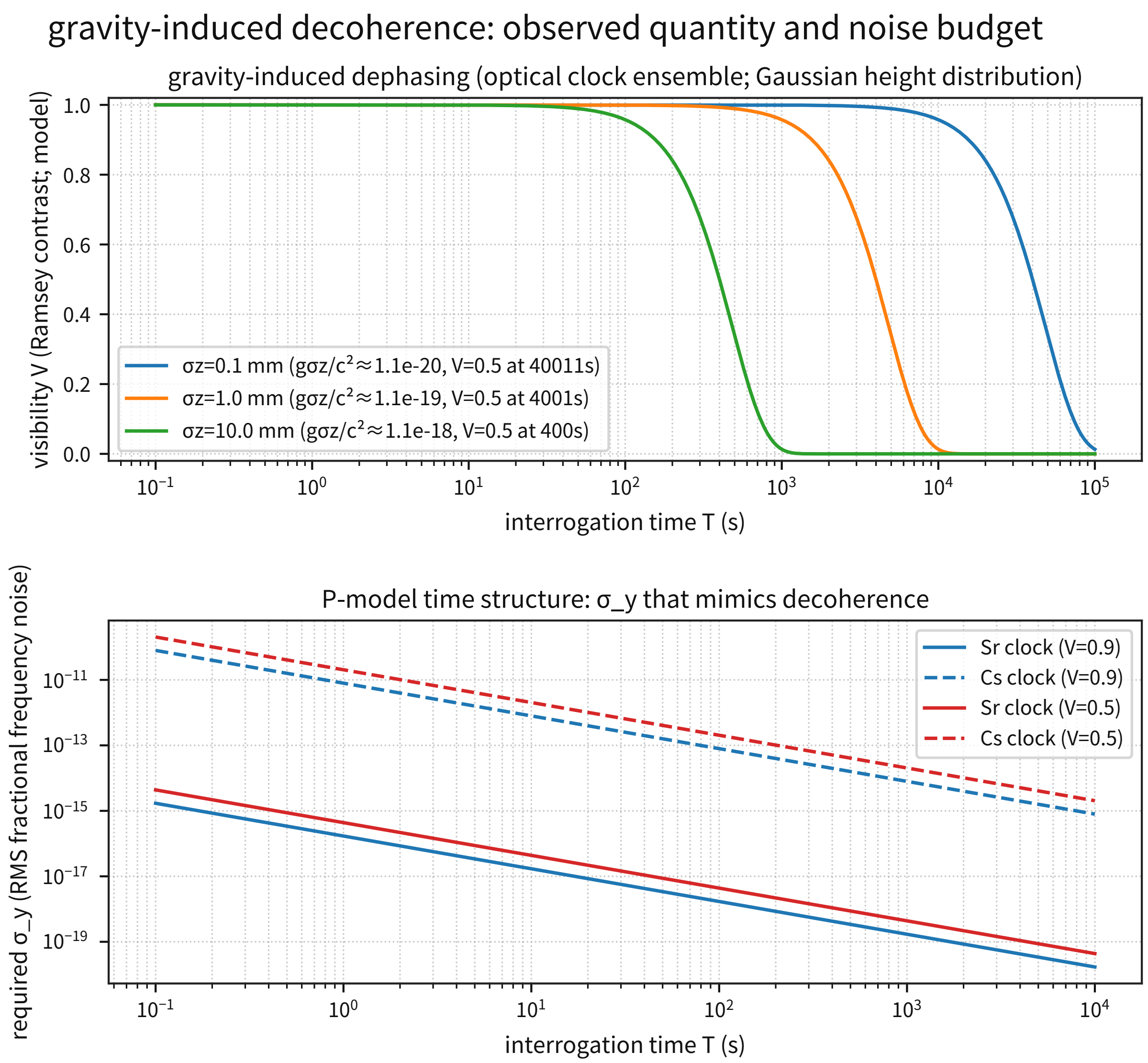

Figure 16: Shows the scale of visibility reduction versus the height distribution (σ_z) together with the required σ_y.

Supplement: confirms that the effect is extremely small under weak-field laboratory conditions.

Frozen: The representative scale of Pikovski-type gravity-induced dephasing is fixed ($t_{1/2}$ and the equivalent σ_y for $\sigma_z = 0.1/1/10$ mm). **Reject**: If the observed visibility reduction exceeds the estimate above by more than 3σ of the total uncertainty, the hypothesis of gravity-induced dephasing alone is rejected. **Next**: Constrain σ_y independently with future high-precision interferometers / clocks, and connect this to a separation audit of dephasing / collapse / environmental noise (which term is actually dominant).

3.6 Quantum Interference of Light (Single Photon / HOM / Squeezed Light)

Objective: Fix, using representative experimental values, the scale at which optical-side phase noise and loss are dominant, and position the result not as a later ”differential prediction” but as a ”required condition (consistency).”

Input: Representative experiments for single-photon interference, HOM, and squeezed light (primary sources).

Processing: Convert visibility and loss (η) into comparable scales, such as the equivalent optical-path-length noise, and compare them.

Metric: Equivalent optical-path-length noise σ_L, HOM visibility V, and the lower bound on loss η.

Results:

Topic	Representative condition	Main value	Primary source
Single-photon interference	V≥0.8, λ=1550 nm	equivalent optical-path-length noise σ_L≈165 nm	[45]
HOM	example: D=13 ns / 1 μs	visibility = 0.982 ± 0.013 / $0.987^{+0.013}_{-0.020}$	[46][47]
Squeezed light	10 dB	loss-only lower bound η≥0.900	[48]
Integrated HOM + squeezed-light I/F	integrated	channels_n=5; HOM z=37.1/24.35; delay-difference z=0.21; squeezing variance ratio=0.10; PSD(10k/100k)=0.84	[46][47][48]

For the HOM noise-PSD-shape watch, an additional audit was performed with low-frequency drift correction and band-pair sensitivity (16 pairs). $PSD(10k/100k)$ is 0.841 raw and 0.851 detrended; the median over band pairs is 0.642 raw / 0.650 detrended; and only 5 of 16 pairs, i.e. 31.25 percent, clear the threshold (>=1). Because the single-point ratio depends strongly on the band definition, it is not promoted to the main gate and is instead fixed as a monitored item under a non-primary watch.

Reject: If HOM visibility remains at the classical limit ($V <= 0.5$), or if the 10 dB condition for squeezed light (loss-only lower bound $\eta >= 0.90$) is not satisfied, then the quantum interference input/output interface of this section does not hold (rejection).

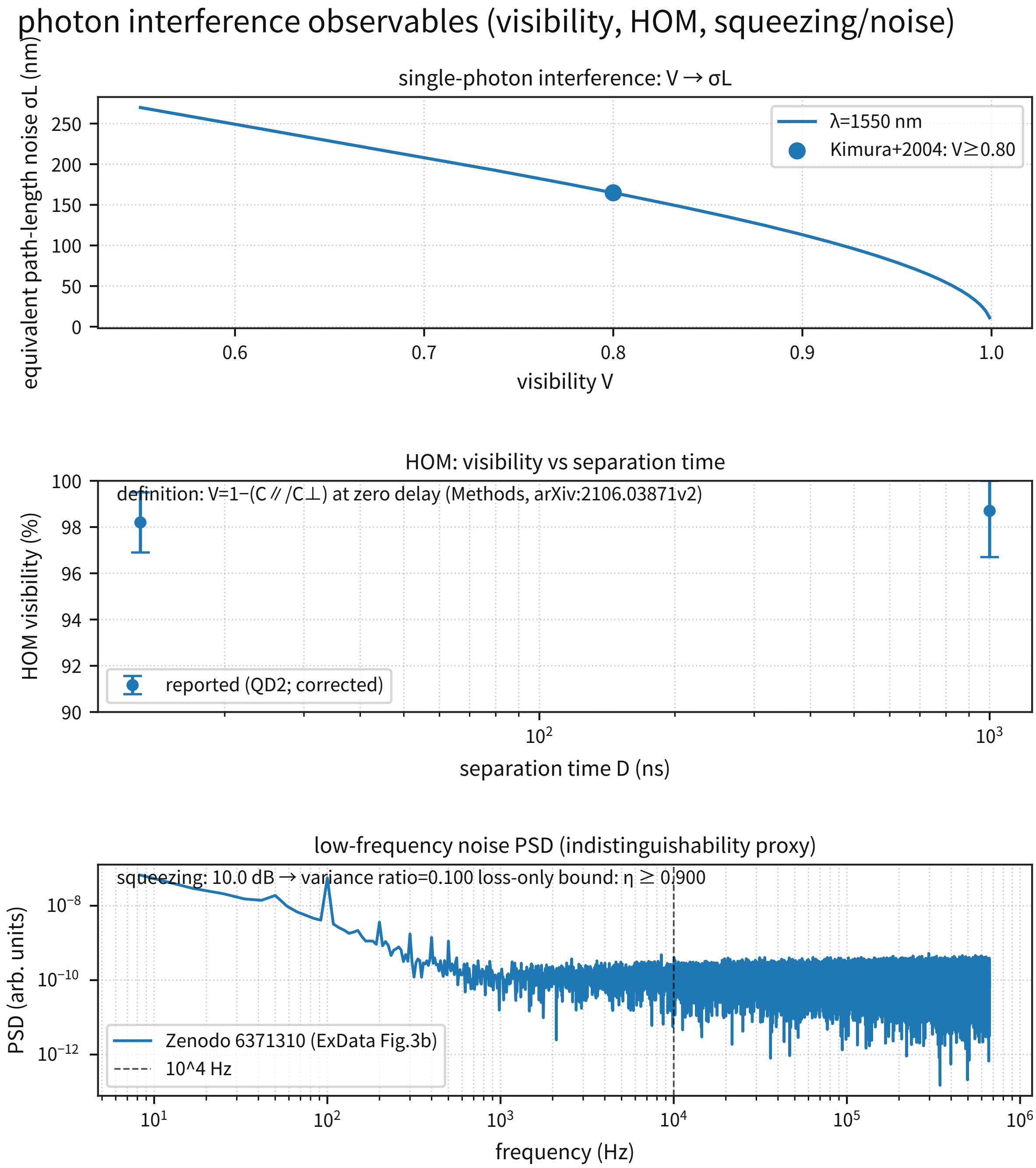

Figure 17: Collects the representative values for single-photon interference, HOM, and squeezed light.

Supplement: fixes the scale on which optical-side phase noise and loss are dominant.

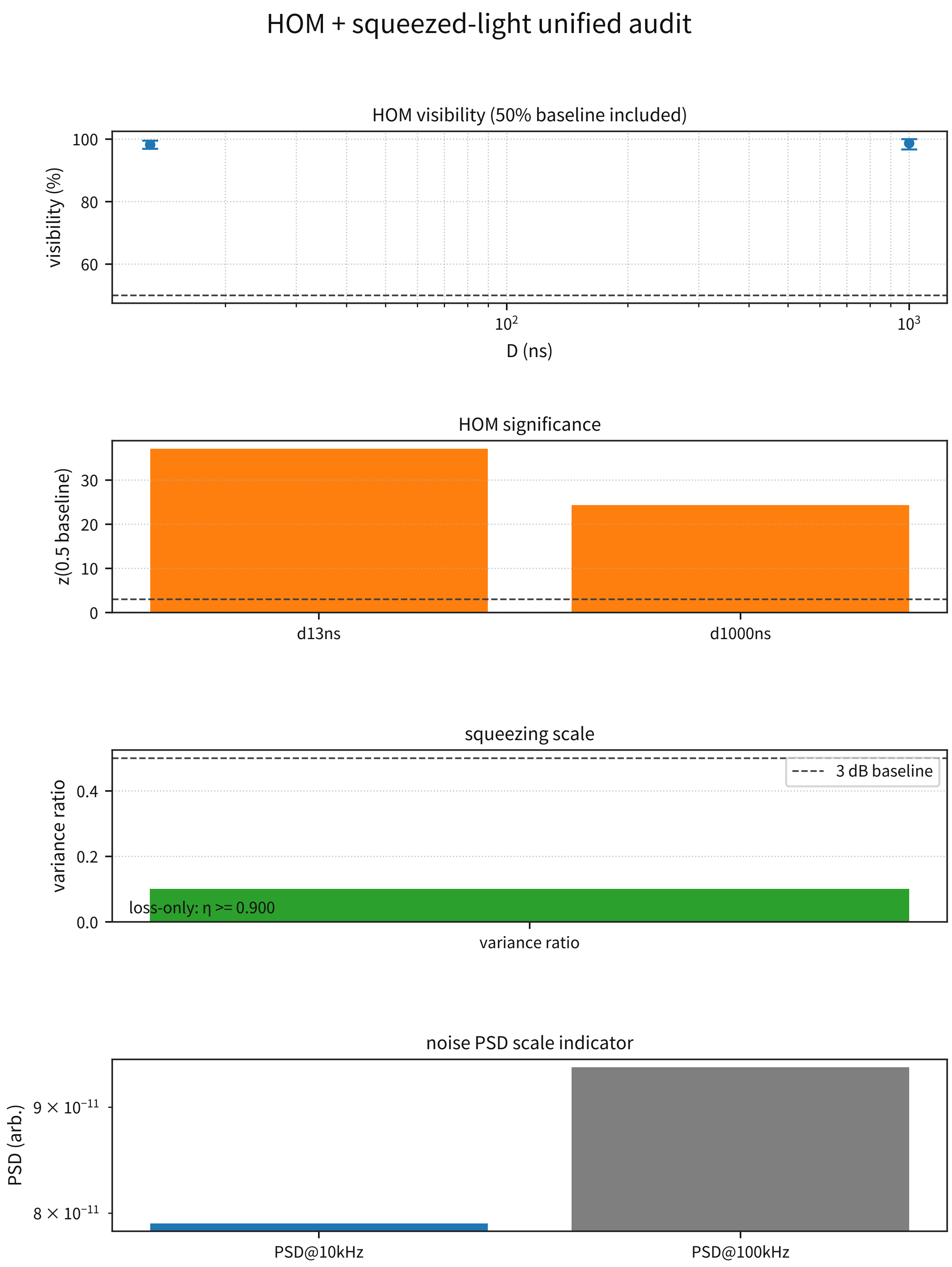

Figure 18: An integrated audit figure that places, on the same input/output interface, the significance of HOM visibility relative to the classical threshold (50%), the delay dependence from 13 ns↔1 μs, and the squeezed-light variance ratio (10 dB→0.10) together with the PSD ratio (10 kHz/100 kHz).

Supplement: treated as the fixed output of this section, connecting to the next independent cross-check.

Frozen: The representative values of single-photon interference / HOM / squeezed light (V, η, σ_L), together with the significance relative to the classical limit (HOM: $V <= 0.5$), are fixed as integrated metrics. **Reject**: If the rejection conditions above hold (HOM stays at the classical limit, or the 10 dB condition is not met), the quantum-interference input/output interface of this section is rejected. **Next**: Extend the full delay sweep curve $V(\tau)$ and the error budget for loss / phase noise, and move the audit closer to a Bell-like "procedure-dependence audit."

3.7 Vacuum / Precision QED (Casimir / Lamb Shift)

Objective: Fix known precision physics (Casimir / Lamb shift / H 1S–2S / α) as the baseline, and show quantitatively, as a safety check, that the minimal coupling of the P-model does not immediately generate a contradiction.

Input: Primary sources for Casimir, Lamb shift, H 1S–2S, and α (atom recoil vs electron g-2).

Processing: Put the representative values into a ledger and compare them with an order-of-magnitude estimate of the contribution induced by the gravitational potential on atomic scales.

Metric: Representative precision (relative / absolute), the z-score of α consistency, and the order estimates $|\phi_{\mathrm{nuc}}|/c^2$ and ΔE.

Results:

Probe	Main value (representative)	Precision / note	Primary source
Casimir	sphere-plane; sphere diameter 201.7 μm	relative precision about 1 percent (near the shortest distance)	[49]
Lamb shift	scaling summary (Z^4, Z^6)	example contributions from non-QED systematics such as finite nuclear size	[50]
H 1S–2S	$2.466061413187035(10) \times 10^{15}$ Hz	fractional 4.2×10^{-15} (including the uncertainty budget)	[51]
α (independent cross-check)	z$\approx$0.59σ	atom recoil vs electron g-2	[35][36]

Safety check (atomic scale): under the minimal coupling of the P-model (treating $\phi = -c^2 \ln(P/P_\infty)$ as the gravitational potential), the nuclear gravitational potential is extremely small on atomic scales, and the additional term it gives to electron levels is many orders of magnitude below current precision-QED measurements (hence it is not in contradiction with known precision physics). The order is

$$\left|\frac{\phi_{\mathrm{nuc}}}{c^2}\right| \sim \frac{GM_{\mathrm{nuc}}}{c^2 r}, \qquad \Delta E \sim m_e \, |\phi_{\mathrm{nuc}}|$$

which gives, representatively, $|\phi_{\mathrm{nuc}}|/c^2 \approx 10^{-44}\,(\mathrm{H}) \sim 10^{-40}\,(\mathrm{heavy})$ and $\Delta E \approx 10^{-38} \sim 10^{-34}\,\mathrm{eV}$. Accordingly, within the scope treated in this paper, no correction is required that would immediately violate precision-QED tests such as the Lamb shift or g-2. If, however, a future model is introduced in which P fluctuations carry an additional short-range coupling (non-gravitational), then an upper-bound constraint from precision-QED observables becomes mandatory.

Reject: If a future model including such an additional coupling predicts a difference that exceeds the uncertainty budget of precision observables such as H 1S–2S by more than 3σ, that additional coupling is rejected (this section serves as the audit entry point).

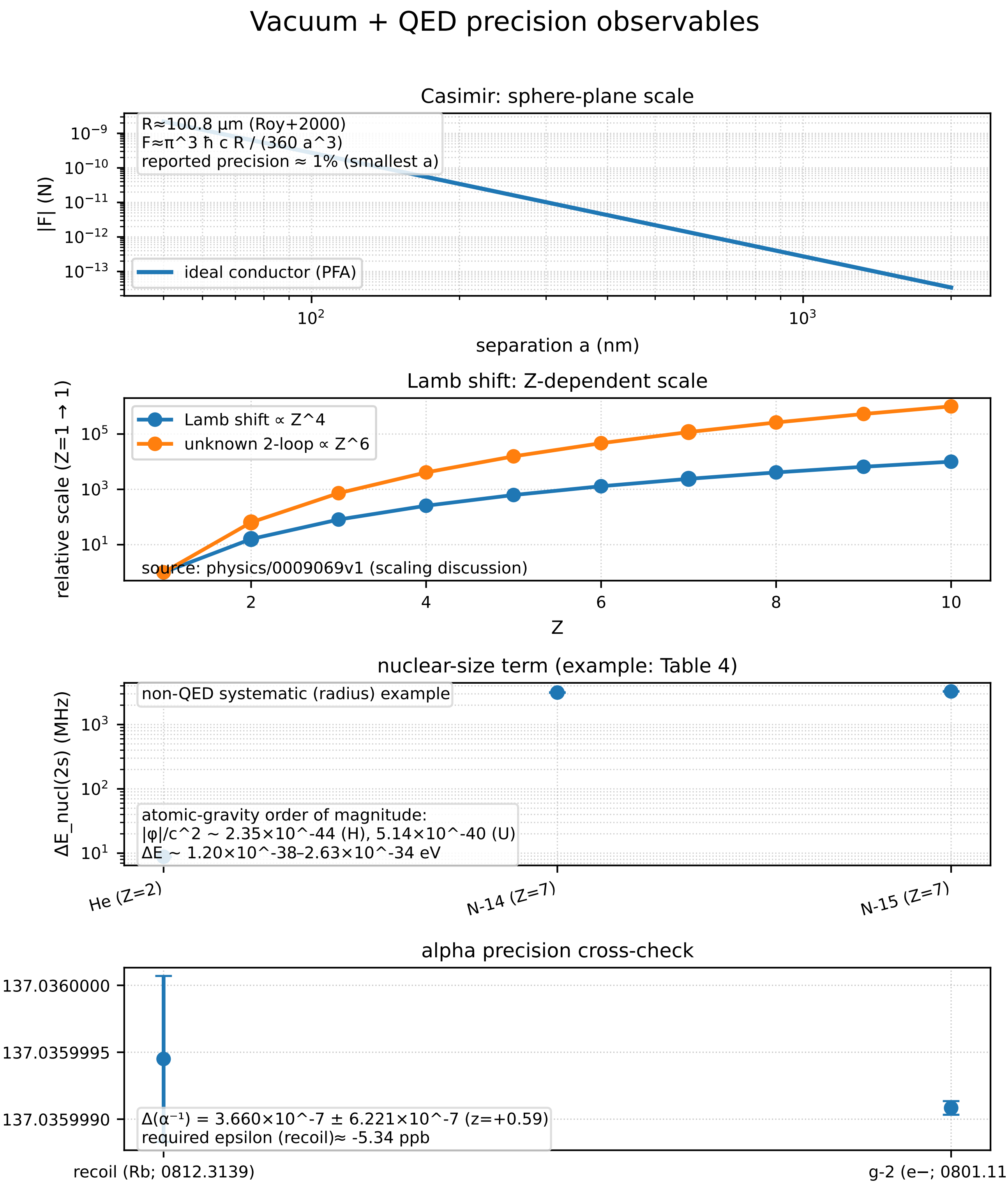

Figure 19: Collects the entry points for Casimir / Lamb shift / H 1S–2S / α and shows that they are not in contradiction with known precision physics.

Supplement: read this as the safety check that the additional correction on the atomic scale is extremely small.

Frozen: The baseline values of Casimir / Lamb shift / H 1S–2S / α and the order upper bounds on the additional correction at atomic scales ($|\phi_{\rm nuc}|/c^2$, ΔE) are fixed as a safety check. **Reject**: If an extension including an additional coupling systematically predicts a difference that exceeds the uncertainty budgets of these precision observables by more than 3σ, that extension is rejected. **Next**: If an additional short-range coupling (non-gravitational) is to be introduced, then upper-bound constraints from precision QED should be built into the error budget from the start so that the parameter space of the P-fluctuation hypothesis is closed first.

3.8 Atomic Nuclei (Overview)

Objective: Fix nuclear-scale verification in two separated layers: (A) effective-model verification as an operational input/output interface (4.8.1-4.8.3), and (B) wave-interference theory as a first-principles interpretation (4.8.4). In this overview section, the rejection / non-rejection of the effective model is fixed first, and then Section 4.8.4 connects the $\Delta\omega \to B.E.$ mapping defined in the nuclear-mapping section of Part I to the observational input/output interface.

The connection rules are fixed as follows. The effective model is treated as an "operational input/output interface that reproduces observables," and its threshold judgment is fixed first. The wave-interference theory is treated as the "first-principles interpretation that generates that input/output interface," and is compared after being projected onto the same observational interface. The region where the two agree is checked around deuteron / He-4 and through the entry diagnostics for $A = 2-16$ in 4.8.4-4.8.6, whereas the region where their divergence grows is judged through the differential predictions of 5.4 (ΔB, σ_{req}). Accordingly, the route through this section is "4.8.2 (minimal fixing of the operational I/F) → 4.8.4-4.8.6 (connection of the interpretation layer) → 5.4 (falsification conditions)."

Input: The deuteron mass defect (CODATA / NIST) and the low-energy parameters of np scattering (a_t, r_t, and so on; eq18-19). The list of primary sources is collected in the appendix "Data Sources (Primary Sources)."

Processing: Fix the deuteron binding energy B and the low-energy parameters of np scattering (scattering length a, effective range r, and shape parameter v_2) separately as "constraints (fit)" and "predictions (pred)." As an analysis-dependent systematic, the difference between eq18 (GWU/SAID) and eq19 (Nijmegen) is treated as an "envelope (min-max)," and rejection / non-rejection is decided by whether the prediction enters that envelope.

Equation: Low-energy scattering is represented by the effective-range expansion (ERE):

$$k \cot \delta_{t,s}(k) = -\frac{1}{a_{t,s}} + \frac{1}{2} r_{t,s} k^2 + v_{2,t,s} k^4 + \cdots \tag{3}$$

Metric: Whether the triplet / singlet predicted values (r_t, $v_{2,t}$, r_s, $v_{2,s}$) enter the eq18-eq19 envelope. In particular, the sign of $v_{2,s}$ is a strong differential prediction, and classes that cannot recover the sign are rejected.

Results:

Deuteron binding energy (entry point): **B≈2.224566 MeV** (from CODATA).

ansatz class	target (constraint / prediction)	key result (summary)	Judgment
square-well (2 param)	fit: B, a_t; pred: r_t	r_t misses by +0.009 to +0.028 fm (eq18 / eq19)	reject (too coarse)
core+well	fit: B, a_t, r_t; pred: $v_{2,t}$ and singlet r_s, $v_{2,s}$	r_t does not enter the envelope, $v_{2,t}$ is too large, and the singlet side is also inconsistent	reject
repulsive core + well	fit: through triplet $v_{2,t}$; pred: singlet	the optimum collapses to $R_c \to 0, V_c \to 0$ and degenerates into a square well	reject

Continued on next page

ansatz class	target (constraint / prediction)	key result (summary)	Judgment
2-range (two-stage; shared geometry)	fit: triplet B, a_t, r_t, $v_{2,t}$; pred: singlet r_s, $v_{2,s}$	singlet $r_s \approx 1.88 - 2.00$ fm (obs 2.626-2.678) and sign mismatch in $v_{2,s}$	reject
2-range (extra singlet freedom)	fit: singlet a_s, r_s; pred: $v_{2,s}$	$v_{2,s} > 0$ ($\approx +0.52, +1.12$), outside the observed envelope (-0.48 to -0.005)	reject (decisive)
repulsive core + 2-range	pred: $v_{2,s}$	even after adding a core, $v_{2,s} > 0$ remains	reject
λ_π constraint (2/3-range)	pred: $v_{2,s}$	even after adding a tail, the sign cannot be rescued and $v_{2,s} > 0$ remains	rejection reinforced
barrier+tail (global k,q)	pred: v_2 (both channels)	$v_{2,s} < 0$ is achieved, but at the same time $v_{2,t}$ exits the envelope	insufficient freedom
barrier+tail (range extension)	pred: triplet $v_{2,t}$ and singlet r_s, $v_{2,s}$	even when $v_{2,t}$ and $v_{2,s}$ enter the envelope, r_s collapses (~ 1.07 fm)	no-go
barrier+tail ((k,q) split)	pred: triplet $v_{2,t}$ and singlet r_s, $v_{2,s}$	r_s always collapses (~ 1.17 fm)	no-go
singlet geometry ($R1_s/\lambda\pi$)	trade-off	r_s and $v_{2,s}$ cannot be satisfied simultaneously	no-go
singlet geometry ($R2_s/\lambda\pi$)	fit: singlet r_s, $v_{2,s}$	at $R2_s/\lambda\pi = 1.5$, both r_s and $v_{2,s}$ enter the envelope simultaneously	released (singlet)
triplet split (barrier ratio / k_t)	fit: triplet a_t, r_t, $v_{2,t}$	minimal extension after freezing the singlet. best: barrier=0.1, $k_t = 0.0$. Both datasets enter the envelope ($v_{2,t} \approx 0.052 - 0.055,\ v_{2,s} < 0$)	released (provisional)

Main values of the canonical solution (triplet split; ERE, $k \to 0$):

dataset	a_t (fm)	r_t (fm)	$v_{2,t}$ (fm^3)	a_s (fm)	r_s (fm)	$v_{2,s}$ (fm^3)
eq18 (GWU/SAID)	5.410	1.750	0.0520	-23.719	2.626	-0.1028
eq19 (Nijmegen)	5.412	1.752	0.0547	-23.739	2.676	-0.0584

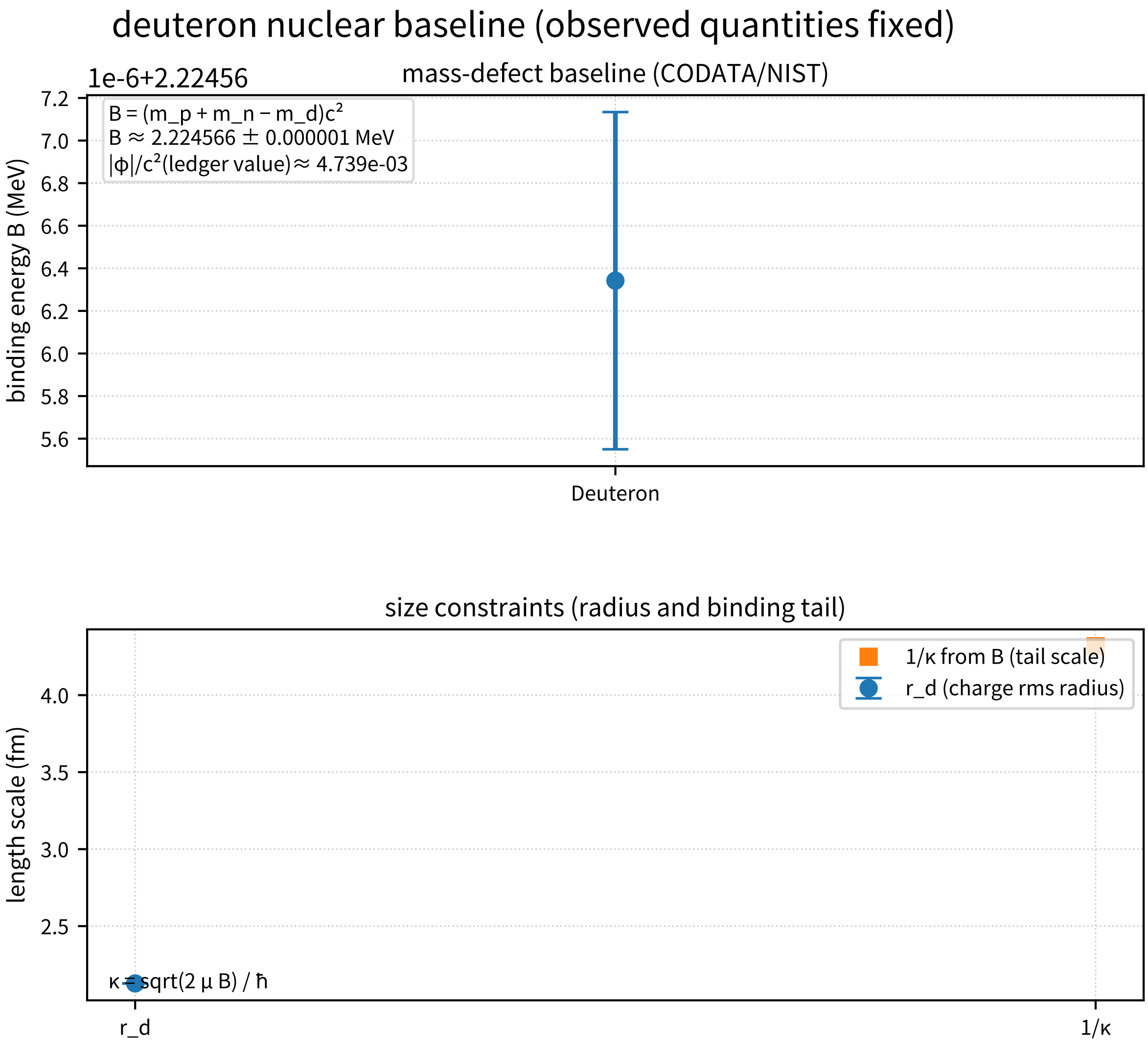

Figure 20: Shows the deuteron binding energy B and the positioning of the size constraint.

Supplement: confirms the minimum condition that the subsequent $u(r)$ ansatz must satisfy.

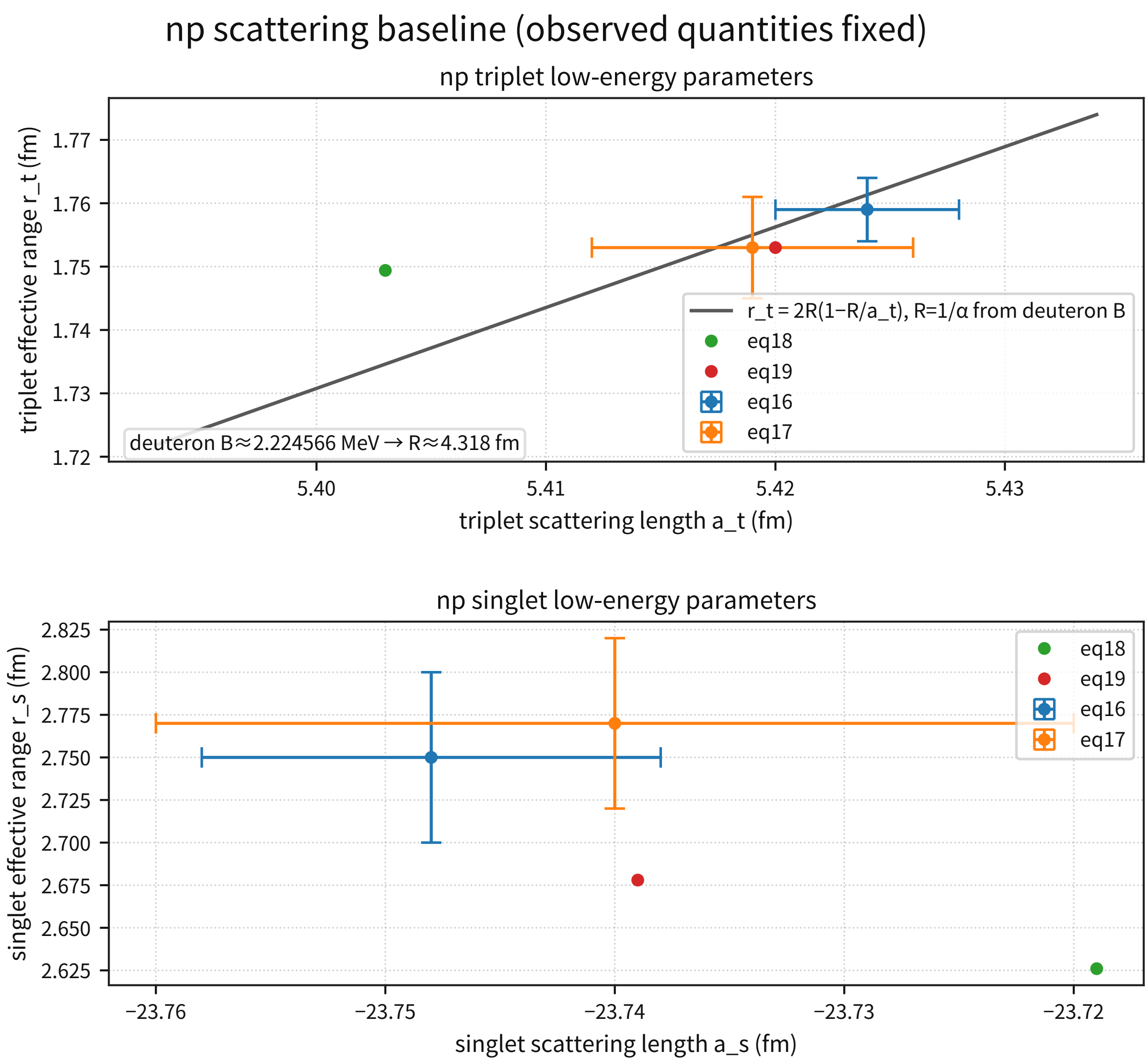

Figure 21: A baseline figure that freezes the low-energy np-scattering parameters (a_t, r_t, a_s, r_s) at first order.

Supplement: compares the eq18 / eq19 dataset systematics on the same input/output interface.

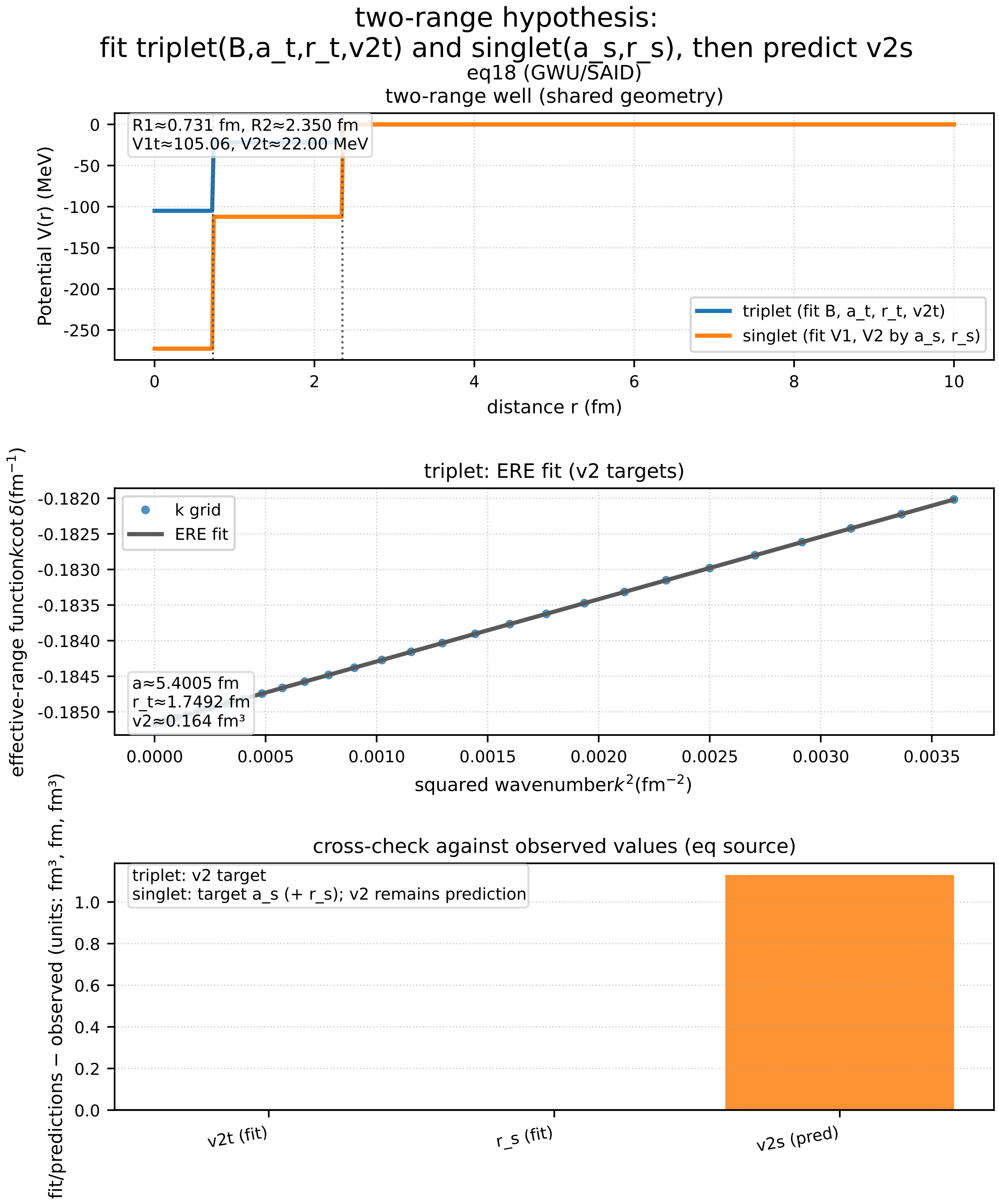

Figure 22: Under the 2-range hypothesis, constrains the singlet with a_s and r_s and evaluates $v_{2,s}$ as a differential prediction (eq18).

Supplement: visualizes that the sign mismatch in $v_{2,s}$ is the rejection point.

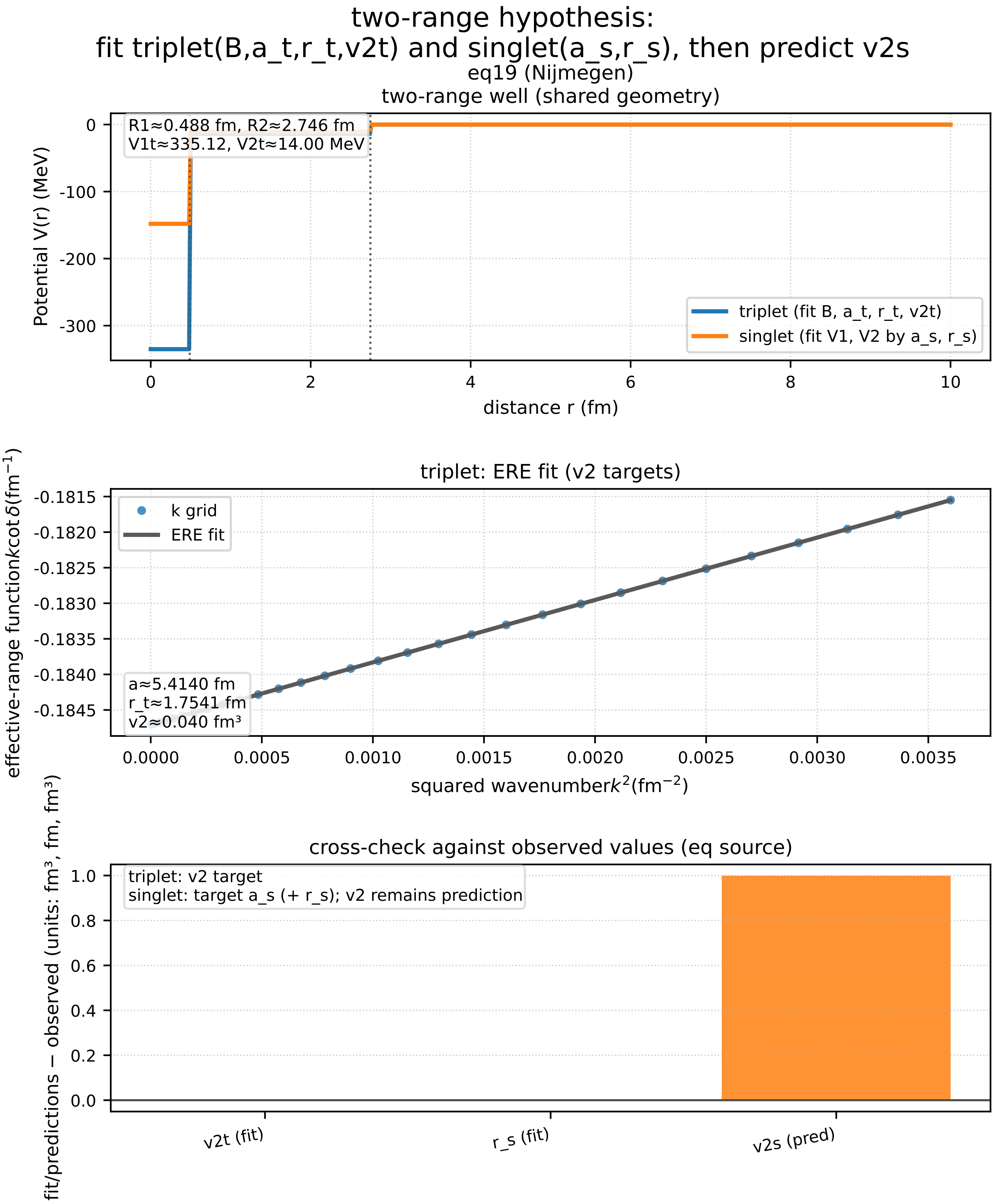

Figure 23: Under the 2-range hypothesis, constrains the singlet with a_s and r_s and evaluates $v_{2,s}$ as a differential prediction (eq19).

Supplement: detailed notes in the figure (geometry, ERE coefficients) are moved into the main text, and the operational rule is fixed as: fit the triplet with $v_{2,t}$, fit the singlet with a_s, r_s, then predict $v_{2,s}$.

λπ-constrained three-range (barrier+tail split, free tail depth) + (k,q) scan

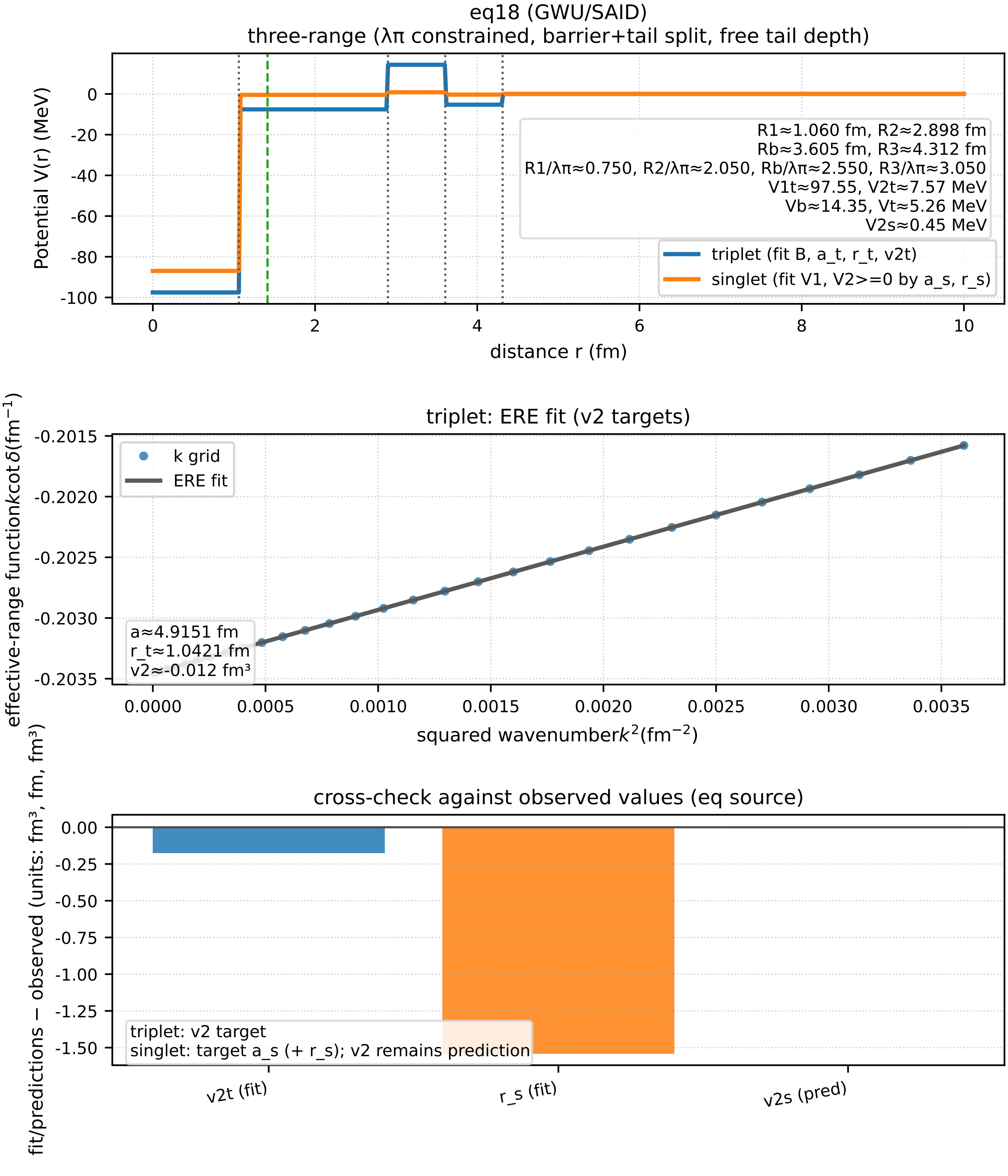

Figure 24: Shows an example in which barrier+tail under the λ_π constraint (global (k,q)) enters the $v_{2,s}$ envelope (eq18).

Supplement: used as the entry point for fixing the "hypothesis class that is not rejected under the minimal extension."

λπ-constrained three-range (barrier+tail split, free tail depth) + (k,q) scan

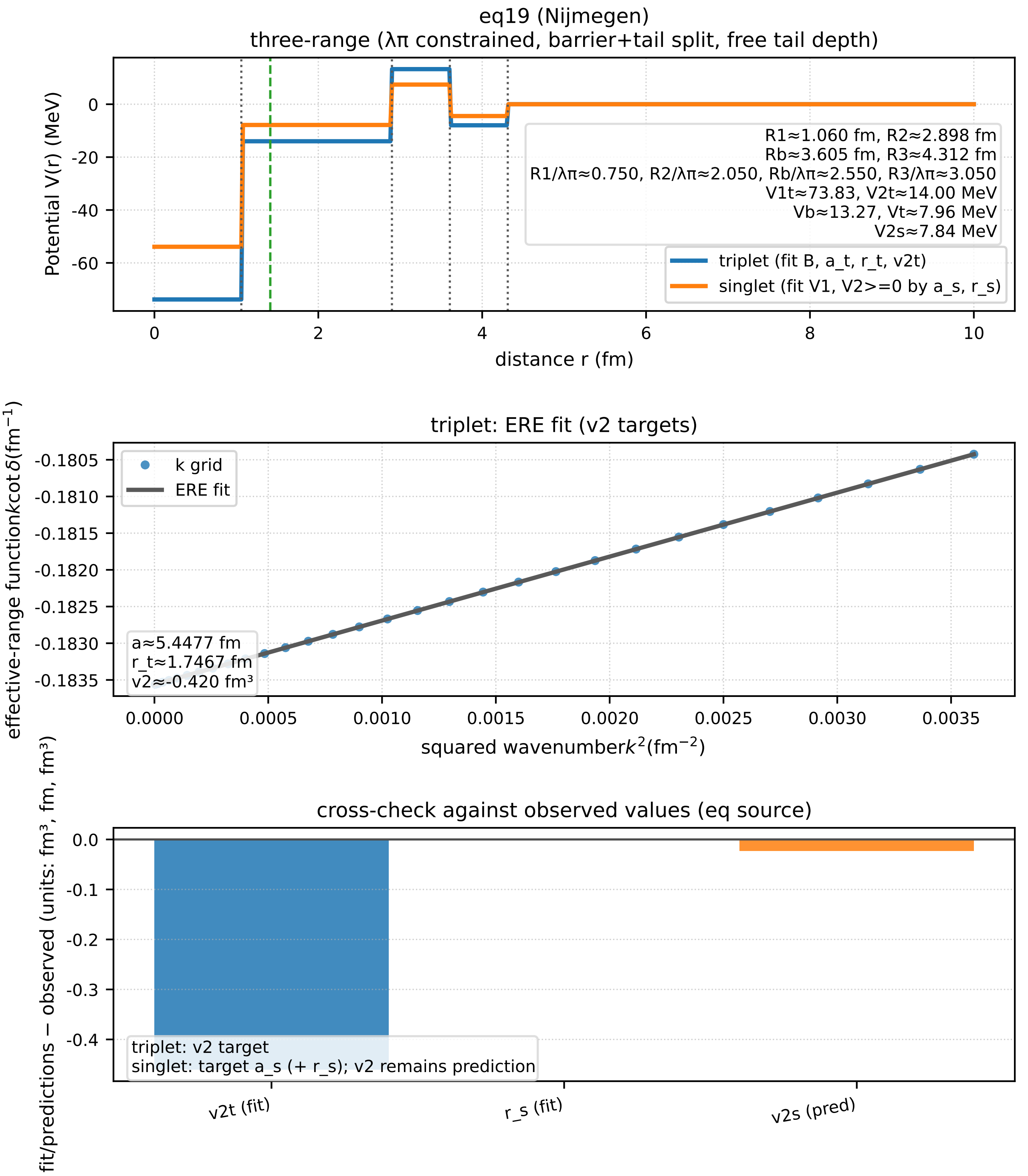

Figure 25: Shows an example in which barrier+tail under the λ_π constraint (global (k,q)) enters the $v_{2,s}$ envelope (eq19).

Supplement: notes formerly placed under the figure are moved into the main text, and only those global (k,q) choices that satisfy the simultaneous envelope condition for both triplet and singlet are adopted.

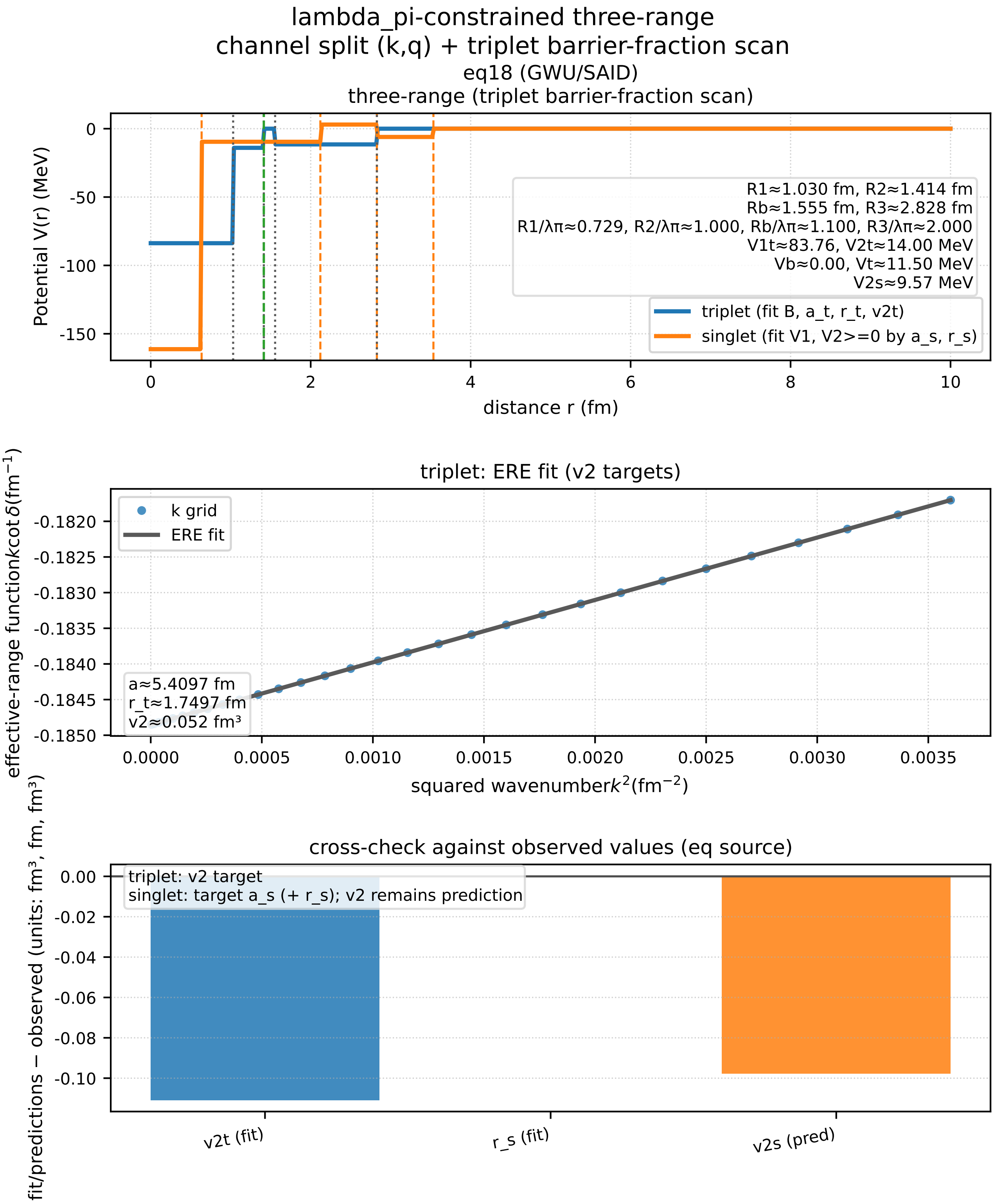

Figure 26: With the singlet geometric split frozen at $R_{1,s}/\lambda_\pi = 0.45$, $R_{2,s}/\lambda_\pi = 1.5$, scans the triplet barrier / tail split ratio and shows that the canonical solution ($barrier\_len\_fraction_t = 0.1$, $k_t = 0.0$) enters the envelope (eq18).

Supplement: fixes the reason for adopting the selected canonical solution (minimum additional freedom; simultaneous envelope for both datasets).

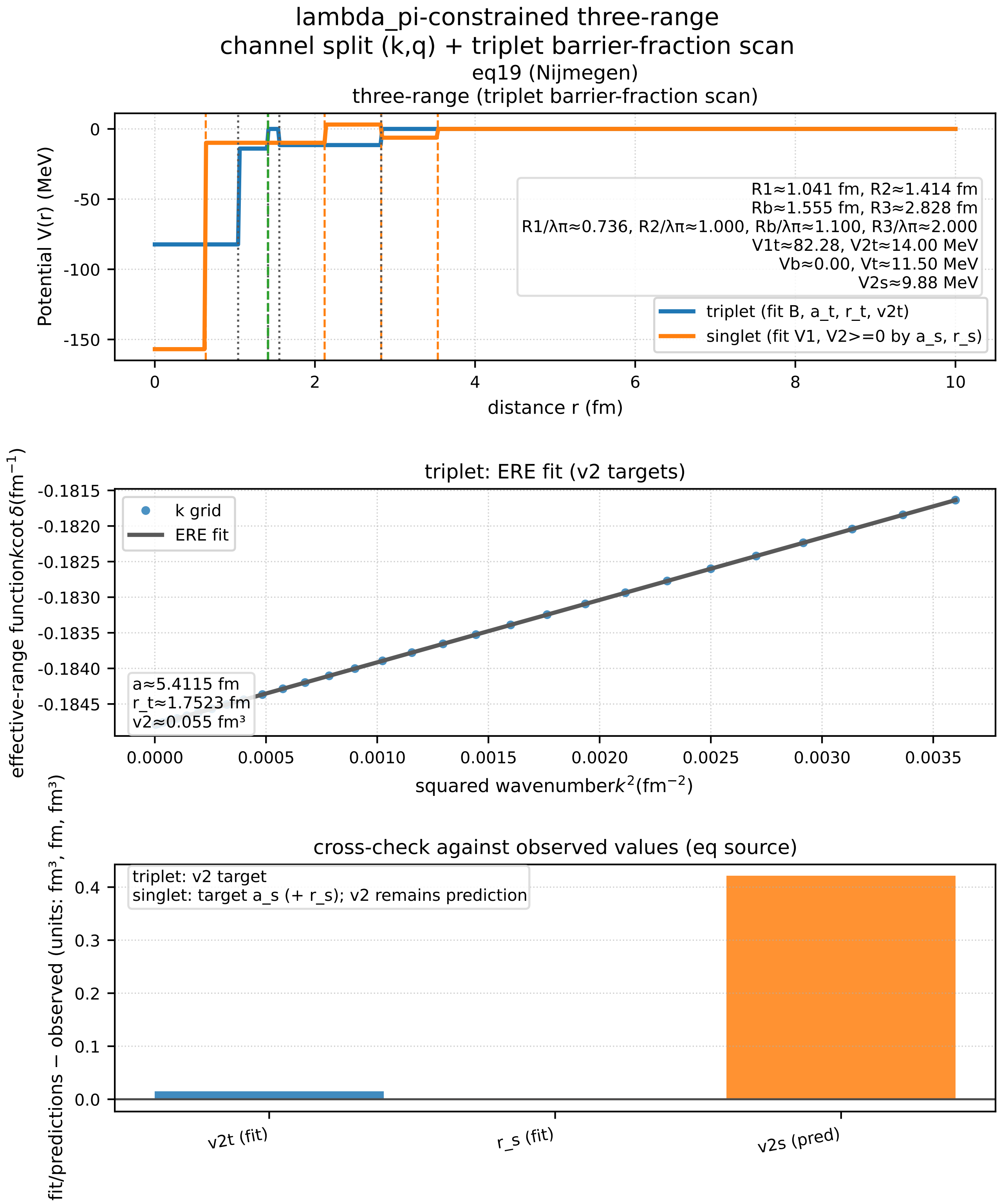

Figure 27: With the singlet geometric split frozen at $R_{1,s}/\lambda_\pi = 0.45$, $R_{2,s}/\lambda_\pi = 1.5$, scans the triplet barrier / tail split ratio and shows that the canonical solution ($barrier\_len\_fraction_t = 0.1$, $k_t = 0.0$) enters the envelope (eq19).

Supplement: notes formerly placed under the figure are moved into the main text, and the reason for adopting the selected canonical solution (minimum additional freedom; simultaneous envelope for both datasets) is fixed.

Discussion: At present, the minimal extension of the 2-range family still cannot rescue the sign of singlet $v_{2,s}$ and is therefore rejected. By contrast, barrier+tail under the λ_π constraint (global (k, q)) can bring singlet $v_{2,s}$ into the envelope, but additional degrees of freedom are required to satisfy the triplet side simultaneously. That remains a continuing task on the strong-interaction side (extension of the effective potential).

Note: This section does not claim a first-principles derivation of the nuclear force. Its purpose is to freeze the **effective constraint (effective model)** that connects the P-model variable $u = \ln(P/P_\infty)$ to nuclear observables (B, a, r, ...), and to distinguish between the ansatz classes that are rejected (the 2-range family) and those that are not rejected under the minimal extension (barrier+tail under the λ_π constraint; global (k,q)).

3.8.1 No-Go Organization for the Shell-Closure Kink of Charge Radii (Δ^2r)

Objective: Compare the shell-closure kink of charge radii directly through the observable Δ^2r, and fix (i) the no-go result for the base model (mean field + radius mapping), (ii) the robustness against series covariance (the statistical-processing side), and (iii) whether a minimal extension on the nuclear-structure side (pairing) can rescue it.

Input: IAEA charge radii (adopted values + σ), AME2020 (binding energies for OES / pairing), NNDC (β2; candidate deformation correction), and the fixed outputs of the nuclear module used in this paper (HF+3-body + radius mapping).

Processing: Evaluate Δ^2r on both the Sn and Sp axes, and fix the strict judgment $\max\left|\Delta^2 r_{\mathrm{pred}} - \Delta^2 r_{\mathrm{obs}}\right|/\sigma \leq 3$ as the gate. For robustness, introduce AR(1) correlation ρ and recompute σ through $\sigma^2 = a^T \Sigma a$ with $a = [1, -2, 1]$, in order to judge whether the issue can be rescued purely by assumptions on statistical processing. As a rescue candidate, introduce the pairing correction $r \to r + k_n \Delta_n + k_p \Delta_p$, freeze the coefficients k_n and k_p by weighted least squares over nonmagic centers ($A >= 40$), and then reevaluate the kink set.

Equation:

$$\Delta^2 r(x) = r(x+2) - 2r(x) + r(x-2) \tag{4}$$

AR(1) covariance (minimal model):

$$\sigma^2_{\Delta^2 r} = a^T \Sigma a, \quad a = [1, -2, 1], \quad \Sigma_{ij} = \rho^{|i-j|} \sigma_i \sigma_j.$$

Metric: For the strict set with $A_{\min} = 100$ (even-even only, center-magic only), $\max|\Delta^2 r_{\mathrm{pred}} - \Delta^2 r_{\mathrm{obs}}|/\sigma_{\Delta^2 r}$ on both the Sn and Sp axes.

Results:

At $A_{\min} = 100$ (independent σ, ρ=0), the worst base-case values are **Sn=3.075σ, Sp=3.723σ**, so the no-go remains. The conclusion of the AR(1) covariance sweep is that "rescue requires anticorrelation"; ρ>=0 does not rescue the case:

ρ (AR(1))	max Sn (σ)	max Sp (σ)	Judgment
0.00 (independent)	3.075	3.723	no-go
-0.40 (closest pass)	2.432	2.964	pass
pass range	$\rho\in$[-0.90, -0.40]	—	anticorrelation required

If pairing is introduced explicitly into the radius as the minimal nuclear-structure extension (without

$\beta 2$ correction), the model reaches a strict pass:

Model ($A_{\min} = 100$)	max Sn (σ)	max Sp (σ)	Judgment
base (def0; no pairing)	3.075	3.723	no-go
deformation only (def1; $\beta 2$)	5.534	9.296	no-go (worse)
pairing (def0_pair_fit; $k_n \approx 0.0116$, $k_p \approx 0.00872$)	0.688	2.014	strict pass
pairing + deformation (def1_pair_fit)	3.185	7.171	no-go

Discussion: To rescue the case through assumptions on statistical processing (covariance) alone, neighboring points would need to be anticorrelated (ρ<=-0.40). For ρ>=0, which is closer to common-mode systematics, σ shrinks and the residuals grow, so rescue does not occur. Accordingly, the no-go at $A_{\min} = 100$ is organized not as a fluctuation of data processing but as a conditional limitation that calls for additional nuclear-structure physics (pairing).

Note: The "rescue" in this section does not refit at the kink points themselves; it uses the coefficients k_n and k_p frozen on nonmagic centers. Accordingly, pairing is not an escape through extra fitting, but an extension that explicitly fixes an additional degree of freedom in nuclear structure.

Physical Basis of the Pairing Correction: Because of the pairing interaction in nuclei, even nuclei are more strongly bound than odd nuclei, and odd-even staggering (OES) appears in mass differences, separation energies, radii, and related observables. Constructing Δ_n, Δ_p from the AME2020 binding energies by the three-point formula is a standard nuclear-physics method for estimating the local pairing gap from data, and the correction introduced here is therefore not a P-model-specific assumption but an explicit introduction of a known degree of freedom in nuclear structure.[12]

The radius r is related approximately to the nuclear density ρ through $A \approx \int \rho d^3 r \approx (4\pi/3) r^3 \rho$, so at fixed A a small variation gives $dr/r \approx -(d\rho)/(3\rho)$. Since pairing can affect the density distribution (and therefore the radius) through occupancy smoothing and surface polarization, the form $r \to r + k_n \Delta_n + k_p \Delta_p$, in which changes in the local gap indicators Δ_n, Δ_p contribute linearly to the radius, is physically natural as a minimal response model.

k_n, k_p are response coefficients representing the radius change per 1 MeV of pairing gap. Here, $k_n \approx 0.0116$ fm/MeV and $k_p \approx 0.00872$ fm/MeV are frozen by weighted LS from **data at nonmagic centers ($A >= 40$)** and are not refit at the kink points (near shell closure). The additional freedom is limited to two parameters (k_n, k_p), and the $\beta 2$ geometric correction is not adopted as the baseline result because it fails under the same freezing procedure. This avoids the arbitrariness of "escaping by extra fitting."

Accordingly, the pairing correction is not a modification of the core mapping of Part I, but is positioned as a minimal extension that explicitly inserts additional nuclear-structure physics.

Adoption (nuclear module of this manuscript; frozen):

- For the evaluation of Δ^2r (shell-closure kink), adopt the pairing correction (def0), i.e. $r \to r + k_n \Delta_n + k_p \Delta_p$. The coefficients k_n and k_p are frozen by fit_protocol (weighted LS over nonmagic centers with $A >= 40$), and are not refit at the kink points in order to avoid overfitting.
- The geometric deformation correction using $\beta 2$ (def1) fails under the same freezing procedure and is therefore **not adopted** as the baseline result of this paper (if adopted, it must be treated as a separate hypothesis with its own independently frozen falsification conditions).

Falsification condition (fixed; Δ^2r shell-closure kink):

- If, for the adopted model (pairing only; def0), the strict set with $A_{\min} = 100$ fails to satisfy $\max\left|\Delta^2 r_{\text{pred}} - \Delta^2 r_{\text{obs}}\right| / \sigma_{\text{obs}} \leq 3$ on both the Sn and Sp axes, then this nuclear module (including its additional freedom and freezing procedure) is rejected.

Output: Strict-decision metrics for the radius kink (Δ^2r) ($A_{\min} = 100$; minimal pairing). (The list of files used for reproduction is collected in Section 8.)

3.8.2 Fixing the Minimal Additional Physics (ν Saturation) in the $\Delta\omega \to$B.E. Mapping

Purpose: Without changing the falsification conditions frozen in the previous step (3σ operation), fix how the decision (pass/fail) changes when minimal additional physics is introduced into the distance I/F (local spacing d).

Input: Differential predictions for the distance I/F (local spacing d; prediction/observation/residual by nuclide), and the rejection conditions (3σ operation; threshold definitions).

Output (for audit):

- A nuclide-by-nuclide prediction/observation/residual table (ledger of the differential predictions).
- The 3σ-operation rejection-condition pack (threshold definitions plus application conditions; the list of files used for reproduction is collected in Section 8).

Processing: Keep the distance I/F as d, but replace the connectivity indicator from $\nu_{\text{base}} = 2(A-1)/A$ to $\nu_{\text{eff}} = \min(\nu_{\text{base}}, \nu_{\text{sat}})$. In this manuscript, $\nu_{\text{sat}} = 1.5$ is frozen, and $C_{\text{eff}} = (\nu_{\text{eff}} \cdot A)/2$ is used.

Equations:

$$\nu_{\text{base}} = \frac{2(A-1)}{A}, \quad \nu_{\text{eff}} = \min(\nu_{\text{base}}, \nu_{\text{sat}}), \quad C_{\text{eff}} = \frac{\nu_{\text{eff}} A}{2}. \tag{5}$$

Metrics: $|z_{\text{median}}| \leq 3$ and $|z_{\Delta\text{median}}| \leq 3$ (the frozen thresholds for 3σ operation).

Results:

Model	med log10(Bp/Bo)	z_{med}	$z_{\Delta\text{med}}$	Decision
local spacing d	0.1557	5.63	2.39	median fail
$d + \nu$ saturation	0.0345	1.31	2.21	pass

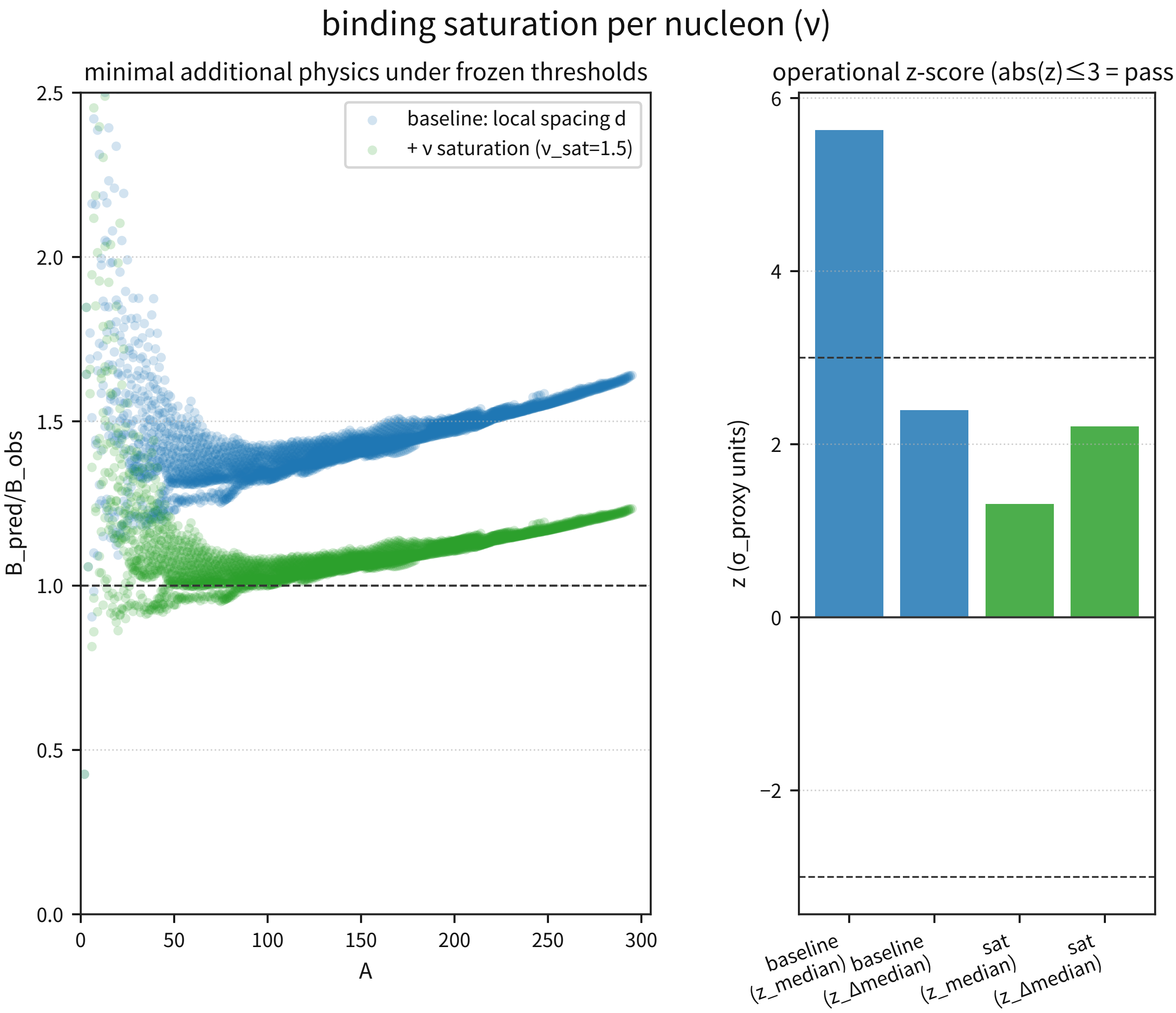

Figure 28: The left panel compares the residuals of the baseline (d only) against $d + \nu$ saturation and shows that the median offset is improved most strongly.

Supplement: the right panel compares the z metrics while keeping the thresholds fixed and shows that z_{median} moves inside the 3σ range. Because the thresholds themselves were not changed, the improvement in the decision is attributable to minimal additional physics on the I/F side.

Discussion: This result suggests that the global offset that dominated at large A depended more strongly on the non-saturation of the connectivity indicator than on a modification of the distance I/F itself. Accordingly, the ν saturation introduced in this section is treated as a structural improvement that preserves the falsification conditions.

Notes: The ν_{eff} used here is not claimed to be a first-principles statement about the nucleon coordination number. It is a minimal indicator introduced to close the operational I/F of the $\Delta\omega \rightarrow$B.E. mapping. Pairing and shell effects continue to be audited separately as secondary structure under their own falsification conditions.

3.8.3 Differential Predictions vs Existing Theories and Required Precision

Purpose: Compare the P-model mapping fixed in the previous subsection with surrogate indicators of standard theory over the same AME2020 nuclide set, and numerically fix which nuclide groups are most likely to resolve the differential prediction.

Input: The prediction/residual table of the P-model mapping including the ν saturation introduced above, and the 3σ-operation rejection conditions (thresholds held fixed).

Output (for audit):

- A prediction/observation/residual table for all nuclides (AME2020).
- The falsification-condition pack (3σ-operation thresholds, precision gate, and application conditions).

(The list of files used for reproduction is collected in Section 8.)

Processing: Using the P-model (local spacing $d + \nu_{\rm sat}$) as the baseline, compare it under the same I/F with the Yukawa-distance surrogate (global R) and the SEMF (fixed coefficients). Next, compute $\Delta B = B_{\rm pred}^{\rm Pmodel} - B_{\rm pred}^{\rm standard}$ for each nuclide, and fix $\sigma_{\rm req} = |\Delta B|/3$ as the precision required for a 3σ resolution.

Equations:

$$\Delta B = B_{\rm pred}^{\rm P} - B_{\rm pred}^{\rm std}, \qquad \sigma_{{\rm req},3\sigma} = \frac{|\Delta B|}{3}, \qquad \sigma_{\rm req,rel} = \frac{\sigma_{{\rm req},3\sigma}}{B_{\rm obs}}.$$

Metrics: $z_{\rm median}$ and $z_{\Delta\rm median}$ (reusing the 3σ-operation thresholds), together with median $|\Delta B|$ and median $\sigma_{\rm req,rel}$.

Results:

Model	med log10(Bp/Bo)	$z_{\rm med}$	$z_{\Delta\rm med}$	med $\lvert\Delta B\rvert$ (MeV)	Decision
P-model (local+d, ν saturation)	0.0345	1.31	2.21	76.73	pass
Yukawa-distance surrogate (global R)	-1.1933	-3.22	-2.61	1072.50	median fail
SEMF (fixed coefficients)	0.0025	1.51	1.19	6.69	pass

Differential channel	median $\lvert\Delta B\rvert$ (MeV)	p84 (MeV)	max (MeV)	median $\sigma_{\rm req,rel}$
P-SEMF	75.15	230.77	472.66	0.0270
P-Yukawa surrogate	1150.06	1866.18	2532.92	0.3343

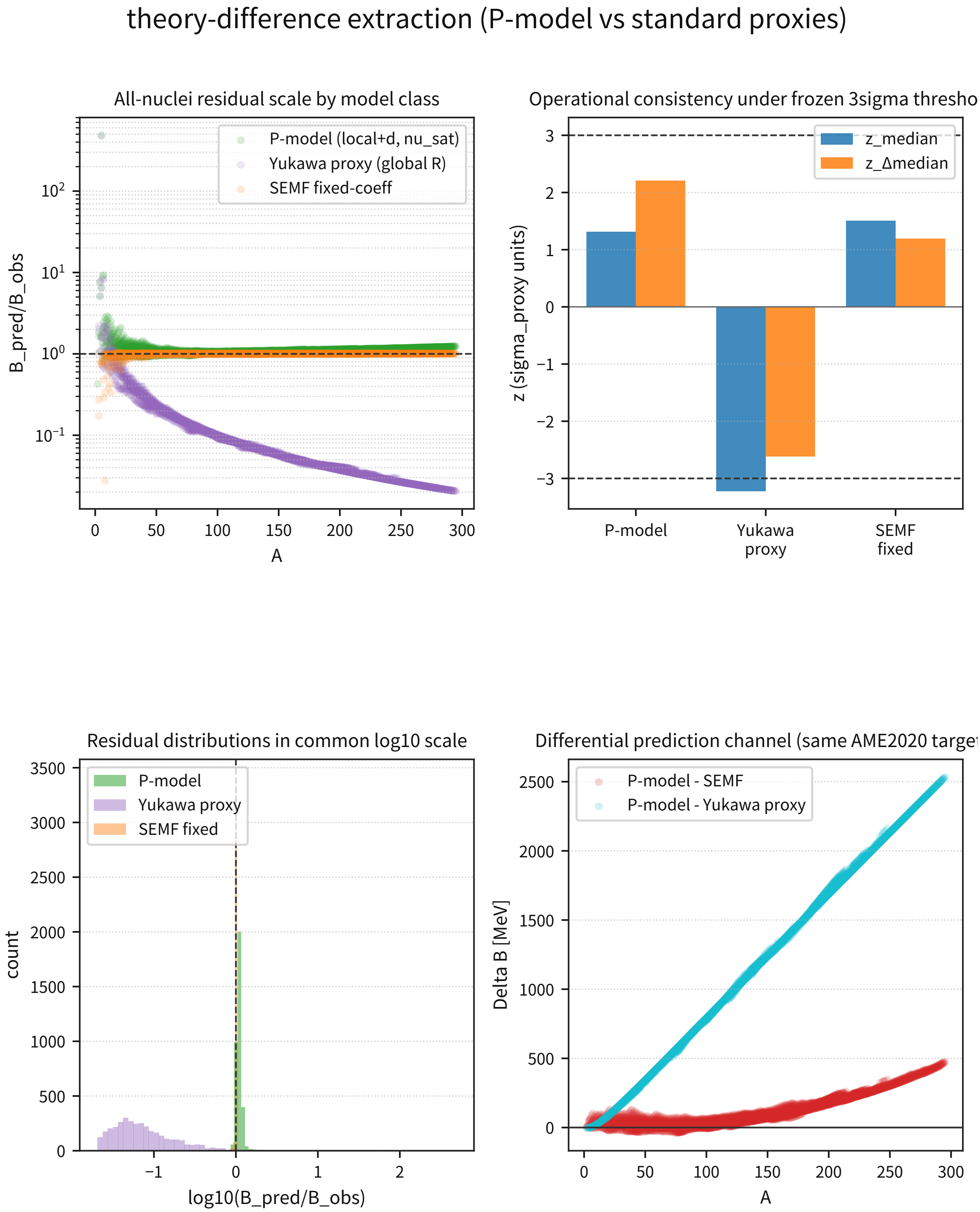

Figure 29: The upper panel compares the distribution of $B_{\mathrm{pred}}/B_{\mathrm{obs}}$ and the 3σ-operation metrics, showing that the P-model and the SEMF remain within the operational thresholds, whereas the Yukawa-distance surrogate falls outside the threshold at the median.

Supplement: the lower panel shows the A dependence of the differential ΔB and fixes the fact that the difference is amplified on the heavy nuclide side.

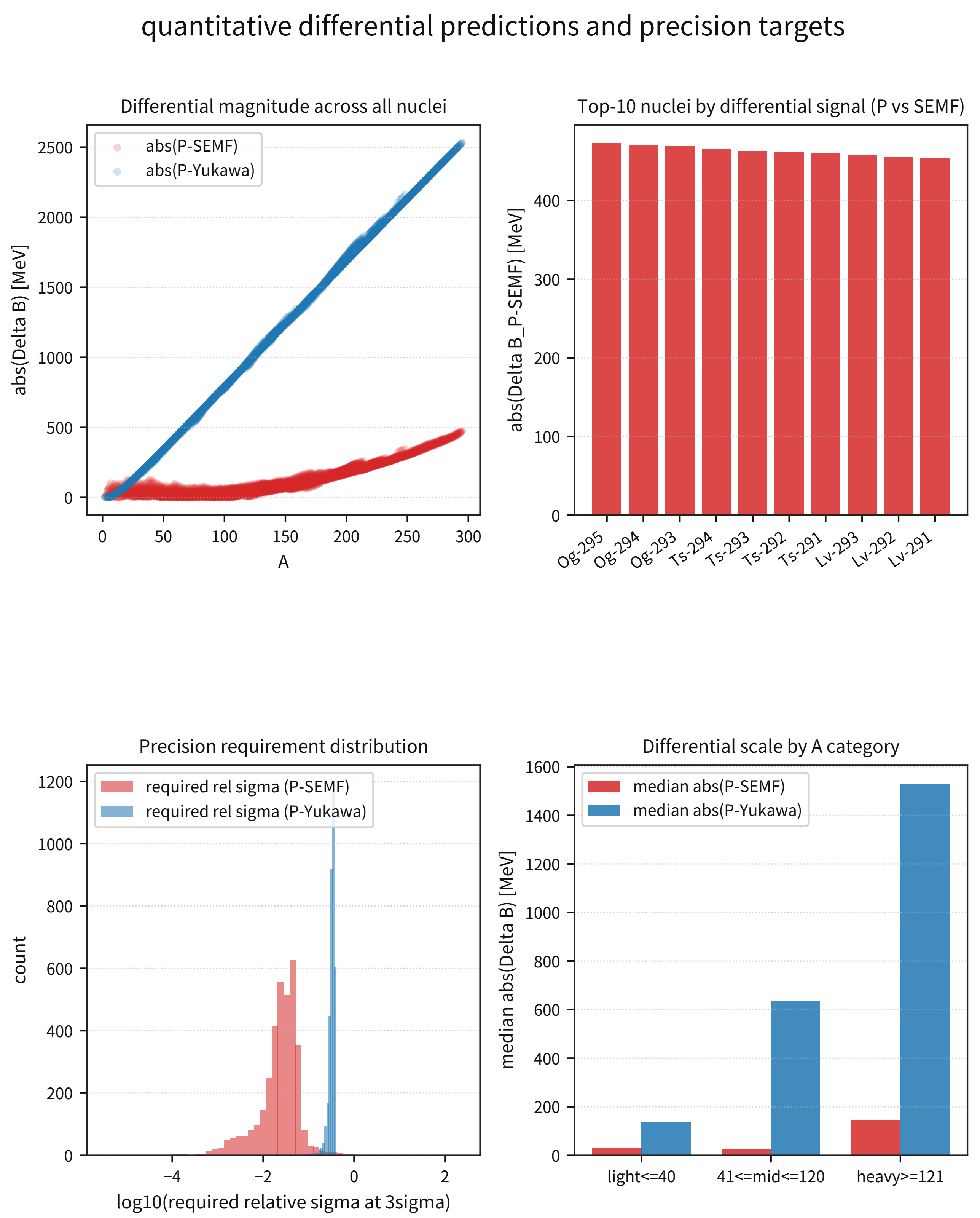

Figure 30: This figure displays simultaneously the nuclide distribution of $|\Delta B|$, the top-ranked nuclides, and the distribution of the required relative precision, thereby fixing the priority nuclides for the differential test as an explicit list and connecting them to the subsequent addition of independent observables (such as separation energies and radii).

Supplement: in the heavy-nuclide group, median $\sigma_{\text{req,rel}}$ for P-SEMF is about 3.4%.

On the binding-energy residual map in the Z-N plane, each nuclide is placed at (Z, N) and colored by $log10(B_{\text{pred}}/B_{\text{obs}})$. Residuals are relatively large near the vertical and horizontal lines of the magic numbers $(N, Z = 2, 8, 20, 28, 50, 82, 126)$, and the fraction of outliers with $|z| > 3$ in the nuclide-by-nuclide residual metric (the robust z of $log10(B_{\text{pred}}/B_{\text{obs}})$) is $21/298 = 7.0\%$ in the magic-number group, higher than $59/3256 = 1.8\%$ in the nonmagic group. This indicates that shell closure cannot be absorbed by a smooth distance mapping alone, and that shell/correlation terms become dominant. By contrast, in the deformed-nucleus region (the subset where observed β_2 is available, with $|\beta_2| \geq 0.20$ and nonmagic), the median of the same metric drops to median $|z| \approx 0.52$ (0.66 for the full nonmagic sample), and the outlier fraction remains at $5/237 = 2.1\%$. Accordingly, the operational P-model mapping of this manuscript can be fixed in words as follows: it captures the bulk binding relatively well for deformed nuclei, whereas near the magic numbers shell-closure effects become the main source of the residuals.

Summary (where z is the robust z of $log10(B_{\text{pred}}/B_{\text{obs}})$ and an outlier means $|z| > 3$):

Group	n	Outliers	Outlier fraction	median $\lvert z\rvert$
magic-number group (N or Z = 2, 8, 20, 28, 50, 82, 126)	298	21	7.0%	0.896
nonmagic	3256	59	1.8%	0.661
deformed nuclei (β_2-available subset; $\lvert\beta_2\rvert \geq 0.20$ and nonmagic)	237	5	2.1%	0.52

In the integrated differential-channel analysis of this manuscript, the above differential channels (P-SEMF / P-Yukawa) were incorporated into the falsification-condition pack. Operationally, only nuclide groups satisfying $\sigma_{\text{rel}}(total) <= \sigma_{\text{req}}, rel$ are adopted as targets of the 3σ decision.

Rejection Criterion: The accept/reject decision for the differential channels is governed by the conditions/thresholds defined in the falsification-condition pack (listed in Section 8); if a nuclide group that satisfies the precision gate exceeds the 3σ threshold, it is rejected.

Discussion: The differential signal in this section is larger for heavy nuclei than for light nuclei, thereby quantifying the operational question: "Which nuclides should be prioritized observationally to separate the P-model from standard surrogates as quickly as possible?"

Notes: The Yukawa/EFT comparison used here is an operational surrogate indicator and not a first-principles EFT calculation itself. The σ_{req} fixed here is treated as the baseline before the introduction of additional degrees of freedom.

3.8.4 Verification of the Wave-Interference Theory (Deuteron)

Purpose: Connect the $\Delta\omega \to B.E.$ mapping fixed in the nuclear mapping section of Part I to the observational I/F in the minimal two-nucleon system (the deuteron), thereby fixing the entry point for the next extension to He-4 and light-nucleus systematics.

Input: The observed $B_{\text{obs}} = 2.224566342 \pm 0.000000792\,\text{MeV}$ (CODATA/NIST), and the triplet np scattering parameters (eq18/eq19: a_t, r_t).

Processing: Solve for the bound-state pole $k = i\kappa$ in the ERE (truncated at second order), and reconstruct B_{pred}. Treat the difference between eq18 (GWU/SAID) and eq19 (Nijmegen) as an analysis-dependent surrogate for the systematic uncertainty, and decide according to whether B_{obs} lies within the predictive envelope (min-max).

Equations:

$$k \cot \delta_t = -\frac{1}{a_t} + \frac{r_t}{2} k^2, \quad \frac{r_t}{2}\kappa^2 - \kappa + \frac{1}{a_t} = 0, \quad B_{\text{pred}} = \frac{(\hbar c)^2 \kappa^2}{2\mu c^2}.$$

Metrics: $|B_{\text{pred}} - B_{\text{obs}}| / B_{\text{obs}}$, ΔB (keV), and whether B_{obs} lies within the envelope (pass within the envelope).

Results:

dataset	a_t (fm)	r_t (fm)	κ (fm^{-1})	B_{pred} (MeV)	ΔB (keV)	Relative error
eq18 (GWU/SAID)	5.403	1.7494	0.23227	2.23740	+12.831	0.577%
eq19 (Nijmegen)	5.420	1.7530	0.23146	2.22173	-2.832	0.127%

Summary metric	Value
B_{pred} envelope	[2.22173, 2.23740] MeV
B_{obs}	2.224566 MeV (inside envelope)
Systematic surrogate (half-range)	7.83 keV
$z_{\text{total,proxy}}$	0.64
Decision	pass within envelope

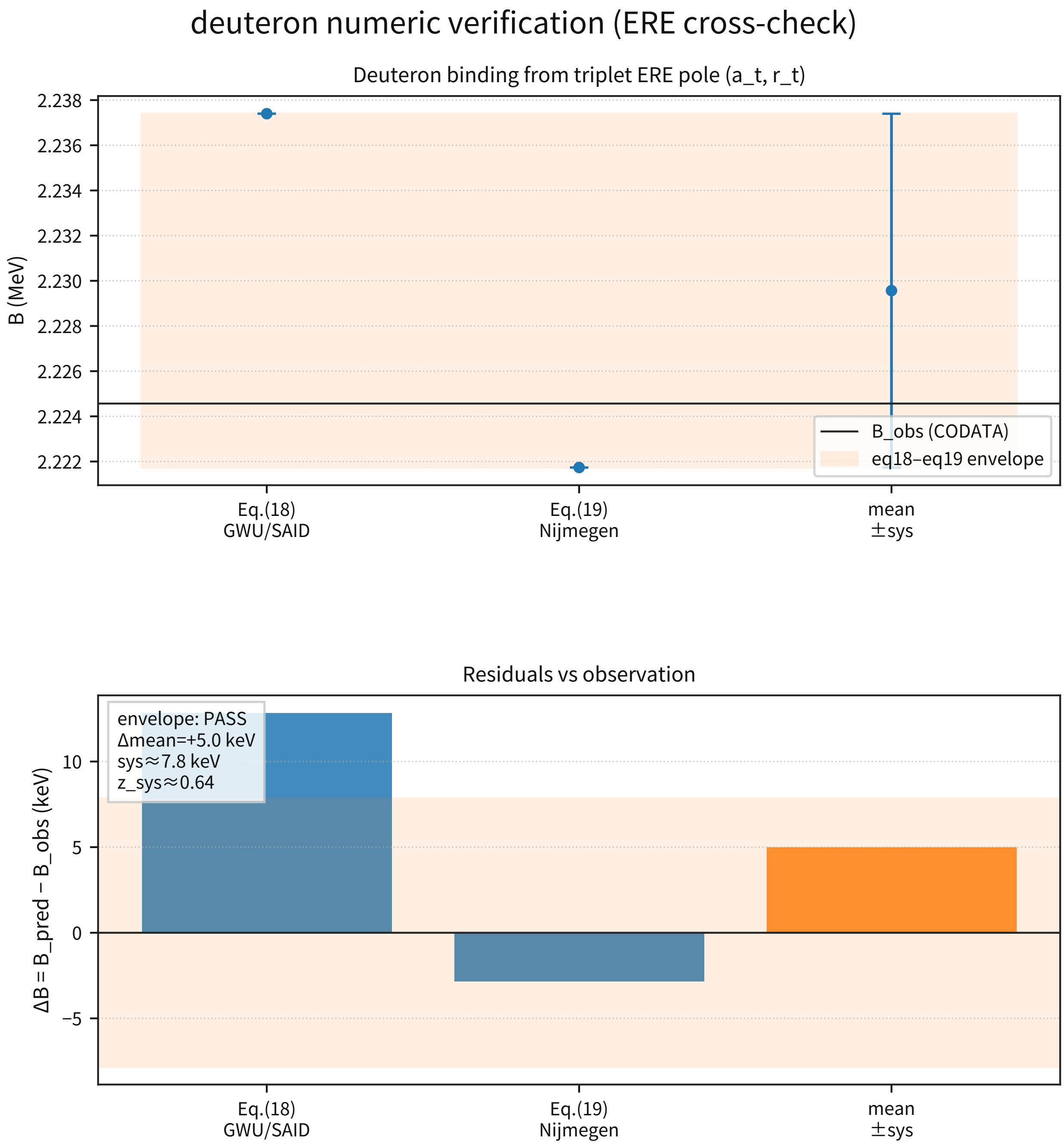

Figure 31: The left panel shows the differences in κ and B_{pred} obtained from eq18/eq19 and visualizes that B_{obs} lies within the predictive envelope.

Supplement: here the difference is dominated not by measurement statistics but by the difference in analysis procedure (eq18/eq19).

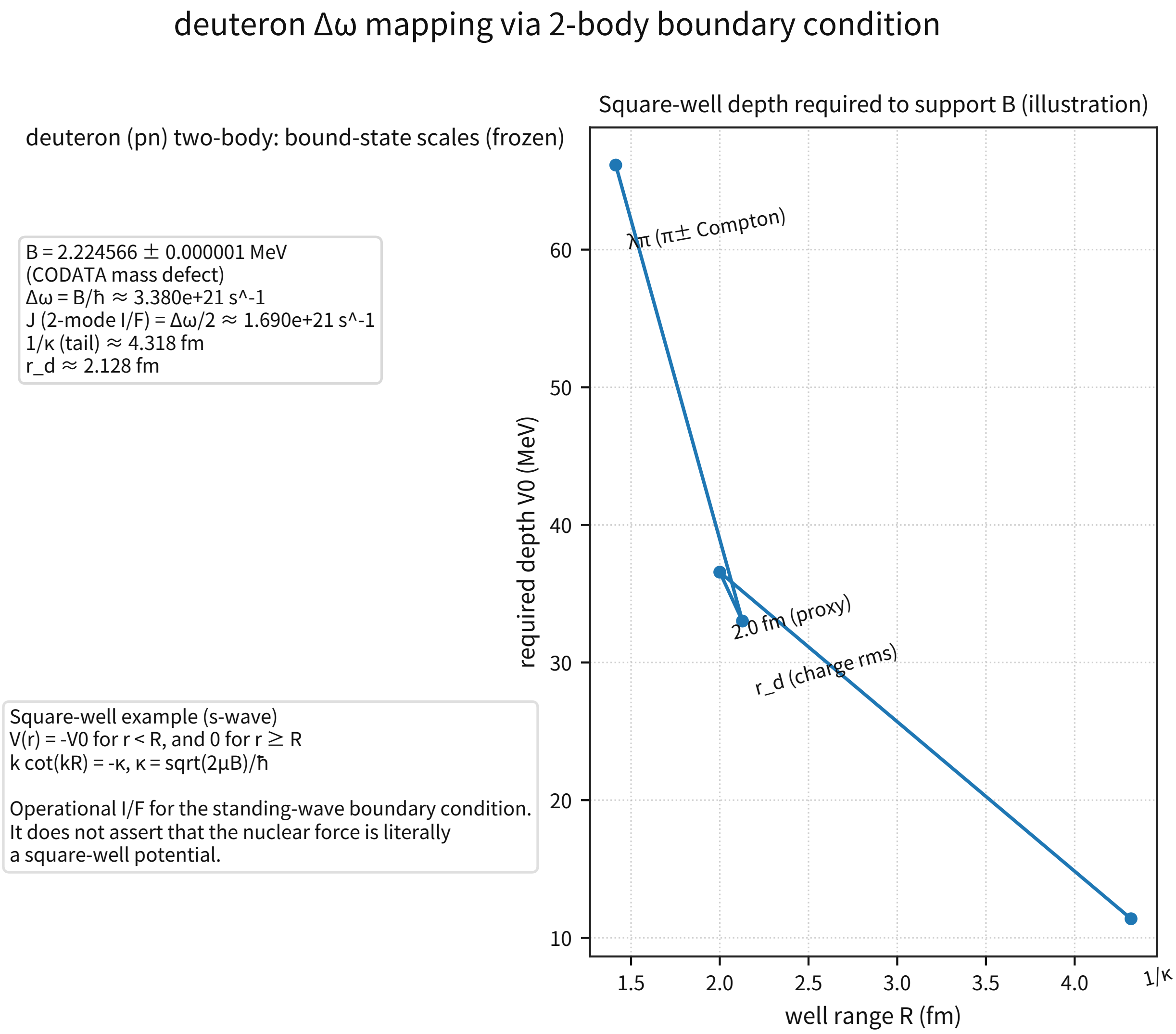

Figure 32: This figure shows the I/F of the two-body boundary condition $kcot(kR) = -\kappa$ for representative radius scales.

Supplement: for representative cases including $R = \lambda_\pi$ and $R = r_d$, it fixes the scale of the required well depth. It is used not as a first-principles derivation of the nuclear force, but as an operational boundary condition for the observational I/F.

Discussion: In the deuteron, the entry point of the $\Delta\omega \to B.E.$ mapping is consistent with the observable, and the analysis difference between eq18 and eq19 can be separated as a surrogate for the systematic uncertainty. Accordingly, the next step is to extend the same I/F to He-4 and the light-nucleus systematics.

Notes: This section is an operational verification using the ERE truncated at second order and does not claim equivalence to first-principles QCD. Higher-order terms such as $v_{2,t}$ are handled in the subsequent systematic evaluation.

Rejection Criterion: If $B_{\rm obs}$ does not lie within the predictive envelope (outside the envelope), this mapping (this ERE-truncated I/F) is rejected.

3.8.5 Verification of the Wave-Interference Theory (He-4)

Purpose: Extend the $\Delta\omega \to B.E.$ mapping fixed by the deuteron to He-4 and compare, under the same I/F, the different many-body reductions (collective / pair-wise).

Input: The observed $B_{\text{obs}} = 28.295610855 \pm 0.000001624$ MeV and $r_{\text{rms}} = 1.6785 \pm 0.0021$ fm, together with the deuteron-side quantities frozen earlier: $R_{\text{ref}} = 1/\kappa_d = 4.3177$ fm, $J_{\text{ref}} = B_d/2 = 1.1123$ MeV, and $L = \lambda_\pi = 1.4138$ fm.

Processing: Use $R_{\text{alpha}} = \sqrt{5/3}\, r_{\text{rms}}$ to evaluate $J_E(R_{\text{alpha}})$, and reconstruct the He-4 binding through $B_{\text{pred}} = 2CJ_E(R_{\text{alpha}})$. For C, compare collective$(A-1) = 3$, pn_only$(ZN) = 4$, and pairwise_all$(A(A-1)/2) = 6$.

Equations:

$$R_\alpha = \sqrt{\frac{5}{3}}\, r_{\text{rms}}, \quad J_E(R) = J_{\text{ref}} \exp\left(\frac{R_{\text{ref}} - R}{L}\right), \quad B_{\text{pred}} = 2CJ_E(R_\alpha), \quad \left(\frac{B}{A}\right)_{\text{pred}} = \frac{B_{\text{pred}}}{4}.$$

Metrics: $|B_{\text{pred}} - B_{\text{obs}}| / B_{\text{obs}}$, B/A, and the radius-uncertainty proxy $z_{r_\alpha,\text{only}}$.

Results:

Method	C	B_{pred} (MeV)	B_{pred}/A (MeV)	ΔB (MeV)	Relative difference
collective	3	30.5512	7.6378	+2.2556	+7.97%
pn_only	4	40.7349	10.1837	+12.4393	+43.96%
pairwise_all	6	61.1024	15.2756	+32.8068	+115.94%

Summary metric	Value
B_{obs}/A	7.0739 MeV
C_{required} $(B_{\text{obs}}/(2J_E))$	2.7785
Minimum-difference method	collective ($C = 3$)
Decision	pair-wise summation is too large; collective is closest

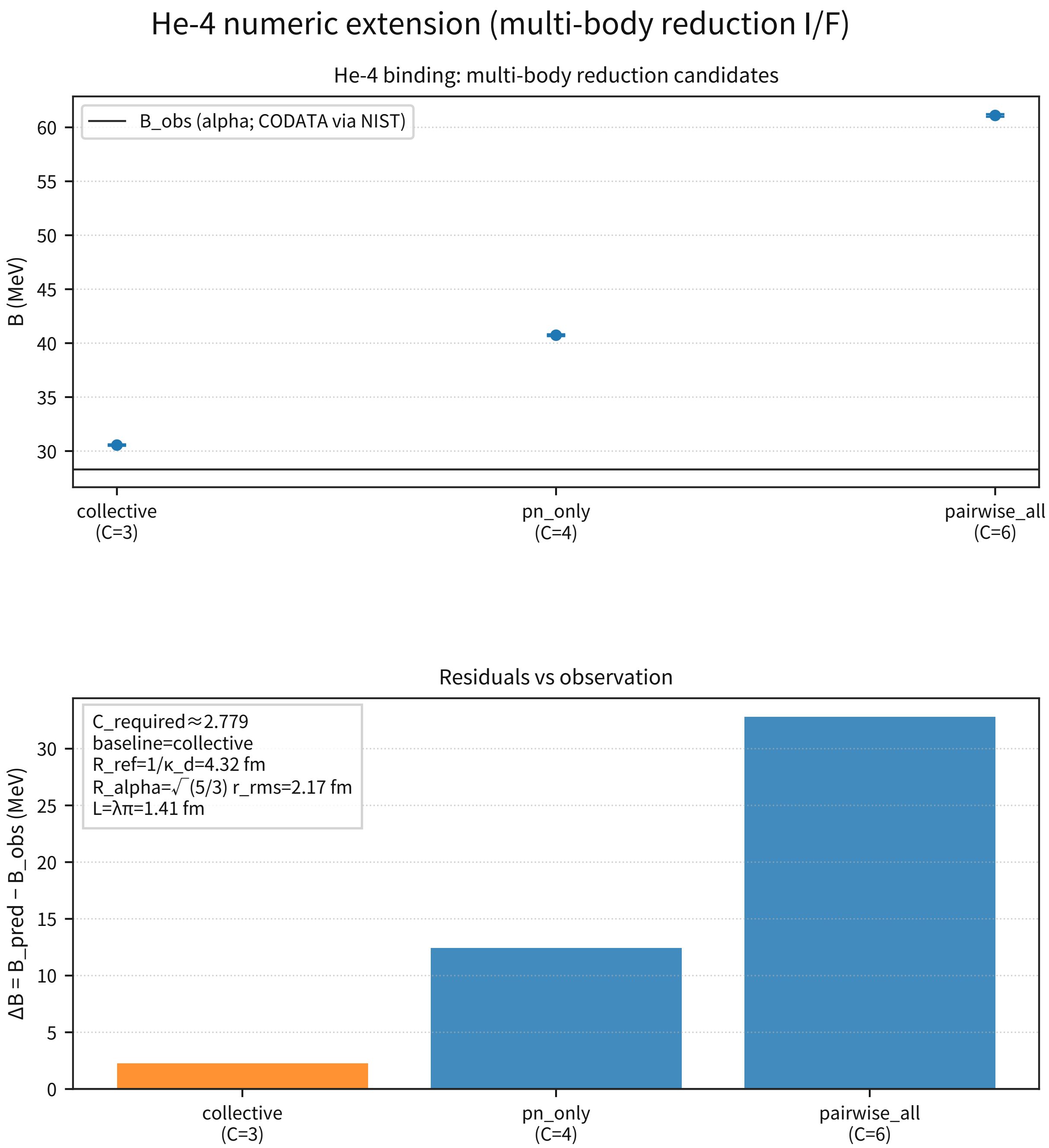

Figure 33: The left panel compares B_{obs} with B_{pred} from each reduction rule and shows that the pair-wise family systematically overpredicts.

Supplement: the right panel shows the relationship between C_{required} and the candidate values of C, where the collective choice gives the smallest residual.

Discussion: For He-4, $C_{\text{required}} \approx 2.78$, close to $A - 1 = 3$. This indicates that a collective reduction is more natural as an effective I/F than a simple pair-wise summation.

Notes: This conclusion does not mean that the collective reduction has been derived from first principles. This section is an extrapolation diagnostic of the mapping frozen by the deuteron, and the next step is to add a systematic audit over the light nuclei with $A = 2 - 16$.

Rejection Criterion: If the radius-only surrogate gives $|z_{r_\alpha,\text{only}}| > 3$, this extrapolation of the mapping is rejected.

3.8.6 Verification of the Wave-Interference Theory (Light-Nucleus Systematics; A=2-16)

Purpose: Extend the deuteron/He-4 checks to the light-nucleus systematics ($B.E./A$ vs A and the A dependence of the residuals), and fix at which mass-number range the same I/F begins to break down.

Input: Observed binding energies of the light nuclei (d/t/h/α), and prediction comparisons for representative nuclides (collective / pn_only / pairwise).

Output (for audit):

- A table of observed values for the light nuclei ($A = 2, 3, 4$), together with aggregated prediction comparisons for representative nuclides.

(The list of files used for reproduction is collected in Section 8.)

Processing: On the observation side, fix B_{obs}/A for $A = 2, 3, 4$. On the prediction side, compare $r_B = B_{\text{pred}}/B_{\text{obs}}$ and $\log_{10}(r_B)$ for $A = 2, 4, 12, 16$ under the collective rule ($C = A - 1$).

Metrics: B_{obs}/A, $r_B = B_{\text{pred}}/B_{\text{obs}}$, $\log_{10}(r_B)$, and ΔB.

Results:

Nuclide	A	B_{obs} (MeV)	B_{obs}/A (MeV)
d	2	2.2246	1.1123
t	3	8.4818	2.8273
h	3	7.7180	2.5727
alpha	4	28.2956	7.0739

A (collective)	$r_B = B_{\text{pred}}/B_{\text{obs}}$	$\log_{10}(r_B)$	ΔB (MeV)
2	1.0000	+0.0000	+0.0000
4	1.0827	+0.0345	+2.3393
12	0.5899	-0.2292	-37.7944
16	0.4714	-0.3267	-67.4655

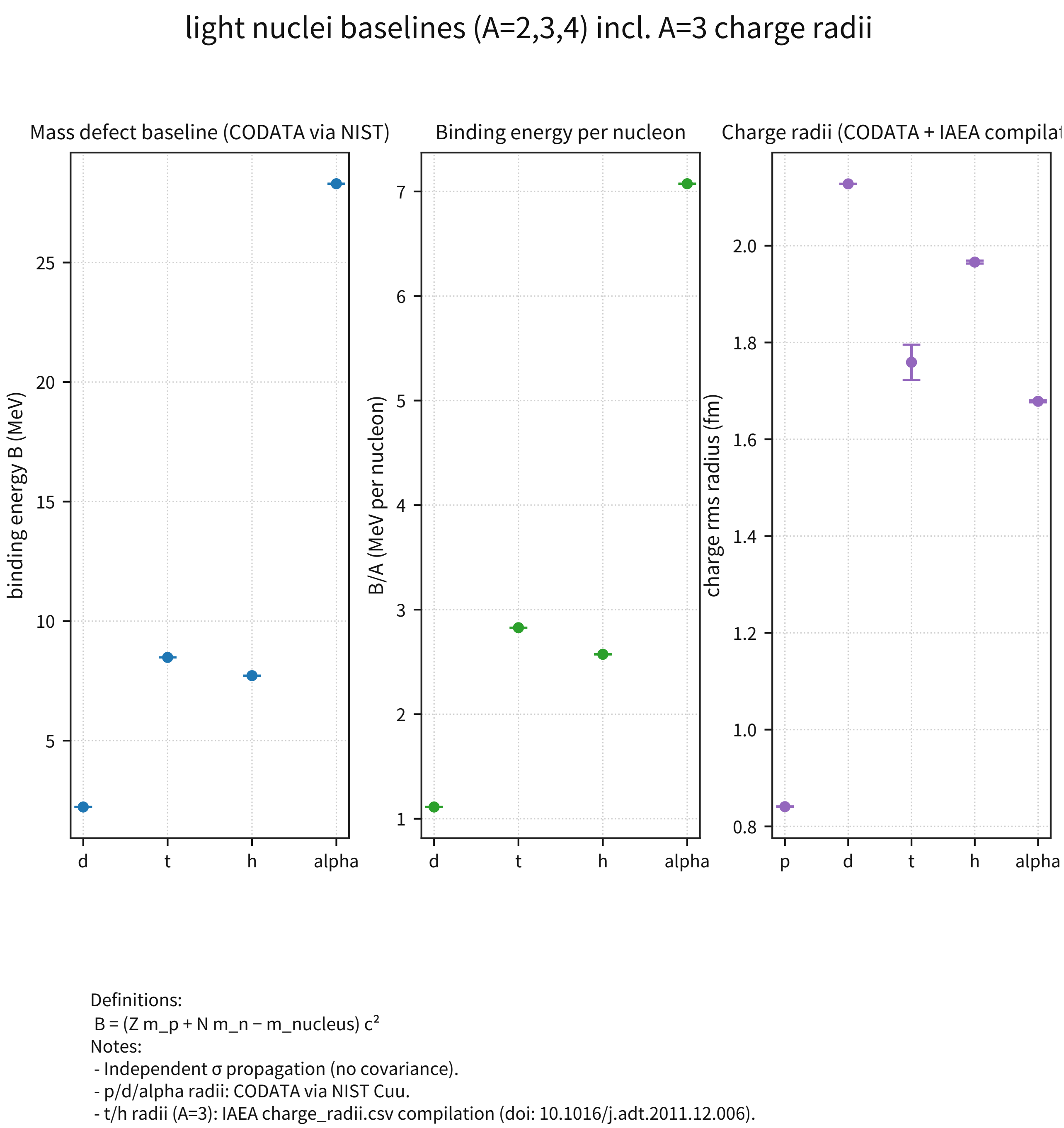

Figure 34: $A = 2 - 4$ fixes the monotonic observational baseline for $B.E./A$.

Supplement: the sharp rise toward He-4 marks a regime that is difficult to explain with a single two-body scale alone. This serves as a baseline for identifying which missing part of the I/F is dominant in the next light-nucleus systematic comparison.

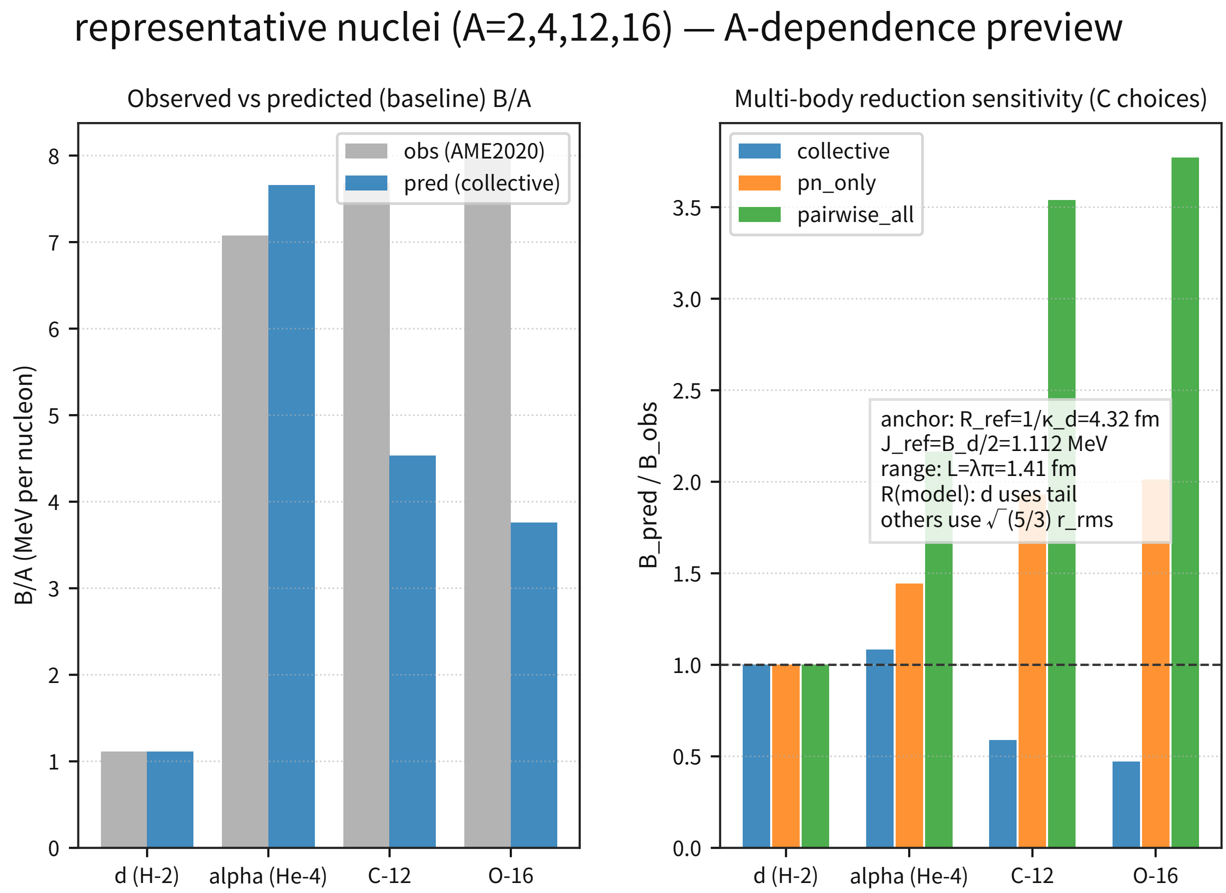

Figure 35: Under the same collective I/F, placing $A = 2, 4, 12, 16$ side by side shows an A trend that is slightly high at $A = 4$ but turns to underprediction at $A >= 12$.

Supplement: this sign reversal functions as an operational indicator that the A dependence of the distance surrogate or of the coordination-number mapping is still insufficient.

Discussion: On the light-nucleus side, the observed $B.E./A$ rises sharply. The collective I/F remains close up to around He-4 but flips to underprediction at $A = 12, 16$. Accordingly, at the present stage we do not adopt the claim that a single mapping explains all A, and instead continue the separated audit of A-dependent corrections (distance surrogate / coordination number / shell effects).

Notes: This section is an operational audit over representative points (2, 3, 4, 12, 16) within $A = 2{-}16$, i.e. the points reproducible with the currently fixed outputs. The all-nuclide evaluation is handled in the differential predictions and falsification conditions of Section 5.

Rejection Criterion: Including the extrapolation diagnostic at the representative points, if the frozen 3σ-operation conditions in the falsification-condition pack (listed in Section 8) are not satisfied, this minimal mapping is rejected (additional physics is required).

3.8.7 Figure/Table Pack (A-E)

Purpose: Fix, as Figures A-E, the minimal figure/table set that connects 4.8 (the theoretical entry point) to 5.4 (the differential prediction), and make explicit the path concept → levels → all-nuclide curve → residuals → differential decision.

Input: The frozen figures A-E (concept, level conditions, all-nuclide curve, residual audit, and differential prediction).

Processing: Regenerate the five figures above and fix their roles in 4.8 and 5.4 as Figure A (concept) → Figure B (levels/boundary) → Figure C (all-nuclide curve) → Figure D (residual audit) → Figure E (differential prediction).

Metrics: Check whether the figure roles are unique (A-E without duplication) and whether the reader can reach the falsification condition of 5.4 (σ_{req}) through a single route.

Results: Figure A fixes the minimal I/F of near-field interference ($\Delta\omega = 2J(R0)$, $B.E. = \hbar\Delta\omega$).

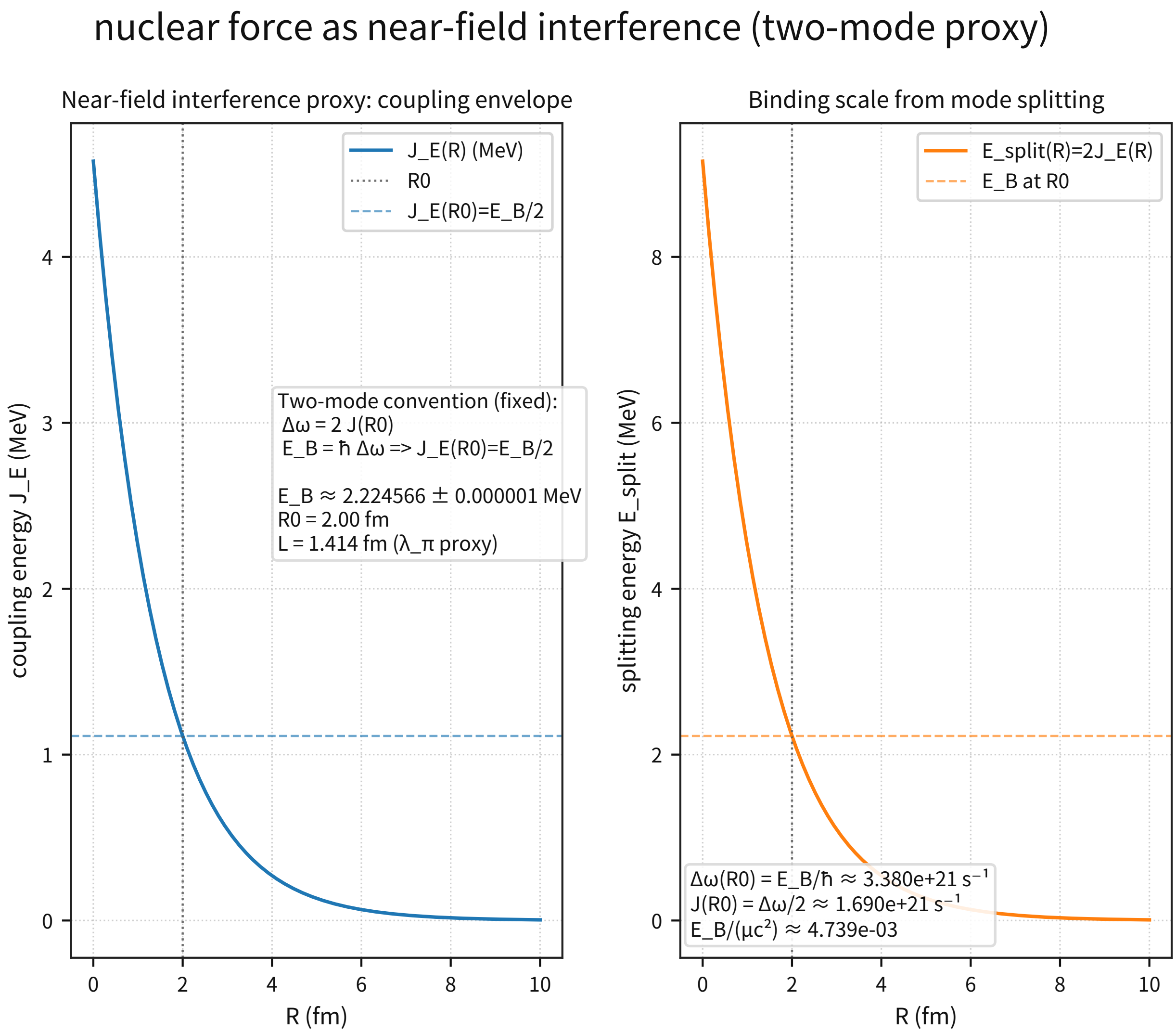

Figure 36: The left panel shows the envelope of $J_E(R)$ and the right panel the split energy $E_{\text{split}} = 2J_E$.

Supplement: it clarifies the entry point that treats nuclear binding through near-field overlap rather than a long-range field.

Figure B visualizes, for representative radii, the level condition satisfying the two-body boundary condition $kcot(kR) = -\kappa$. (This is the same figure as in 4.8.4, reproduced here to fix the route.)

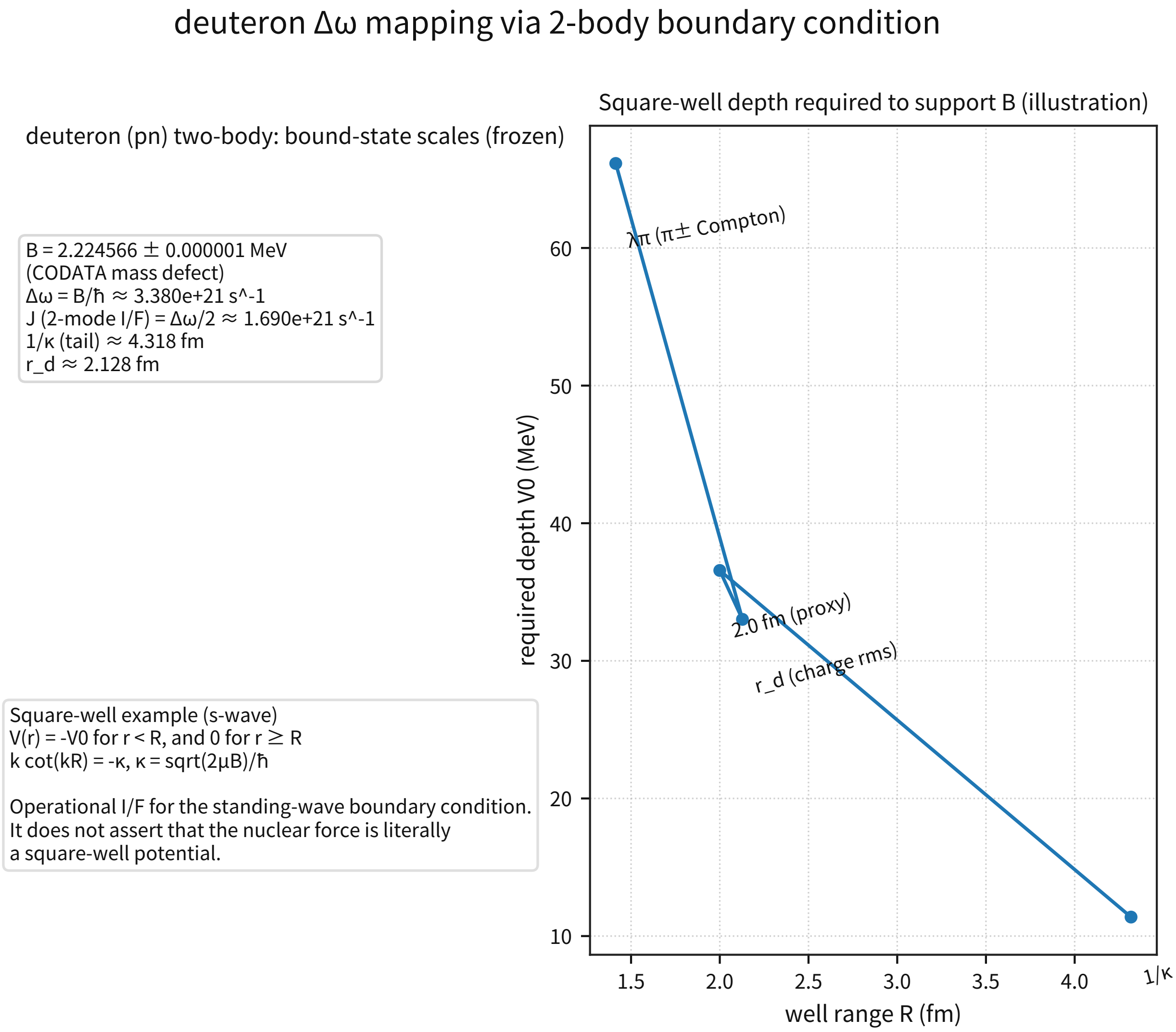

Figure 37: It shows how the scale of the required well depth changes for $R = \lambda_\pi$, $R = r_d$, and $R = 1/\kappa$, thereby fixing the operational I/F used for the deuteron (the two-nucleon system).

Figure C surveys the A dependence of $B_{\text{pred}}/B_{\text{obs}}$ over all AME2020 nuclides and shows the global trend of the baseline mapping.

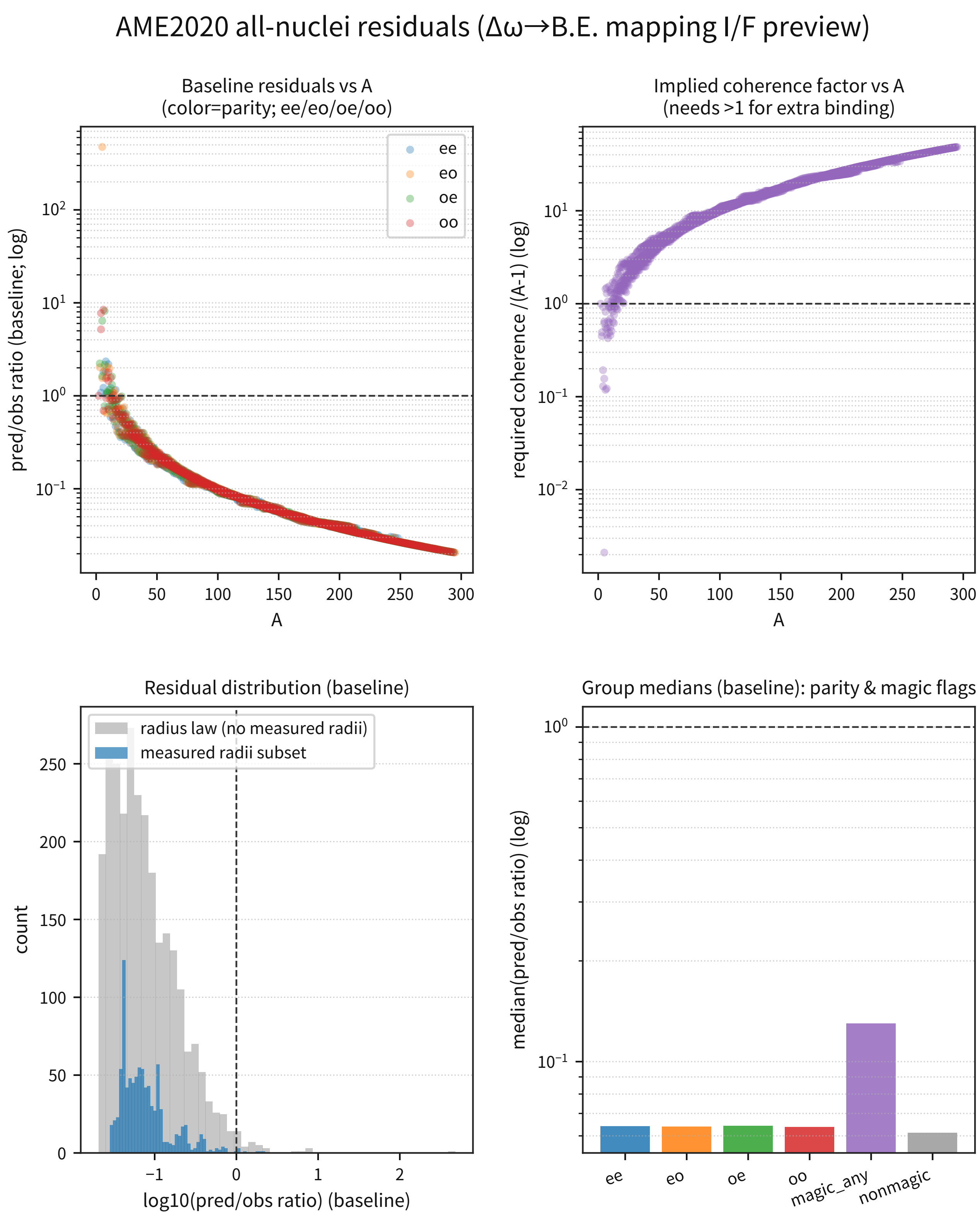

Figure 38: The distribution panel of the same figure fixes the initial state of the residuals, including the magic/parity groups.

Supplement: it is used as the baseline before introducing any additional physics.

Figure D compares the residuals obtained when the distance surrogate is switched between global R

and local spacing d.

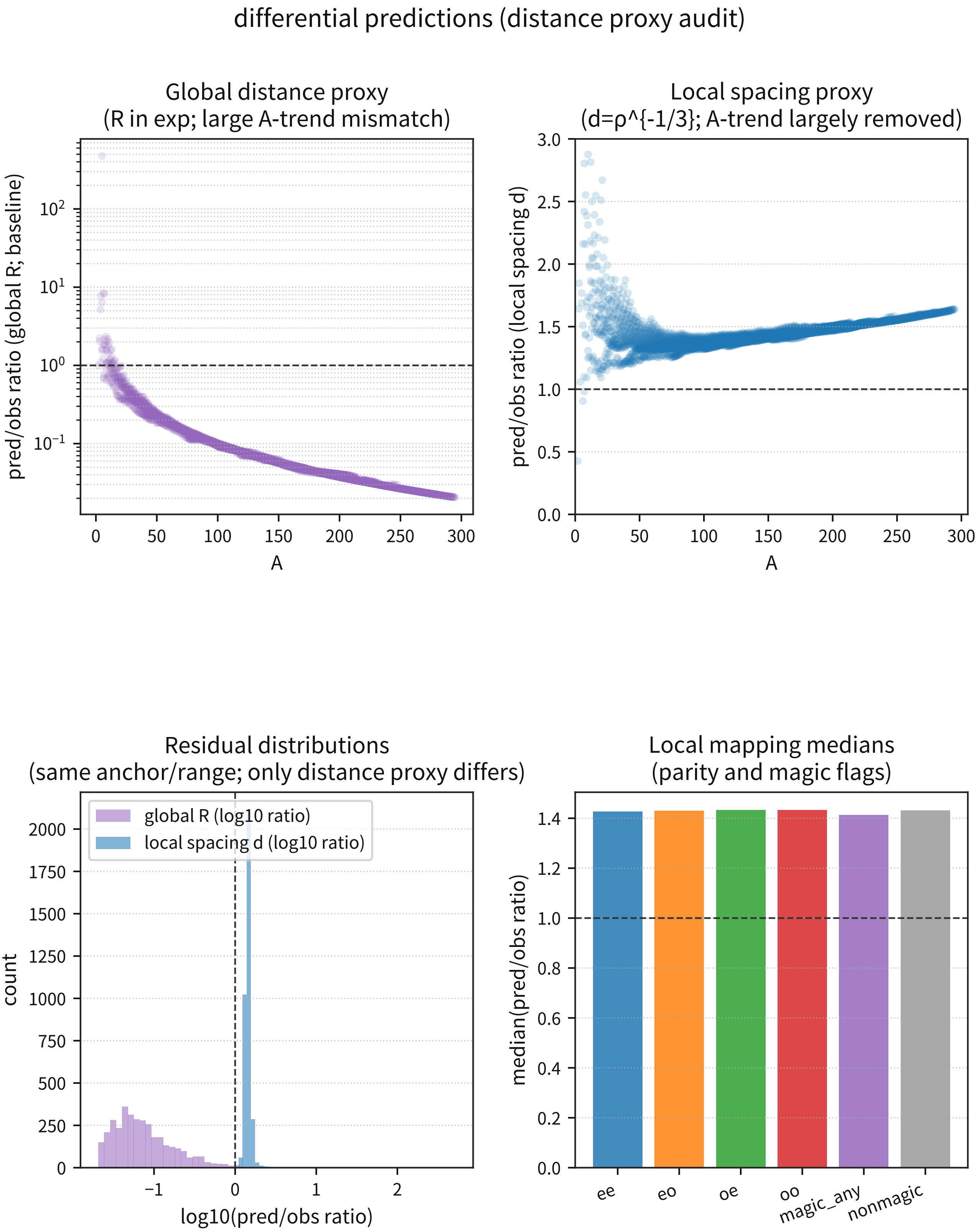

Figure 39: It directly audits how far the A trend is reduced by the local surrogate.

Supplement: it separates whether the main cause of the residuals lies in the ”distance I/F” or in ”secondary structure (pairing/shell).”

Figure E fixes, nuclide by nuclide, the differential signal from the standard surrogate, ΔB, together with $\sigma_{\text{req}} = |\Delta B|/3$ and σ_{req}, rel. (This is the same figure as in 4.8.1, reproduced here to fix the route.)

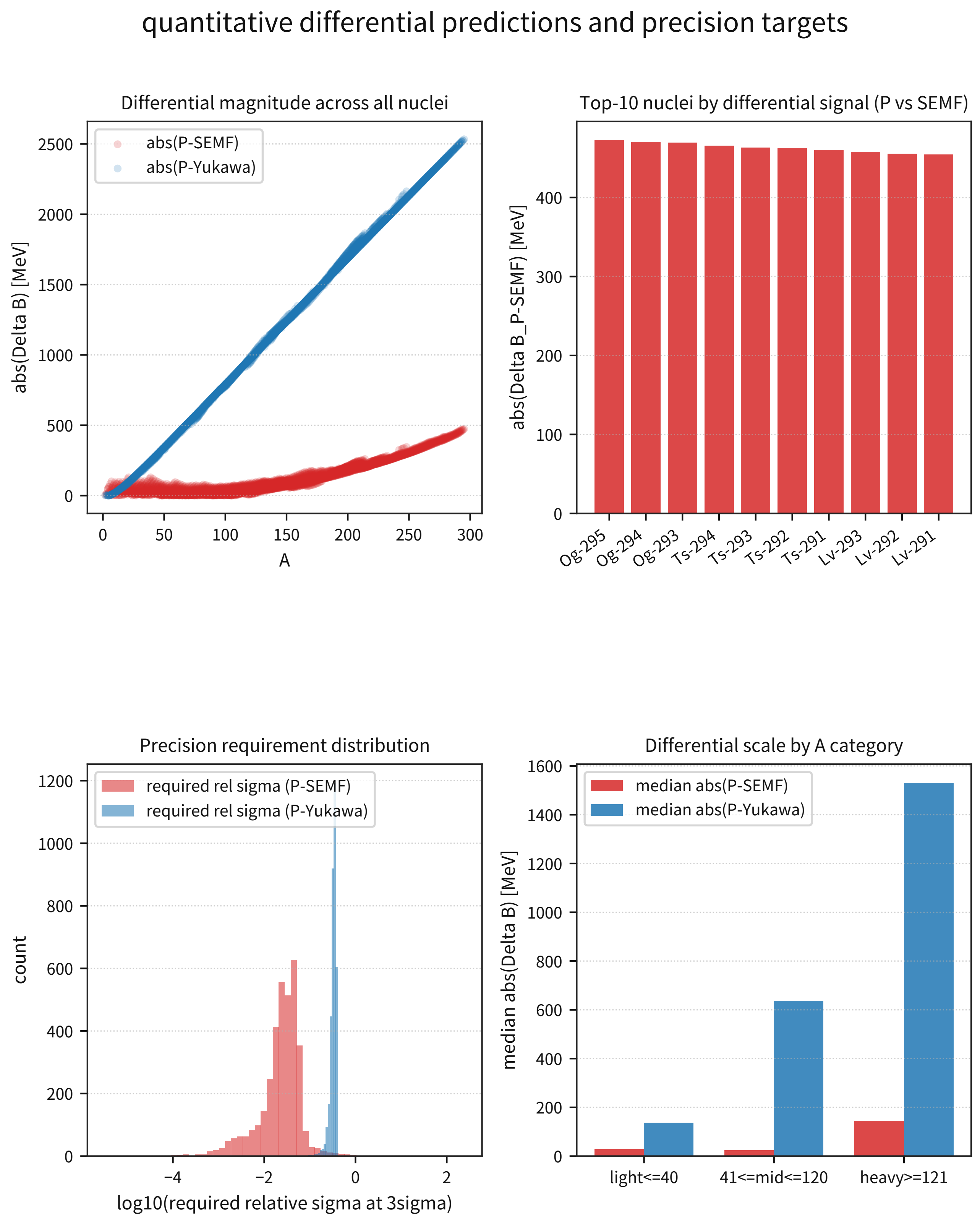

Figure 40: This figure is connected to the top-nuclide selection and the category thresholds (light/mid/heavy) of 5.4 and is used as an input figure to the falsification-condition pack.

Discussion: By fixing Figures A-E, the theoretical mapping of 4.9 and the falsification conditions of 5.4 become readable as a continuous line of a single I/F. This makes it possible to decide step by step whether additional degrees of freedom are needed, depending on where improvement appears in Figures C/D/E.

Notes: This pack is the minimal set for fixing the route and does not claim to replace first-principles QCD calculations. The accept/reject judgment is made using the 3σ gate of 5.4 and an independent bundle of cross-checks.

Rejection Criterion: The decision is governed by the 3σ gate of 5.4 ($\sigma_{\rm rel}(total) <= \sigma_{\rm req}, rel$) and by the falsification-condition pack (listed in Section 8).

Freeze: The $\Delta\omega \to B.E.$ mapping based on the distance I/F (local spacing $d + \nu$ saturation), the all-nuclide residual surface (Z-N residual map), and the differential predictions (ΔB, $\sigma_{\rm req}$) were fixed as the route A-E. **Reject**: If the same mapping does not satisfy the 3σ operational metrics (z) and the precision gate ($\sigma_{\rm rel}(total) <= \sigma_{\rm req}, rel$), this minimal mapping is rejected, and shell closure/correlation (magic) and pairing must be audited separately as additional degrees of freedom. **Next**: Without waiting for public release, extend the same I/F to test how far the dominant residuals remaining near the magic numbers (the outlier fraction) can be reduced with minimal pairing/shell terms while preserving the contrast against nonmagic (deformed) nuclei.

3.8.8 Vorticity of P_μ and Nuclear Spin Coupling

Purpose: Fix whether the residual triplet/singlet asymmetry left after 4.8.1-4.8.7 (in particular, the sign difference between $v_{2,t}$ and $v_{2,s}$) can be derived from the spin coupling generated by the spatial components P_i of P_μ, without inserting the sign "by hand." At the same time, constrain the strength of the microscopic nucleon coupling to have the same origin as the rotational indicator $\lambda_{\rm rot}$ of Part II 4.12, thereby prohibiting the introduction of excessive parameters.

Input: (i) the ghost-free action of P_μ fixed in Part I §2.7.1B, (ii) $\lambda_{\rm rot}$ frozen in Part II 4.12, and (iii) the operational thresholds of the nuclear I/F (a_t, r_t, $v_{2,t}$, a_s, r_s, $v_{2,s}$).

Processing: For the nucleon doublet $N = (p,n)^T$, write the minimal coupling of P_μ and the Pauli-type spin coupling with the same coefficient set. Then take the nonrelativistic limit and obtain the effective two-body potential $V = V_0 + V_{SS}\,\sigma_1\cdot\sigma_2 + V_T S_{12} + V_{LS}\,L\cdot S$. The triplet/singlet difference is then fixed automatically by the eigenvalues $\langle\sigma_1\cdot\sigma_2\rangle_{S=1} = +1$ and $\langle\sigma_1\cdot\sigma_2\rangle_{S=0} = -3$, together with the fact that S_{12} vanishes in the singlet, $\langle\,^1S_0|S_{12}|\,^1S_0\rangle = 0$.

Equations: Fix the interaction Lagrangian as

$$\mathcal{L}_{N,P} := -g_P\,\bar{N}\gamma^\mu N\,P_\mu - \frac{\lambda_{\rm rot}g_P}{4m_N}\,\bar{N}\sigma^{\mu\nu}N\,F^{(P)}_{\mu\nu}$$

without introducing any new independent constants. The effective two-nucleon potential is

$$V_{NP}(r) = V_0(r) + V_{SS}(r)\,\sigma_1\cdot\sigma_2 + V_T(r)\,S_{12} + V_{LS}(r)\,L\cdot S$$

with the nuclear kernel $W(r)$ written as

$$V_{SS}(r) = -\frac{2\lambda_{\rm rot}g_P^2}{3m_N^2}\nabla^2 W(r), \quad V_T(r) = -\frac{\lambda_{\rm rot}g_P^2}{m_N^2}\left(\partial_r^2 - \frac{1}{r}\partial_r\right)W(r),$$

$$V_{LS}(r) = -\frac{\lambda_{\rm rot}g_P^2}{2m_N^2 r}\partial_r W(r), \quad S_{12} := 3(\sigma_1\cdot\hat{r})(\sigma_2\cdot\hat{r}) - \sigma_1\cdot\sigma_2$$

where $W(r)$ is the nuclear kernel determined by the propagator of Part I §2.7.1B ($W \propto 1/r$ in the limit $m_P \to 0$).

Results: From the same Lagrangian, (i) on the triplet side, V_{SS} and V_T naturally act as effective attraction, while (ii) on the singlet side, S_{12} vanishes and $\sigma_1 \cdot \sigma_2 = -3$ increases the repulsive contribution. Thus the asymmetry emerges automatically. Accordingly, the sign difference between $v_{2,t}$ and $v_{2,s}$ is fixed not by inserting signs channel by channel, but by the eigenvalue difference of the spin projections.

Macroscopic connection (cross-scale freeze of the rotational coupling coefficient): Fix the spin-coupling strength used on the nuclear side as

$$g_{\text{spin}} \equiv \lambda_{\text{rot}} g_P$$

and do not change it from the λ_{rot} of Part II 4.12 (GP-B/LAGEOS). Therefore frame dragging in rotating astronomical bodies and nuclear spin coupling are connected as scale-separated manifestations of the same equation. This connection is an operational freeze based on the cross-scale-freeze hypothesis (ignoring running), while the introduction of energy-scale dependence is deferred to a future, separate audit.

Falsification Conditions:

1. Spin-sign gate: reject if the same λ_{rot} cannot reproduce both $\text{sign}(v_{2,t}) > 0$ and $\text{sign}(v_{2,s}) < 0$.
2. No-new-free-coupling gate: do not introduce any free coupling on the nuclear side other than λ_{rot}.
3. Macro-micro bridging gate: reject if the optimal nuclear-side value $\lambda_{\text{rot}}^{\text{nuc}}$ deviates by more than 3σ from the frozen GP-B/LAGEOS range.

3.9 Material Properties (Atoms, Molecules, and Condensed Systems)

In this section, 4.9.1-4.9.6 are grouped into the single class of "material properties." From atomic and molecular spectroscopic baseline values to the lattice constant, heat capacity, resistivity, and thermal expansion of condensed systems (Si), all targets are fixed under the same audit I/F (freeze criteria plus rejection conditions).

3.9.1 Atoms and Molecules (Atomic Spectroscopic Baselines: H I, He I; Molecular Constants: H2/HD/D2)

Purpose: We do not claim that atomic and molecular binding or level structure has been derived from the P-model from first principles. Instead, we freeze the spectroscopic **baseline values (targets)** as primary reference points and connect them to future rederivations and rejection decisions.

Input: Spectroscopic baseline values for atoms and molecules (primary sources).

- Atoms: NIST ASD (H I / He I) and the NIST AtSpec handbook (H I 21 cm hyperfine).[16][17]
- Molecular constants: NIST WebBook (Huber & Herzberg compilation; H2/HD/D2).[19]
- Dissociation energy D_0 (0 K): high-precision spectroscopy (ionization $\rightarrow D_0$).[23][24][25]
- Molecular transitions (primary line lists): ExoMol (H2/HD) and MOLAT (D2). [26][27][28][29]
- Isotopic masses: NIST Atomic Weights and Isotopic Compositions (Hydrogen).[18]

Metrics: Treat the targets frozen in this section (λ, $\tilde{\nu}$, A, ω_e, B_e, D_0, ΔfH°gas, etc.) as the reference baseline, and in future rederivations connect the rejection decision to deviations from those targets (e.g. z scores).

Results:

Atomic baseline lines (vacuum λ):

Species	Target	Value
H I	Lyα	121.567 nm
H I	Hγ	434.169 nm
H I	Hβ	486.266 nm
H I	Hα	656.466 nm
H I	Hα multiplet (obs, E1)	656.452-656.466 nm (N=5)
H I	Hβ multiplet (obs, E1)	486.264-486.270 nm (N=3)
H I	21 cm hyperfine (f)	1420.4057517667(10) MHz ($\lambda \approx 21.106114$ cm)
He I	representative lines	447.273 / 501.708 / 587.725 / 668.000 nm

Molecular constants (ground state; NIST WebBook):

Molecule	ω_e (cm^{-1})	$\omega_e x_e$ (cm^{-1})	B_e (cm^{-1})	α_e (cm^{-1})	D_e (cm^{-1})	r_e (Å)
H2	4401.213	121.336	60.8530	3.0622	0.0471	0.74144

Continued on next page

Molecule	ω_e (cm^{-1})	$\omega_e x_e$ (cm^{-1})	B_e (cm^{-1})	α_e (cm^{-1})	D_e (cm^{-1})	r_e (Å)
HD	3813.15	91.65	45.655	1.986	0.02605	0.74142
D2	3115.5	61.82	30.4436	1.0786	0.01141	0.74152

Spectroscopic dissociation energy D_0 (0 K; spectroscopic):

Molecule	D_0 (cm^{-1})	D_0 (eV)	Reference
H2 (ortho; N=1)	35999.582834(11)	4.46338	[23]
HD (N=0)	36405.78253(7)	4.51374	[25]
D2 (N=0)	36748.362282(26)	4.55622	[24]

Molecular transitions (primary line lists): from the ExoMol line lists (H2/HD), extract the top 10 lines ranked by Einstein A and freeze the wavenumber $\tilde{\nu}$ and A as baseline values (the selection rule is fixed to avoid arbitrariness). Since the ExoMol line list for D2 could not be confirmed in the current work environment of this manuscript, it is supplemented by another source (MOLAT). [26][27]

(Notation) This manuscript uses vacuum wavelength; the air/vacuum conversion is treated separately as another systematic.

Isotopic dependence (reduced-mass scaling; reference figure): The molecular constants of the WebBook (H2/HD/D2) are consistent with the scaling $\omega_e \propto_\mu^{-1/2}$ and $B_e \propto_\mu^{-1}$ with respect to the reduced mass μ (for the NIST isotopic-mass ratios, $|1-\text{ratio}| \lesssim 10^{-3}$). This is the minimal systematic expected for "binding by waves," and when rederiving molecular binding in the P-model, not violating this isotopic systematic becomes a necessary condition (used as a constraint to avoid arbitrary fitting by extra degrees of freedom).

Thermochemistry (dissociation enthalpy; 298 K): For the molecular "binding energy," use the gas-phase thermochemistry of the NIST WebBook (ΔfH°gas) as an independent primary source from the spectroscopic constants and freeze the dissociation enthalpy (298 K). The purpose here is to freeze baseline values, not to replace D_0 (the spectroscopic dissociation energy at 0 K).[20] Results (NIST WebBook; eV per molecule / kJ/mol):

Molecule	Dissociation (298 K) eV	kJ/mol
H2→2H	4.5188	435.996
HD→H+D	4.5540	439.398
D2→2D	4.5959	443.44

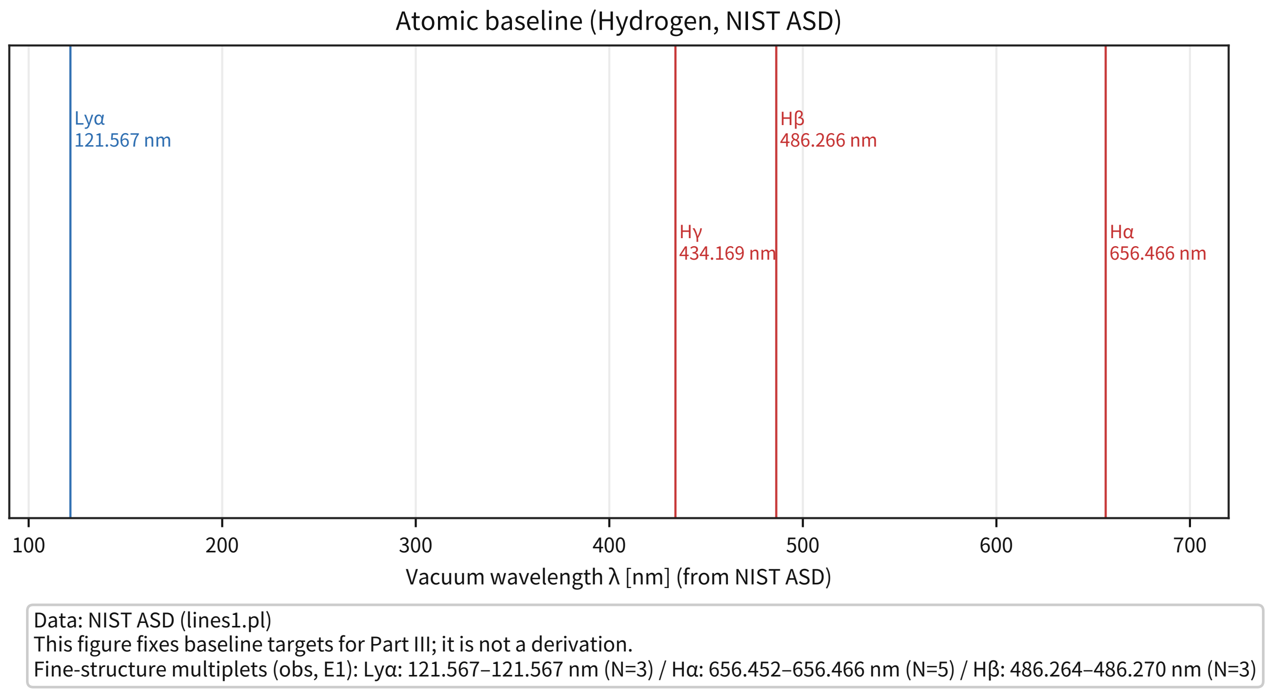

Figure 41: This figure summarizes the representative H I lines (Lyα to Hα) and the multiplets.

Supplement: it fixes the spectroscopic targets of this section (vacuum λ).

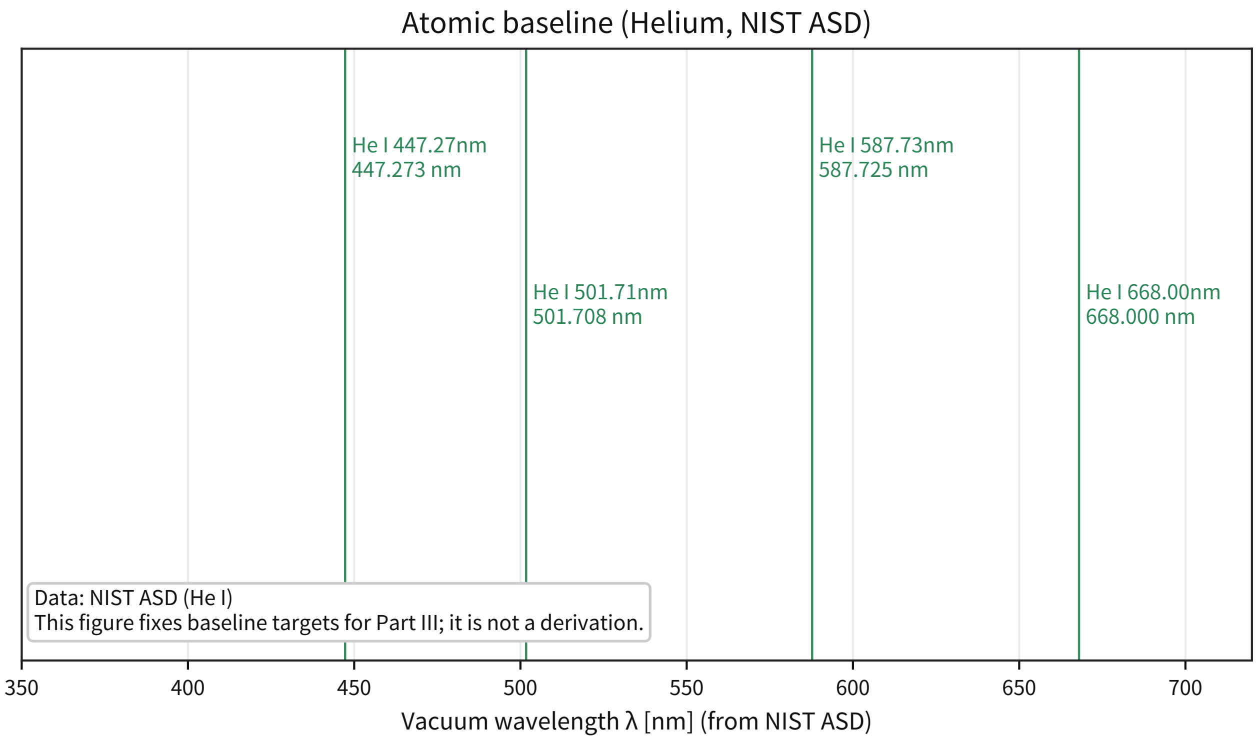

Figure 42: This figure shows representative He I lines.

Supplement: it is used as the entry point for freezing baseline values in a multielectron system.

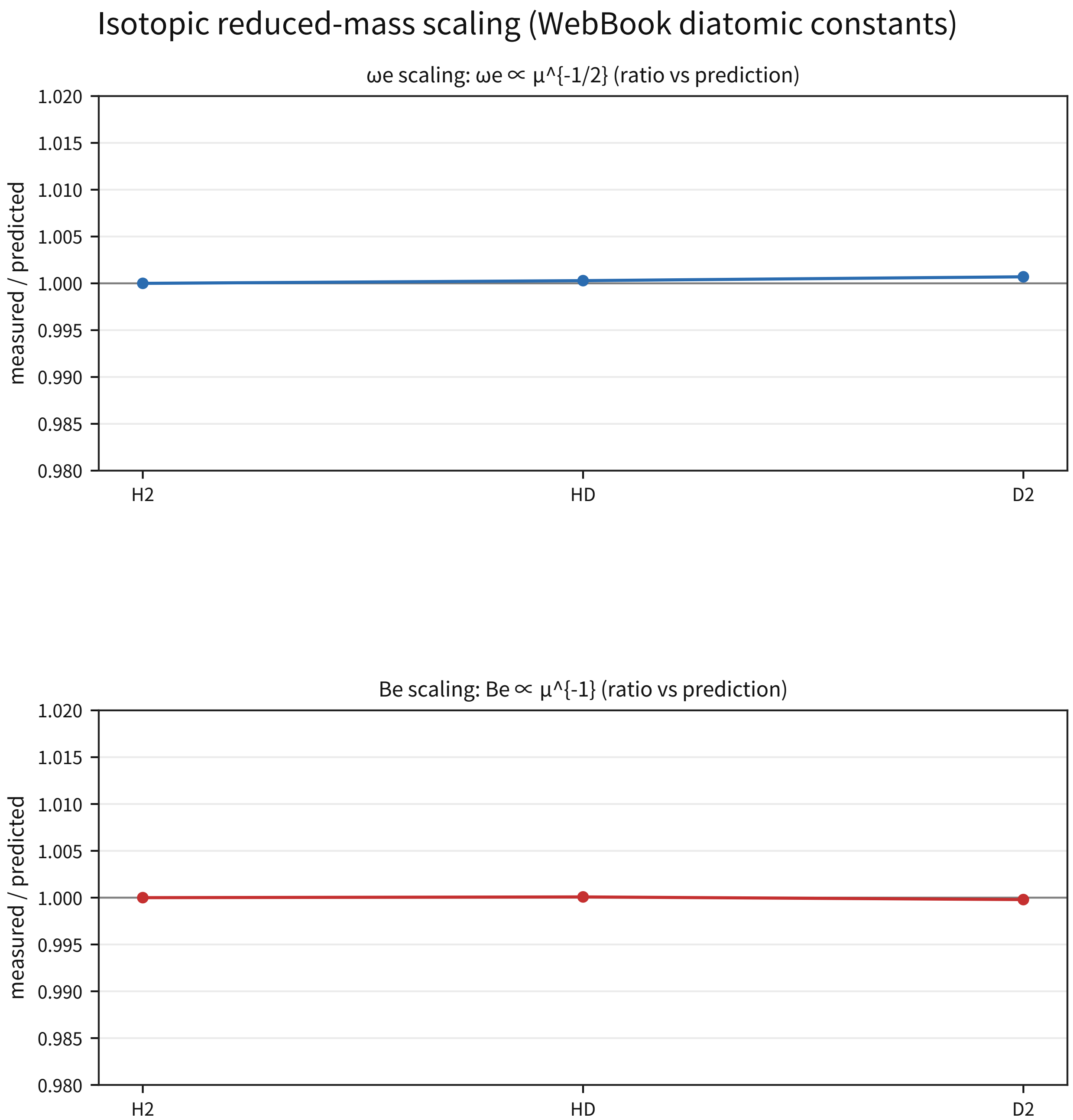

Figure 43: This figure shows that the molecular constants of H2/HD/D2 are consistent with scaling by the reduced mass μ.

Supplement: it visualizes the isotopic systematic that a future rederivation must not violate (a necessary condition). Figure-footnote remarks are moved into the main text; μ is computed from NIST isotopic masses (or, when needed, by mass-number approximation), and the reference molecule is fixed as H2.

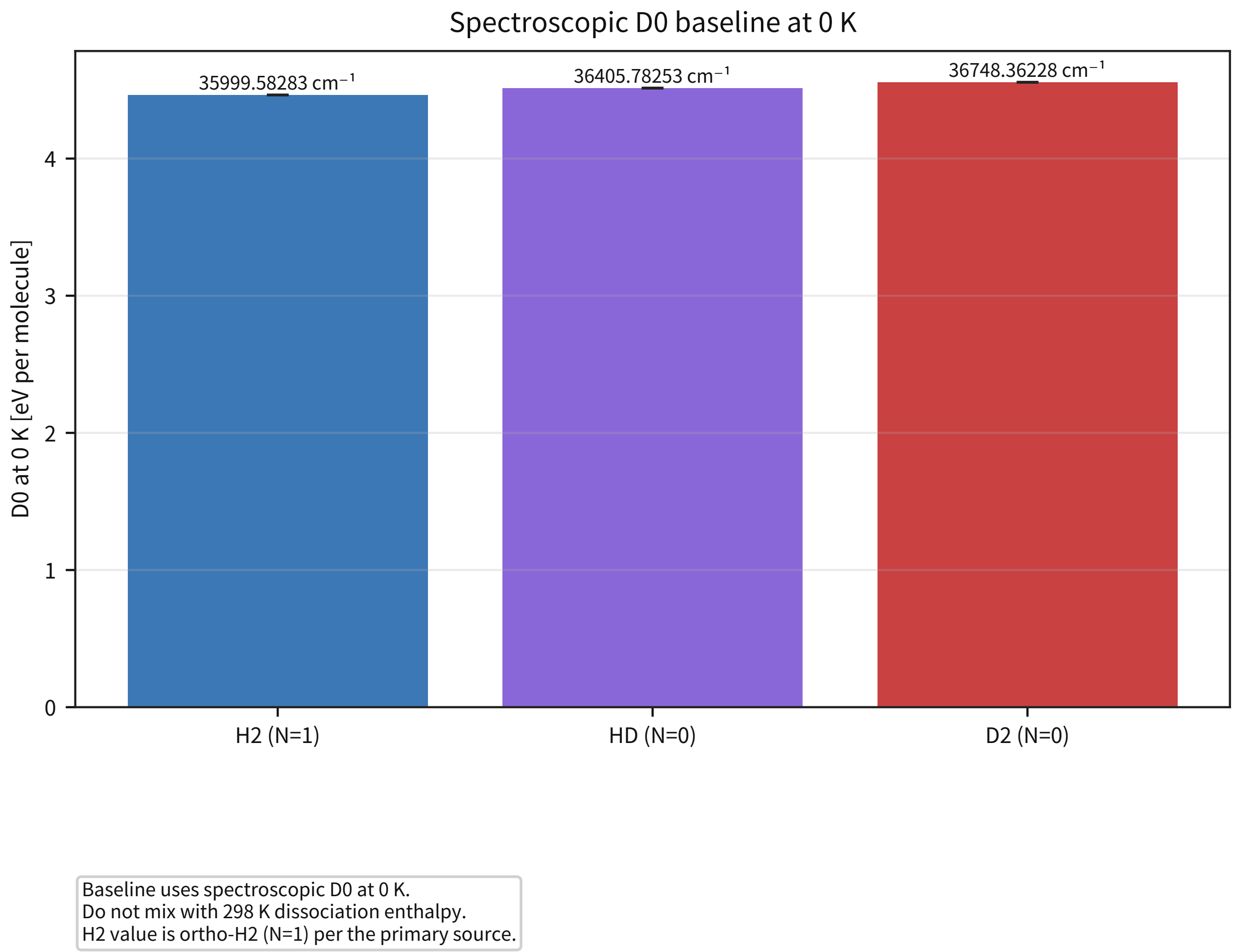

Figure 44: This figure shows the baseline values of D_0 (0 K) obtained from high-precision spectroscopy.

Supplement: including its independence from thermochemistry (298 K), it is fixed as a target for the binding energy.

representative molecular-transition baseline

molecule	representative transition 1	representative transition 2
H2	v=0,J=28 → v=0,J=26 nu = 2903.823 cm^-1 A = 5.51e-06 s^-1	v=0,J=27 → v=0,J=25 nu = 2931.587 cm^-1 A = 5.50e-06 s^-1
HD	v=9,J=23,qb → v=7,J=22,b nu = 3246.098 cm^-1 A = 1.19e-03 s^-1	v=3,J=33,b → v=2,J=32,b nu = 2657.953 cm^-1 A = 1.13e-03 s^-1
D2	C−(v=11,J=1) → X(v=17,J=1) nu = 79256.690 cm^-1 A = 5.27e+08 s^-1	C−(v=11,J=2) → X(v=17,J=2) nu = 79251.560 cm^-1 A = 5.27e+08 s^-1

Figure 45: This figure compresses representative transitions of H2 / HD / D2 into a three-row table.

Supplement: in the main text, only two transitions per molecule are shown, while the remaining primary line lists are kept in the public CSV and the Part IV registry. The selection rule (top-N) is preserved to avoid arbitrary line picking.

Notes: (Next stage) The main body including molecules (chemical bonding) must proceed toward rederivation while keeping the extension of the primary data (transitions, binding energies, isotopic dependence) and consistency with precision QED as safety checks, all while minimizing additional degrees of freedom.

Freeze: The spectroscopic and binding baseline values of atoms (H I/He I) and molecules (H2/HD/D2) were frozen as primary targets, while the isotopic systematic (reduced-mass scaling) and thermochemistry (298 K) were frozen as independent cross-checks. **Reject**: If a future rederivation departs systematically from these targets by more than 3σ of the total uncertainty, or violates the isotopic systematics (ω_e inversely proportional to the square root of the reduced mass and B_e inversely proportional to the reduced mass), that rederivation is rejected (arbitrary line picking is not allowed). **Next**: Supplement unresolved primary line lists (such as D2) from primary sources, expand the target set, and search for an ansatz that simultaneously satisfies rederivation with minimal degrees of freedom and consistency with precision QED (4.7).

3.9.2 Condensed Matter (Condensed Systems; Baseline Silicon Lattice Constants)

For the condensed-system block (4.9.2-4.9.6), the main text first summarizes the minimum set needed for the conclusion (baseline + rejection conditions), while the details are separated into the individual subsections (and the appendix).

Item (condensed systems)	Frozen reference (representative value)	Rejection condition (key point)
Si lattice constant / lattice spacing	$a = 5.431020511$ A; $d(220) = 1.920155716$ A	Reject if $\lvert a_{\text{pred}} - a_{\text{target}}\rvert > 3\sigma_a$ or $\lvert d_{\text{pred}} - d_{\text{target}}\rvert > 3\sigma_d$
Silicon heat capacity $C_p^\circ(T)$ (solid/liquid; Shomate)	$C_p^\circ(298.15\text{K}) = 19.99$; $C_p^\circ(1000\text{K}) = 26.32$; $C_p^\circ(2000\text{K}; \text{liquid}) = 27.20$ (J/mol K)	Reject if $\lvert C_{p,\text{pred}} - C_{p,\text{target}}\rvert > 3\sigma_{\text{proxy}}$ with $\sigma_{\text{proxy}} = \max(0.02\, C_p, 0.2)$
Low-temperature silicon $C_p^\circ(T)$ (JANAF) + Debye	Freeze $\theta_D = 622.1$ K (fit) as the representative value	Reject any claim whose low-temperature $C_p^\circ(T)$ falls outside the table uncertainty scale
Silicon resistivity $\rho(T)$ and temperature coefficient	Necessary condition $d\ln\rho/dT > 0$ near room temperature + observed envelope (0.0205-0.957 %/K)	Reject $d\ln\rho/dT \le 0$; reject out-of-envelope claims unless they provide a primary-source uncertainty basis
Silicon thermal expansion $\alpha(T)$	Sign change at $T \approx 123.7$ K (frozen) + fit-error scale	A constant model $\alpha \approx A \cdot C_v$ is rejected by the sign mismatch; predictive claims are rejected if $\lvert\alpha_{\text{pred}} - \alpha_{\text{target}}\rvert > 3\sigma_{\text{fit}}$ at representative points
OFHC copper $\kappa(T)$ (RRR-dependent)	Curve with $\kappa(4\text{K}) \sim \kappa_{\max} \sim \kappa(300\text{K})$ for RRR=50-500	Reject if $\lvert\kappa_{\text{pred}} - \kappa_{\text{target}}\rvert > 3\sigma_{\text{fit}}$ at representative points and the RRR ordering is broken

Objective: As the first endpoint for condensed-matter systems, freeze the geometric reference values of a solid (lattice constants) from primary sources and connect them to future re-derivations and rejection tests.

Input: The silicon lattice constant a from CODATA 2022 (NIST Cuu) and the ideal Si(220) lattice spacing d.

Processing: Cache the primary-source HTML (NIST Cuu) locally and freeze the extracted values directly as the targets. No extra fitting is performed; the main purpose is to catch transcription mistakes in the fixed values.

Metric: a and d (Si(220)). Rejection condition: if a first-principles calculation or similar approach claims to "predict" these values, reject it when $\lvert a_{\text{pred}} - a_{\text{target}}\rvert > 3\sigma_a$ or $\lvert d_{\text{pred}} - d_{\text{target}}\rvert > 3\sigma_d$, where σ is the CODATA uncertainty.

Results:

Quantity	Value	σ
Silicon lattice constant a	5.431020511 A	8.9×10^{-8} A
Si (220) lattice spacing d	1.920155716 A	3.2×10^{-8} A

Supplement: the frozen values for silicon a and d (Si(220)) are the values in the table above. For individual plots, refer to the public artifact ledger in Part IV.

Note: This subsection fixes a condensed-matter target only. It does not claim that the P-model has already predicted a from first principles.

3.9.3 Condensed Matter (Condensed Systems; Silicon Solid/Liquid Heat Capacity $C_p(T)$ via Shomate)

Objective: Fix the baseline heat-capacity curve $C_p(T)$ from primary data as the minimum target for condensed-matter thermodynamics, and connect it to future re-derivations with minimal degrees of freedom and to rejection tests.

Input: Condensed-phase thermochemistry from the NIST Chemistry WebBook (Si; Thermo-Condensed; Shomate coefficients).

Processing: Reconstruct $C_p^\circ(T)$ from the WebBook Shomate coefficients (solid/liquid) and freeze it as the reference curve.

Equation:

$$Cp^\circ(T) = A + Bt + Ct^2 + Dt^3 + \frac{E}{t^2}, \qquad t \equiv T/1000.$$

Metric: $C_p^\circ(T)$ at representative temperatures (table below) and the phase switch (solid to liquid). Rejection condition: because the WebBook does not state an uncertainty explicitly, adopt the conservative proxy $\sigma_{\text{proxy}} = \max(0.02\,C_p,\ 0.2\ \mathrm{J\,mol^{-1}\,K^{-1}})$ and reject when $|C_{p,\text{pred}} - C_{p,\text{target}}| > 3\sigma_{\text{proxy}}$. This is an operational proxy and should be replaced if a primary uncertainty becomes available later.

Results:

Phase	T (K)	C_p° (J/mol K)
solid	298.15	19.99
solid	1000	26.32
liquid	2000	27.20

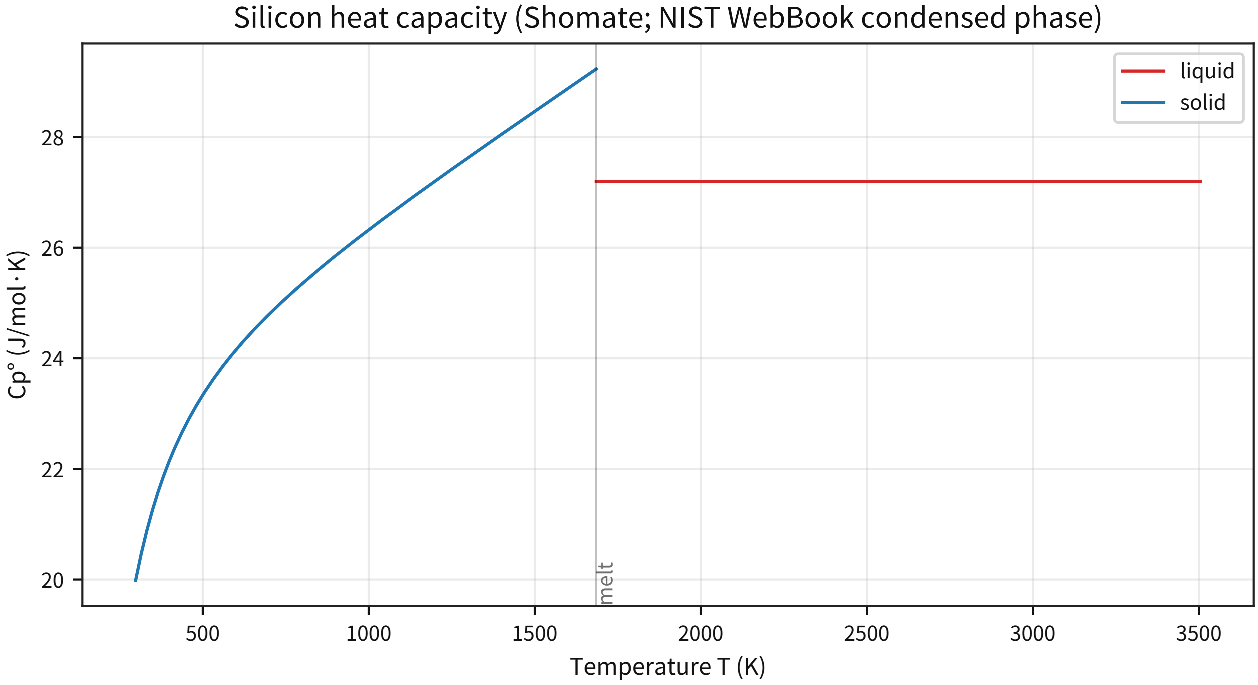

Figure 46: Reconstructs solid/liquid $C_p^\circ(T)$ from the Shomate coefficients and fixes the temperature-dependent shape.

Supplement: it also shows visually that the curve connects smoothly from 298 K to high temperature while still missing the low-temperature Debye regime.

Note: The condensed-phase C_p table in the WebBook starts at 298 K, so the low-temperature primary baseline (the Debye T^3 regime) must be added separately.

3.9.4 Condensed Matter (Condensed Systems; Low-Temperature Silicon Heat Capacity $C_p(T)$ via JANAF + Debye θ_D)

Objective: Complement the WebBook baseline (298 K and above) by freezing the low-temperature $C_p(T)$ side (at least 100-300 K) from primary data, and compress it into a Debye one-parameter target (θ_D) that can be used for future minimal re-derivations and rejection tests.

Input: The NIST-JANAF table (Si-004; primary table of C_p°).

Processing: Fit the JANAF $C_p^\circ(T)$ values over 100-300 K with the Debye one-parameter form (θ_D) by least squares. At low temperature, approximate $C_p \approx C_v$ and use the Debye C_v formula as a proxy for C_p.

Equation:

$$C_V(T;\theta_D) = 9R\left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D/T} \frac{x^4 e^x}{(e^x-1)^2}\,dx.$$

Metric: Debye θ_D and the proxy uncertainty σ_{proxy} estimated from the residual scatter. Rejection condition: since JANAF does not provide an explicit σ, use the proxy indicator and reject when $|\theta_{D,\text{pred}} - \theta_{D,\text{target}}| > \max(3\sigma_{\text{proxy}},\ 5\ \text{K})$.

Results:

Metric	Value
C_p°(100 K)	7.268 J/mol K
C_p°(200 K)	15.636 J/mol K
C_p°(298.15 K)	20.000 J/mol K
Debye θ_D (100-300 K; proxy with $C_v \approx C_p$)	622.1 K ($\sigma_{\text{proxy}} \approx 17.4$ K)
Cross-check (298.15 K; WebBook - JANAF)	-0.0094 J/mol K

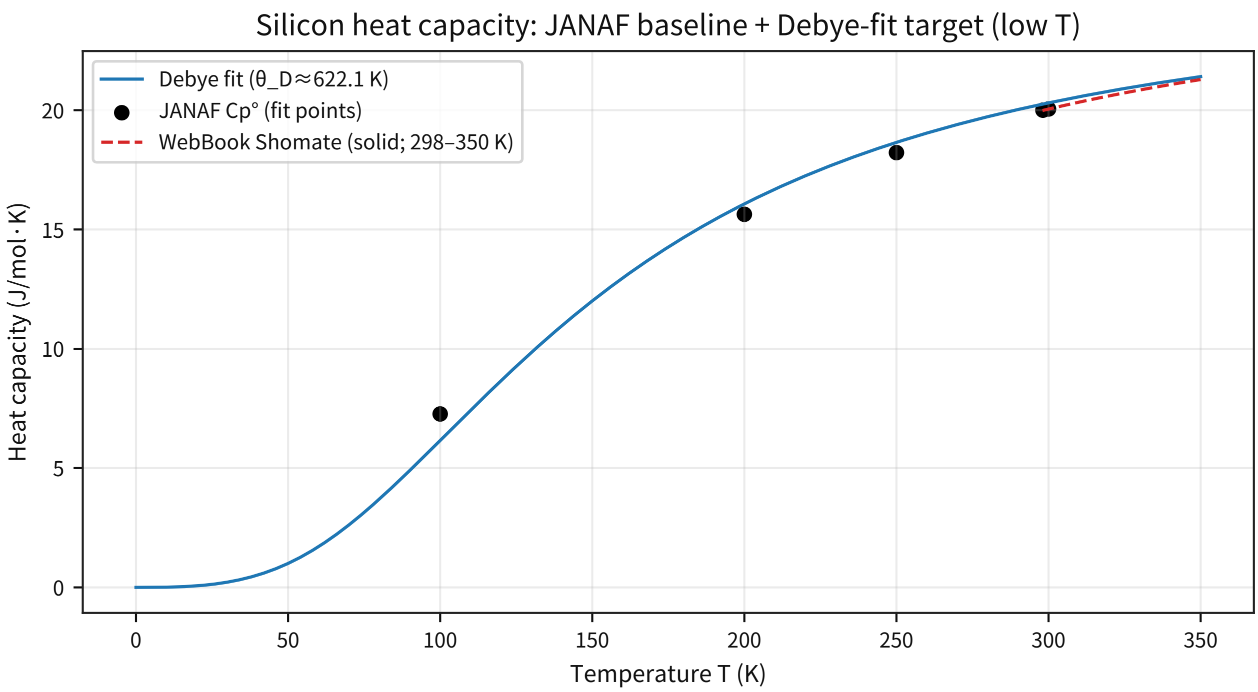

Figure 47: Approximates JANAF (100-300 K) with a one-parameter Debye fit and freezes θ_D as the target.

Supplement: it also shows the 298 K cross-check against the WebBook to confirm data consistency.

3.9.5 Condensed Matter (Condensed Systems; Silicon Resistivity $\rho(T)$ and the Near-Room-Temperature Coefficient)

Objective: Within the range consistent with the adopted electromagnetic sector, freeze $\rho(T)$ as a representative transport observable for a solid, and fix the near-room-temperature coefficient ($d \ln \rho / dT$) as an observed target for future re-derivations and rejection tests.

Input: NBS IR 74-496 (legacy IR; Appendix E values of $\rho(T)$ over 0-50°C).

Processing: Extract $\rho(20°\text{C})$, $\rho(23°\text{C})$, and $\rho(30°\text{C})$ from the table and estimate the near-room-temperature coefficient by finite differences. Because the distribution varies strongly with doping and type, this subsection freezes not a precise $\pm\sigma$ but an envelope of necessary conditions.

Equation:

$$\left.\frac{d\ln\rho}{dT}\right|_{23^\circ\mathrm{C}} \approx \frac{\ln\rho(30^\circ\mathrm{C}) - \ln\rho(20^\circ\mathrm{C})}{10\,\mathrm{K}}.$$

Metric: $\rho(23^\circ\mathrm{C})$ and $d\ln\rho/dT$ (proxy indicator). Rejection condition: over this measurement range the coefficient is positive, so $d\ln\rho/dT \le 0$ is rejected as a necessary-condition failure. Claims outside the observed envelope (0.0205-0.957 %/K) are also rejected unless they first provide a primary-source uncertainty basis.

Results:

Metric	Value
$\rho(23^\circ\mathrm{C})$ ($n = 21$)	7.761×10^{-4} to 1.159×10^{3} Ω·cm
$d\ln\rho/dT$ (20-30°C; proxy)	0.0205 to 0.957 %/K

Supplement: for the detailed scatter plots of the extracted $\rho(T)$ over 0-50°C and the proxy $d\ln\rho/dT$ near room temperature, refer to the public artifact ledger in Part IV.

3.9.6 Condensed Matter (Condensed Systems; Silicon Thermal Expansion Coefficient $\alpha(T)$)

Objective: Following the lattice constant and heat capacity, freeze the temperature response of the length scale (thermal expansion) from primary data and connect it to future re-derivations and rejection tests. This is the entry point at which the placement of phonon degrees of freedom is closed operationally.

Input: NIST TRC Cryogenics (Material Properties: Silicon; thermal expansion coefficient $\alpha(T)$).

Processing: Evaluate $\alpha(T)$ over 13-600 K using the NIST empirical fit and freeze the representative points (table below) as the targets. Also record the fit standard-error scale shown on the page for $T < 50$ K and $T \ge 50$ K as a systematic estimate.

Metric: $\alpha(T)$ at representative temperatures and the fit standard-error scale. Rejection condition: reject if $|\alpha_{\mathrm{pred}}(T) - \alpha_{\mathrm{target}}(T)| > 3\sigma_{\mathrm{fit}}(T)$ at representative points, where σ_{fit} is the page-reported standard error.

Results:

Metric	Value
Data range	13-600 K
Curve-fit standard error (reported on page)	0.03×10^{-8}/K for $T < 50$ K, 0.5×10^{-8}/K for $T \ge 50$ K
Sign change of α (negative to positive)	$T \approx 123.7$ K

T (K)	α (10^{-8}/K)
20	-0.34
50	-29.22
100	-33.88
293.15	255.47

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T (K)	α (10^{-8}/K)
600	383.91

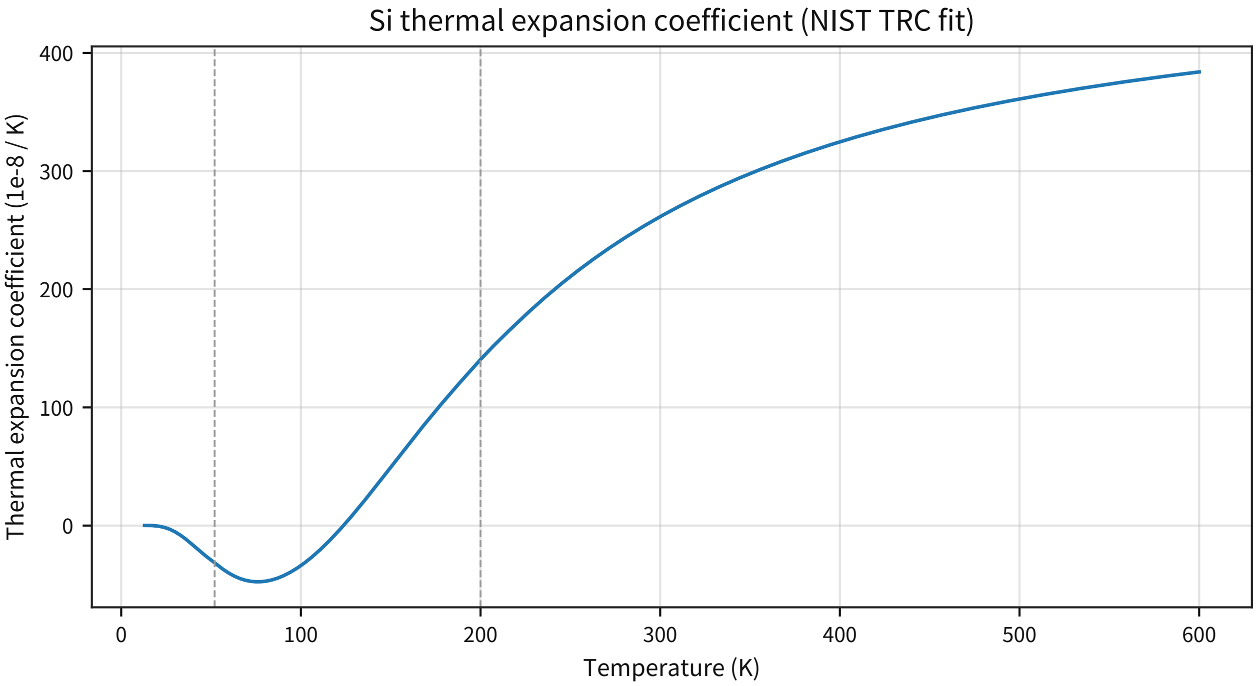

Figure 48: Freezes, in a single plot, the sign reversal of $\alpha(T)$ at low temperature and its growth through and above room temperature.

Supplement: the fit-error scale reported on the page is shown in parallel as a guide to the required precision.

Discussion: Because the frozen baseline $\alpha(T)$ includes a negative-to-positive sign reversal, a constant model such as $\alpha \approx A \cdot C_v$, which keeps α always positive, is rejected. The comparisons below integrate both the trial-model comparisons that are required to satisfy the sign reversal together with the σ_{fit} scale (as reported by NIST), and the audit log for the temperature-band split procedure systematics. The main text keeps only four representative figures (primary baseline, minimal model, temperature-split audit, and DOS-constrained model), while the rest of the ansatz-sweep figures are collected in Section 9, "Appendix A."

Minimal Grüneisen Model ($\alpha \approx A_{\text{eff}} \cdot C_v$) First, evaluate the minimal Debye-Grüneisen model ($\alpha \approx A_{\text{eff}} \cdot C_v$ with $A_{\text{eff}} = \gamma/(BV_m)$ treated as a constant). In this model α is always positive, just like C_v, so it cannot be made consistent with the frozen $\alpha(T)$ that includes a sign change and is therefore rejected.

Metric	Value
Debye θ_D (already fixed; 4.9.4)	622.1 K
A_{eff} (least squares over $T \geq 200$ K)	1.44×10^{-7} mol/J
Relative residual RMSE ($T \geq 200$ K)	0.182

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Metric	Value
Sign mismatches (full range)	106 points

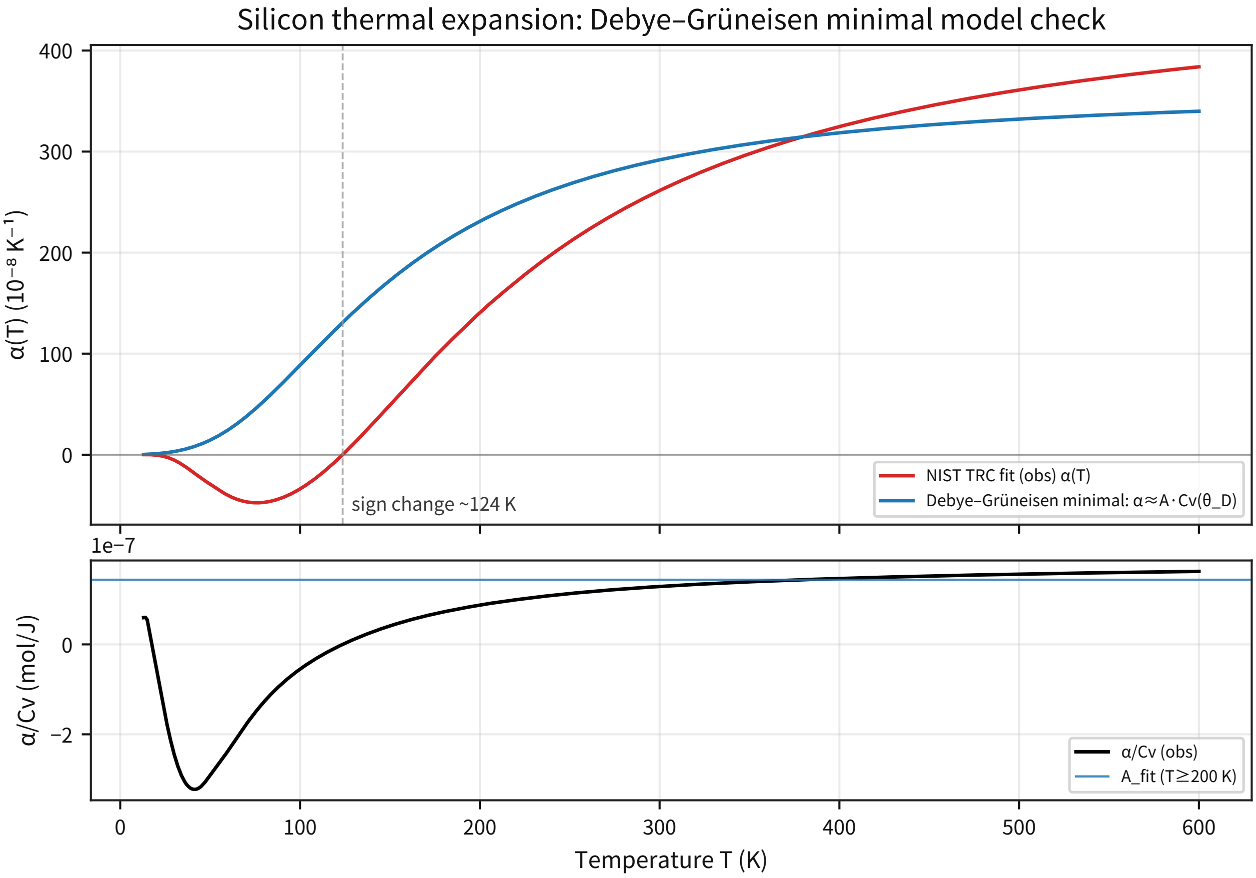

Figure 49: Top: comparison of the frozen $\alpha(T)$ and the minimal Debye-Grüneisen model ($\alpha \approx A \cdot C_v$).

Supplement: bottom panel shows the temperature dependence of α/C_v (demonstrating that it is not constant) together with the A_{fit} reference line for $T \geq 200$ K. Therefore, any realistic description of $\alpha(T)$ requires extensions such as mode-dependent γ or anharmonicity.

Minimal Extension of $\gamma(T)$ (tanh) Next, introduce the minimal temperature dependence of $A_{\text{eff}}(T) = \gamma(T)/(BV_m)$ through the ansatz $A_{\text{eff}}(T) = A_\infty \cdot \tanh((T - T_0)/\Delta T)$, where T_0 is frozen at the observed negative-to-positive crossing of $\alpha(T)$, and determine ΔT and A_∞ by least squares. The sign mismatch disappears for $T \geq 50$ K, but the residual remains large relative to the fit-standard-error scale (reported by NIST), so this ansatz class is rejected.

Metric	Value
T_0 (negative to positive; frozen)	123.7 K
A_∞ (tanh; fit over $T \geq 50$ K)	1.56×10^{-7} mol/J
ΔT (tanh; fit over $T \geq 50$ K)	136.4 K
Sign mismatches ($T \geq 50$ K)	0

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Metric	Value
Relative residual RMSE ($T \geq 50$ K; proxy)	0.237
$\lvert\text{resid}\rvert > 3\sigma_{\text{fit}}$ ($T \geq 50$ K)	514 points

Minimal Implementation of Mode-Dependent γ (Two Branches) As a minimal implementation of mode-dependent γ, freeze the two-branch (two-mode) model $\alpha(T) \approx A_1 \cdot C_v(T; \theta_1) + A_2 \cdot C_v(T; \theta_2)$ and evaluate whether it can satisfy the sign reversal and the precision requirement (σ_{fit}) at the same time. Here θ_1 is frozen at the Debye fit value (4.9.4), while θ_2 is scanned and A_1/A_2 are determined by σ_{fit}-weighted least squares.

Metric	Value
θ_1 (frozen; Debye)	622.1 K
θ_2 (best scan)	577.0 K
A_1	1.60×10^{-6} mol/J
A_2	-1.43×10^{-6} mol/J
$T_{0,\text{pred}}$ (negative to positive)	126.8 K
Sign mismatches ($T \geq 50$ K)	3 points
$\max\lvert z\rvert$ ($T \geq 50$ K)	23.8
Reduced χ^2 ($T \geq 50$ K)	56.8
$\lvert z\rvert > 3$ ($T \geq 50$ K)	423 points

Debye + Einstein (One Branch) Next, fix the minimal model that adds a single Einstein (optical) branch to the Debye (acoustic) term, $\alpha \approx A_D \cdot C_{v,D}(\theta_D) + A_E \cdot C_{v,E}(\theta_E)$, and evaluate its viability (with θ_D frozen and θ_E scanned).

Metric	Value
θ_D (frozen)	622.1 K
θ_E (best scan)	598.6 K
A_D	-1.94×10^{-7} mol/J
A_E	3.64×10^{-7} mol/J
$T_{0,\text{pred}}$ (negative to positive)	126.9 K
Sign mismatches ($T \geq 50$ K)	3 points
$\max\lvert z\rvert$ ($T \geq 50$ K)	23.1
Reduced χ^2 ($T \geq 50$ K)	57.8
$\lvert z\rvert > 3$ ($T \geq 50$ K)	428 points

Diagnostic: Polynomial $A_{\text{eff}}(T)$ with Frozen $C_{v,\text{Debye}}$ Finally, keep the form $\alpha \approx A_{\text{eff}}(T) \cdot C_{v,\text{Debye}}$ but allow smooth temperature dependence through $A_{\text{eff}}(T) = \Sigma c_k T^k$ for $n = 0..3$ as a diagnostic. This is not a derivation; it is a minimal test of whether Debye C_v alone can reproduce the shape. Even after extending to $n = 3$, the model is still rejected at the σ_{fit} scale, so an approximation based on a single Debye C_v is insufficient.

Metric	Value
Degrees scanned n	0, 1, 2, 3
Best case (minimum reduced χ^2)	$n = 3$
$\max\lvert z\rvert$ ($T \geq 50$ K)	52.6
Reduced χ^2 ($T \geq 50$ K)	380
$\lvert z\rvert > 3$ ($T \geq 50$ K)	500 points

Diagnostic: Replace C_v by $C_{p,\text{proxy}}$ to Isolate Shape Deficiency To test whether the problem is only the shape deficiency of C_v, replace $C_{v,\text{Debye}}$ by a proxy built from the primary $C_p(T)$ (JANAF). For $T \geq 100$ K, linearly interpolate the JANAF $C_p(T)$ values; for $T < 100$ K, connect the Debye C_v branch after scaling it by $C_p(100\text{ K})$ to construct $C_{p,\text{proxy}}(T)$. Then fit $A_{\text{eff}}(T) = A_\infty \cdot \tanh((T - T_0)/\Delta T)$ by least squares, with T_0 frozen at the observed sign reversal.

Metric	Value
$C_{p,\text{proxy}}$ (connection rule)	Debye $\times$ scale for $T < 100$ K; JANAF interpolation for $T \geq 100$ K
Scale factor ($C_p(100)/C_{v,\text{Debye}}(100)$)	1.181
T_0 (negative to positive; frozen)	123.7 K
ΔT (tanh; fit over $T \geq 50$ K)	126.3 K
A_∞	1.54×10^{-7} mol/J
Sign mismatches ($T \geq 50$ K)	0
$\max\lvert z\rvert$ ($T \geq 50$ K)	55.2
Reduced χ^2 ($T \geq 50$ K)	499
$\lvert z\rvert > 3$ ($T \geq 50$ K)	512 points

Isolation Check: Temperature Dependence of $B(T)$ (Bulk Modulus) At this stage, B in $A_{\text{eff}}(T) = \gamma(T)/(BV_m)$ is still treated implicitly as a constant. To isolate how much $B(T)$ matters, use it as an independent input. Constructing $B(T)$ from the silicon elastic constants in the Ioffe database shows that the change from 400 to 600 K is only about -1.6%, so it is unlikely to be the dominant effect.

Increasing the Branch Count: Debye + Einstein $\times 2$ Next, fix the minimal model that extends the optical side to two Einstein branches, $\alpha \approx A_D \cdot C_{v,D}(\theta_D) + A_{E1} \cdot C_{v,E}(\theta_{E1}) + A_{E2} \cdot C_{v,E}(\theta_{E2})$, with θ_D frozen and $\theta_{E1} < \theta_{E2}$ scanned, and evaluate its viability. The two-branch extension improves the residual substantially and lowers $\max\lvert z\rvert$ to 3.23, but it still misses the strict criterion (3σ + reduced $\chi^2 < 2$) by a small margin and is rejected.

Metric	Value
θ_D (frozen)	622.1 K
θ_{E1} (best scan)	503.5 K
θ_{E2} (best scan)	821.7 K
A_D	-3.22×10^{-7} mol/J
A_{E1}	3.81×10^{-7} mol/J
A_{E2}	1.16×10^{-7} mol/J

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Metric	Value
$T_{0,\mathrm{pred}}$ (negative to positive)	123.0 K
Sign mismatches ($T \geq 50$ K)	1 point
max$\lvert z \rvert$ ($T \geq 50$ K)	3.23
Reduced χ^2 ($T \geq 50$ K)	2.32
$\lvert z \rvert > 3$ ($T \geq 50$ K)	6 points

Temperature-Band Split Audit (Procedure Systematics) To quantify how strongly the choice of fit range moves the statistics as a procedure systematic, perform temperature-split holdout tests. Freeze the parameters estimated on the train range and extrapolate them to the test range, then compare max$\lvert z \rvert$ and reduced χ^2 (with dof=n on the test side). The Einstein×2 model can reach max $\lvert z \rvert < 1$ on train, but degrades sharply on test, showing that range selection can become the dominant systematic.

Split (train → test)	Model	max$\lvert z \rvert$ train	max$\lvert z \rvert$ test	Reduced χ^2 train	Reduced χ^2 test
A (50-300 → 300-600 K)	Einstein×1	14.5	50.1	29.5	1060
A (50-300 → 300-600 K)	Einstein×2	0.94	15.4	0.188	63.1
B (200-600 → 50-200 K)	Einstein×1	3.56	85.9	1.67	3030
B (200-600 → 50-200 K)	Einstein×2	0.306	45.3	0.0058	562

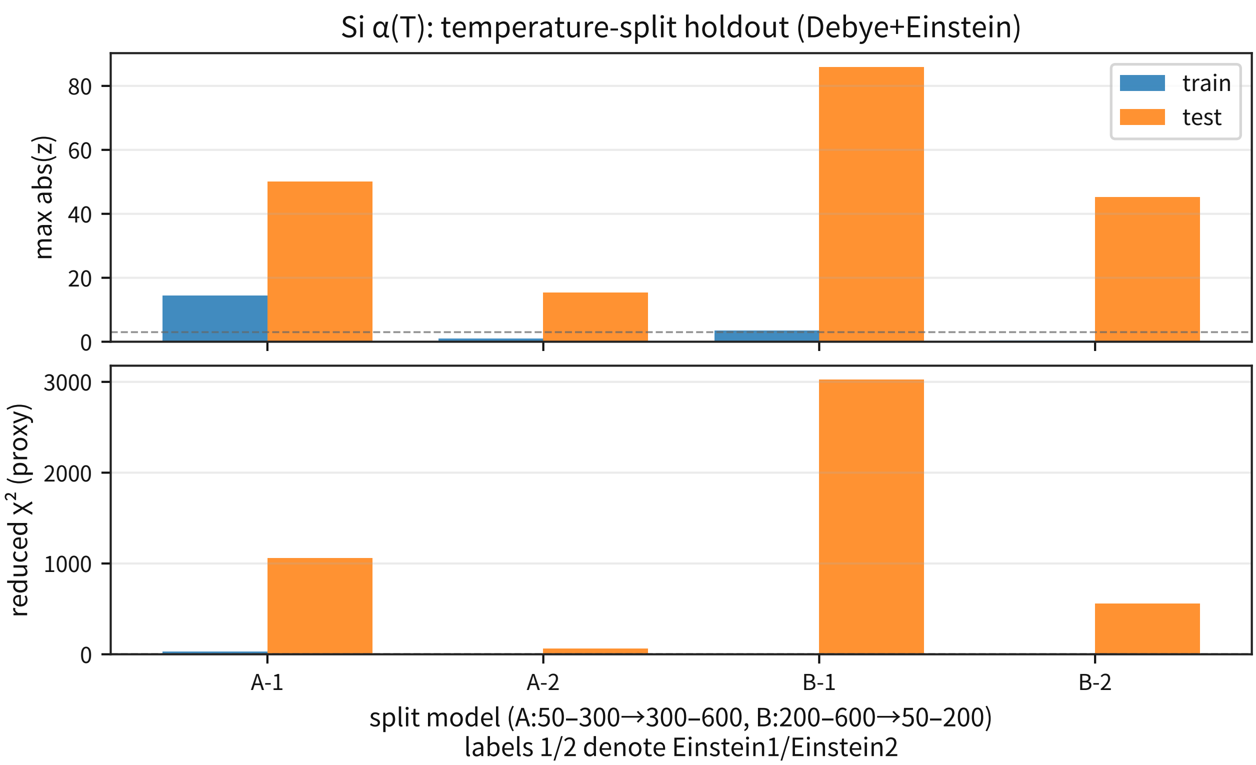

Figure 50: Top: compares train/test max$\lvert z \rvert$ for split A/B (1 = Einstein×1, 2 = Einstein×2).

Supplement: bottom panel compares reduced χ^2 for the same splits to make visible that range selection itself can become a systematic.

Phonon-Frequency Anchors (Ioffe) Finally, as the entry point for a primary constraint on the phonon side, freeze the Einstein temperatures θ_E to discrete anchors taken from the "representative phonon frequencies" table in the Ioffe database. Build a candidate set with TA on the low-frequency side and optical modes on the high-frequency side, select the two points that minimize the weighted SSE over the fit range ($T \geq 50$ K), and evaluate Debye + Einstein×2. This is not a derivation; it is a diagnostic of how much range dependence remains after suppressing the continuous scan of θ_E and its associated growth in free parameters.

Metric	Value
θ_{E1} (best Ioffe anchor; TA)	216.0 K
θ_{E2} (best Ioffe anchor; optical)	743.9 K
$\max\|z\|$ ($T \geq 50$ K)	5.17
Reduced χ^2 ($T \geq 50$ K)	4.16
Holdout A: $\max\|z\|$ (train → test)	1.94 → 7.81

Likewise, extend the optical-side split from two to three branches (Einstein×3), choose the three Ioffe-derived discrete candidates ($\theta_{E1} < \theta_{E2} < \theta_{E3}$) that minimize the SSE over $T \geq 50$ K, and evaluate the result. The fit-range $\max|z|$ improves to 4.75, but the model is still rejected under the strict criterion, and under holdout A (50-300 → 300-600 K) the test-side $\max|z|$ worsens to 15.9, so the range dependence remains.

Metric	Value
θ_{E1} (best Ioffe anchor; TA)	216.0 K
θ_{E2} (best Ioffe anchor; optical)	667.1 K
θ_{E3} (best Ioffe anchor; optical)	705.5 K
$\max\|z\|$ ($T \geq 50$ K)	4.75
Reduced χ^2 ($T \geq 50$ K)	4.13
Holdout A: $\max\|z\|$ (train → test)	0.90 → 15.90

Phonon DOS and Softening Proxies As a further attempt to freeze the mode weighting (the weights in C_v) from a primary or near-primary input, cache the numerical table of the phonon DOS (ω-$D(\omega)$) embedded in the public HTML and evaluate the Grüneisen ansatz using mode-count-based splits (half-integral / 2:1 split). This input is only a proxy because it lacks an explicit primary-literature statement; it should be replaced once primary INS/IXS raw or supplementary data are obtained.

Model (DOS-constrained)	$\max\|z\|$ ($T \geq 50$ K)	Reduced χ^2 ($T \geq 50$ K)	Holdout A: $\max\|z\|$ (train → test)
2-group (acoustic/optical)	26.5	65.0	22.7 → 19.8
3-group (TA/LA/optical)	7.45	11.3	1.48 → 20.5
4-group (TA/LA/TO/LO)	6.05	5.97	1.12 → 16.95

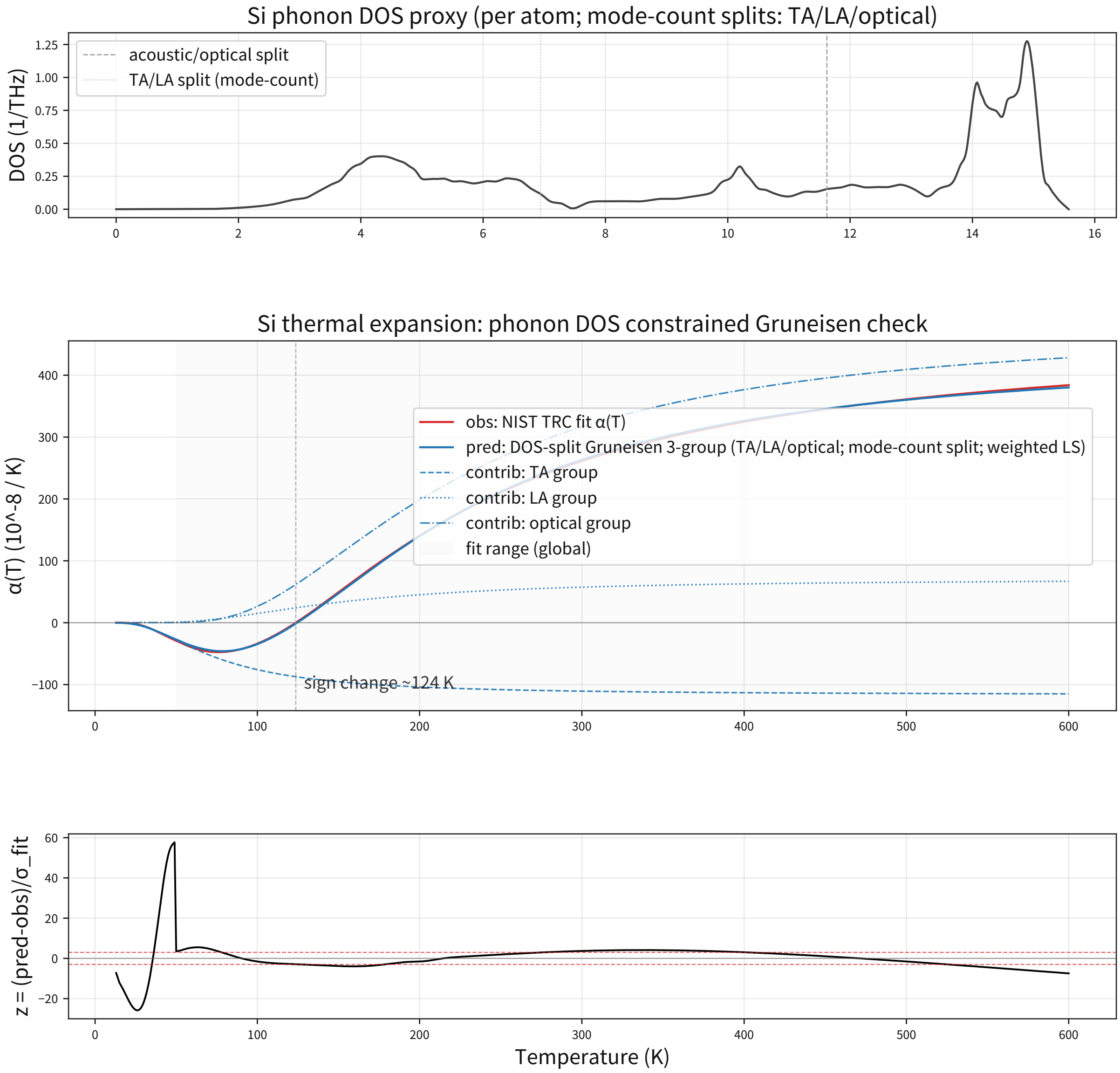

Figure 51: Top: phonon DOS (per atom) and the locations of the mode-count split (TA/LA/optical).

Supplement: middle panel compares the frozen $\alpha(T)$ with the DOS-constrained Grüneisen model (fitting only the A coefficients). Bottom panel shows $z = (\alpha_{\text{pred}} - \alpha_{\text{obs}})/\sigma_{\text{fit}}$ and makes it visible that the extrapolation from low-T train to high-T test breaks down. Therefore, freezing the phonon-DOS shape alone is not sufficient; primary constraints on temperature-dependent phonon softening $\omega(T)$ and mode-dependent γ are still required.

Next, as the minimal proxy for anharmonicity, weakly vary the optical frequency scale using $s(T) = \max(s_{\min}, 1 - f \cdot (T/T_{\max}))$, where $T_{\max}$ is the upper bound of the evaluation temperature and f is chosen by grid scan to minimize the train-side SSE, and test whether this "linear optical softening" improves holdout performance. It does not: with only one softening degree of freedom, the holdout collapse is not resolved, and primary constraints on mode-dependent γ remain necessary.

Model (softening)	$f(T_{\max})$	$\max\lvert z\rvert$ ($T \geq 50$ K)	Holdout A: $\max\lvert z\rvert$ (train $\rightarrow$ test)
2-group + optical softening	0.060	22.8	21.1 $\rightarrow$ 12.2
3-group + optical softening	0.000	7.45	1.48 $\rightarrow$ 20.5

Then, as an attempt to constrain the shape of the softening itself from primary data, digitize the silicon Raman measurements of $\omega(T)$ over 4-623 K from the primary PDF, freeze $g(T) \in [0, 1]$ (0 at low temperature, 1 at high temperature), and evaluate $s(T) = 1 - f \cdot g(T)$ with only f scanned on a grid. The result still does not resolve the holdout collapse: for the 2-group model, even the best $f \approx 0.04$ leaves $\max\lvert z\rvert = 24.3$, while the 3-group model selects $f = 0$ and shows no improvement.

Model (Raman-shape softening)	$f(T_{\max})$	$\max\lvert z\rvert$ ($T \geq 50$ K)	Holdout A: $\max\lvert z\rvert$ (train $\rightarrow$ test)
2-group + Raman shape	0.040	24.3	22.48 $\rightarrow$ 18.13
3-group + Raman shape	0.000	7.45	1.48 $\rightarrow$ 20.52

Finally, freeze the mean shift reported by the INS-based primary source (Kim et al., PRB 91, 014307 (2015)), $\langle \Delta\epsilon/\epsilon \rangle \approx -0.07$ from 100 to 1500 K, as a linear proxy for a common $\omega(T)$ scale over all modes, and apply it to the DOS basis. Even with this global softening, the holdout collapse is not resolved, and in the 3-group case the high-temperature mismatch becomes worse.

Model (global softening; Kim2015)	s(1500K)	$\max\lvert z\rvert$ ($T \geq 50$ K)	Holdout A: $\max\lvert z\rvert$ (train $\rightarrow$ test)
2-group + Kim2015	-0.07 (fixed)	26.46	23.08 $\rightarrow$ 18.10
3-group + Kim2015	-0.07 (fixed)	8.29	1.69 $\rightarrow$ 23.59

3.10 Statistical Mechanics and Thermodynamics (Transport and Blackbody)

This section groups 4.10.1-4.10.3 as one class of statistical-mechanical and thermodynamic tests. It links transport ($\kappa(T)$) and blackbody equilibrium (radiation and entropy) and unifies the frozen baselines for thermal statistics.

3.10.1 Transport (OFHC Copper Thermal Conductivity $\kappa(T)$)

Objective: As a primary target for low-temperature transport, freeze the RRR-dependent $\kappa(T)$ curves and connect them to future re-derivations and rejection tests.

Input: NIST TRC Cryogenics (Material Properties: OFHC Copper; rational fit for $\kappa(T)$; RRR = 50/100/150/300/500).

Processing: Compute $\kappa(T)$ from the NIST rational fit for $\log_{10}\kappa(T)$ and project it onto a common temperature grid (4-300 K) as the frozen baseline. Treat the fit relative error for each RRR (1-2%) as a proxy 1σ indicator and define the rejection condition at representative points.

Equation:

$$\log_{10}\kappa(T) = \frac{a + cT^{1/2} + eT + gT^{3/2} + iT^2}{1 + bT^{1/2} + dT + fT^{3/2} + hT^2}.$$

Metric: $\kappa(4\text{ K})$, $\kappa_{\max}$, the peak temperature ($T_{\text{at\_max}}$), $\kappa(300\text{ K})$, and monotonicity with respect to RRR. Rejection condition: with $\sigma_{\text{fit}} \approx \epsilon_{\text{fit}}\kappa$ ($\epsilon_{\text{fit}} \in [0.01, 0.02]$), reject if $|\kappa_{\text{pred}} - \kappa_{\text{target}}| > 3\sigma_{\text{fit}}$ at representative temperatures.

Results:

Metric	Value
$\kappa(4\text{ K})$	320 (RRR 50) to 3182 (RRR 500) W/(m K)
$\kappa_{\max}$	1476 (at $T \approx 26$ K; RRR 50) to 7614 (at $T \approx 14$ K; RRR 500) W/(m K)
$\kappa(300\text{ K})$	392 to 401 W/(m K) (weak RRR dependence)
NIST fit relative error	1-2%

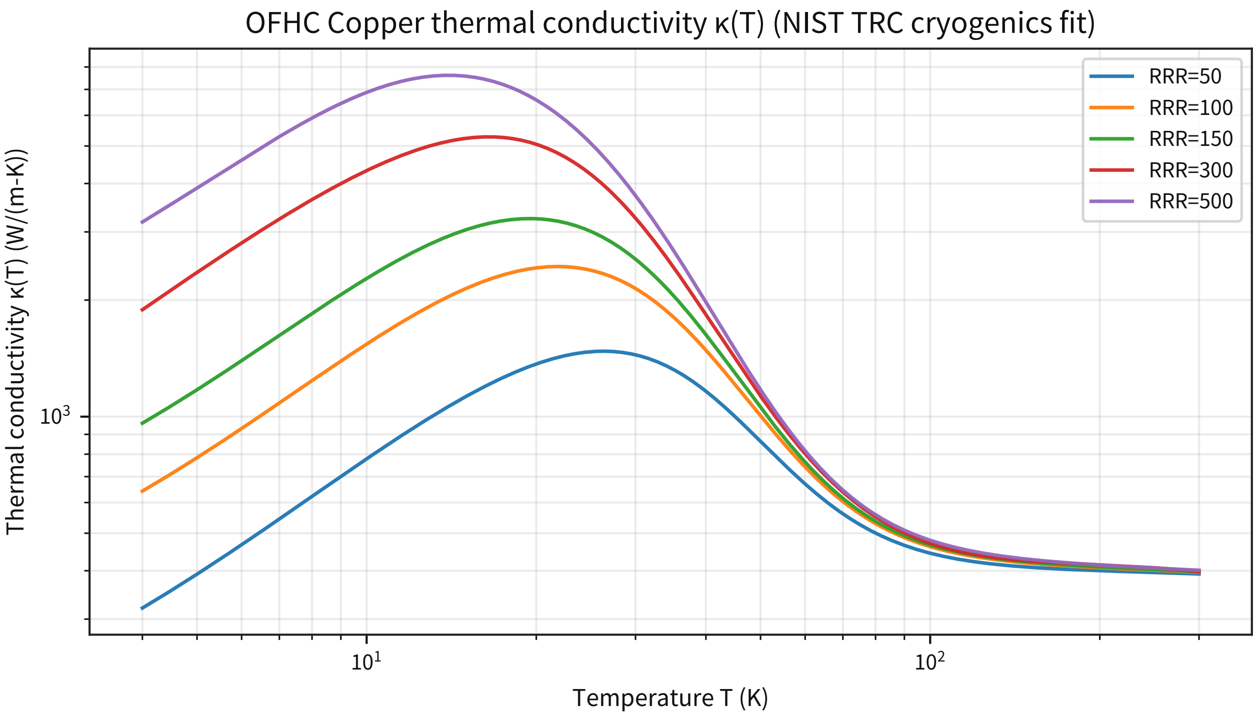

Figure 52: Compares $\kappa(T)$ for each RRR on the same 4-300 K temperature grid.

Supplement: it fixes visually that the low-temperature peak around 14-26 K rises with RRR and that the curves converge again near room temperature.

Freeze: For condensed systems (Si/Cu), the primary targets for lattice constants, heat capacity, resistivity, thermal expansion, and thermal conductivity (representative points plus error scales) are now frozen, giving an entry point at which re-derivations can be judged numerically as pass or fail. In addition, the temperature-band split holdout (train $\rightarrow$ test extrapolation) has been applied to $\alpha(T)$, $C_p(T)$, and $\kappa(T)$, and the residual growth under extrapolation has been fixed in the audit output as a procedure systematic (see the Section 8 audit summary). **Reject**: Claims that fail these targets (3σ deviation at representative points, violation of necessary conditions, or falling outside the frozen envelope) are rejected. In particular, because silicon $\alpha(T)$ includes a sign reversal, a constant model $\alpha \approx A \cdot C_v$ is rejected across the full range. **Holdout**: For the temperature-band split holdout (train $\rightarrow$ test extrapolation), the operational gate is max $|z| \leq 3$ on the test side (with $\chi^2_\nu \leq 9$ as an auxiliary metric). Examples satisfying this are called holdout-ok. With only the minimal Debye + Einstein (two-branch) model, silicon $\alpha(T)$ collapses (test max $|z| \approx$ 15-45), showing that range selection can become the dominant systematic. By contrast, when the phonon-DOS weights are frozen and a low-dimensional $\gamma(\omega)$ basis (three coefficients + fixed overlap) includes primary constraints on mode-dependent $\omega(T)$ (INS; Kim2015 Figure 2) and the bulk modulus $B(T)$, examples are obtained that satisfy both strict and holdout (test max $|z| \approx$ 1.9-2.5, reduced $\chi^2 \leq 2$). Silicon $C_p(T)$ collapses in holdout when Debye 1p or Shomate 5p is used alone (test max $|z| \approx 6$ to 8, and for Shomate the low-temperature side reaches max $|z| \approx 90$), so range selection can again dominate. By contrast, the "hybrid frozen" construction, in which the low-temperature (100-300 K) Debye θ_D is frozen by minimizing max $|z|$ and the high-temperature branch (298-1685 K) is frozen using the WebBook Shomate coefficients as primary constraints, satisfies test max $|z| \leq 3$ on both splits (holdout-ok). For copper $\kappa(T)$, the difference in ansatz class appears directly as a holdout pass/fail decision: the rational form is stable, whereas a log-polynomial collapses when extrapolated around the low-temperature peak. **Next**: Search for minimal ansatz classes with stronger primary constraints while keeping the cross-checks in condensed-matter/thermal systems (consistency between $\alpha(T)$ and $C_p(T)$, the RRR dependence of $\kappa(T)$, and so on), and fix them in the audit output as pass/fail.

3.10.2 Statistical Mechanics and Thermodynamics (Baseline Blackbody-Radiation Quantities)

Objective: As the prerequisite for connecting temperature and equilibrium distributions (thermal statistics) to the P-model, freeze the baseline blackbody quantities (σ, a, $n_\gamma \propto T^3$) from primary sources (SI/CODATA) and connect them to the rejection conditions for future extensions.

Input: NIST Cuu (CODATA) constants (c, h, k_B, σ) and the mathematical constants π and $\zeta(3)$.

Processing: Compute σ from the standard blackbody formula and derive the radiation constant a and the coefficient of the photon number density.

Equation:

$$\sigma = \frac{2\pi^5 k_B^4}{15\, h^3 c^2}, \qquad u(T) = aT^4, \quad a = \frac{4\sigma}{c}, \qquad n_\gamma(T) = \frac{2\zeta(3)}{\pi^2}\left(\frac{k_B T}{\hbar c}\right)^3.$$

Metric: σ (baseline quantity), a (energy-density scale), and the coefficient of n_γ (the prefactor of T^3).

Results:

Metric	Value
σ	$5.670374419 \times 10^{-8}$ W $(\mathrm{m}^{-2}\,\mathrm{K}^{-4})$
a	$7.56573325 \times 10^{-16}$ J $(\mathrm{m}^{-3}\,\mathrm{K}^{-4})$
n_γ coefficient	2.0286846×10^{7} $\mathrm{m}^{-3}\,\mathrm{K}^{-3}$
$\langle E\rangle/(k_B T)$	2.70118

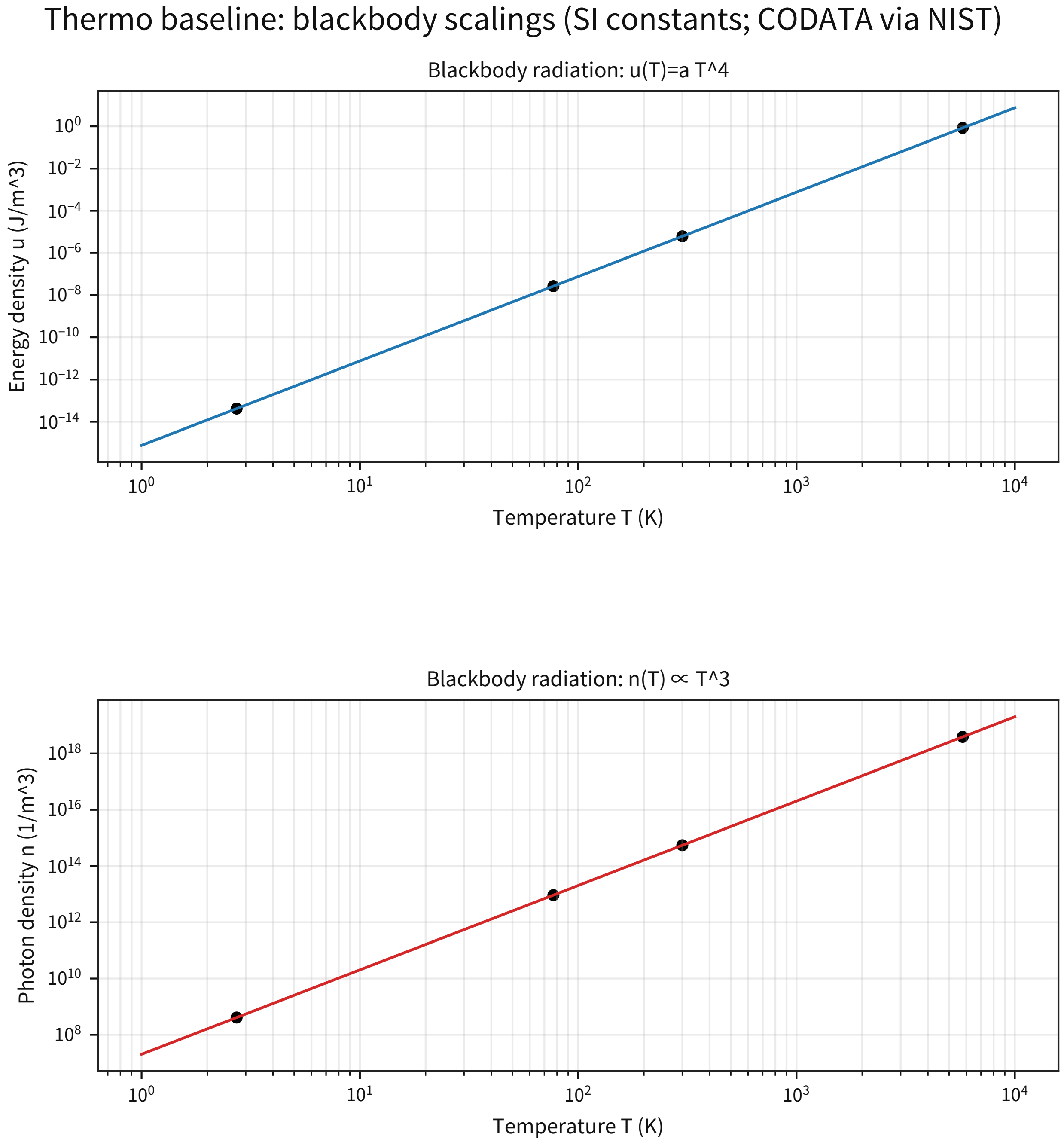

Figure 53: Left: the T^4 scaling of the energy density $u(T)$. Right: the T^3 scaling of the photon number density $n_\gamma(T)$.

Supplement: any future extension (for example, hypotheses about fluctuations or couplings of P) is rejected if it contradicts these baseline quantities or scaling relations.

Freeze: The baseline blackbody-radiation quantities (σ, a, $n_\gamma \propto T^3$) and $\langle E\rangle/(k_B T)$ are derived from primary constants and frozen as reference quantities. **Reject**: If an extension of the P-model systematically violates the equilibrium blackbody T^4/T^3 scaling or the baseline quantities above by more than 3σ of the total uncertainty, that extension is rejected. **Next**: Even when introducing extensions involving temperature or fluctuations of P, keep this blackbody baseline as the temperature calibration point first. If non-equilibrium effects (such as spectral distortions) are claimed, measurable extra parameters and the energy budget must be stated in parallel.

3.10.3 Statistical Mechanics and Thermodynamics (Blackbody: Entropy and Consistency with the Second Law)

Objective: As an operational baseline for temperature and entropy, freeze the equilibrium thermodynamics of a blackbody (photon gas), $u(T)$, $p(T)$, $s(T)$, and $s/(n_\gamma k_B) = \text{const}$, and provide the rejection condition for any future P-model extension that contradicts the second law or the equilibrium scaling relations.

Input: SI/CODATA constants (c, h, k_B) and the mathematical constants π and $\zeta(3)$.

Processing: Starting from $u = aT^4$, derive $p = u/3$ and $s = (4/3)aT^3$, and freeze as baseline quantities both the photon number density $n_\gamma(T)$ and the fact that the ratio $s/(n_\gamma k_B)$ is temperature-independent.

Equation:

$$u = aT^4, \quad p = \frac{u}{3}, \quad s = \frac{4}{3}aT^3, \qquad n_\gamma = \frac{2\zeta(3)}{\pi^2}\left(\frac{k_B T}{\hbar c}\right)^3, \qquad \frac{s}{n_\gamma k_B} = \frac{2\pi^4}{45\zeta(3)}.$$

Metric: $s/(n_\gamma k_B)$ (dimensionless) and the temperature scaling of u, p, s, and n_γ (T^4, T^4, T^3, T^3).

Results:

Metric	Value
$s/(n_\gamma k_B)$	3.60157
Coefficient of $s(T)$ (=(4/3)a)	$1.0087644 \times 10^{-15}$ J (m^{-3} K^{-4})

Representative temperatures:

T (K)	u (J/m^3)	p (Pa)	s (J/(m^3 K))	$s/(n_\gamma k_B)$
2.725	4.17×10^{-14}	1.39×10^{-14}	2.04×10^{-14}	3.60157
77	2.66×10^{-8}	8.87×10^{-9}	4.61×10^{-10}	3.60157
300	6.13×10^{-6}	2.04×10^{-6}	2.72×10^{-8}	3.60157
5772	8.40×10^{-1}	2.80×10^{-1}	1.94×10^{-4}	3.60157

Thermo baseline: blackbody entropy and second-law consistency (SI constants)

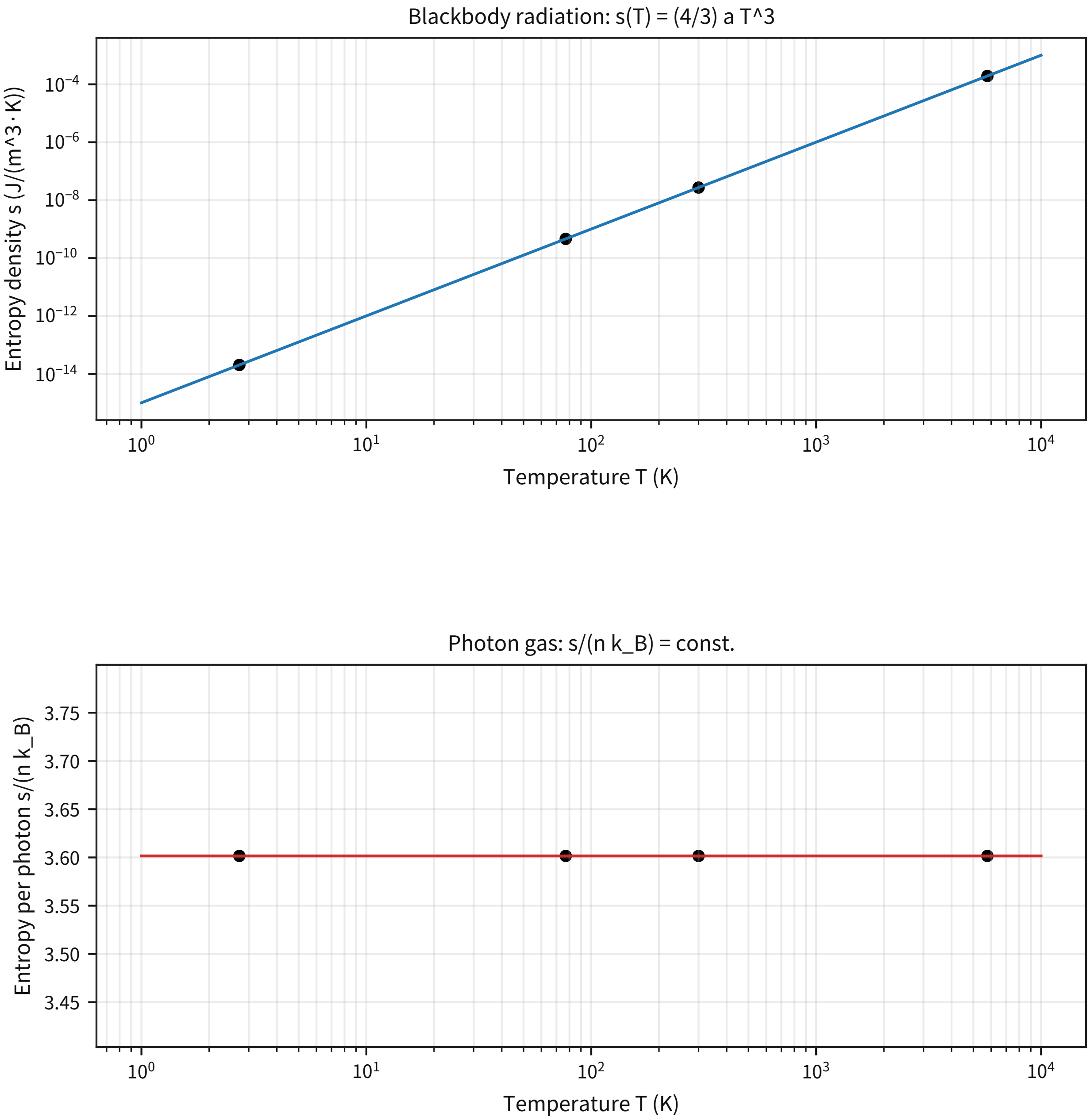

Figure 54: Left: fixes the scaling $s(T) = (4/3)aT^3$ on one plot from the CMB, through room temperature, up to the solar-photosphere temperature.

Supplement: right panel confirms that $s/(n_\gamma k_B)$ stays constant independently of temperature and freezes it as a baseline quantity of the equilibrium photon gas.

Discussion: Before introducing any P-model-specific definition of temperature, this subsection first freezes the operational baseline that "temperature and entropy in equilibrium can be calibrated by a blackbody (photon gas)." Therefore, if an extension of the P-model (for example, involving fluctuations or couplings of P) systematically violates the equilibrium T^3/T^4 scaling or $p = u/3$, it is rejected as inconsistent with the second law (and with established thermal statistics).

Note: The rejection condition here only requires that the scale of equilibrium thermodynamics not be altered. It is not a general statement that already includes non-equilibrium or open systems (heat baths, measurement, information, and so on). If the P-model claims genuinely new non-equilibrium effects, it must present a separate measurable mechanism and energy budget.

(Condensed matter / thermal: fixing the falsification-condition pack; for audit)

- Treat the falsification-condition pack (JSON; listed in Section 8) as the authoritative source.
- Audit condition: adopt the rejection conditions included in the metric outputs of each baseline ($\pm 3\sigma$, $\pm 3\sigma_{\rm proxy}$, necessary-condition envelopes, and so on) as the thresholds, and do not refit the targets or change the thresholds after the fact.

Freeze: Freeze the equilibrium relations of the blackbody (photon gas), $u = aT^4$, $p = u/3$, $s = (4/3)aT^3$, $n_\gamma \propto T^3$, and the dimensionless ratio $s/(n_\gamma k_B)$ as operational baselines. **Reject**: If an extension of the P-model systematically violates, at equilibrium, $p = u/3$, the T^3/T^4 scaling, or $s/(n_\gamma k_B) = \text{const}$ by more than 3σ of the total uncertainty, it is rejected. **Next**: If extensions that include non-equilibrium or open-system behavior are introduced, define measurable mechanisms (noise, dissipation, information) and the energy budget, and only then add extra degrees of freedom within a range that does not contradict the equilibrium-blackbody baseline first.

3.11 Quantum "Dynamic Collapse": Process Visualization of the Measurement Problem

Objective: Provide a minimal demonstration, as a time evolution with nonlinear terms, of how measurement would occur in the P-model beyond what is covered by the Bell-selection audit alone, and quantify the relation between branch convergence and pointer-record correlations.

Input: A two-state system (initial superposition) and a minimal detector model with three pointer channels plus an environmental dissipation channel.

Processing: Integrate a continuous-measurement-type nonlinear stochastic equation with Euler-Maruyama, and over many trajectories evaluate the convergence time of $z(t) = \langle M \rangle$, the pointer-consensus rate, and branch stability (reversal rate and sign consistency).

Equation:

$$d|\psi\rangle = \left[-iH\,dt - \frac{\gamma_m}{2}\left(M - \langle M\rangle\right)^2 dt + \sqrt{\gamma_m}\left(M - \langle M\rangle\right) dW_t\right] |\psi\rangle, \qquad H = \frac{\omega}{2}\sigma_x + \frac{\omega_{\mathrm{env}}}{2} E\,\sigma_z,\ M = \sigma_z \tag{6}$$

$$dx_k = \left(-\Gamma_k x_k + \chi_k \langle M\rangle + \eta_k E\right) dt + \sigma_{\xi,k} dW'_{k,t}, \qquad dE = \left(-\lambda_E E\right) dt + \sigma_E dW_{E,t}$$

Metric: Collapse fraction, convergence time (median / p90), pointer-consensus rate, branch-reversal rate, post-collapse sign consistency, final mean $|z(T)|$, and the coherence-suppression ratio.

Results:

Metric	Value
Collapse fraction	1.000
Collapse time, median	39.6 ms
Collapse time, p90	102.5 ms
Pointer-consensus rate	0.9656
Branch-reversal rate	0.075
Branch-stability rate	0.925
Post-collapse sign consistency, median	1.000
mean $\lvert z(T)\rvert$	0.994
Coherence-suppression ratio (final median / initial)	0.0276

Dynamic-collapse simulation figure: the upper panel shows the time evolution of $z(t)$ and the three pointer degrees of freedom $x_k(t)$, and the lower panel shows the collapse-time distribution and the final-state correlations. Visually, one can confirm simultaneously both the convergence from superposition to branching and the multi-degree-of-freedom pointer records that track the branch.

Output: fix as outputs both the time-series visualization of the dynamic collapse (state expectation value, pointer degrees of freedom, convergence-time distribution, and final-state correlations) and the audit-metric table that can be recomputed under the same conditions (collapse fraction / collapse time / consensus rate / reversal rate / coherence-suppression ratio).

Discussion: This result is the minimal demonstration required to write measurement as a physical process, and it complements the Bell audit (the entry audit) in Part III. At the same time, the present model is still a reduced system of two states + three pointer channels + an OU environment; it is not a full description of Schrödinger-cat-scale measurement (many degrees of freedom, dissipation, and first-principles amplification chains).

Note: This subsection does not claim to reject standard quantum mechanics; it is an operational step that provides a constructive demonstration on the P-model side. For this model to remain valid, future experimental systems must reproduce, under an isomorphic gate, the distribution of τ_{coll}, the pointer-consensus rate, and the branch stability.

Freeze: In this paper, the coefficient set of the continuous-measurement equation and the collapse threshold ($|z| \geq 0.95$) are frozen, and reproducibility under the same settings is treated as the audit criterion. **Reject**: If, under the same settings, the collapse fraction drops significantly, the pointer-consensus rate decreases, or the branch-reversal rate stays high, this minimal dynamic-collapse model is rejected. **Next**: Extend the model to a many-degree-of-freedom detector (amplification chains, dissipative channels, threshold detection) and move to an isomorphic audit that can be compared directly with experimental time-series logs.

4 Differential Predictions

This chapter fixes, for the quantum themes of Part III, both the differential predictions (what can be distinguished observationally) and the rejection conditions (3σ), so that the reader can track, under future data updates, which precision and which procedure become the actual decision point. The overall picture is summarized in the following table, and Bell and the Born rule are formalized in 5.1-5.2 as "packs" of falsification procedures.

Theme	Observable	Standard-QM prediction	P-model prediction	Current precision	Required precision	Rejection condition
Bell (CHSH)	$\lvert S\rvert$	Ideal $2\sqrt{2} \approx 2.828$ (LR bound 2)	Selection-dependent: $\lvert S\rvert \approx 1.10$-$2.90$ (Weihs + Delft; moves with procedure)	$\sigma_{\text{stat}} \approx 0.02$-$0.20$; $R_{\text{sel}} \approx 3.65$-$19.3$ (median 7.60); $z_{\text{delay}}(dt) \approx 5.54$ (time tags only)	Procedure systematics $\ll$ statistics ($R_{\text{sel}} < 1$), and $\min(\lvert S\rvert) - 2 > 3\sigma_{\text{total}}$	Reject the selection entry point once a setting-independent parent population (w_{ab} independent of a, b) is established and $\lvert S\rvert > 2$ is reproduced consistently above 3σ across multiple datasets
Bell (CH)	J_{prob}	$J_{\text{prob}} > 0$ (LR bound 0)	Selection-dependent: J_{prob} moves under window sweeps (crosses 0 for NIST; does not cross 0 over the scanned range for Kwiat)	$\sigma_{\text{stat}} \approx (0.9$-$3.9) \times 10^{-5}$; $\Delta_{\text{sys}} \approx (0.12$-$3.5) \times 10^{-3}$; $R_{\text{sel}} \approx 5.82$-$654$ (median ≈ 330)	Procedure systematics $\ll$ statistics ($R_{\text{sel}} < 1$), and $\min(J_{\text{prob}}) - 0 > 3\sigma_{\text{total}}$	Reject the selection entry point once, under a setting-independent procedure, $J_{\text{prob}} > 0$ is reproduced consistently above 3σ across multiple datasets
Born-rule deviation (self-gravity extension)	$\chi \equiv (E_G/\hbar)T$	$\delta = 0$ (Born)	$\chi \approx (m/m_{\text{crit}})^2$; $m_{\text{crit}} \equiv \sqrt{\hbar R/(GT)}$	At present: $\chi \lesssim 10^{-8}$ for $m \lesssim 10^8$ amu (outside detectable range)	$\chi \gtrsim 0.1$-1 ($m \gtrsim 0.3$-$1\, m_{\text{crit}}$) with phase / visibility error $\lesssim \chi/3$	Reject the extension if, in the regime $\chi \gtrsim 1$, no deviation is seen at 3σ (upper bound below the predicted signal)

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Theme	Observable	Standard-QM prediction	P-model prediction	Current precision	Required precision	Rejection condition
Dynamic collapse of measurement (environment coupling)	τ_{coll}, pointer-consensus rate, branch stability	Conditional update (apparatus-model dependent)	Nonlinear measurement equation with explicit branch convergence; here collapse fraction = 1.00, $\tau_{50} = 39.6$ ms, pointer-consensus rate = 0.966, branch-reversal rate = 0.075	Reduced model with 2 states + 3 pointer channels + environmental dissipation (320 trajectories)	Reproduce the same class of indicators within 3σ in a many-degree-of-freedom detector	Reject if the collapse fraction drops significantly, the pointer-consensus rate decreases, or the branch-reversal rate remains high
COW	$\Delta\phi$	$\Delta\phi = -mgH^2/(\hbar v_0)$ (example: -70.09 rad)	Weak-field same; $\delta\phi/\phi \approx 1.39 \times 10^{-9}$ (example: $\delta\phi \approx -9.75 \times 10^{-8}$ rad)	Relative phase precision $\approx 10^{-2}$ (assumed)	Relative $\lesssim 4.64 \times 10^{-10}$ (3σ)	Reject if a future large-ΔU test disagrees above 3σ
Atom interferometer	ϕ	$\phi \approx k_{\mathrm{eff}} g T^2$ (example: 2.31×10^7 rad)	Same in the weak field + $\delta\phi/\phi \approx 1.39 \times 10^{-9}$ (example: $\delta\phi \approx -3.22 \times 10^{-2}$ rad)	Relative $\approx 10^{-9}$ (assumed)	Relative $\lesssim 4.64 \times 10^{-10}$ (3σ)	Differential detection is difficult at current precision. Reject if a long-free-fall test shows disagreement above 3σ
Quantum clock	$\Delta f/f$ (or ϵ)	$\Delta f/f \approx \Delta U/c^2$ ($\epsilon = 0$)	Same in the weak field; P-GR differential term is $\delta(\Delta f/f) \approx -6.05 \times 10^{-23}$ ($\Delta h \approx 399$ m)	$\sigma(\Delta f/f) \approx 2.89 \times 10^{-17}$	Absolute $\lesssim 2.02 \times 10^{-23}$ (3σ)	Reject if a deviation from $\epsilon = 0$ exceeds 3σ, or if the long-baseline $\delta(\Delta f/f)$ disagrees above 3σ
de Broglie precision	z (α consistency)	$z = 0$	$z = 0$	$z = 0.588$	Reject region is $\lvert z\rvert > 3$	$\lvert z\rvert > 3$ (inconsistency with independent α)

Continued on next page

Theme	Observable	Standard-QM prediction	P-model prediction	Current precision	Required precision	Rejection condition
Gravity-induced decoherence	$V(t), \sigma_y$	Pikovski formula (ensemble dephasing)	Added noise σ_y; with $\sigma_z = 0.1/1/10\,\mathrm{mm}$, $t \approx 4 \times 10^4/4 \times 10^3/4 \times 10^2$ s gives $V = 0.5$	Not yet detected	Constrain σ_y to $\lesssim 10^{-20}$-10^{-18}	Reject if the measured $V(t)$ disagrees with the frozen σ_y model by more than 3σ

4.1 Bell: Falsification-Condition Pack

Objective: Instead of making the existence or absence of a Bell "violation" the conclusion, freeze, against public primary data, how strongly the selection knobs (coincidence window / offset / trial definition) move the statistics, and formalize the conditions under which the selection entry point (procedure dependence) is rejected as an operational threshold pack.

Input: Public time-tag datasets (or event-ready data): Weihs 1998, NIST 2015, Kwiat/Christensen 2013, and Delft 2015/2016. [5][2][6][3][4]

Processing: For the natural window, define an "accidental-coincidence proxy" statistically from the peak of the dt histogram, and recommend the smallest window that contains at least 99% of the signal after subtracting the off-peak background (plateau fraction). Snap the recommended window to the grid and freeze it; treat window / offset sweeps not as an optimization but as the evaluation of the systematic width Δ_{sys}.

Equation: selection sensitivity (ratio of systematic width to statistics)

$$\Delta_{\mathrm{sys}}(T) \equiv \max_k T(k) - \min_k T(k), \qquad R_{\mathrm{sel}} \equiv \frac{\Delta_{\mathrm{sys}}(T)}{\sigma_{\mathrm{stat,med}}} \tag{7}$$

Here $\sigma_{\mathrm{stat,med}}$ is the median of the statistical errors $\sigma_{\mathrm{stat}}(k)$ obtained over the sweep of each k (bootstrap estimate). The variable k denotes the selection knob (window, offset, trial definition, and so on), and T denotes the statistic (for example, CHSH $|S|$ or CH J_{prob}). As the adopted definitions (sign convention), fix CHSH and CH (probability form) as

$$E(a,b) \equiv \frac{N_{++} + N_{--} - N_{+-} - N_{-+}}{N_{++} + N_{--} + N_{+-} + N_{-+}}, \qquad S \equiv E(a,b) - E(a,b') + E(a',b) + E(a',b') \tag{8}$$

The sign placement of CHSH is convention-dependent; for example,

$$S' \equiv E(a,b) + E(a,b') + E(a',b) - E(a',b')$$

is equivalent under relabeling of the settings. Therefore, the decision metric is fixed to $|S|$ (absolute value).

$$J_{\mathrm{prob}} \equiv P_{++}(a,b) - P_{++}(a,b') + P_{++}(a',b) + P_{++}(a',b') - P_{+}(a') - P_{+}(b) \tag{9}$$

Under this convention, the local-realist bounds are $|S| \leq 2$ and $J_{\mathrm{prob}} \leq 0$. In addition, define the delay signature (a proxy for setting dependence) on the post-pairing dt distribution by

$$dt \equiv t_B - t_A - \text{offset}, \qquad z_{\mathrm{delay}} \equiv \frac{|\Delta\text{median}|}{\sigma_{\Delta\text{median}}} \tag{10}$$

where Δmedian is the difference in median between setting 0 and setting 1. Freeze z_{delay} as the "detection strength of setting dependence."

Metric: R_{sel} (selection sensitivity), z_{delay} (delay signature), and the dataset-crossing minimum values on the conservative side are adopted as the "falsification-condition pack." The operational thresholds are fixed to $R_{\mathrm{sel}} \geq 1$ (the systematic width moves by at least $1\sigma_{\mathrm{stat}}$) and $z_{\mathrm{delay}} \geq 3$ (3σ-equivalent detection).

Results:

Dataset	Statistic	Selection knob	R_{sel}	z_{delay} (max)
Weihs 1998	CHSH $\|S\|$	window	19.3	5.54
Delft 2015	CHSH S	event-ready offset	3.65	n/a
Delft 2016	CHSH S	event-ready offset	7.60	n/a
NIST 2015	CH J_{prob}	window	654	43.9
Kwiat/Christensen 2013	CH	window	5.82	n/a

Table 64: Bell cross-dataset integration (summary of the falsification-condition pack; operational)

Metric (cross-dataset)	Representative value	Meaning (key point)
R_{sel} (min/median/max)	3.65 / 7.60 / 654	Degree to which the procedure systematic (sweep width) exceeds the statistical error (selection sensitivity). Operationally the conservative side adopts the minimum.
z_{delay} (dt; min)	5.54	Under the dt definition for fast switching / time tags, the proxy for setting-dependent delay (weighting) exceeds 3σ.
Profile-correlation rank ($\epsilon = 10^{-10}$)	5	Degree to which the cross-dataset correlation of the sweep profiles is dominated by a small number of modes (existence of common procedure modes).
Strongest profile-correlation pair	Weihs 1998 and NIST: corr ≈ -0.888	Strongest correlation between two datasets, indicating shared procedure modes.
Top 15-item budget entries (median)	offset 3.88, window 3.13, eff drift 0.35	Which systematic knobs dominate and by how much (median scale).
Top item-correlation pair	window-accidental corr ≈ 0.99995	Strongly correlated error sources, identifying pairs that cannot be treated as independent.

Continued on next page

Metric (cross-dataset)	Representative value	Meaning (key point)
Pairing cross-check max ($\Delta_{\mathrm{impl}}/\sigma_{\mathrm{boot}}$)	0.086	The statistical difference caused by implementation differences (pairing rules and so on) remains below 1σ of the statistical error, reducing suspicion of implementation dependence.

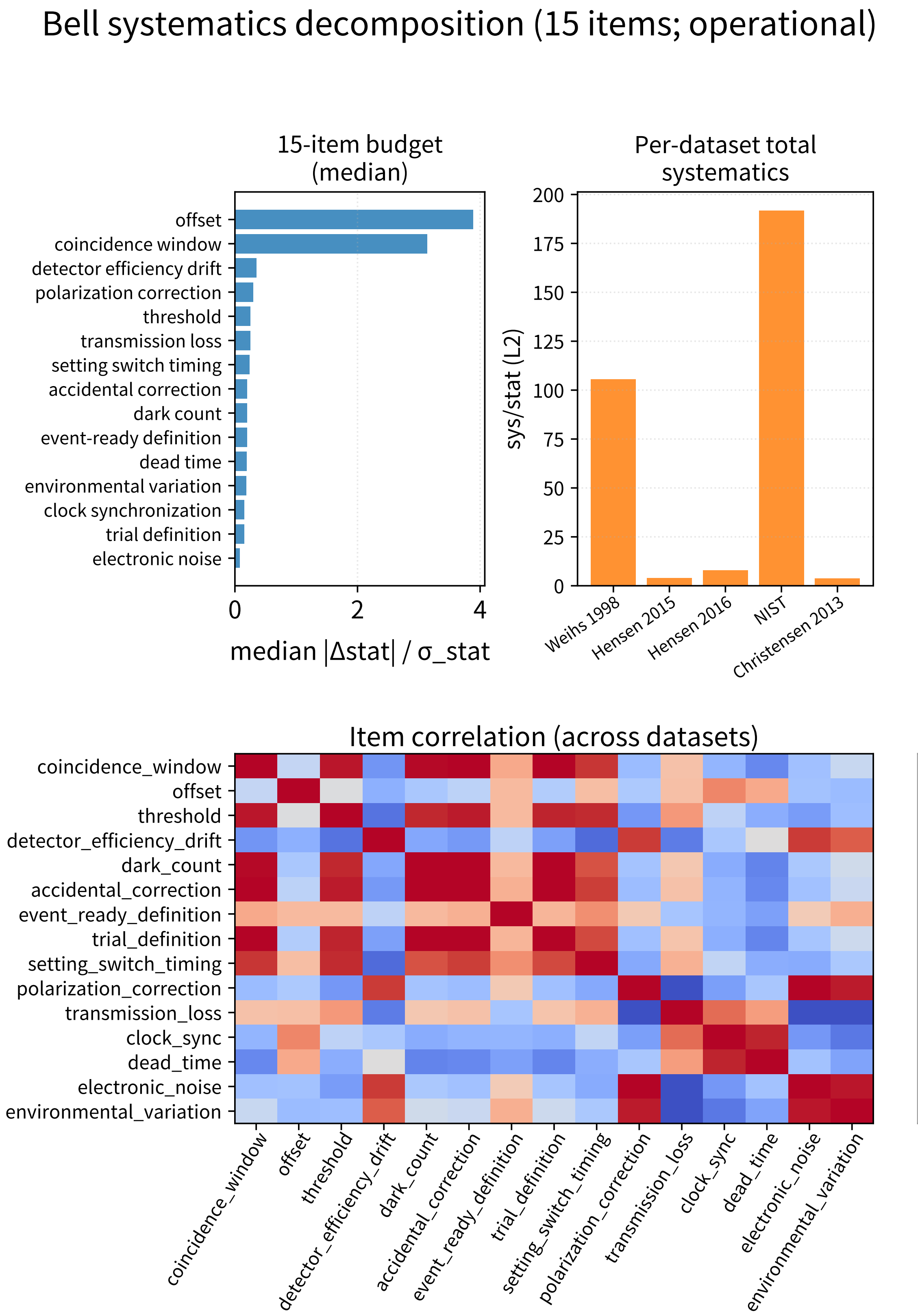

Figure 55: Shows the 15-item systematic budget (medians), the per-dataset totals, and the item-by-item correlation heatmap.

Supplement: fixes both ”which knobs dominate” and ”which pairs cannot be regarded as independent.”

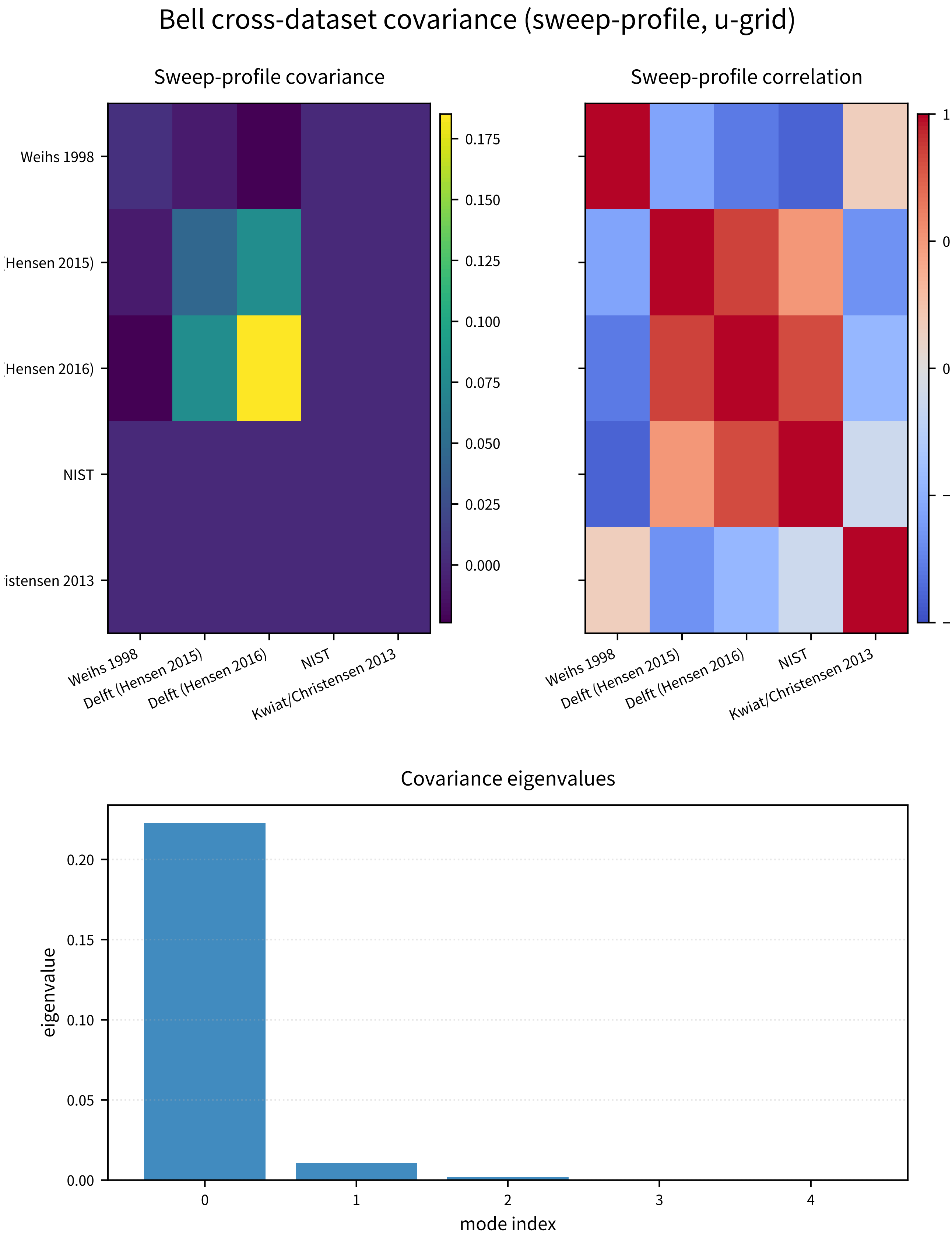

Figure 56: Projects the sweep profiles across datasets onto a common grid and visualizes the covariance (correlation) structure.

Supplement: this serves as the entry point for unified updating of the correlations even when future datasets are added.

Across datasets, the selection sensitivity is $\min(R_{\mathrm{sel}})$=3.65, median 7.60 (maximum 654), so at least over the range treated in this paper, the fact that "the selection knobs move the statistic" itself is quantified well above 1σ. For the datasets where z_{delay} can be defined from photon time tags, $\min(z_{\mathrm{delay}})$=5.54 (Weihs) and $\max(z_{\mathrm{delay}})$=43.9 (NIST), so the setting dependence of the delay distribution (proxy indicator) is detected above 3σ.

For Kwiat, do not apply the main delay-signature gate of the fast-switch family. Instead, after confirming by resweep that $\max|z| = 2.73 < 3$, keep it as a watch in the non-primary gate and manage it separately as a selection-system diagnostic.

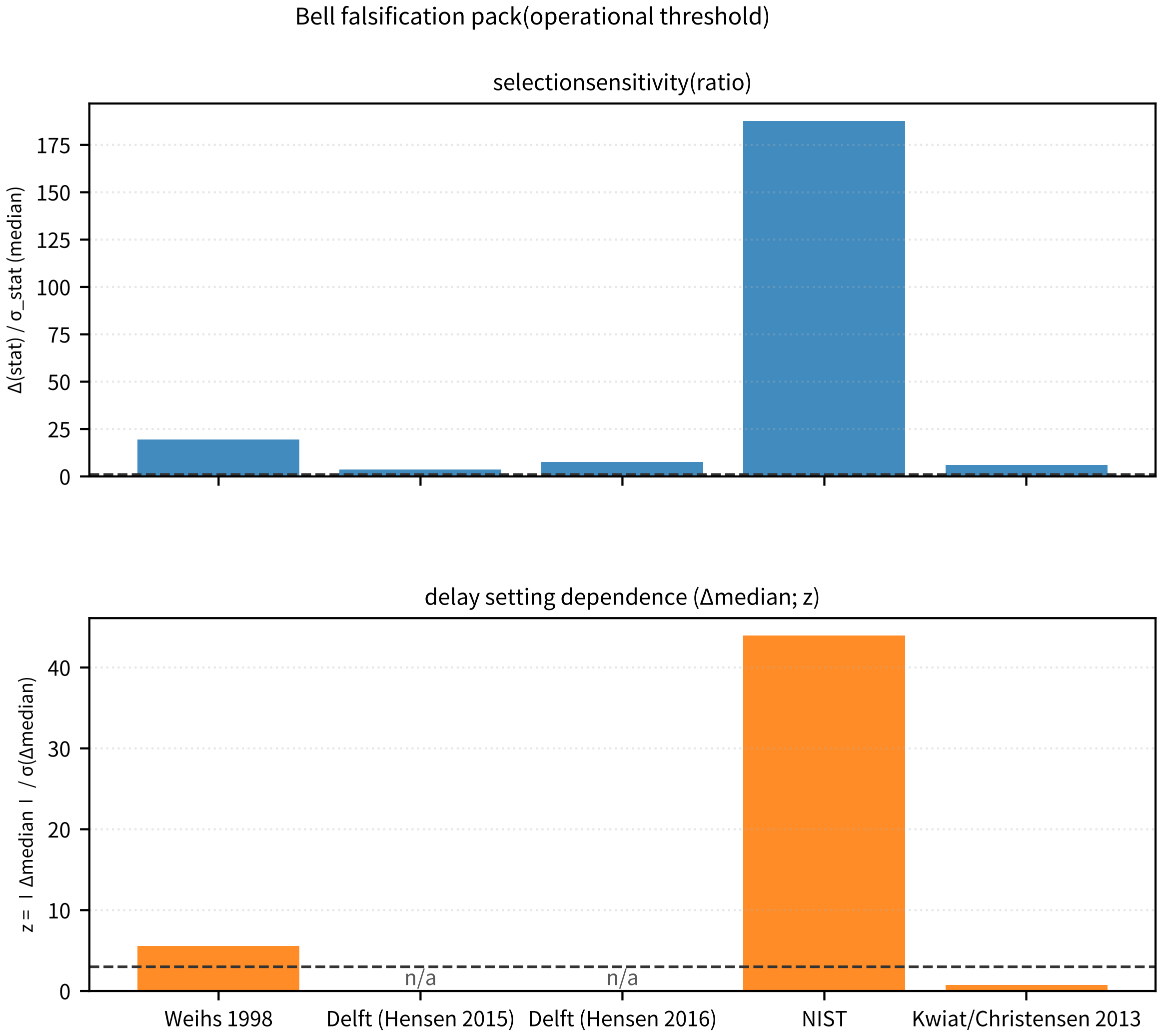

Figure 57: Left: compares R_{sel} (selection sensitivity) across datasets and fixes how far above the operational threshold ($R_{\mathrm{sel}} = 1$) each one lies (reproduced from Section 3.2, Figure 3).

Supplement: right panel shows, for the datasets where the delay signature can be defined, the margin relative to the 3σ threshold ($z_{\mathrm{delay}} = 3$).

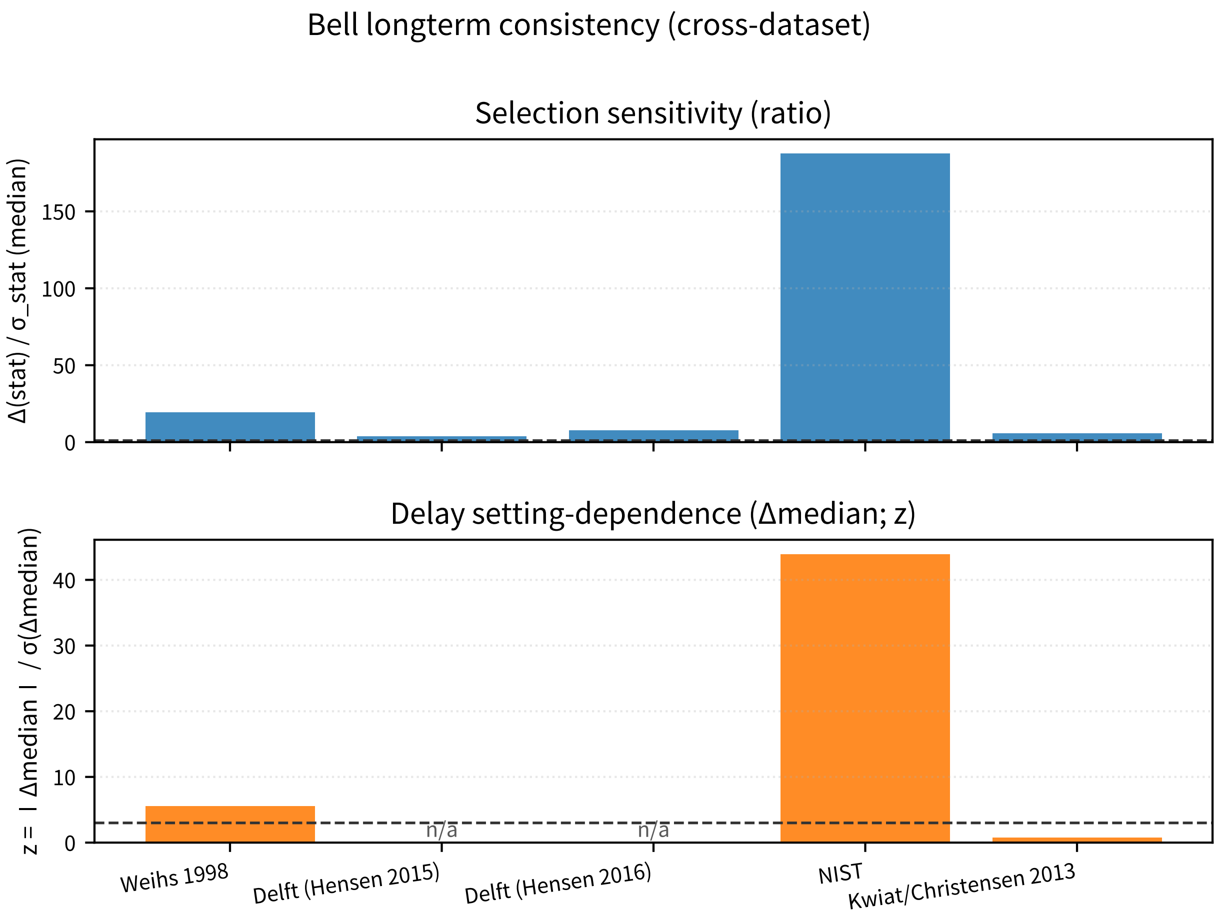

Figure 58: Provides a cross-dataset overview of the Bell "entry indicators" and freezes the conservative lower bounds of R_{sel} and z_{delay}.

Supplement: this ensures that pass/fail can be updated through the same I/F when future datasets are added.

Discussion: The claim of this paper is not to newly claim the existence of loopholes. Rather, it is to freeze, on primary data, the quantitative sensitivity of selection (which knob moves the statistic and by how much) and connect it to explicit falsification conditions (under which procedure the selection entry point is rejected). Accordingly, the falsification of Bell here depends not on the bare existence of a violation, but on the establishment of a setting-independent parent population and the robust reproduction of the result under that procedure.

Note: The thresholds used here ($R_{\text{sel}} = 1$, $z_{\text{delay}} = 3$) are operational thresholds of this repository, not universal physical constants. For event-ready datasets (Delft) and trial-based datasets (Kwiat/Christensen), the definition of z_{delay} is not isomorphic, so it is kept as n/a and updated only after projection onto the same I/F where possible.

(Fixing the falsification-condition pack; for audit)

- Treat the falsification-condition pack (JSON; listed in Section 8) as the authoritative source.
- Audit condition: apply the same I/F to the datasets listed in the pack, and do not change, after the fact, either the natural window (plateau fraction 0.99; exceptions only by override) or the sweep grid.

- Rejection thresholds (operational; the pack thresholds are authoritative): the lower bound of the selection-origin ratio is 1, and the lower bound of the delay signature is $z_{\mathrm{delay}} = 3$ (the legacy KS threshold remains 0.01).
- Supplement: the pack closes the audit entry point into a single JSON, including the cross-dataset integration summary (covariance / 15 systematics / loophole diagnostics) and the referenced outputs.

4.2 Born-Rule Deviation: The m_{crit} Threshold and Precision Requirements for Future Experiments (MAQRO / OTIMA)

Objective: Without claiming that the Born rule has been derived from first principles, freeze, for self-gravity (mean-field) or similar extension hypotheses, the scale at which a differential signal would become visible by reducing it to the threshold m_{crit} and the required precision (3σ).

Input: Wavepacket size R, interference time T, and mass m (the experimentally reachable value). For molecular interferometry (such as OTIMA), assume the $m \lesssim 10^6 \dots 10^8$ amu range; for space-based macro-interference concepts (such as MAQRO), assume that long T can be reached. Concrete instrument design is outside the scope of this paper.

Processing: Use the scale of the gravitational self-energy $E_G \sim Gm^2/R$ to define the dimensionless control parameter $\chi \equiv (E_G/\hbar)T$. For $\chi \ll 1$, the differential signal is effectively zero; for $\chi \sim 1$, a difference may become manifest. Freeze the 3σ detection condition to require that (i) the phase / visibility measurement error be at most on the order of $\chi/3$, and (ii) the count statistics satisfy $3/\sqrt{N} \lesssim \chi$.

Equation:

$$\chi \equiv \frac{E_G}{\hbar}T \sim \frac{Gm^2}{\hbar R}T = \left(\frac{m}{m_{\text{crit}}}\right)^2, \qquad m_{\text{crit}} \equiv \sqrt{\frac{\hbar R}{GT}}$$

The 3σ guide for count-statistics detection is

$$|\delta| \gtrsim \frac{3}{\sqrt{N}}$$

where δ is a representative relative deviation from the Born rule and N is the effective trial count.

Metric: The reach ratio m/m_{crit}, the differential scale χ, and the 3σ requirements on $(N, \sigma_{\text{phase}}/\sigma_V)$.

Results:

Scenario	R	T	m	m_{crit}	$\chi = (m/m_{\text{crit}})^2$	3σ requirement (guide)
OTIMA-like (example)	1 nm	1 ms	10^8 amu	7.6×10^{11} amu	1.7×10^{-8}	$\sigma_{\text{phase}} \lesssim 6 \times 10^{-9}$ rad and $N \gtrsim 3 \times 10^{16}$ (not realistic)
MAQRO-like (example)	100 nm	100 s	10^{10} amu	2.4×10^{10} amu	0.17	$\sigma_{\text{phase}}/\sigma_V \lesssim 0.06$ and $N \gtrsim 3 \times 10^2$
MAQRO-like (threshold)	100 nm	100 s	2.4×10^{10} amu	2.4×10^{10} amu	1	$\sigma_{\text{phase}}/\sigma_V \lesssim 0.33$ and $N \gtrsim 9$

Under the position adopted in this paper, current molecular interferometry lies at $m \ll m_{\text{crit}}$ ($\chi \ll 1$), so no Born-rule deviation is predicted there. The actual decision point for direct testing therefore depends on space-based experiments that can push χ into the 0.1-1 range by combining long T with sufficiently large m.

Output: A flowchart of the Born-rule accept / deviation conditions and metric outputs such as

m_{crit}/χ.

Output (for audit):

- The flowchart of the Born-rule accept / deviation conditions (Figure 1 of Part III-A).
- Machine-readable tables of metrics such as m_{crit}/χ. The authoritative file list is summarized in Section 8.

Rejection condition: If, in the regime $\chi \gtrsim 1$ ($m \gtrsim m_{\mathrm{crit}}$), no deviation is detected at 3σ (i.e. the observational upper bound falls below the prediction of the extension hypothesis), then that extension hypothesis is rejected.

Note: χ is only a scale indicator for the regime in which self-gravity contributes a phase of order 1 rad. The actual prefactor and the observable (phase or visibility) depend on the adopted extension model (Schrödinger-Newton, environment coupling, non-Markovianity, and so on). Accordingly, the numerical values in this subsection are used as baselines for connecting the experimental design variables (R, T, m, N) to the differential prediction.

4.3 Atomic Nuclei ($\Delta\omega \to B.E.$): Integrating the Falsification Pack with Independent Cross-Checks

Objective: Connect the differential-channel precision gate ($\sigma_{\mathrm{rel}}(\mathrm{total}) \leq \sigma_{\mathrm{req,rel}}$) to independent observables (separation energies and the charge-radius kink), and freeze as an operational condition that the verdict is not based on channel differences alone.

Input: The differential-channel indicators (difference scale and required precision for P-SEMF / P-Yukawa) and the independent cross-check indicators (separation-energy gap and strict radius-kink audit).

Output (for audit):

- Quantitative results for the differential channels (difference scales and required precision for P-SEMF / P-Yukawa).
- Quantitative results for the independent cross-checks (separation-energy gap and strict radius kink). The authoritative file list is summarized in Section 8.

Processing: Keep the differential-channel $\sigma_{\mathrm{req,rel}}$ fixed and bind the independent cross-check set (Sn/S2n gap plus strict $\Delta^2 r$ kink) into the same JSON, thereby adding an independent-cross-check-required gate.

Equation:

$$\sigma_{\mathrm{req,rel}} = \frac{|\Delta B|}{3B_{\mathrm{obs}}}, \qquad \mathrm{gate} : \sigma_{\mathrm{rel}}(\mathrm{total}) \leq \sigma_{\mathrm{req,rel}}$$

On the independent cross-check side, adopt the strict radius-kink gate

$$\max\left(|\Delta^2 r_{\mathrm{pred}} - \Delta^2 r_{\mathrm{obs}}|/\sigma_{\Delta^2 r}\right) \leq 3.$$

Metric: z_{median}, $z_{\Delta\mathrm{median}}$, median $\sigma_{\mathrm{req,rel}}$ (P-SEMF / P-Yukawa), and the independent cross-checks Sn/S2n RMS and the strict pass/fail of the radius kink at $A_{\mathrm{min}} = 100$.

Results: Both global-R and local-d keep failing in the median indicators, while the A-trend indicators remain within 3σ. The required relative precision of the differential channels is about 2.70% for P-SEMF and 33.43% for P-Yukawa, so the high-precision gate is governed by the P-SEMF side. On the independent cross-check side, the separation-energy gap indicator has valid points, and the radius kink reaches a strict pass with pairing frozen at $A_{\mathrm{min}} = 100$ (Sn $\approx 0.688\sigma$, Sp $\approx 2.014\sigma$).

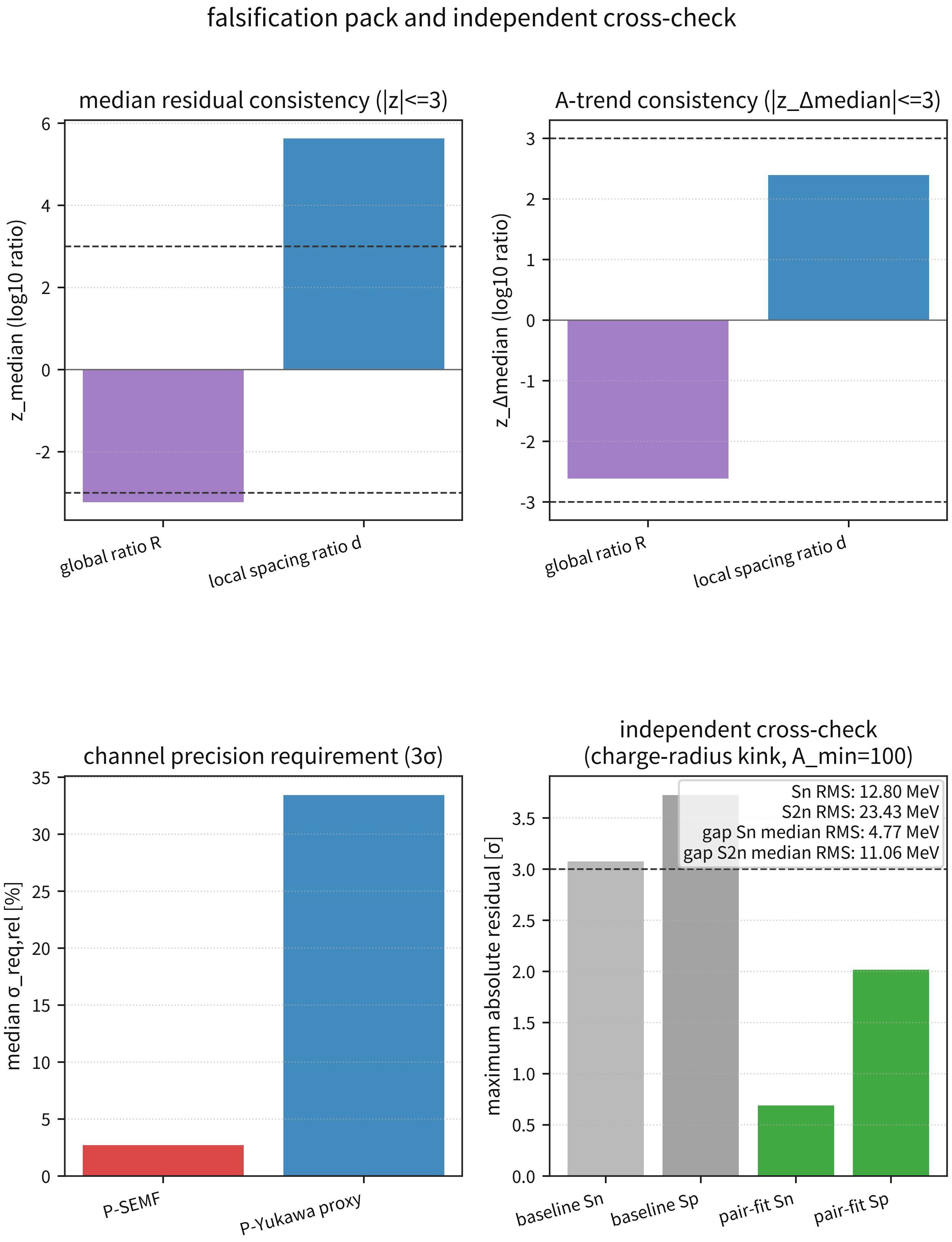

Figure 59: The upper panel keeps the conventional z indicators (median and A-trend), and the lower panel integrates the differential-channel precision demand with the independent cross-checks (Sn/S2n and the radius kink) into a single falsification pack.

Supplement: this fixes, as an operational I/F, that the verdict of the pack does not depend on the convenience of a single-channel fit alone.

Discussion: This update does not change the threshold itself (the 3σ operation remains fixed). Instead, it enlarges the bundle of information used by the decision, making the acceptance condition for channel differences more stringent. Accordingly, even when future additional physics is evaluated, it remains comparable under the same falsification thresholds and the same bundle of independent cross-checks.

Note: The cross-check in this subsection is not a sufficient condition; it is a necessary guardrail. Even if the independent cross-checks pass, nuclide groups that fail the differential-channel condition $\sigma_{\rm rel}(\text{total}) \le \sigma_{\rm req,rel}$ are not promoted to the 3σ decision target and remain pending.

(Falsification-pack freeze; for audit)

- Use the falsification pack (JSON; listed in Section 8) as the authoritative source.
- Application (audit conditions): in addition to the 3σ thresholds for $z_{\rm median}$ and $z_{\Delta{\rm median}}$, freeze the differential-channel precision gate $\sigma_{\rm rel}(total) \le \sigma_{\rm req,rel}$ and the independent cross-checks (Sn/S2n gaps and the strict $\Delta^2 r$ kink test with pairing frozen at $A_{\rm min} = 100$) as required conditions.
- Rejection thresholds (operational; use the pack thresholds as authoritative): $|z_{\rm median}|_{\rm max} = 3$ and $|z_{\Delta{\rm median}}|_{\rm max} = 3$ with fixed A bins.

4.4 Nuclear-Force Wave Interference: Differential Predictions and Rejection Conditions

Objective: Reduce the $\Delta\omega \to B.E.$ mapping fixed in 4.8.4-4.8.6 to the channel difference ΔB relative to standard-theory proxies (SEMF and the Yukawa-distance proxy), and to the required precision $\sigma_{\rm req} = |\Delta B|/3$, thereby freezing the decision difficulty for each nuclide as a falsification I/F for Chapter 5.

Input: The differential quantification results for all AME2020 nuclides ($n = 3554$) and the ranked list of top decision-point candidate nuclides.

Output (for audit):

- Differential quantification over all nuclides (difference scale, required precision, and A-band statistics).
- Ranked list of the top decision-point candidates. The authoritative file list is summarized in Section 8.

Processing: Take the P-model (with local spacing d and frozen $\nu_{\rm sat}$) as the baseline and compute, for each nuclide, the channel differences P-SEMF and P-Yukawa. Then define $\sigma_{\rm req}$ and $\sigma_{\rm req,rel}$ as the 3σ decision thresholds, and freeze the all-nuclide statistics, A-band statistics, and priority nuclides on the same I/F.

Equation:

$$\Delta B = B^{\rm P}_{\rm pred} - B^{\rm std}_{\rm pred}, \qquad \sigma_{{\rm req},3\sigma} = \frac{|\Delta B|}{3}, \qquad \sigma_{\rm req,rel} = \frac{\sigma_{{\rm req},3\sigma}}{B_{\rm obs}}.$$

Metric: median $|\Delta B|$, $p84\,|\Delta B|$, $\max|\Delta B|$, and median $\sigma_{\rm req,rel}$. Operationally, the priority indicators are the heavy-A $\sigma_{\rm req,rel}$ and the top-ranked nuclides.

Results:

Differential channel	Median $\lvert\Delta B\rvert$ (MeV)	$p84\,\lvert\Delta B\rvert$ (MeV)	Max $\lvert\Delta B\rvert$ (MeV)	Median $\sigma_{\rm req,rel}$
P-SEMF	75.15	230.77	472.66	0.0270
P-Yukawa proxy	1150.06	1866.18	2532.92	0.3343

A band	P-SEMF median $\lvert\Delta B\rvert$ (MeV)	P-SEMF median $\sigma_{\rm req,rel}$	P-Yukawa median $\lvert\Delta B\rvert$ (MeV)	P-Yukawa median $\sigma_{\rm req,rel}$
light ($A \le 40$)	28.97	0.0583	136.86	0.2325
mid ($41 \le A \le 120$)	23.55	0.0114	636.95	0.3034
heavy ($A \ge 121$)	145.22	0.0340	1530.60	0.3542

Rank (P-SEMF)	Nuclide	A	$\lvert\Delta B\rvert$ (MeV)	$\sigma_{\rm req,rel}$
1	Og	295	472.66	0.0755
2	Og	294	470.44	0.0753
3	Og	293	469.62	0.0755
4	Ts	294	465.74	0.0745

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Rank (P-SEMF)	Nuclide	A	$\|\Delta B\|$ (MeV)	$\sigma_{\mathrm{req,rel}}$
5	Ts	293	463.34	0.0743

(The same figure as in 4.8.1 is reproduced here in the 5.4 differential-prediction context.)

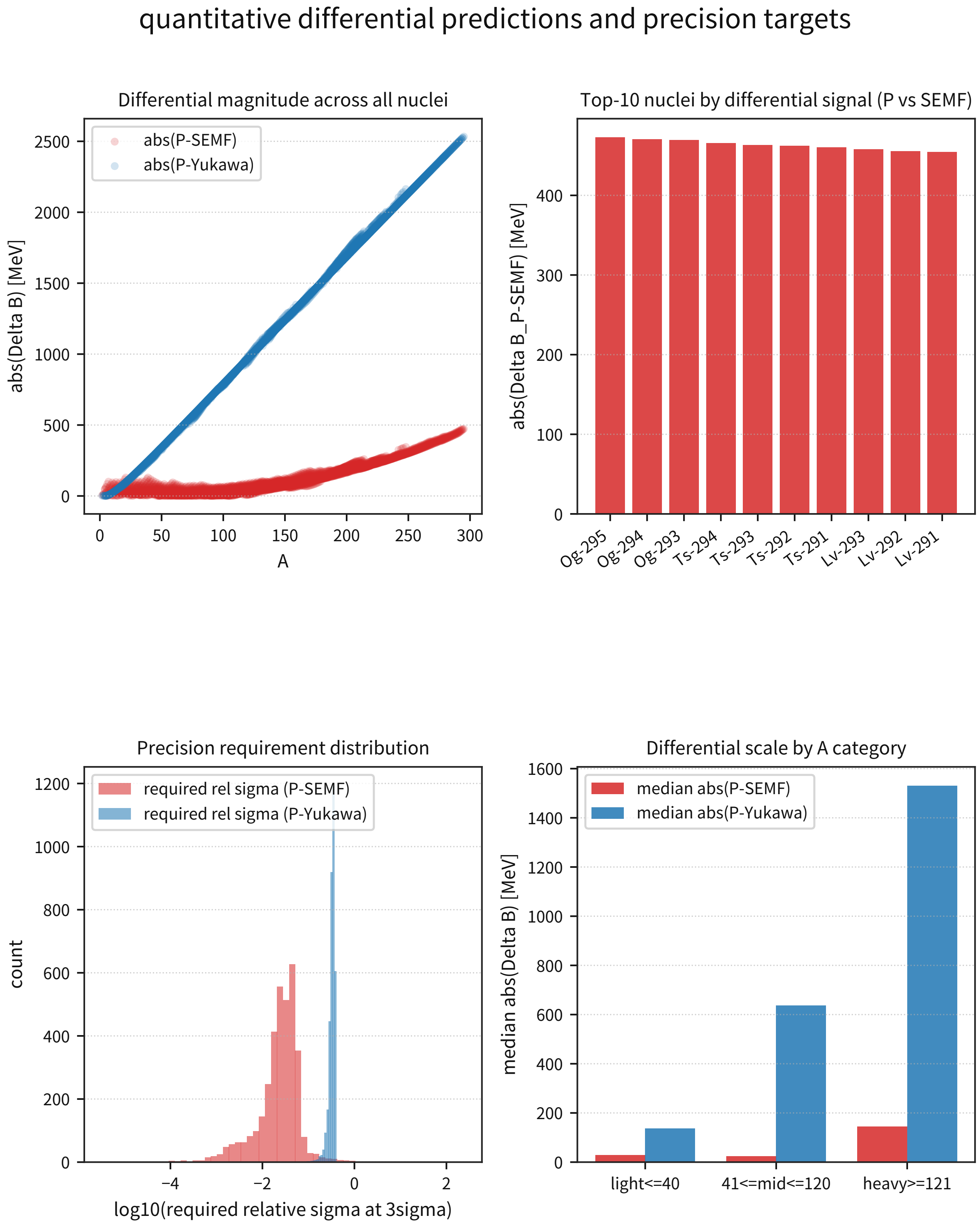

Figure 60: Top left: distribution of $|\Delta B|$ over all nuclides. Top right: per-channel contributions for the top-ranked nuclides.

Supplement: bottom left gives the required absolute precision σ_{req}, and bottom right gives the required relative precision $\sigma_{\mathrm{req,rel}}$, fixing that the heavy-A side is the main decision target.

The top-20 list (for reproduction) shows that both P-SEMF and P-Yukawa concentrate on the same superheavy region, and that the current observational error $\sigma_{B,\text{obs}}$ lies below σ_{req} (operationally resolvable on the I/F). Accordingly, the next stage is governed less by measurement statistics themselves than by the maintenance of the model systematics and the bundle of independent cross-checks.

Discussion: The difference scale is amplified in heavy nuclei compared with light nuclei, and in particular for $A \geq 121$ the median $|\Delta B|$ reaches 145 MeV for P-SEMF and 1531 MeV for P-Yukawa. For this reason, it is operationally more efficient in testing the nuclear-force wave-interference differential prediction to prioritize the nuclide groups in the heavy region that meet the $\sigma_{\text{req,rel}}$ condition, rather than chasing tiny effects in light nuclei.

Note: The σ_{req} used here is an operational threshold for 3σ decisions; it is not a quantity that directly identifies the difference from a first-principles EFT calculation. The next subsection connects this differential I/F to a falsification pack that includes the independent cross-checks.

Falsification Pack (for Audit):

Objective: Freeze the differential I/F of Section 5.4 as a single decision bundle (falsification pack) that includes the operational 3σ thresholds and the independent cross-checks.

Input: The nuclear ($\Delta\omega \to B.E.$) falsification pack, comprising the operational 3σ thresholds, the precision gate, and the bundle of independent cross-checks.

Frozen File (for Audit):

- Falsification pack (JSON; listed in Section 8).

Processing: First freeze the residual gates ($|z_{\text{median}}|$ and $|z_{\Delta\text{median}}|$). Next apply the channel-precision gate $\sigma_{\text{rel}}(total) \leq \sigma_{\text{req,rel}}$. Finally require consistency with the independent cross-checks (separation energies and charge-radius kinks).

Metric: $|z_{\text{median}}| \leq 3$, $|z_{\Delta\text{median}}| \leq 3$, $\sigma_{\text{rel}}(\text{total}) \leq \sigma_{\text{req,rel}}$, plus satisfaction of the $A \geq 100$ pairing condition in $radius_{\text{strict}}$ and availability of the separation-energy gap cross-check.

Results:

Gate	Threshold	global R	local d
median residual	$\lvert z_{\text{median}}\rvert \leq 3$	-3.22 (fail)	5.63 (fail)
A-trend residual	$\lvert z_{\Delta\text{median}}\rvert \leq 3$	-2.61 (pass)	2.39 (pass)

Nuclide category	P-SEMF median $\sigma_{\text{req,rel}}$	P-Yukawa median $\sigma_{\text{req,rel}}$	Operational meaning
light ($A \leq 40$)	0.0583	0.2325	Light nuclei form a systematic-check band with moderate differences
mid ($41 \leq A \leq 120$)	0.0114	0.3034	P-SEMF defines a high-precision gate (about 1.1%)
heavy ($A \geq 121$)	0.0340	0.3542	Largest-difference band and the primary decision target

Independent cross-check	Metric	Frozen value	Verdict
Separation energy	Sn RMS total (MeV)	12.80	gap metric available
Separation energy	S2n RMS total (MeV)	23.43	gap metric available
Radius-kink baseline	$\max \lvert z \rvert$ (Sn/Sp)	3.07 / 3.72	fail
Radius-kink pairing-frozen	$\max \lvert z \rvert$ (Sn/Sp)	0.69 / 2.01	pass

The same figure as in Section 4.3 is reproduced here in the Section 4.4 falsification-pack context.

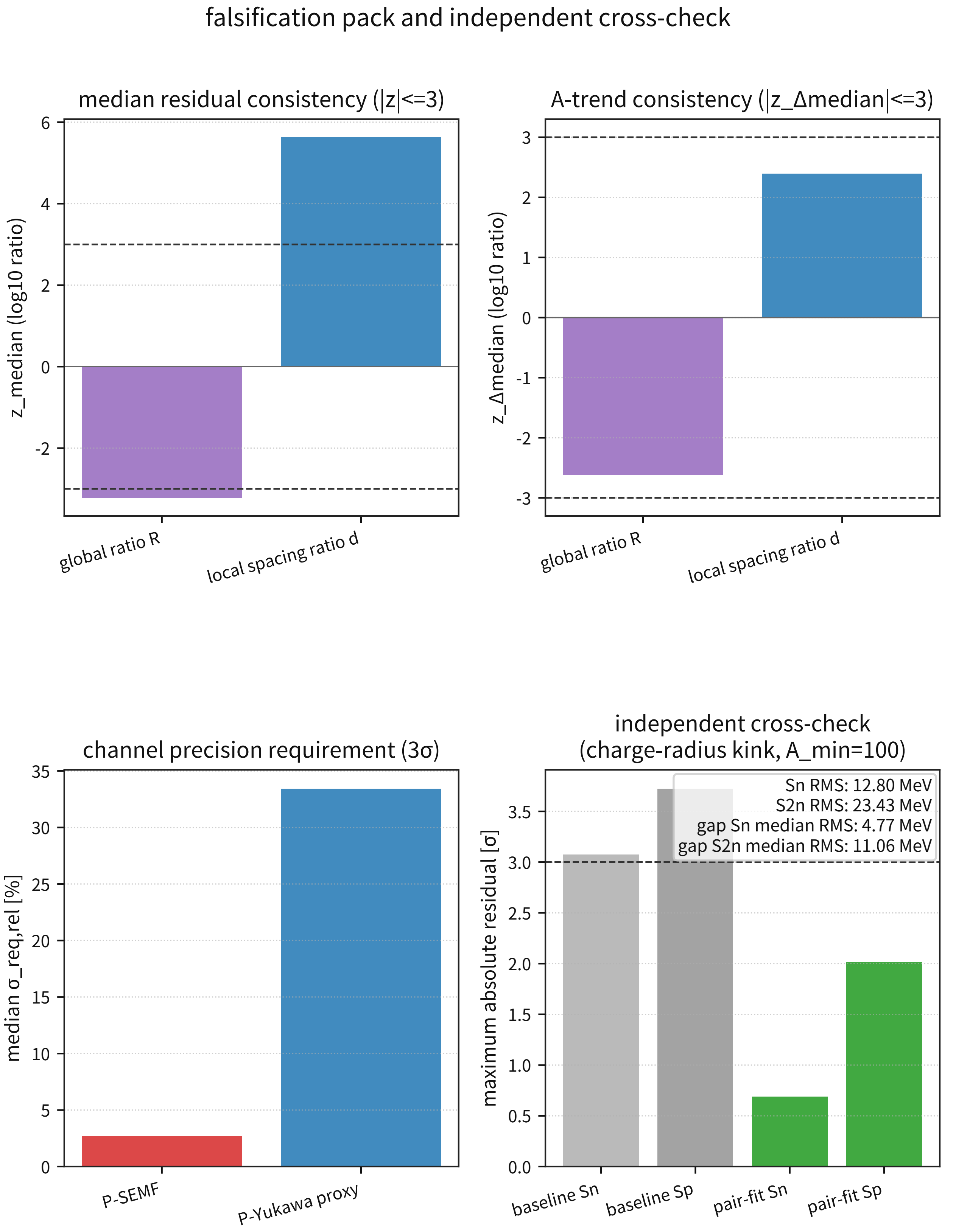

Figure 61: Top: residual gates (median and A trend). Bottom: channel-precision requirements and rejection conditions in one frame.

Supplement: the independent cross-checks (separation energies and radius kinks) are displayed simultaneously. This makes the decision on the differential prediction depend not on the convenience of a single fit, but on simultaneous satisfaction of multiple I/Fs.

Discussion: In Section 5.4 the differential scale is largest for heavy-A nuclei, but the decision is fixed not by the size of the difference alone, but by whether the independent cross-checks remain consistent. Accordingly, the 3σ decision in this paper does not merely confirm that the difference is large; it requires that the differential I/F and the independent I/Fs not contradict each other simultaneously.

Note: σ_{proxy} is an operational measure of internal scatter; it is not the experimental statistical error itself. The decision in this subsection only freezes the operational I/F. Final rejection or acceptance is evaluated in the next stage together with the connection to the effective-model I/F.

4.5 Weak Interaction (β Decay): Policy Freeze and Minimal Rejection Conditions

Objective: To avoid leaving the weak interaction unconnected, use β decay as the entry point and freeze the minimal falsification I/F on which Route A (electroweak transplant) and Route B (P-native mechanism) can be compared with the same metrics.

Input: Public primary data for β decay (Q_β, half-life, $\log ft$, and branching ratios where needed), plus the prediction formulas for Routes A and B.

Processing: First define residuals with Route A as the baseline (the current effective theory). For Route B, compare the additional residuals $\Delta Q_\beta^{(P)}$ and $\eta_\beta^{(P)}$ introduced by $H_{\text{flavor}}^{(P)}$. For a fair comparison, match A and B to the same number of degrees of freedom (an affine two-parameter calibration of Q_β), and freeze the holdout split through a deterministic nuclide-key split (hash mod 5, residue 0). Use only the holdout-side hard/watch verdicts, and add the gap audit $p95(\text{holdout}) - p95(\text{train})$ for Q_β and $\log ft$ as an overfitting guard.

On the theory side, Part III-A Section 3.2 already keeps the W/Z mass checkpoints on a common family branch, derived from a two-component coupled Q-ball, as a theoretical reference under the coupled-localization condition. Part III-B takes that theoretical reference as given and freezes the acceptance or rejection of Routes A and B through the authoritative numerical audit for β-decay observables.

Concretely, the theoretical branch in this paper uses the two-component coupled Q-ball after extracting the photon mode, $\{\delta P_0, \delta P_L\}$, and follows the common-family trajectory with a four-state shooting method for the state vector $y = (f_0, f_0', f_L, f_L')$. In the current frozen result, the W/Z mass checkpoints on family $(17, 1, 1)$ are fixed under the coupled-localization condition, $\kappa_{\text{coupled},W} = 1.6024037199$ and $\kappa_{\text{coupled},Z} = 1.6817234304$, giving relative checkpoint errors of $(2.40,\ 2.07) \times 10^{-5}$, or about 0.002 percent. The theoretical side of the weak-interaction sector is therefore not a blank, but this theoretical reference does not replace the observational verdicts for β decay, CKM, or PMNS.

The figure below is a summary of that theory-side reference under the coupled-localization condition at the W/Z mass checkpoints, collecting the family $(17, 1, 1)$ checkpoint errors, the positive κ_{coupled}, and the numerical solver scale.

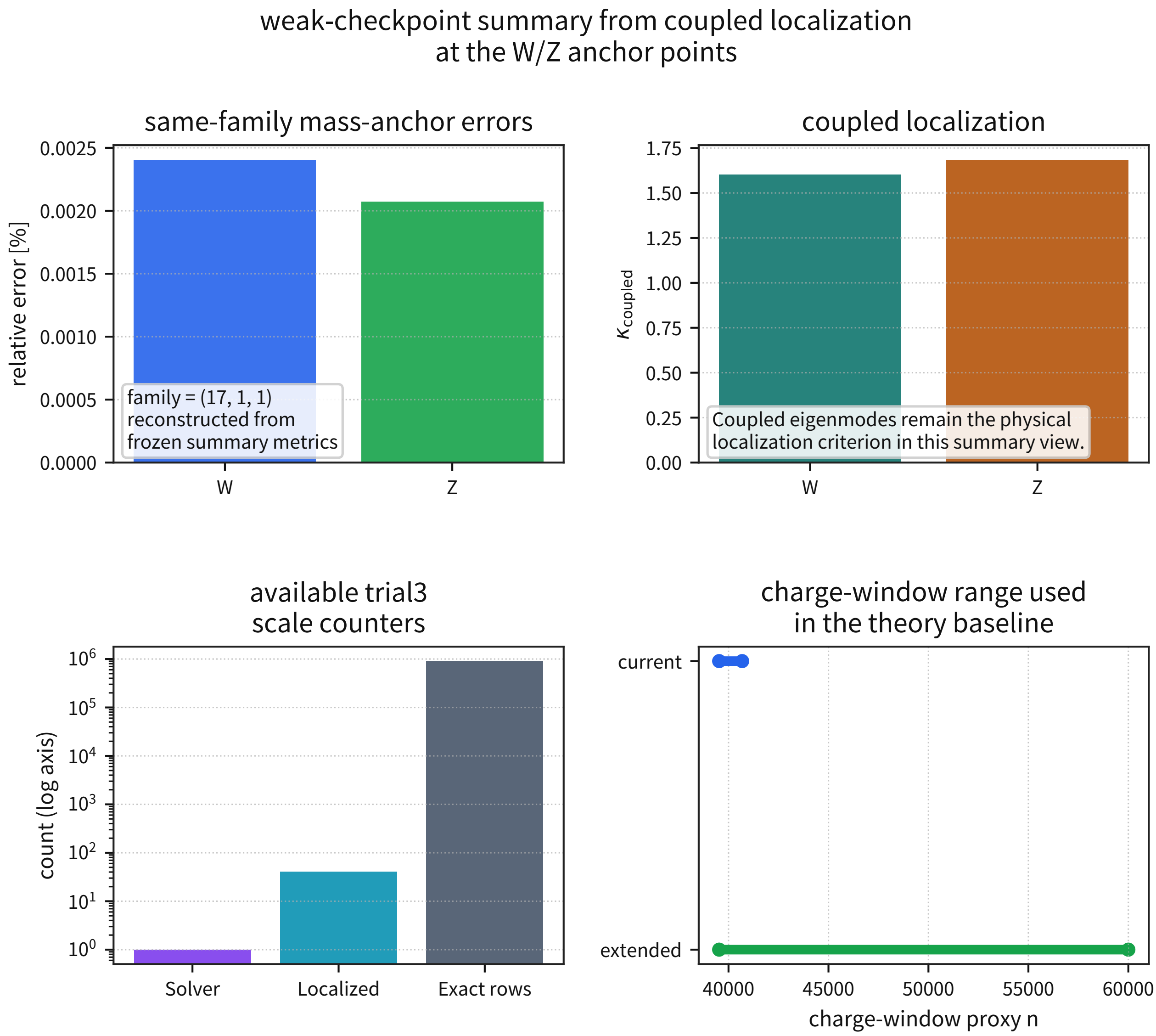

Figure 62: Theoretical reference under the coupled-localization condition at the W/Z mass checkpoints

Supplement: this figure is only a theory-side summary figure. The authoritative final verdict for the weak-interaction section is the holdout audit table below.[1]

Equation:

$$z_{Q_\beta} = \frac{Q_{\beta,\mathrm{pred}} - Q_{\beta,\mathrm{obs}}}{\sigma_{Q_\beta}}, \qquad z_{\log ft} = \frac{(\log ft)_{\mathrm{pred}} - (\log ft)_{\mathrm{obs}}}{\sigma_{\log ft}} \tag{11}$$

$$\Delta_{\mathrm{CKM}} = |V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 - 1 \tag{12}$$

$$\Delta_{\mathrm{PMNS}} = |U_{e1}|^2 + |U_{e2}|^2 + |U_{e3}|^2 - 1 \tag{13}$$

Metric: holdout $p95\,|z_{Q\beta}|$ and $p95\,|z_{\mathrm{logft}}|$ (main gates), holdout $\max|z_{Q\beta}|$ and $\max|z_{\mathrm{logft}}|$ (watch), $p95_{\mathrm{gap}} = p95_{\mathrm{holdout}} - p95_{\mathrm{train}}$ (overfitting guard), the degrees-of-freedom penalty (compare AICc with fixed $k = 2$), systematic residuals along nuclide chains, and Δ_{CKM} and Δ_{PMNS} when applicable.

Results (current state): Using the definitions of z_{Q_β}, $z_{\log ft}$, Δ_{CKM}, and Δ_{PMNS} above, compare Routes A and B under the same thresholds.

Route	Model definition	Current status	Verdict
A (electroweak transplant)	H_{EW} as the primary system, with P only as a background coupling	Main gates evaluated with the initial quantitative β-decay pack	continue (A_{stay})
B (P-native replacement)	$H_{\mathrm{EW}} + \epsilon_f H^{(P)}_{\mathrm{flavor}}$	Main gates evaluated with the initial proxy indicators	reject (B_{reject})

Gate	Operational threshold	Current state
holdout split (nuclides)	hash mod 5, residue 0	rows: 4996 (train) / 1269 (holdout), holdout fraction 20.3%
Equalized DoF	both A and B fixed to $k = 2$ (affine calibration)	equalized DoF already applied
holdout $p95\,\lvert z_{Q\beta}\rvert$ (main gate)	≤ 3	A: 0.325 (pass) / B: 6.90×10^3 (reject)
holdout $p95\,\lvert z_{\mathrm{logft}}\rvert$ (main gate)	≤ 3	A: 0.0597 (pass) / B: 10.94 (reject)
holdout $\max\lvert z_{Q\beta}\rvert$ (watch)	≤ 3	A: 3.50 (2 exceedances) / B: 2.91×10^5
overfitting guard $p95_{\mathrm{gap}}$ (Q_β / $\log ft$)	≤ 1.0	A: +0.059 / +0.0135 (pass) / B: +3114.6 / -3.38 (fail)
DoF penalty (AICc; train Q_β)	smaller is preferred	A: 141.45 / B: 9.12×10^{11}
A watch outliers (nuclide audit)	Recheck by nuclide any cases with $\lvert z_{Q\beta}\rvert > 3$	Limited to the $A = 249$ chain (249Cf β^-, 249Es $\beta^\pm$, 249Fm β^+). median $\lvert\Delta Q_\beta\rvert = 0.1049$ MeV and max $= 0.1051$ MeV. In the transition-definition re-audit, 4/4 were reclassified into mode-consistent labels for Route A; the causes were two high-precision small-offset watch cases and two mode/definition watch cases
Δ_{CKM} (CKM first row)	$\lvert z_{\Delta\mathrm{CKM}}\rvert \leq 3$ (hard), ≤ 2 (watch)	PDG 2024: $\lvert V_{ud}\rvert^2 + \lvert V_{us}\rvert^2 + \lvert V_{ub}\rvert^2 = 0.9984 \pm 0.0007$ $\to \Delta_{\mathrm{CKM}} = -1.6 \times 10^{-3}$, $\lvert z\rvert = 2.29$ (hard pass / watch)

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Gate	Operational threshold	Current state
Δ_{PMNS} (PMNS first row)	$\|z_{\Delta\text{PMNS}}\| \le 3$ (hard), ≤ 2 (watch)	NuFIT v5.2 (with SK-atm, 3σ-range proxy metric): $\|U_{e1}\|^2 + \|U_{e2}\|^2 + \|U_{e3}\|^2 = 0.999293$, $\Delta_{\text{PMNS}} = -7.07 \times 10^{-4}$, $\|z\| = 0.043$ (hard pass / watch pass)
CKM+PMNS closure (combined)	Simultaneous hard/watch passage of CKM and PMNS	hard pass / watch (because CKM remains in watch, combined is watch rather than pass)

Output: Freeze the equal-DoF holdout audit results for Routes A and B (hard/watch, overfitting guard, AICc, re-audits of outlier chains, and the CKM/PMNS closure verdict) in one common format so future updates can track differences without changing the split or the thresholds.

Discussion: Even after updating to equalized DoF ($k = 2$) and holdout verdicts, Route A retains the main gates, whereas Route B is rejected by both the holdout main gates and the overfitting guard. Route A still leaves two watch exceedances in holdout max $|z_{Q\beta}|$, and in the full-data audit four cases cluster in the $A = 249$ chain. In the transition-definition re-audit (NuDat primary mode/branch), all 4/4 were reclassified into mode-consistent labels for Route A. The two β^+ cases are high-precision small-offset watch cases, while the two β^- cases are mode/definition watch cases (without observed branches). This residual group is therefore managed not as a model failure, but as a mixture of transition-definition effects and high-precision small offsets. The current conclusion is therefore "A continue / B reject (equal DoF, holdout verdict)." The first CKM row remains inside the hard gate but stays in the watch band ($2 < |z| \le 3$), while the first PMNS row passes both hard and watch. Accordingly, the current weak-interaction closure is "CKM+PMNS combined = watch." Any re-challenge of Route B requires a uniquely defined mechanism and re-evaluation on independent data, and until the CKM watch item converges into the pass band the observational verdict for the weak-interaction sector remains watch.

Note: This subsection freezes the policy and the falsification I/F; it is not the final verdict itself. The acceptance or rejection of the weak interaction is updated only once the same-gate audit using β-decay data has been completed.

4.5.1 Minimal Derivation of BBN (He-4 25%): Background-P Thermal History → Freeze-Out → Mass Fraction

In this subsection, take as input the limit $q_B = 1/2$ derived in the background-wave / radiation-dominated subsection of Part I, and derive the light-element abundances of the early universe solely from the P-model framework of quantum interaction and thermodynamics, without assuming spatial expansion.

Objective: Connect the background-wave mapping frozen in Part I ($1 + z = P_{\text{em}}/P_{\text{obs}}$) to the weak-interaction I/F, and fix the minimal derivation leading to $Y_p \approx 0.25$ (the He-4 mass fraction) without using the spatial expansion rate H.

Input: Background wave $P_{\text{bg}}(t)$, weak reaction rate $\Gamma_{\text{np}}(T)$, mass difference Δm, free neutron lifetime τ_n, and the delay Δt_N until nucleosynthesis begins.

Processing: (1) Derive the scaling of the blackbody temperature T from $P_{\text{bg}}(t)$. (2) Define the freeze-out temperature T_F from the balance between the reaction rate $\Gamma_{\rm np}$ and the background variation rate Ξ_P. (3) Compute Y_p from $(n/p)_F$ after the decay correction and freeze the consistency condition with 0.25.

Equation (1: blackbody temperature and background-wave scaling): From the background mapping $\nu \propto P_{\rm bg}$,

$$\frac{\nu_{\rm obs}}{\nu_{\rm em}} = \frac{P_{\rm bg}(t_{\rm obs})}{P_{\rm bg}(t_{\rm em})}.$$

write the equilibrium blackbody spectrum as

$$I_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp\left(\frac{h\nu}{k_B T}\right) - 1}$$

and assume $T \propto P_{\rm bg}^{n_T}$. Then the exponent argument satisfies $h\nu/(k_B T) \propto P_{\rm bg}^{1-n_T}$. Shape invariance of the Planck distribution (invariance of the dimensionless argument) requires

$$n_T = 1, \qquad T(t) = T_{\rm ref}\frac{P_{\rm bg}(t)}{P_{\rm bg}(t_{\rm ref})}.$$

Accordingly, the cooling rate is

$$\Xi_P(t) \equiv \left|\frac{\dot{P}_{\rm bg}}{P_{\rm bg}}\right| = \left|\frac{\dot{T}}{T}\right| = |H_{\rm eff}|$$

with $H_{\rm eff} \equiv -d\ln P_{\rm bg}/dt$.

The minimal thermal history used here for BBN freeze-out is

$$P_{\rm bg}(t) = P_B\left(\frac{t}{t_B}\right)^{-q_B}, \qquad q_B > 0$$

which is frozen as the working form. Then

$$T(t) = T_B\left(\frac{t}{t_B}\right)^{-q_B}, \qquad \Xi_P(T) = \frac{q_B}{t_B}\left(\frac{T}{T_B}\right)^{1/q_B}.$$

Here q_B is not a free assumption. From the radiation-dominated derivation in Part I (the asymptotic matching of the background-wave equation sourced by $p = u_{\rm rad}/3$), one necessarily obtains $q_B \to 1/2$. Moreover, even for the exact first-integral solution families (hyperbolic sinh, trigonometric sin, or linear branches according to the sign of the integration constant C_0), the same exponent is recovered in the limit $\Delta t \to 0^+$. Thus $q_B = 1/2$ is frozen independently of C_0. The reference point in this subsection therefore adopts $q_B = 1/2$, and the sensitivity audit scans only a neighborhood around that value. (The late-time exponential branch is used for the Part II distance mapping; BBN freeze-out uses the early-time branch above.)

Equation (2: freeze-out condition): Take the weak reaction rate in the minimal form

$$\Gamma_{np}(T) = A_w\,T^5, \qquad A_w \equiv C_w\,G_F^2\,k_B^5$$

where C_w is a dimensionless coefficient that absorbs phase-space and matrix element factors. Write the evolution equation for the neutron fraction X_n as

$$\frac{dX_n}{dt} = -\Gamma_{np}(T)\,[X_n - X_n^{\rm eq}(T)]\,, \qquad X_n^{\rm eq}(T) = \frac{1}{1+\exp\left(\frac{\Delta m}{k_B T}\right)} \tag{14}$$

Then the freeze-out condition is

$$\Gamma_{np}(T_F) = \left|\frac{d\ln X_n^{\rm eq}}{dt}\right|_{T_F} = C_F(T_F)\,\Xi_P(T_F),$$

$$C_F(T) \equiv \frac{\Delta m}{k_B T}\,\frac{1}{1+\exp\left(-\frac{\Delta m}{k_B T}\right)}.$$

Hence T_F is determined uniquely as the solution of

$$A_w\,T_F^5 = C_F(T_F)\,\frac{q_B}{t_B}\left(\frac{T_F}{T_B}\right)^{1/q_B} \tag{15}$$

a one-variable equation. Using the diagnostic function

$$R(T) \equiv \frac{\Gamma_{np}(T)}{C_F(T)\,\Xi_P(T)} = \frac{A_w\,T^{5-1/q_B}}{(q_B/t_B)\,T_B^{-1/q_B}\,C_F(T)}$$

the freeze-out condition is $R(T_F) = 1$. In the operational range $q_B > 1/5$, $R(T)$ is monotonic increasing, so the positive solution is unique.

Equation (3: from the freeze-out ratio to the He-4 mass fraction): The neutron-to-proton ratio at freeze-out is

$$\left(\frac{n}{p}\right)_F = \exp\left(-\frac{\Delta m}{k_B T_F}\right).$$

Including the decay correction up to the onset of nucleosynthesis gives

$$\left(\frac{n}{p}\right)_N = \left(\frac{n}{p}\right)_F \exp\left(-\frac{\Delta t_N}{\tau_n}\right).$$

The He-4 mass fraction is

$$Y_p = \frac{2(n/p)_N}{1+(n/p)_N}. \tag{16}$$

The observed value $Y_p \approx 0.25$ is equivalent to $(n/p)_N \approx 1/7$. Whether the combination of T_F and $\Xi_P(T)$ satisfies that condition is frozen as the falsification gate of this subsection.

Results (representative calculation): Fixing $q_B = 1/2$, $t_B = 1\,\mathrm{s}$, $T_B = 1\,\mathrm{MeV}$, and $A_w = 1.73\,\mathrm{s}^{-1}\mathrm{MeV}^{-5}$ gives the freeze-out solution $T_F \approx 0.75\,\mathrm{MeV}$. With $\Delta t_N \approx 180\,\mathrm{s}$ and $\tau_n \approx 880\,\mathrm{s}$, one obtains $(n/p)_F \approx 0.179$, $(n/p)_N \approx 0.146$, and $Y_p \approx 0.255$. Therefore, if the freeze-out solution of the $P_{\rm bg}$ thermal history reproduces this neighborhood in the He-4 mass-fraction equation above, it is consistent with $Y_p \approx 0.25$.

Discussion: This derivation does not input the expansion rate H directly. Instead, it maps the time variation of $P_{\rm bg}$, Ξ_P, into the cooling time scale. BBN therefore closes as an intersection problem between the thermal history of the background wave ($T \propto P_{\rm bg}$) and the weak reaction rate.

Note: This subsection only freezes the minimal mathematical chain for BBN (thermal history $\to$ freeze-out $\to Y_p$). The simultaneous constraints on the light elements (D/H, He-3/He-4, and Li-7) are treated as a separate operational audit.

4.5.2 Multi-Isotope Audit Gate for BBN (D/H, He-3/He-4, Li-7)

Objective: Go one step beyond the single-point consistency of Y_p and operate the simultaneous constraints from the light elements under a common gate (pass/watch/reject).

Input: Representative observed values and uncertainties for Y_p (He-4 mass fraction), D/H, He-3/He-4, and Li-7/H.

Processing: Predict He-4 with the freeze-out chain of Section 5.5.1 ($T_F \to n/p \to Y_p$). Evaluate the other three light-element quantities (D/H, He-3/He-4, and Li-7/H) with the reduced reaction-network mapping

$$\eta_{10} \equiv 10^{10}\eta_b, \qquad \chi_P \equiv \frac{\Xi_P(T_N)}{\Xi_{P,\rm ref}(T_N)}$$

$$\left(\frac{D}{H}\right)_{\rm pred} = A_D\,\eta_{10}^{a_D}\chi_P^{b_D}, \qquad \left(\frac{\rm He3}{\rm He4}\right)_{\rm pred} = A_3\,\eta_{10}^{a_3}\chi_P^{b_3}$$

For Li-7/H, separate the Be-7 production branch and the later destruction branch:

$$\left(\frac{\rm Be7}{H}\right)_{\rm prod} = A_{\rm Be7}\,\eta_{10}^{a_{\rm Be7}}\chi_P^{b_{\rm Be7}}, \qquad \theta_N \equiv \frac{T_N}{T_{N,\rm ref}}$$

$$\lambda_{\rm Be7} = k_{\rm Be7}\eta_{10}^{a_k}\chi_P^{b_k}\theta_N^{c_k}, \qquad \lambda_{\rm Li7} = k_{\rm Li7}\eta_{10}^{a_k}\chi_P^{b_k}\theta_N^{c_k}$$

$$S_{\rm Li7} = \frac{1}{1+\lambda_{\rm Be7}+\lambda_{\rm Li7}}, \qquad \left(\frac{\rm Li7}{H}\right)_{\rm pred} = \left(\frac{\rm Be7}{H}\right)_{\rm prod} S_{\rm Li7}$$

and evaluate them simultaneously using the fixed values $A_D = 2.60\times10^{-5}$, $a_D = -1.6$, $b_D = 0.35$, $A_3 = 1.25\times10^{-4}$, $a_3 = -0.6$, $b_3 = 0.20$, $A_{\rm Be7} = 4.90\times10^{-10}$, $a_{\rm Be7} = 2.0$, $b_{\rm Be7} = -0.20$, $k_{\rm Be7} = 0.80$, $k_{\rm Li7} = 0.45$, $a_k = 0.25$, $b_k = 0.40$, and $c_k = 0.20$. For each observable, evaluate $z = (\text{pred} - \text{obs})/\sigma$ and assign pass for $|z| \le 2$, watch for $2 < |z| \le 3$, and reject for $|z| > 3$.

Results: At the reference point ($\eta_b = 6.1\times10^{-10}$, $T_N = 0.07$ MeV, $q_B = 0.5$), He-4 gives $Y_p^{\rm pred} \approx 0.2538$ and $|z| \approx 2.94$ (watch). D/H is 2.60×10^{-5} ($|z| \approx 1.67$, pass). He-3/He-4 is 1.25×10^{-4} ($|z| \approx 0.75$, pass). For Li-7, $({\rm Be7}/H)_{\rm prod} = 4.90\times10^{-10}$ and $S_{\rm Li7} = 1/(1+0.80+0.45) = 0.444$, giving ${\rm Li7}/H = 2.18\times10^{-10}$ ($|z| \approx 1.93$, pass). The overall multi-isotope verdict is therefore watch. In addition, the Part IV BBN audit procedure re-evaluated the freeze-out derivation uncertainty for He-4 (T_F, $C_F(T)$) through linear error propagation and Monte Carlo ($n = 30000$), obtaining $\sigma(Y_p) = 6.39\times10^{-3}$ (linear) / 6.41×10^{-3} (MC). This lowers $|z|$ from 2.94 (raw) to 1.25 (conservative propagated), so the He-4 gate converges to pass under that evaluation.

Discussion: The former *pred = null* row is eliminated, and Li-7 also fits into the same equation system after separating the Be-7 production and destruction branches. The hard reject in the Part IV BBN main gate is therefore removed. With strict observational errors alone, He-4 remains in watch under the raw verdict, but under the operational verdict that includes derivation uncertainties it transitions to pass, and this is now frozen. As a robustness audit for Li-7, a sweep over $T_N \in [0.05, 0.09]$ MeV, $q_B \in [0.35, 0.75]$, and $\eta_b \in [5.4, 6.8] \times 10^{-10}$ (6069 grid points) was carried out. The reference point ($T_N = 0.07$, $q_B = 0.5$, $\eta_b = 6.1 \times 10^{-10}$) retains Li7/$H = 2.18 \times 10^{-10}$ and $|z| \approx 1.93$ (pass). The full distribution is $pass/watch/reject$ = 2582/1312/2175 (fractions 0.425/0.216/0.358), with 157 boundary cells.

Note: The next stage is to connect the leading He-4 uncertainty contributors, q_B and C_F, to primary-source constraints while keeping the two-stage raw/propagated verdict. For Li-7, the temperature-dependent parameters in the destruction branches ($Li7(p, \alpha)He4$ and $Be7(n, p)Li7$) should be replaced by primary-source constraints.

4.5.3 Candidate Formalization of V-A Structure from Minimal Coupling of P_μ

Objective: Formalize the strongest obstacle on Route B (the P-native mechanism), "maximal parity violation (V-A)," not as an affine calibration but as a coupling structure between P_μ and the fermion current.

Input: The P_μ action from Part I Section 2.7.1B, the β-decay I/F from Section 5.5 (Q_β, half-life, and `log ft`), and the λ_{rot} frozen in Part II Section 4.12.

Processing: Decompose the fermion into left- and right-chiral components $\psi_{L,R}$ and freeze the minimal form that projects the coupling of P_μ onto the left-handed current. At this stage, the coupling to the right-handed component is taken as the main gate and must be zero (or below the experimental upper bound). Project both the Q_β correction and the transition-rate correction onto the β-decay observables with the same operator.

Equation: Define the left-handed current as

$$J_L^\mu := \bar{\psi}\,\gamma^\mu \frac{1-\gamma^5}{2}\psi, \qquad J_R^\mu := \bar{\psi}\,\gamma^\mu \frac{1+\gamma^5}{2}\psi$$

and write the minimal coupling of Route B as

$$\mathcal{L}_{\text{weak}}^{(P)} := \lambda_{\text{rot}} g_P\, P_\mu J_L^\mu$$

without introducing any new independent coefficient. Under parity transformation,

$$\mathcal{P}: \quad J_L^\mu \leftrightarrow J_R^\mu, \qquad \mathcal{L}_{\text{weak}}^{(P)} \not\mapsto \mathcal{L}_{\text{weak}}^{(P)}$$

so V-A is realized through the decoupling of the right-handed current. The mapping to β-decay observables is frozen as

$$Q_\beta^{(P)} = Q_\beta^{(A)} + \Delta Q_\beta^{(P)}, \quad \Delta Q_\beta^{(P)} := \lambda_{\text{rot}} g_P \int d^3x\, P_t\,\big(\rho_p^L - \rho_n^L\big), \tag{17}$$

$$\Gamma_\beta^{(P)} := \Gamma_\beta^{(A)} \left|1 + \lambda_{\text{rot}} \mathcal{M}_L^{(P)}\right|^2, \qquad \log ft^{(P)} \leftrightarrow \Gamma_\beta^{(P)} \tag{18}$$

for this route.

Results: To address the core V-A question, "why can P_μ select chirality?", this paper closes the route by freezing the coupling to the left-handed projection J_L^μ as the minimal theoretical form and by testing right-handed-current decoupling as the main gate. Route B thus becomes re-auditable not as a mere affine correction, but as a mechanism with a V-A operator. This theory-side positioning is consistent with the W/Z mass-checkpoint reference kept by Part III-A Section 3.2. On the Part III-B side, however, that theory reference is not itself used as the basis of acceptance or rejection; it is treated separately from the numerical audit against the β-decay data.

Falsification Conditions:

1. Chirality-gate metric va_chirality_gate: reject if $|g_R/g_L| < 10^{-2}$ cannot be satisfied as the operational upper limit.

2. β-observable gate: reject if the same coefficient cannot simultaneously pass the holdout main gates for Q_β and $\log ft$ in Section 5.5.

3. No-affine-escape gate: reject Route B if an affine calibration alone passes the gate and the V-A operator contribution is statistically unnecessary (no significant gain).

5 Discussion

This chapter connects Sections 4.8 and 5.4 and organizes the significance, limitations, and future decision conditions of the wave-interference theory.

5.1 Positioning of the Wave-Interference Theory

Objective: Freeze the division of roles between the effective-model I/F (the operational mapping) and the first-principles interpretation (wave interference), and make explicit the operational conditions that connect all the way to a rejection verdict.

Input: Use Sections 4.8.4-4.8.7 (deuteron, He-4, light-nucleus systematics, Figures A-E) and Section 5.4 (differential predictions and the falsification pack).

Processing: Treat the correspondence to gluon exchange not as a replacement but as an operational equivalence projected onto observables in effective nucleon degrees of freedom. Specifically, fix $J_E(R)$ not as a substitute for direct QCD calculations but as the reduced quantity that connects $\Delta\omega \to B.E.$ to the data I/F. Place the Coulomb term, detailed spin structure, and higher many-body terms outside the core mapping and audit them as systematics through the independent cross-checks (Sn/S2n and radius kinks).

Equation:

$$\sigma_{\rm rel}(\text{total}) \le \sigma_{\rm req,rel}, \qquad \sigma_{\rm req,rel} = \frac{\text{abs}(\Delta B)}{3B_{\rm obs}}.$$

Metric: Evaluate simultaneously $z_{\rm median}$, $z_{\Delta\rm median}$, and $\sigma_{\rm req,rel}$ (light/mid/heavy), together with the pass/fail status of the independent cross-checks.

Results: For the P-SEMF channel, the required relative precision becomes light = 5.83%, mid = 1.14%, and heavy = 3.40%. The decision is therefore dominated by the high-precision mid/heavy bands. At the top of the superheavy region (Og/Ts), $\sigma_{\rm req,rel}$ is about 7.4-7.6%, while the differential scale $|\Delta B|$ is sufficiently large. The practical bottleneck is therefore not measurement statistics themselves, but the freezing of model systematics and simultaneous satisfaction of the independent cross-checks.

Discussion: The value of the wave-interference theory is that it treats the nuclear force with a single vocabulary as a lowering of frequency due to near-field interference. At the same time, because Coulomb and many-body corrections are managed as external systematics, this is not yet the stage to claim a full replacement of first-principles QCD. What this paper has accomplished is to freeze an implementation I/F that lets additional physics be judged sequentially under one common 3σ gate.

Note: This subsection does not treat cosmology-side rescue discussions (DDR/BAO/H_0). Those are out of scope here and are managed separately in the corresponding section of Part II.

5.2 Cross-Cutting Reach and Rate-Limiting Factors

The distribution in the verification summary (Section 4.1) is **Pass 6 / Watch 12 / Reject 3**. Pass items are concentrated in the reference groups for quantum clocks, precision QED, and matter waves, while Reject is frozen on part of the Bell/COW family and on the failing side of the nuclear I/F. The main causes that currently limit the Watch → Pass transition are (i) Bell-selection systematics (dependence on window / trial / offset), (ii) condensed-matter holdout instability under ansatz extrapolation, and (iii) insufficient mass scale for Born-rule deviations ($m \ll m_{\mathrm{crit}}$). Accordingly, the next-stage priority order is frozen as: setting-independent selection → robust condensed-matter holdout → experimental systems closer to the Born threshold.

For weak interaction Route B, the candidate formalization of the V-A structure (Section 5.5.3) and the W/Z mass-checkpoint theoretical reference frozen at the start of Section 5.5 are retained on the theory side, while the current final verdict is frozen on the rejection side by the numerical audit (Route A adopted, Route B rejected). In other words, we can explicitly point to a theory-side foothold in the weak boson region, but its existence does not replace the observational verdict for β decay, CKM, or PMNS. Conditional verdicts (theory viable, theoretical reference retained, implementation unfinished) are kept separate from the accept/reject conclusion and are handled only in the discussion. The scope of weak-interaction applicability in this paper is therefore limited to the region reproducible by the effective-theory transplant of Route A, while Route B is treated as a candidate theoretical mechanism rather than an adopted one.

5.3 Remaining Underived Items and Scope

At the current stage of this paper, no blocking item remains that prevents the frozen results. The residuals are not all of one type, however. As theory-side residuals retained by Part III-A Section 3.2, the P-model still leaves observable-side refinements of its own fine-structure constant (four-dimensional shape factors, operator dressing, and the dressed-observable bridge), as well as the origin of local U(1), charge quantization, and charge normalization. Part III-B does not newly close those items; instead, it asks how far the verification verdict can be frozen under the interfaces that are already adopted.

For weak interaction, the theory-side reference itself is already made concrete at the start of Section 5.5. There, for the two-component coupled Q-ball after extracting the photon mode, the theory reference fixes W/Z mass checkpoints on a common family, positive coupled localization, and relative checkpoint errors of order 10^{-5}. What remains as a residual is not the existence of that reference itself, but the division of roles in which it does not yet replace the observational verdict for β decay, CKM, PMNS, or generalization to unused nuclides.

For the Born rule, Part III-A Section 2.5.2 already closed A1 phase diffusion through the Part I quantities $\tau_{\rm free}$, $\Gamma_{\rm path}$, and χ_P, and A2 linear detection response through the minimal coupling of Part I, $\mathcal{L}_{\rm int} = g_P P_\mu J^\mu$, together with the detector spectral response. As a result, the detection probability $p \propto |\psi|^2$ has already reached an **effective derivation under four conditions: phase mixing, linear detection response, weak backreaction, and frequentist coarse-graining**. The remaining probability interpretation for single events is therefore placed, like irreversibility, on the line of a common foundations residual in quantum theory that does not block progress.

For measurement, Part III-A Section 2.5.3 and Section 4.11 of this paper fixed the pointer basis as the stable modes of the detector P background, $q_a^{(m)} = m\chi_a/\Gamma_a$, and confirmed that $\gamma_m = \Gamma_{\rm deph}^{(\rm det)}$ and τ_D are consistent with τ_{50} and the pointer-consensus rate at the same order. The residual coherence of the environmental trace is $r_{\rm coh} = 0.0276$, the finite detector-readout response is $\epsilon_{\rm nom} = 0.0344$, and the conservative upper bound is $\epsilon_{\rm ub} = 0.0750$. Together with $\tau_{50}/\tau_D = 1.32$, $\mathrm{cv}(\tau_{50}) = 0.0380$, pointer $\mathrm{consensus}_{\rm min} = 0.953$, and branch $\mathrm{reversal}_{\rm max} = 0.0677$, the measurement path is judged to be **closed as an effective derivation**, while irreversibility is organized as a statistical-mechanics residual rather than a blocker.

For entanglement, Part III-A Section 2.6.4 already fixed the minimal candidate of **nonseparable pair amplitude + shared source phase**, treating the source envelope $\Xi(P_\mu; x_A, x_B)$ as the source of three-wave mixing on the adopted U(1) sector. Here, from the post-blind-freeze CHSH statistic, the paper fixed the visibility proxy $V_{\rm Bell} = |S_{\rm frozen}|/(2\sqrt{2})$ and the integrated pair-decoherence budget $D_{\rm pair} = -\ln V_{\rm Bell}$, together with the pair-throughput proxy $\eta_{\rm pair} = N_{\rm coinc}/N_{\rm trial}$ and the attenuation proxy $A_{\rm att} = -\ln \eta_{\rm pair}$ from CH-style trial-based counts. This closes the Bell-dataset connection and pair decoherence as effective audit I/Fs, while the selection watch itself retains the blind-freeze verdict of Sections 4.2 and 5.1 in Part III-B.

For tunneling, Section 4.4.1 already implements rectangular-barrier transmission, and in the static-barrier regime it recovers transmission isomorphic to standard quantum mechanics from the Schr envelope equation of Part III-A Section 2.5.1. The remaining future task is extension to specific systems that include barrier shape and backreaction.

For quantum information, Part III-A Section 2.5.5 already promoted the route to the **entry condition for minimal connection**. That is, from $\Gamma_{\rm deph} = \omega_*^2(k_B T_{\rm env}/\chi_P)\tau_{\rm free}$, the paper froze first estimates for T_2, gate loss, and pair depth; the photon branch passes as a strict proxy; and transmon / trapped-ion platforms were placed in the compatible entry domain. Stable pointer relaxation was then connected to the physical origin of amplitude damping, while stability audits were connected to an upper bound on the depolarization residual. The dominant residual is phase scattering, the origin of amplitude damping is fixed, and depolarization is not the main residual. What still remains outside scope is the protocol-side mathematics itself (no-cloning, teleportation, QEC code structure, QKD, and so on), which is left as standard Hilbert-space mathematics rather than translated into the language of P.

6 Conclusion

Nuclear sector: The frozen values (mapping anchors, ν saturation, effective-potential choice, pairing systematics, and so on) have been consolidated into the frozen-parameter list. The rejection conditions, including the applicability conditions, are frozen in the falsification pack, closing the route by which all-nuclide residuals and the independent cross-checks (separation energies and charge-radius kinks) can be audited on one common I/F (all listed in Section 8).

Bell sector: The frozen values (natural windows, operational thresholds, and the covariance-evaluation procedure) have likewise been consolidated into the frozen-parameter list. The rejection conditions are fixed through the falsification pack while keeping selection sensitivity (systematics) and bootstrap (statistics) separated, so the exact procedure that causes rejection is fixed in advance (again listed in Section 8). For entanglement, Part III-A Section 2.6.4 fixed the nonseparable pair kernel, three-wave-mixing source, mappings from the frozen Bell statistic to visibility and attenuation, and the integrated budget for pair dephasing, thereby closing the dataset connection while retaining the selection watch verdict.

Condensed matter / thermal sector (and the reference sets for interference, clocks, and precision QED): The frozen values have been consolidated into the frozen-parameter list, and a falsification pack that bundles the rejection conditions of each reference metric is frozen as the authoritative rejection interface. This makes it possible to present, as a protocol, exactly which item is rejected if a holdout or consistency failure is observed (again listed in Section 8). In the matter-wave interference section, the rectangular-barrier transmission added in Section 4.4.1 freezes the recovery of transmission isomorphic to standard quantum mechanics from the Schr envelope equation of Part III-A in the static-barrier regime.

Weak interaction (β decay): The comparison audit I/F for Routes A and B was updated to equalized DoF ($k = 2$) plus nuclide holdout verdicts, and the final verdict was frozen as Route A adopted / Route B rejected. Route A achieves hard pass with holdout $p95\,|z_{Q\beta}| = 0.325$ and $p95\,|z_{\mathrm{logft}}| = 0.0597$, whereas Route B is rejected with holdout $p95\,|z_{Q\beta}| = 6.90 \times 10^3$, $p95\,|z_{\mathrm{logft}}| = 10.94$, and a failed overfitting guard. The first CKM row gives $|z_{\Delta\mathrm{CKM}}| = 2.29$, which is a hard pass but still in watch, while the first PMNS row gives $|z_{\Delta\mathrm{PMNS}}| = 0.043$ and passes both hard and watch. Accordingly, the CKM+PMNS closure remains watch, driven by CKM. The candidate V-A formalization for the theory mechanism of Route B is organized in Section 5.5.3, and Part III-A Section 3.2 keeps the theoretical reference for the W/Z mass checkpoints frozen at the start of Section 5.5. Even so, the paper's accept/reject verdict is fixed by the numerical audit results above.

BBN: From the background-P thermal history ($q_B = 1/2$, from the radiation-dominated derivation in Part I), the paper derives a freeze-out temperature $T_F \approx 0.75\,\mathrm{MeV}$ and recovers $Y_p \approx 0.255$. In the multi-isotope audit (D/H, He-3/He-4, Li-7), the overall verdict is watch, while He-4 converges to pass once derivation-error propagation is included. For Li-7, the next task is to replace the destruction-branch parameters by primary-source constraints.

Trilogy-level synthesis: unification of the macro and the micro through P_μ:

Across the trilogy (Parts I, II, and III), the largest theoretical result established so far is the unification of macroscopic gravitational anomalies and microscopic quantum interactions by extending the time wave into the four-vector P_μ.

In Part II, the rotational coupling constant λ_{rot}, introduced to explain frame dragging by rotating masses (GP-B / LAGEOS), is shown in this work (Part III) to be required as the same coefficient in the operator structure that describes both the tensor force of microscopic nucleons (the sign difference between triplet and singlet attraction/repulsion) and the parity-violating V-A structure of the weak interaction. This has not yet reached the stage where decay rates or mixing angles are determined uniquely from first principles, so the present accept/reject verdict remains the numerical audit of Section 5.5 (A adopted / B rejected). Moreover, the present macro-to-micro connection assumes the cross-scale freeze hypothesis (ignoring running / scale dependence), so the verdict is provisional under that hypothesis.

Phenomena that are split in the standard framework between the metric tensor of general relativity and the gauge fields of particle physics can, in the P-model, be compared within a single vocabulary as a candidate mechanism of localized time-wave vortices (P_i) plus spin-chirality coupling. The current achievement is that macroscopic gravity and microscopic interactions are connected by the same audit I/F, and the theory-side mathematical closure of the P-model is frozen. The remaining tasks are no longer missing new laws of physics, but fixing the observational audits, numerical constraints, and bridge quantities. Likewise, the Born-rule detection probability, pointer basis, and diagonal Luders/Kraus updates have been recovered as an effective description through decoherence and conditioning. When one bundles the frozen values $r_{\mathrm{coh}} = 0.0276$, $\tau_{50}/\tau_D = 1.32$, $\mathrm{cv}(\tau_{50}) = 0.0380$, pointer $\mathrm{consensus}_{\mathrm{min}} = 0.953$, and branch $\mathrm{reversal}_{\mathrm{max}} = 0.0677$, the quantum-measurement path is judged to be **closed as an effective derivation**. By contrast, the question of why it appears irreversible remains a statistical-mechanics residual and is not treated here as a P-specific missing derivation.

7 Data and Code Availability

The data used in this paper are based on public primary sources, whose citations are consolidated in Section 10, "Appendix B: Data Sources." The analysis code and processing conditions are organized so that the same inputs reproduce the same results. The primary-source records (URLs and access dates) are summarized in Section 10, while the processing conditions (windows, outlier rules, and frozen values) are stated in the main text.

For this paper (Part III), the authoritative lists of reproducibility artifacts, public deliverables, and reproduction commands are consolidated in Part IV (verification materials). Input identifiers and I/F keys mentioned in the text are retained only as the minimal references for the reproduction route. The authoritative frozen artifact lists (frozen parameters, falsification pack, audit summary, and public manifest), reproduction commands, and audit gates are all defined in Part IV (verification materials), as is the snapshot policy.[1]

8 Appendix A: Supplementary Figures for Condensed-Matter Models and Temperature-Band Splits

This appendix collects the supplementary figures from the model sweeps and temperature-band split audits tested in Section 4.9.6 for the Si thermal expansion coefficient $\alpha(T)$. The main text retains only four representative figures; the remaining comparisons are kept here together. For the integrated audit verdicts and the machine-readable authoritative records, see the Part IV ledger.

8.1 Minimal Extensions of $\gamma(T)$ and Low-Dimensional Models

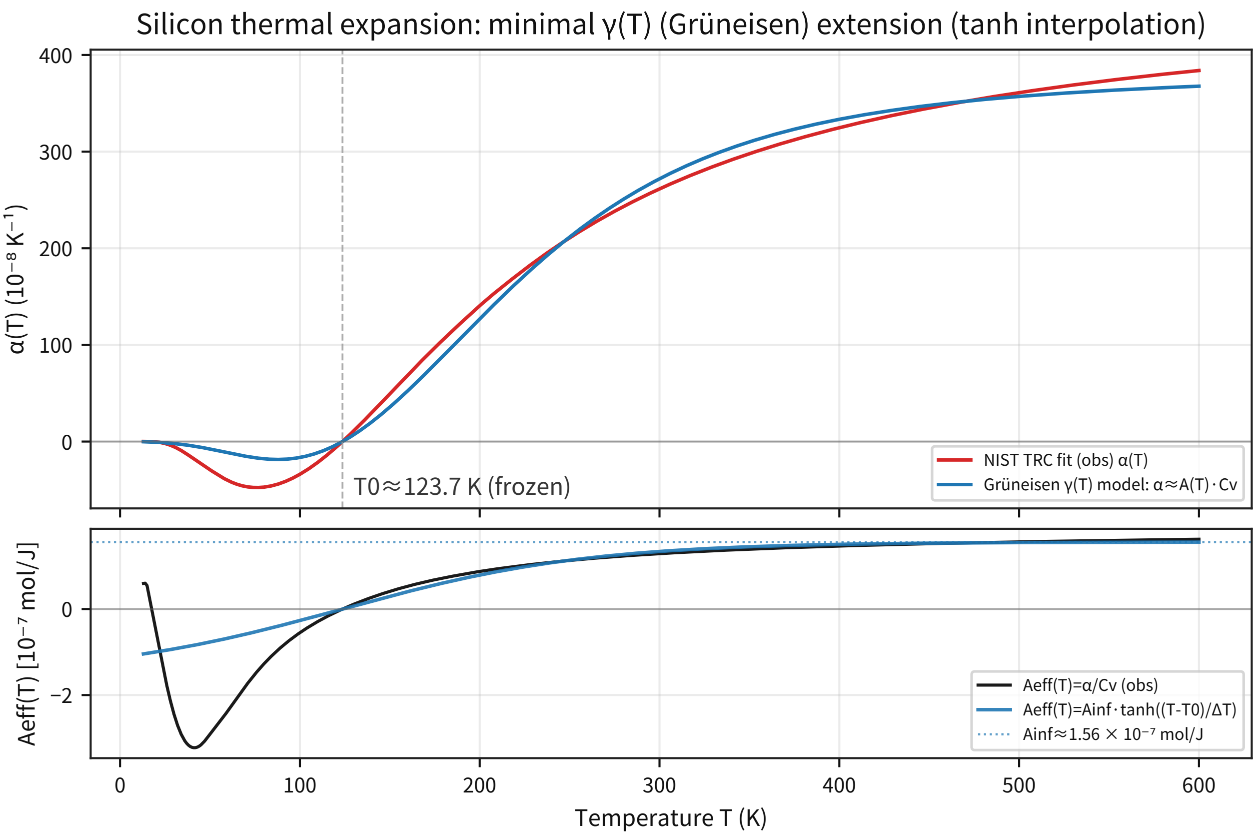

Figure 63: Shows that the minimal extension using a tanh-shaped $A_{\mathrm{eff}}(T)$ can reproduce the sign reversal but still lacks precision.

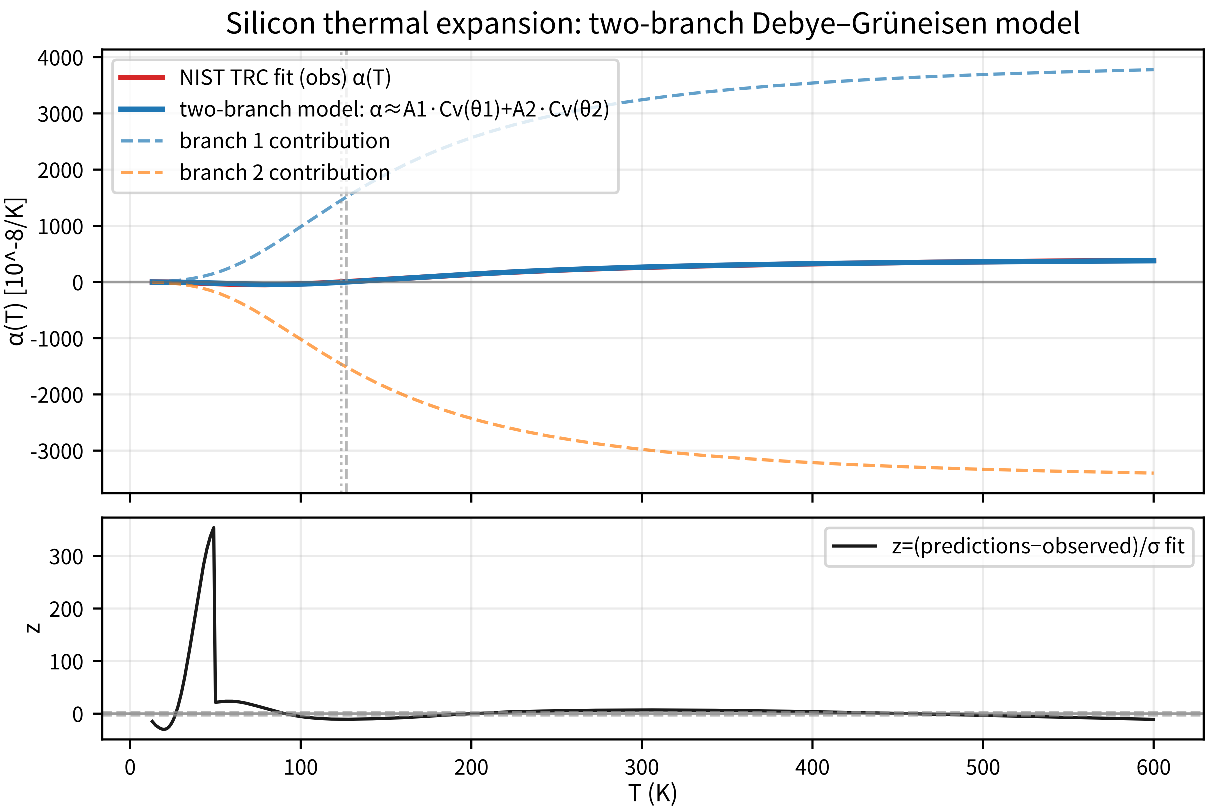

Figure 64: Shows that, even after separating the dashed branch contribution in a two-branch model, the mismatch remains too large on the σ_{fit} scale.

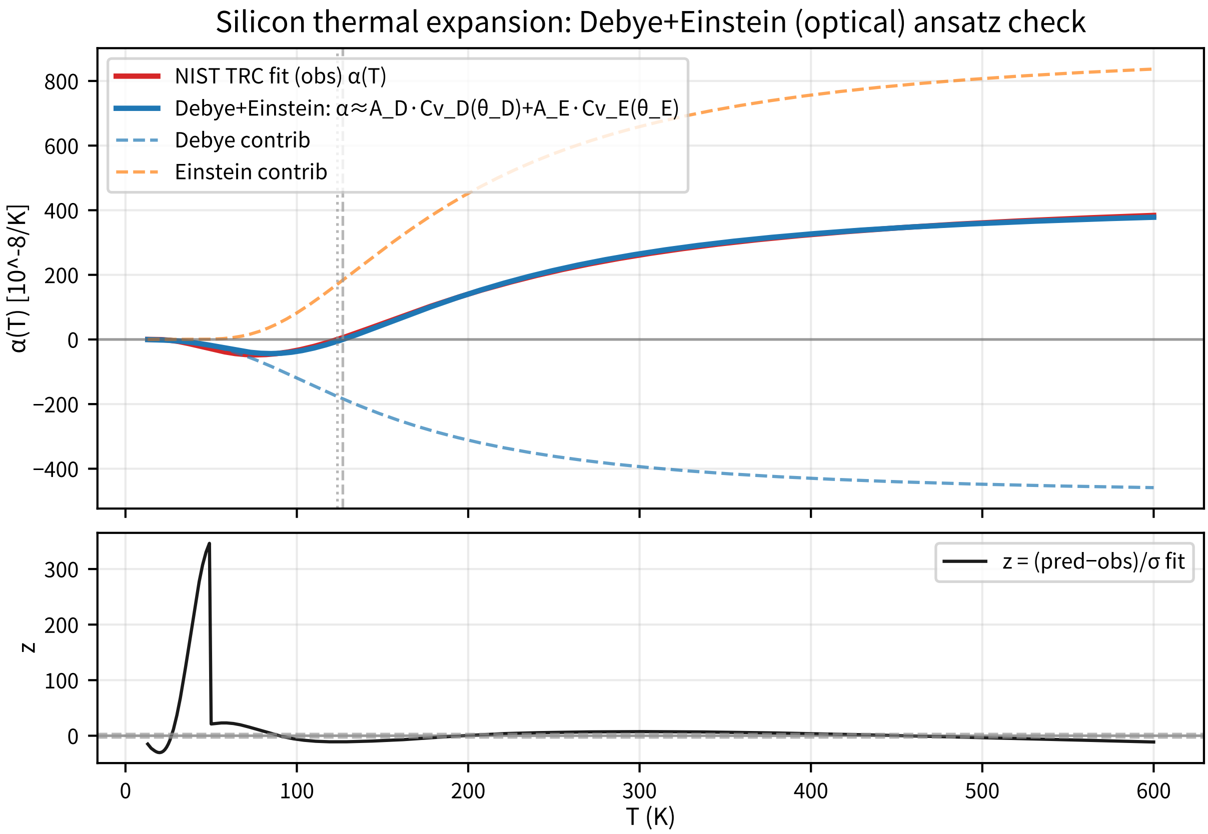

Figure 65: Shows that even a one-branch Debye + Einstein extension does not reach agreement by merely adding the optical branch.

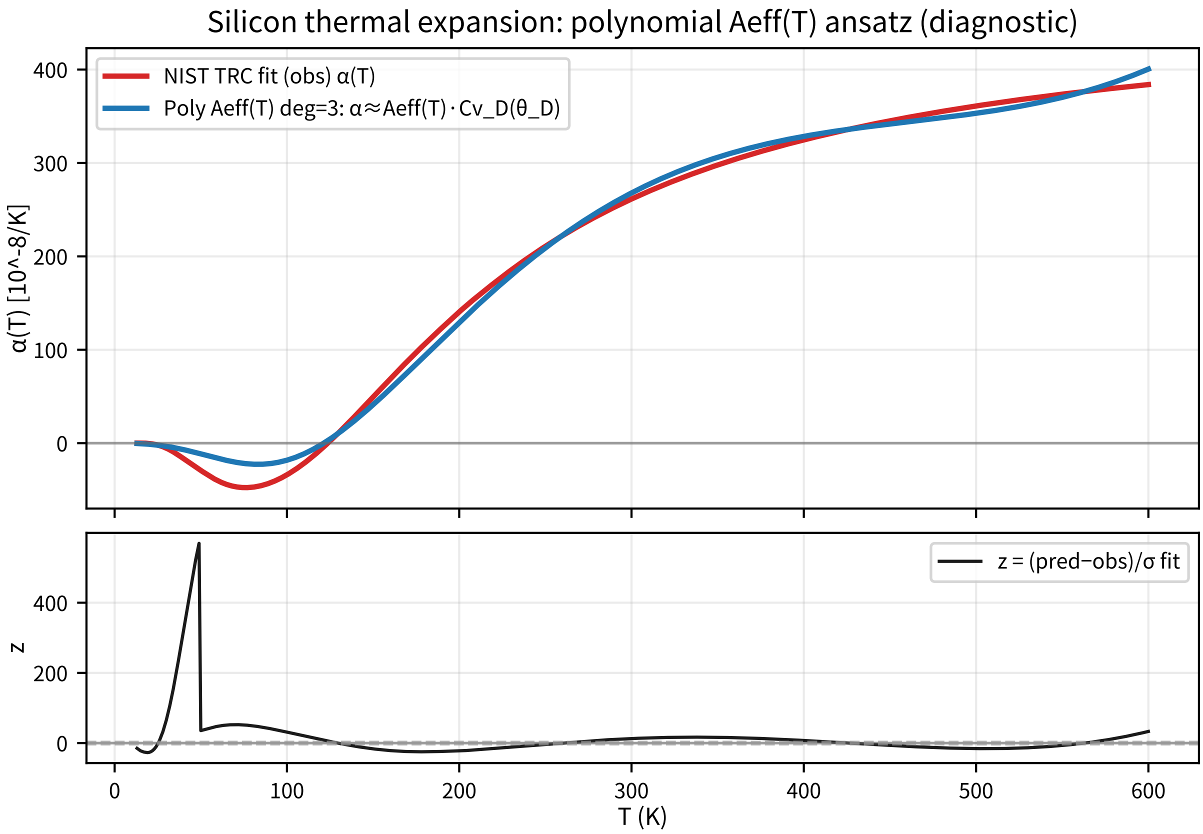

Figure 66: Shows that even with a polynomial $A_{\text{eff}}(T)$, the model does not reach the σ_{fit} scale within the framework of a single-Debye C_v.

8.2 Separating Heat-Capacity Approximations and the Bulk Modulus

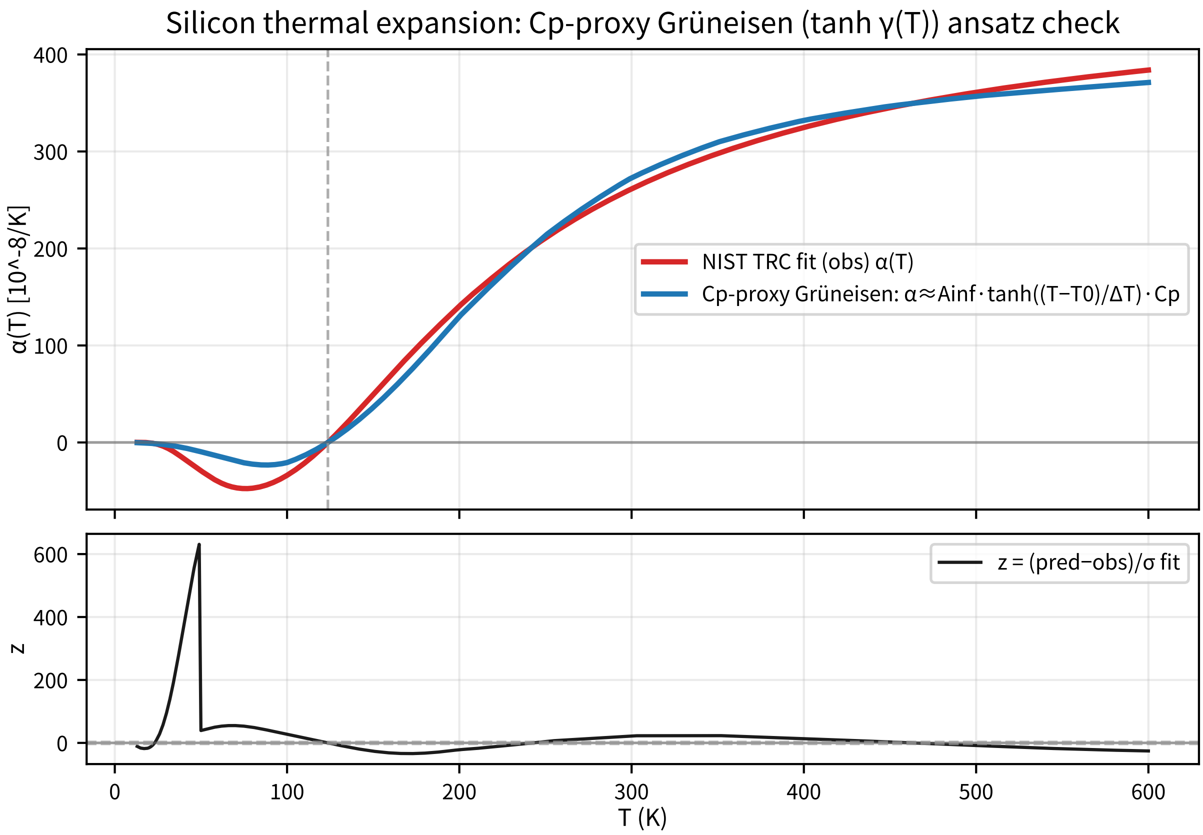

Figure 67: Shows that replacing C_v by a first-order C_p proxy still leaves a shape deficit.

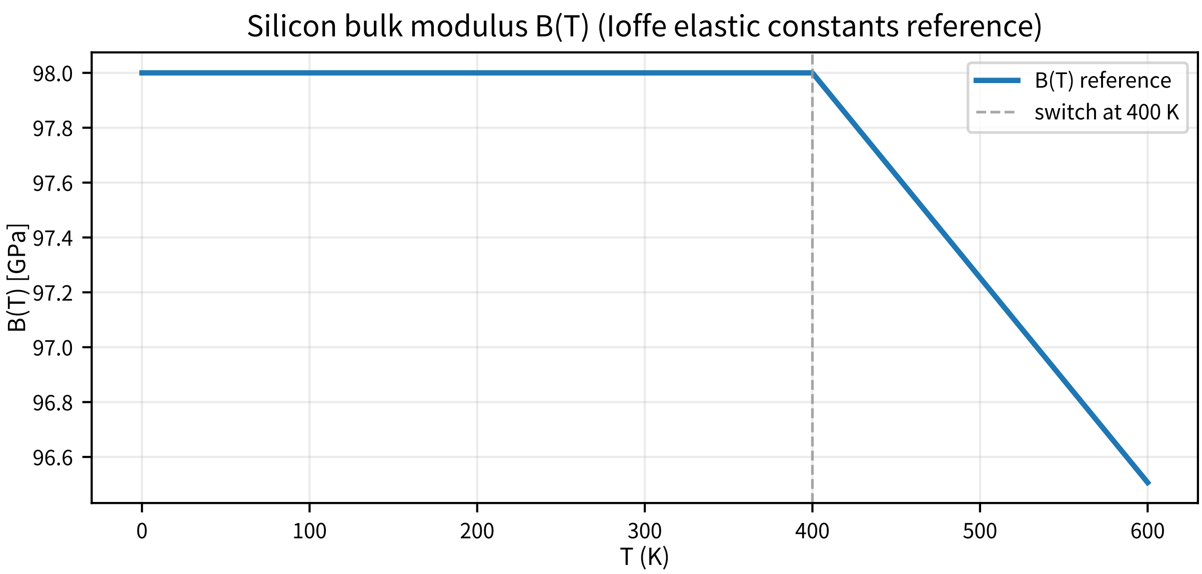

Figure 68: Fixes the $B(T)$ constructed from Ioffe elastic constants as an independent input for isolation tests.

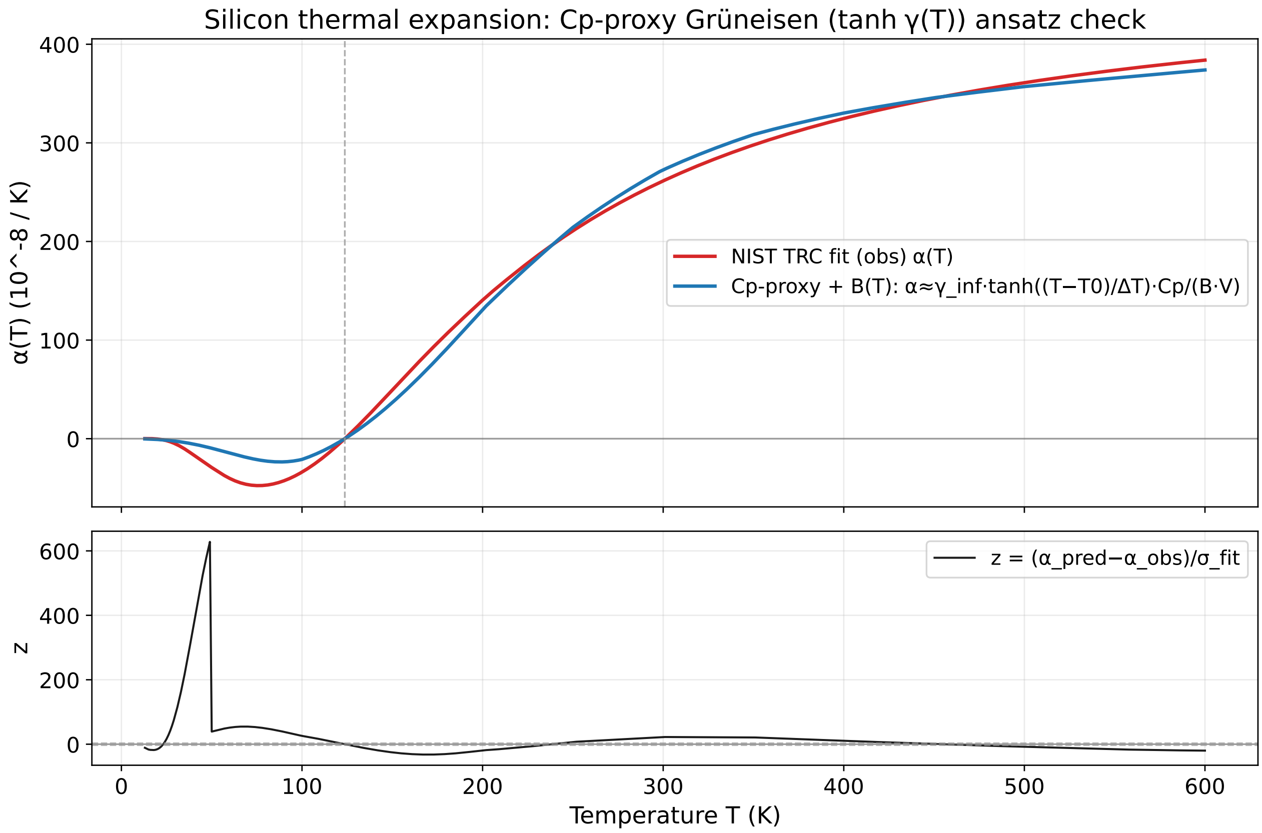

Figure 69: Shows that introducing $B(T)$ does not remove the dominant mismatch.

8.3 Supplementary Figures for Branch-Count Extension and Temperature-Band Splits

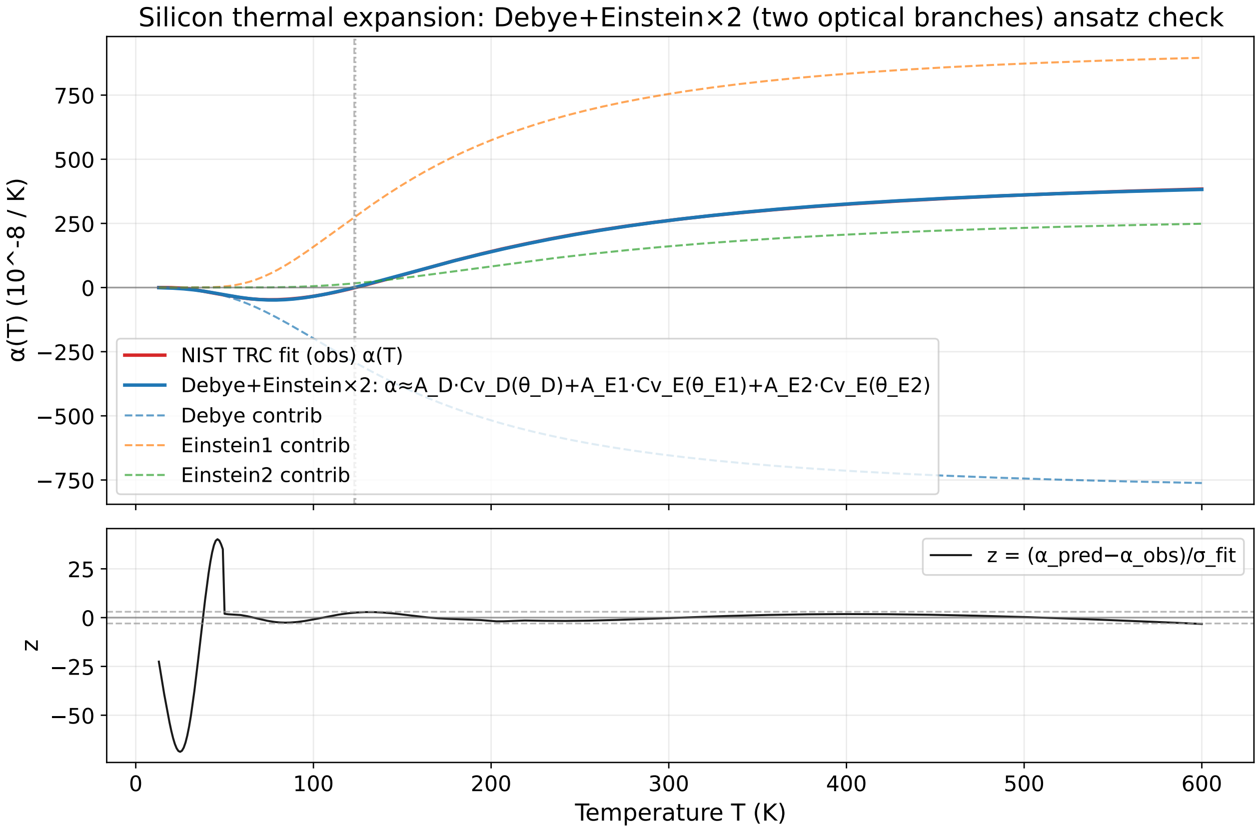

Figure 70: Shows both the improvement gained by extending to two Einstein branches and the fact that the strict target is still missed.

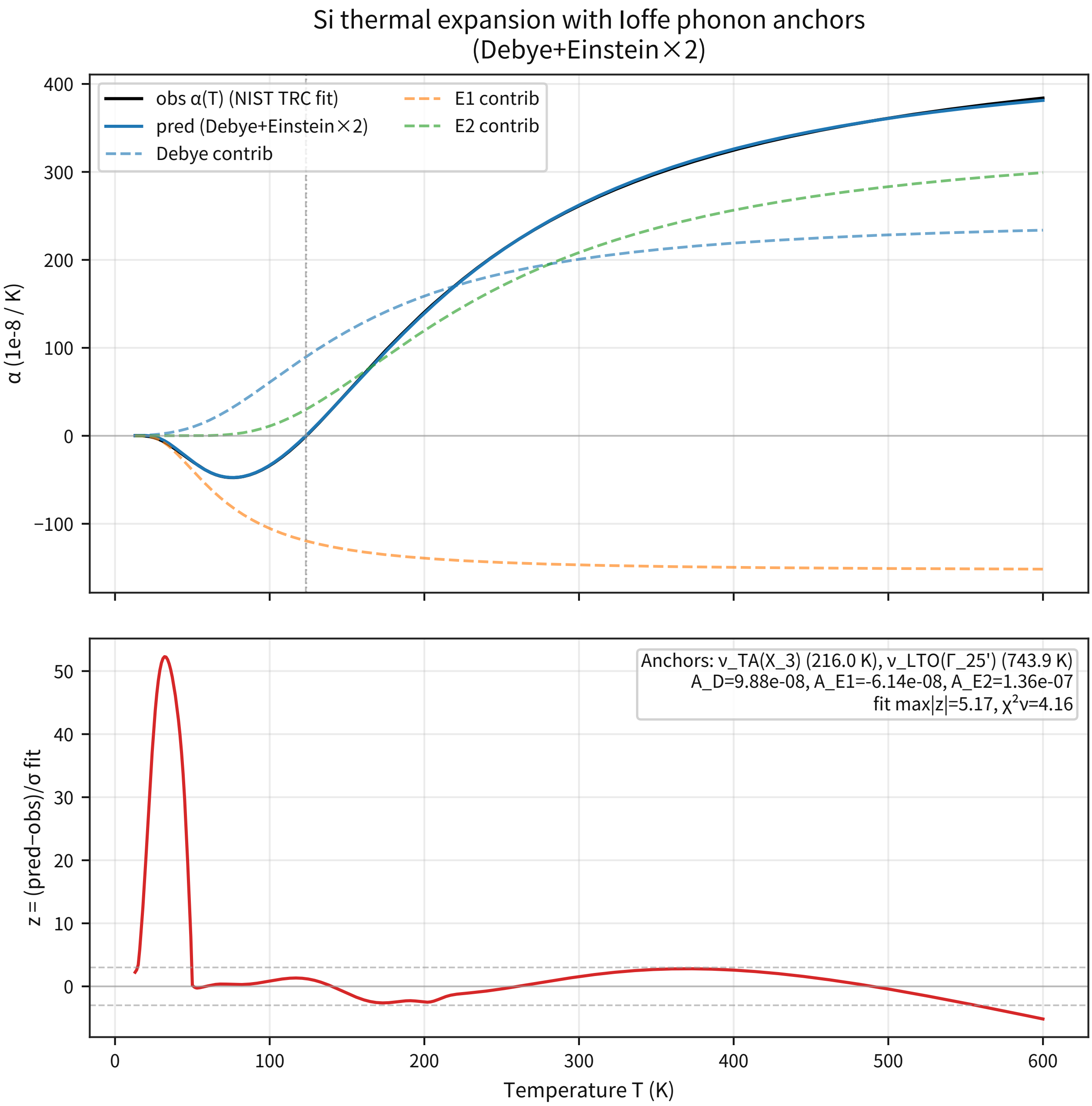

Figure 71: Two-branch fixed figure using discrete phonon-frequency anchors derived from Ioffe.

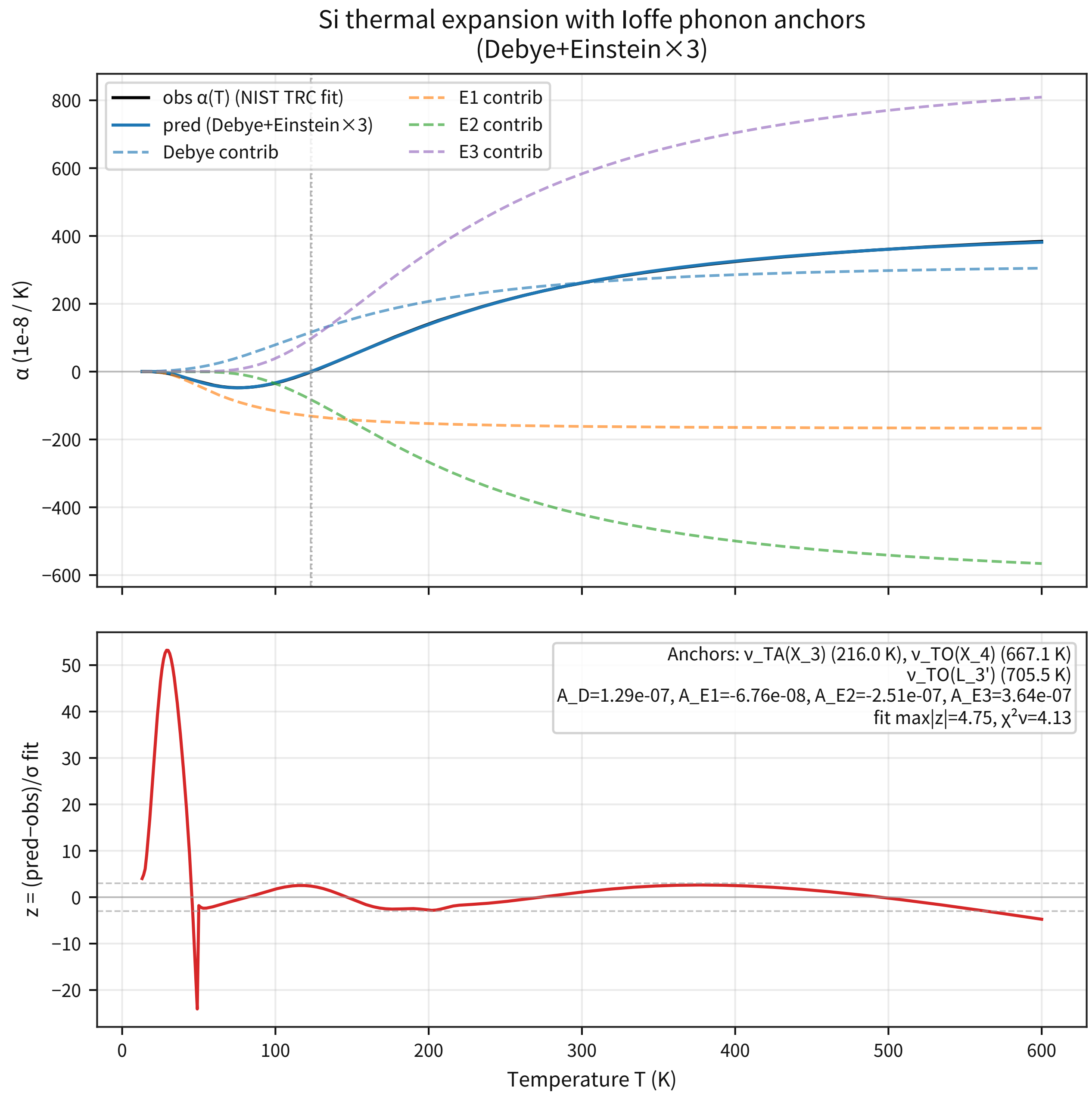

Figure 72: Shows that the temperature-band extrapolation still breaks down even after extending to three Einstein branches.

8.4 Supplementary Figures for DOS Constraints and Softening Approximations

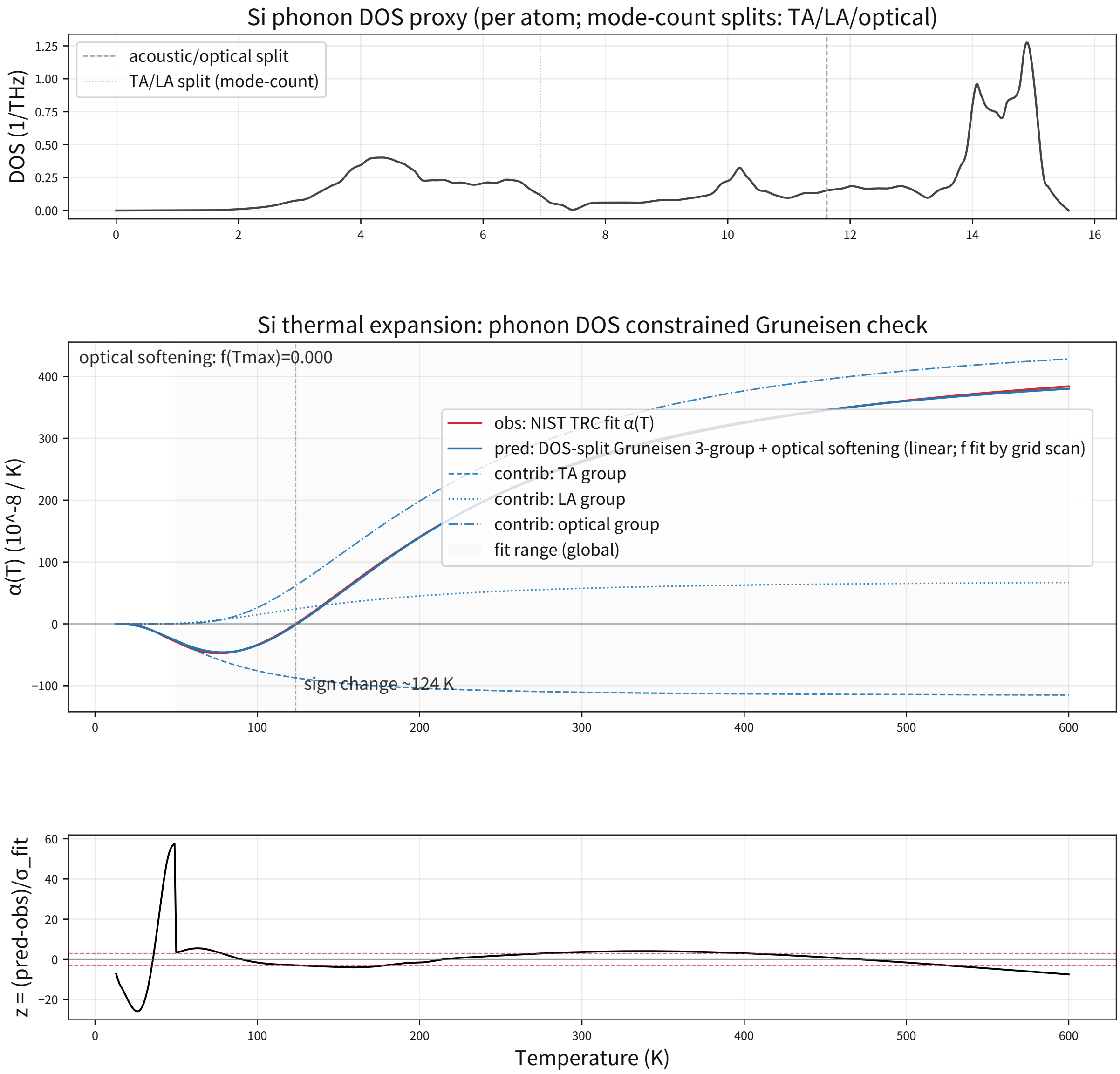

Figure 73: Shows that, even with linear softening added to the DOS-constrained three-group model, the test-side breakdown remains.

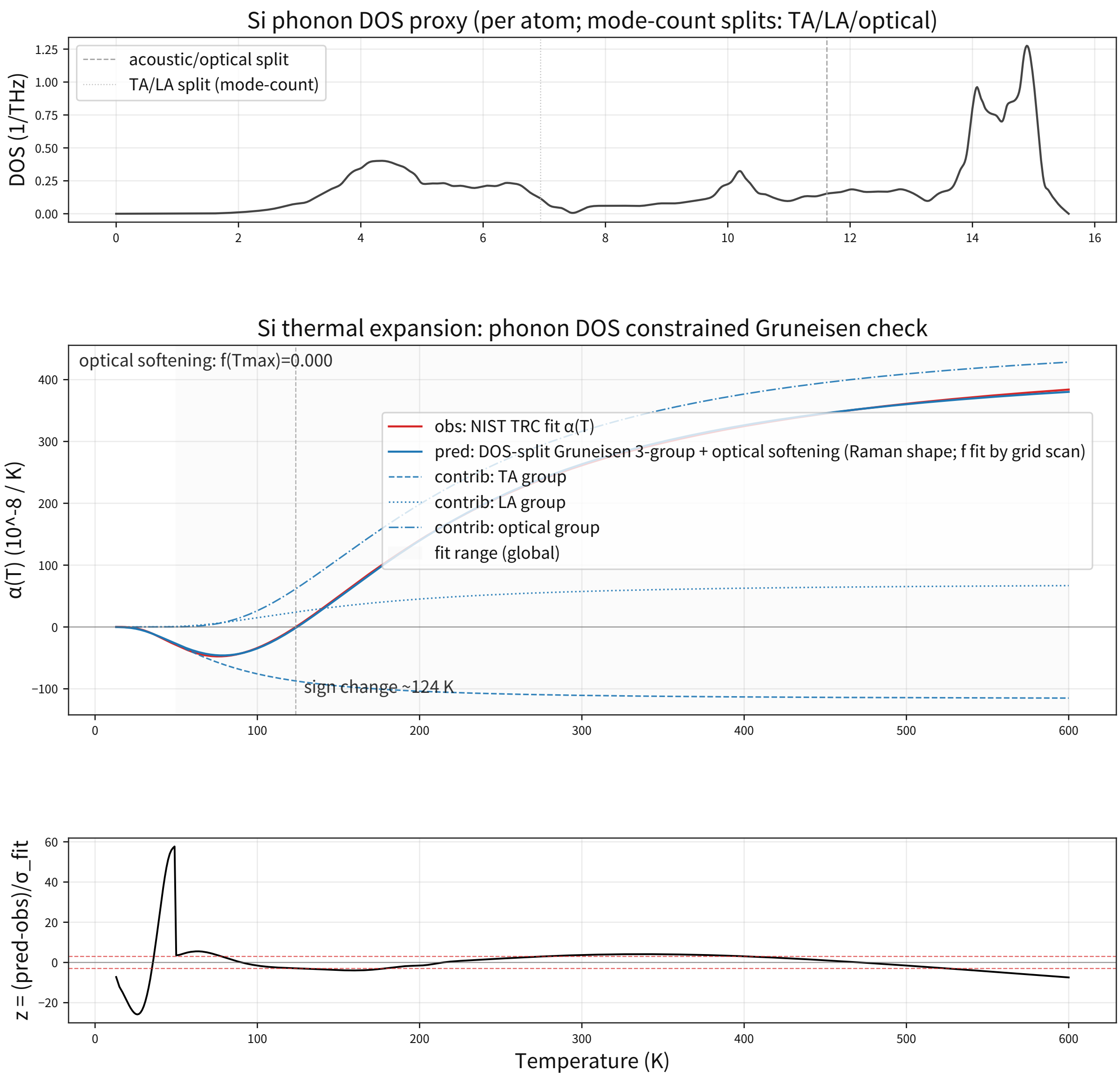

Figure 74: Shows that a Raman-derived softening shape still leaves the high-temperature breakdown.

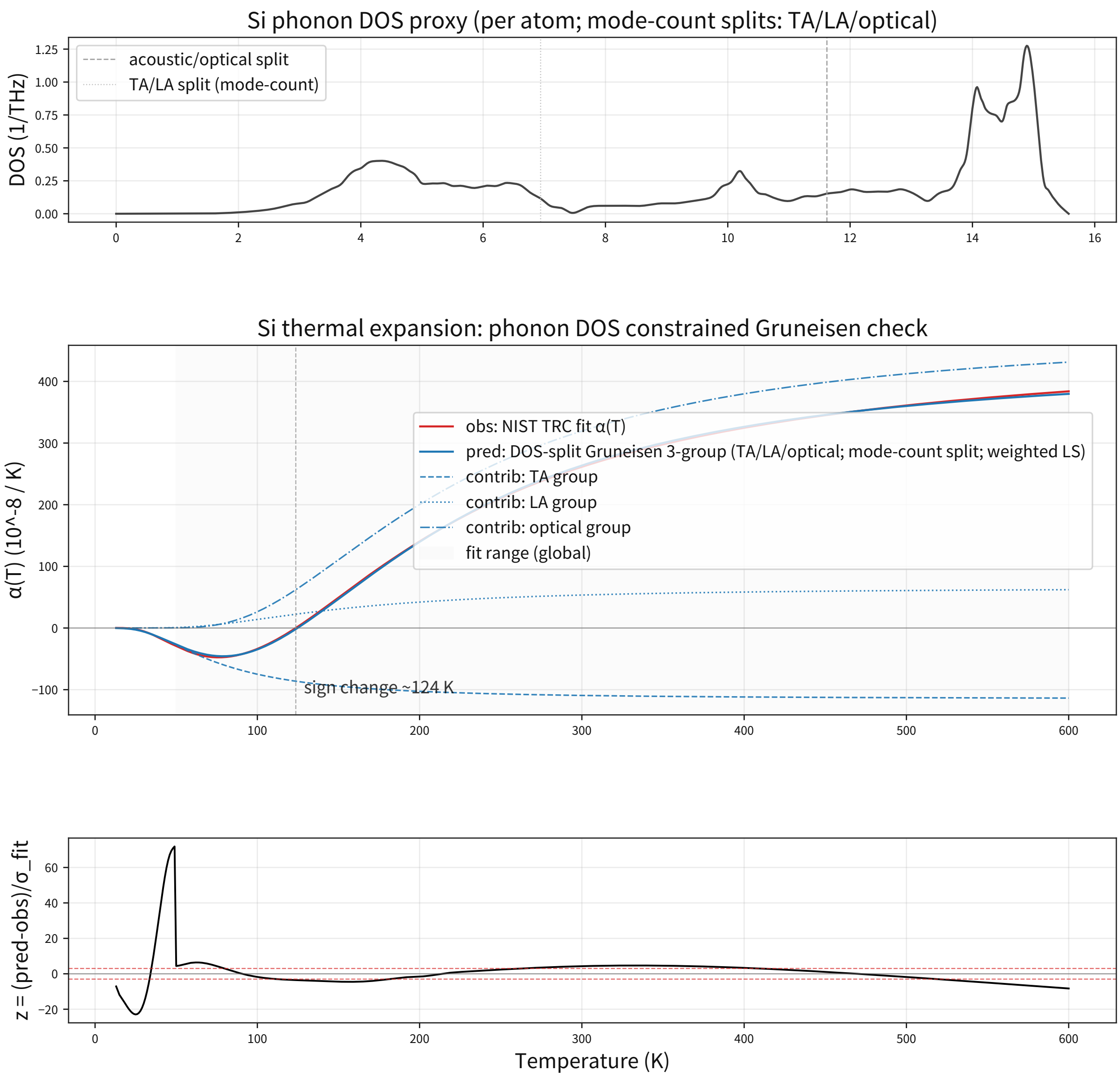

Figure 75: Shows that the Kim2015 global softening proxy alone does not eliminate the holdout breakdown.

9 Appendix B: Data Sources (Primary Sources)

This appendix is the thematic list of the primary sources cited in the main text. Source citations in this appendix are unified through the citation keys used in the manuscript, while the URLs and access dates are authoritative in the Part IV verification ledger. The mapping to the registry keys used for internal management is also kept in the Part IV ledger; the manuscript here freezes only which primary-source group supports which section.

Theme	Representative primary sources	Sections in main text
Bell tests	NIST Bell test data, Delft event-ready data, public primary data from Weihs/Kwiat	4.2, 5.1
Gravity × quantum interference	COW, atom interferometers, primary papers on optical lattice clocks	4.3
Matter-wave interference	Primary papers on the electron double slit and atom-recoil / α consistency	4.4
Optical quantum interference	Single-photon interference, HOM, squeezed light	4.6
Vacuum / precision QED	Casimir, Lamb shift, H 1S-2S, and independent α measurements	4.7
Nuclear physics (binding / scattering / radii)	Primary ledgers from AME2020, ENSDF, and IAEA NDS	4.8, 5.3, 5.4
Atomic / molecular / condensed-matter properties	NIST ASD / WebBook / JANAF / TRC Cryogenics	4.9, 4.10
Weak interaction / BBN	β-decay data, CKM / PMNS, and light-element abundances	5.5

Bell-test sources: [2][3][4][5][6]

Gravity × quantum-interference sources: [31][32][33]

Matter-wave-interference sources: [34][35][36]

Optical-quantum-interference sources: [45][46][48]

Vacuum / precision-QED sources: [49][50][51][35][36]

Nuclear-physics (binding / scattering / radii) sources: [13][14][15]

Atomic / molecular / condensed-matter-property sources: [16][19][21][22]

Weak-interaction / BBN sources: [52][53][54]

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[2] NIST belltestdata (public time-tag data for Shalm et al. 2015). Bucket: https://s3.amazonaws.com/nist-belltestdata/ (prefix: belldata/). Accessed: 2026-01-25.

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